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| D           | <div>Module Parts</div> <div>CPU</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>337S00456</td><td>1</td><td>CPU,CFLU,QMT8,ES,D0,2.3,28W,1.05,BGA1528</td><td>U0500</td><td>CRITICAL</td><td>CPU:2.3G</td></tr><tr><td>337S00456</td><td>1</td><td>CPU,CFLU,QMT8,ES,D0,2.3,28W,1.05,BGA1528</td><td>U0500</td><td>CRITICAL</td><td>CPU:2.5G</td></tr><tr><td>337S00456</td><td>1</td><td>CPU,CFLU,QMT8,ES,D0,2.3,28W,1.05,BGA1528</td><td>U0500</td><td>CRITICAL</td><td>CPU:2.6G</td></tr><tr><td>998-10522</td><td>1</td><td>SOCKET,CPU,CFL-U</td><td>U0500</td><td>CRITICAL</td><td>CPU:SOCKET</td></tr></table> <div>SOC</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>339S00370</td><td>1</td><td>POP,SOC,GIBRALTAR+1GB 20NM,M,B0,CSP1406</td><td>U3900</td><td>CRITICAL</td><td>SOC:B0_1G</td></tr><tr><td>339S00372</td><td>1</td><td>POP,SOC,GIBRALTAR+2GB 20NM,M,B0,CSP1406</td><td>U3900</td><td>CRITICAL</td><td>SOC:B0_2G</td></tr></table> <div>SOC Alternate Parts</div> <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>339S00371</td><td>339S00370</td><td>SOC:B0_1G</td><td>ALL</td><td>Hynix 1GB SCK</td></tr><tr><td>339S00375</td><td>339S00370</td><td>SOC:B0_1G</td><td>ALL</td><td>Micron 1GB ATK</td></tr><tr><td>339S00376</td><td>339S00370</td><td>SOC:B0_1G</td><td>ALL</td><td>Hynix 1GB ATK</td></tr><tr><td>339S00373</td><td>339S00372</td><td>SOC:B0_2G</td><td>ALL</td><td>Hynix 2GB SCK</td></tr><tr><td>339S00377</td><td>339S00372</td><td>SOC:B0_2G</td><td>ALL</td><td>Micron 2GB ATK</td></tr><tr><td>339S00378</td><td>339S00372</td><td>SOC:B0_2G</td><td>ALL</td><td>Hynix 2GB ATK</td></tr></table> <div>SE</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>998-11132</td><td>1</td><td>IC,RTM4,ES2.1,PN80V,DEV,UFLGA64</td><td>U5000</td><td>CRITICAL</td><td>SE:DEV_2017</td></tr><tr><td>338S00253</td><td>1</td><td>IC,RTM4,ES2.1,PN80V,PRD,UFLGA64</td><td>U5000</td><td>CRITICAL</td><td>SE:PROD_2017</td></tr></table> <div>TBT</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>338S00356</td><td>2</td><td>IC,TBT,TITAN RIDGE DP,QT22,ES2,B0,CSP337</td><td>U2800,UB000</td><td>CRITICAL</td><td>TBT_TR:B0</td></tr><tr><td>338S00399</td><td>2</td><td>IC,TBT,TITAN RIDGE DP,QT22,ES2,C0,CSP337</td><td>U2800,UB000</td><td>CRITICAL</td><td>TBT_TR:C0</td></tr><tr><td>338S00408</td><td>2</td><td>IC,TBT,TITAN RIDGE DP,QUJK,QS,C1,CSP337</td><td>U2800,UB000</td><td>CRITICAL</td><td>TBT_TR:C1</td></tr></table> <div>ACE</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>353S00961</td><td>4</td><td>IC,CD3215,ACE,C00,USB FWR SW,BLANK,BGA96</td><td>U3100,U3200,UB300,UB400</td><td>CRITICAL</td><td>ACE:C0_BGA</td></tr><tr><td>353S01442</td><td>4</td><td>IC,CD3215,ACE,C00,USB FWRSW,BLANK,NFBGA96</td><td>U3100,U3200,UB300,UB400</td><td>CRITICAL</td><td>ACE:C0_NFBGA</td></tr></table> <div>WIRELESS</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>339S00458</td><td>1</td><td>IC,MODULE,HARPOON,ES7.7,LGA385</td><td>U3730</td><td>CRITICAL</td><td></td></tr></table> <div>PMU</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>338S00267</td><td>1</td><td>IC,PMU-A,D2449A0C,CALPE,CSP324,4X28X7.32</td><td>U7800</td><td>CRITICAL</td><td></td></tr></table> <div>CHARGER IC</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>353S01525</td><td>1</td><td>IC,ISL92440HIB02,PMU,SUONA,WCPSP40,2.1X3.3</td><td>U7000</td><td>CRITICAL</td><td></td></tr></table> <div>AMR</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>677-10581</td><td>2</td><td>SUBASSY (T&amp;R) PCBA,AMR INTERPOSER,X941</td><td>J4800, J4801</td><td>CRITICAL</td><td></td></tr></table> <div>DRAM</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>333S00125</td><td>4</td><td>IC,SORAM,LPODR3-2133,16GBIT,21NM,H,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:HY_8G</td></tr><tr><td>333S00069</td><td>4</td><td>IC,SORAM,LPODR3-2133,16GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:MI_8G</td></tr><tr><td>333S00166</td><td>4</td><td>IC,SORAM,LPODR3-2133,16GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:SS_8G</td></tr><tr><td>333S00126</td><td>4</td><td>IC,SORAM,LPODR3-2133,32GBIT,21NM,H,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:HY_16G</td></tr><tr><td>333S00070</td><td>4</td><td>IC,SORAM,LPODR3-2133,32GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:MI_16G</td></tr><tr><td>333S00167</td><td>4</td><td>IC,SORAM,LPODR3-2133,32GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:SS_16G</td></tr></table> |                                             |                         |                    |                  |   |   |   |  | PART NUMBER | QTY                       | DESCRIPTION | REFERENCE DES | CRITICAL  | BOM OPTION | 337S00456 | 1            | CPU,CFLU,QMT8,ES,D0,2.3,28W,1.05,BGA1528 | U0500                   | CRITICAL  | CPU:2.3G     | 337S00456    | 1   | CPU,CFLU,QMT8,ES,D0,2.3,28W,1.05,BGA1528 | U0500                   | CRITICAL  | CPU:2.5G     | 337S00456 | 1                  | CPU,CFLU,QMT8,ES,D0,2.3,28W,1.05,BGA1528 | U0500                   | CRITICAL     | CPU:2.6G     | 998-10522          | 1         | SOCKET,CPU,CFL-U                        | U0500                   | CRITICAL | CPU:SOCKET       | PART NUMBER | QTY       | DESCRIPTION                              | REFERENCE DES           | CRITICAL         | BOM OPTION | 339S00370 | 1            | POP,SOC,GIBRALTAR+1GB 20NM,M,B0,CSP1406 | U3900                   | CRITICAL  | SOC:B0_1G  | 339S00372    | 1   | POP,SOC,GIBRALTAR+2GB 20NM,M,B0,CSP1406 | U3900                   | CRITICAL  | SOC:B0_2G  | PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION                           | REF DES                 | COMMENTS:       | 339S00371  | 339S00370      | SOC:B0_1G | ALL                                  | Hynix 1GB SCK           | 339S00375 | 339S00370        | SOC:B0_1G | ALL       | Micron 1GB ATK                          | 339S00376               | 339S00370        | SOC:B0_1G        | ALL       | Hynix 1GB ATK | 339S00373                              | 339S00372               | SOC:B0_2G | ALL              | Hynix 2GB SCK | 339S00377 | 339S00372   | SOC:B0_2G     | ALL      | Micron 2GB ATK | 339S00378 | 339S00372 | SOC:B0_2G                             | ALL         | Hynix 2GB ATK | PART NUMBER   | QTY       | DESCRIPTION | REFERENCE DES               | CRITICAL  | BOM OPTION | 998-11132      | 1         | IC,RTM4,ES2.1,PN80V,DEV,UFLGA64 | U5000                       | CRITICAL  | SE:DEV_2017 | 338S00253     | 1         | IC,RTM4,ES2.1,PN80V,PRD,UFLGA64 | U5000                   | CRITICAL | SE:PROD_2017 | PART NUMBER    | QTY       | DESCRIPTION | REFERENCE DES           | CRITICAL | BOM OPTION | 338S00356      | 2         | IC,TBT,TITAN RIDGE DP,QT22,ES2,B0,CSP337 | U2800,UB000                  | CRITICAL | TBT_TR:B0 | 338S00399     | 2         | IC,TBT,TITAN RIDGE DP,QT22,ES2,C0,CSP337 | U2800,UB000                  | CRITICAL | TBT_TR:C0 | 338S00408     | 2         | IC,TBT,TITAN RIDGE DP,QUJK,QS,C1,CSP337 | U2800,UB000              | CRITICAL | TBT_TR:C1 | PART NUMBER    | QTY       | DESCRIPTION | REFERENCE DES            | CRITICAL | BOM OPTION | 353S00961      | 4           | IC,CD3215,ACE,C00,USB FWR SW,BLANK,BGA96 | U3100,U3200,UB300,UB400 | CRITICAL      | ACE:C0_BGA | 353S01442  | 4         | IC,CD3215,ACE,C00,USB FWRSW,BLANK,NFBGA96 | U3100,U3200,UB300,UB400             | CRITICAL | ACE:C0_NFBGA | PART NUMBER  | QTY      | DESCRIPTION | REFERENCE DES                               | CRITICAL | BOM OPTION | 339S00458 | 1        | IC,MODULE,HARPOON,ES7.7,LGA385 | U3730                                | CRITICAL |          | PART NUMBER | QTY       | DESCRIPTION | REFERENCE DES                    | CRITICAL | BOM OPTION | 338S00267  | 1           | IC,PMU-A,D2449A0C,CALPE,CSP324,4X28X7.32 | U7800       | CRITICAL      |          | PART NUMBER | QTY       | DESCRIPTION | REFERENCE DES                     | CRITICAL | BOM OPTION | 353S01525      | 1         | IC,ISL92440HIB02,PMU,SUONA,WCPSP40,2.1X3.3 | U7000                      | CRITICAL |          | PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION | 677-10581 | 2 | SUBASSY (T&R) PCBA,AMR INTERPOSER,X941 | J4800, J4801 | CRITICAL |  | PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION | 333S00125 | 4 | IC,SORAM,LPODR3-2133,16GBIT,21NM,H,BGA178 | U2300,U2400,U2500,U2600 | CRITICAL | DRAM:HY_8G | 333S00069 | 4 | IC,SORAM,LPODR3-2133,16GBIT,20NM,BGA178 | U2300,U2400,U2500,U2600 | CRITICAL | DRAM:MI_8G | 333S00166 | 4 | IC,SORAM,LPODR3-2133,16GBIT,20NM,BGA178 | U2300,U2400,U2500,U2600 | CRITICAL | DRAM:SS_8G | 333S00126 | 4 | IC,SORAM,LPODR3-2133,32GBIT,21NM,H,BGA178 | U2300,U2400,U2500,U2600 | CRITICAL | DRAM:HY_16G | 333S00070 | 4 | IC,SORAM,LPODR3-2133,32GBIT,20NM,BGA178 | U2300,U2400,U2500,U2600 | CRITICAL | DRAM:MI_16G | 333S00167 | 4 | IC,SORAM,LPODR3-2133,32GBIT,20NM,BGA178 | U2300,U2400,U2500,U2600 | CRITICAL | DRAM:SS_16G |  |  |  |  |  |  |  |  |  |
| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DESCRIPTION                                 | REFERENCE DES           | CRITICAL           | BOM OPTION       |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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|  |  |
| 337S00456   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPU,CFLU,QMT8,ES,D0,2.3,28W,1.05,BGA1528    | U0500                   | CRITICAL           | CPU:2.3G         |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |               |                                        |                         |           |                  |               |           |             |               |          |                |           |           |                                       |             |               |               |           |             |                             |           |            |                |           |                                 |                             |           |             |               |           |                                 |                         |          |              |                |           |             |                         |          |            |                |           |                                          |                              |          |           |               |           |                                          |                              |          |           |               |           |                                         |                          |          |           |                |           |             |                          |          |            |                |             |                                          |                         |               |            |            |           |                                           |                                     |          |              |              |          |             |                                             |          |            |           |          |                                |                                      |          |          |             |           |             |                                  |          |            |            |             |                                          |             |               |          |             |           |             |                                   |          |            |                |           |                                            |                            |          |          |             |     |             |               |          |            |           |   |                                        |              |          |  |             |     |             |               |          |            |           |   |                                           |                         |          |            |           |   |                                         |                         |          |            |           |   |                                         |                         |          |            |           |   |                                           |                         |          |             |           |   |                                         |                         |          |             |           |   |                                         |                         |          |             |  |  |  |  |  |  |  |  |  |
| 337S00456   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPU,CFLU,QMT8,ES,D0,2.3,28W,1.05,BGA1528    | U0500                   | CRITICAL           | CPU:2.5G         |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 337S00456   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 998-10522   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| 339S00370   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 339S00372   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| PART NUMBER | ALTERNATE FOR PART NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| 339S00371   | 339S00370                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 339S00375   | 339S00370                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| 339S00376   | 339S00370                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 339S00373   | 339S00372                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 339S00377   | 339S00372                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 339S00378   | 339S00372                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| 998-11132   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 338S00253   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 338S00356   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 338S00399   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 338S00408   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 353S00961   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 353S01442   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 339S00458   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 338S00267   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | IC,PMU-A,D2449A0C,CALPE,CSP324,4X28X7.32    | U7800                   | CRITICAL           |                  |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 353S01525   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | IC,ISL92440HIB02,PMU,SUONA,WCPSP40,2.1X3.3  | U7000                   | CRITICAL           |                  |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 677-10581   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| 333S00125   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 333S00069   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 333S00166   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 333S00126   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 333S00070   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 333S00167   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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|             | <div>NAND Parts</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00324</td><td>4</td><td>NAND,3DV3,85GBP,XXX,S4E,170G,SD,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:SD_256G</td></tr><tr><td>335S00329</td><td>4</td><td>NAND,3DV3,85GBP,XXX,S4E,170G,T,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:TO_256G</td></tr><tr><td>335S00325</td><td>4</td><td>NAND,3DV3,128GBP,XXX,S4E,170G,SD,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:SD_512G</td></tr><tr><td>335S00330</td><td>4</td><td>NAND,3DV3,128GBP,XXX,S4E,170G,T,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:TO_512G</td></tr><tr><td>335S00327</td><td>4</td><td>NAND,3DV3,256GBP,XXX,S4E,170G,SD,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:SD_1T</td></tr><tr><td>335S00321</td><td>4</td><td>NAND,3DV4,256GBM,S4E,256G,SS,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:SS_1T</td></tr><tr><td>335S00332</td><td>4</td><td>NAND,3DV3,256GBP,XXX,S4E,170G,T,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:TO_1T</td></tr><tr><td>335S00322</td><td>4</td><td>NAND,3DV4,512GBM,S4E,256G,SS,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:SS_2T</td></tr><tr><td>335S00244</td><td>4</td><td>NAND,3DV3,512GBM,S4E,256G,SS,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:SS_2T_DOE</td></tr><tr><td>335S00302</td><td>4</td><td>NAND,3DV3,85GBP,XXX,S4E,170G,SD,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:SD_256G_DOE</td></tr><tr><td>998-11757</td><td>4</td><td>NAND,3DV3,85GBP,XXX,S4E,170G,T,ULGA110</td><td>U8600,U8700,U8800,U8900</td><td>CRITICAL</td><td>NAND:TO_256G_DOE</td></tr></table> <div>Programmables (All Builds)</div> <div>TBT ROMs</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00133</td><td>2</td><td>IC,SPI SERIAL FLASH,8MBITS,3.0V,USON8</td><td>U2890,UB090</td><td>CRITICAL</td><td>TBT_ROM:BLANK</td></tr><tr><td>341S00931</td><td>1</td><td>T29,TR1 (V1.1) PROTO-0,X941</td><td>U2890</td><td>CRITICAL</td><td>TBT_LT_ROM:P0</td></tr><tr><td>341S00932</td><td>1</td><td>T29,TR2 (V1.1) PROTO-1,X941</td><td>UB090</td><td>CRITICAL</td><td>TBT_RT_ROM:P0</td></tr><tr><td>341S00959</td><td>1</td><td>T29,TR1 (V5.1) DCB,X941</td><td>U2890</td><td>CRITICAL</td><td>TBT_LT_ROM:DCB</td></tr><tr><td>341S00960</td><td>1</td><td>T29,TR2 (V5.1) DCB,X941</td><td>UB090</td><td>CRITICAL</td><td>TBT_RT_ROM:DCB</td></tr><tr><td>341S00967</td><td>1</td><td>T29,TR1 (V6.52) PROTO-1,X941</td><td>U2890</td><td>CRITICAL</td><td>TBT_LT_ROM:P1</td></tr><tr><td>341S00968</td><td>1</td><td>T29,TR2 (V6.52) PROTO-1,X941</td><td>UB090</td><td>CRITICAL</td><td>TBT_RT_ROM:P1</td></tr><tr><td>341S01028</td><td>1</td><td>T29,TR1 (VXXXX) EVT,X941</td><td>U2890</td><td>CRITICAL</td><td>TBT_LT_ROM:EVT</td></tr><tr><td>341S01029</td><td>1</td><td>T29,TR2 (VXXXX) EVT,X941</td><td>UB090</td><td>CRITICAL</td><td>TBT_RT_ROM:EVT</td></tr></table> <div>BT ROM</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00256</td><td>1</td><td>IC,SPI SERIAL FLASH,2MBIT,1.8V,DFN8</td><td>U3750</td><td>CRITICAL</td><td>BT_ROM:BLANK</td></tr><tr><td>341S0965</td><td>1</td><td>IC,BT SFLASH ROM (V10.53.122) PROTO-0,X1190</td><td>U3750</td><td>CRITICAL</td><td>BT_ROM:P0</td></tr><tr><td>341S0965</td><td>1</td><td>IC,BT SFLASH ROM (V21) PROTO-1,X1190</td><td>U3750</td><td>CRITICAL</td><td>BT_ROM:P1</td></tr><tr><td>341S01025</td><td>1</td><td>IC,BT SFLASH ROM (VXX) EVT,X1190</td><td>U3750</td><td>CRITICAL</td><td>BT_ROM:EVT</td></tr></table> <div>WIFI ROM</div> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00214</td><td>1</td><td>IC,EEPROM,SER,UWIRE,16K,1.8V,DFN8</td><td>U3710</td><td>CRITICAL</td><td>WIFI_ROM:BLANK</td></tr><tr><td>341S00725</td><td>1</td><td>IC,WIFI ROM (V01) WW1,X665</td><td>U3710</td><td>CRITICAL</td><td>WIFI_ROM:P0</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                             |                         |                    |                  |   |   |   |  | PART NUMBER | QTY                       | DESCRIPTION | REFERENCE DES | CRITICAL  | BOM OPTION | 335S00324 | 4            | NAND,3DV3,85GBP,XXX,S4E,170G,SD,ULGA110  | U8600,U8700,U8800,U8900 | CRITICAL  | NAND:SD_256G | 335S00329    | 4   | NAND,3DV3,85GBP,XXX,S4E,170G,T,ULGA110   | U8600,U8700,U8800,U8900 | CRITICAL  | NAND:TO_256G | 335S00325 | 4                  | NAND,3DV3,128GBP,XXX,S4E,170G,SD,ULGA110 | U8600,U8700,U8800,U8900 | CRITICAL     | NAND:SD_512G | 335S00330          | 4         | NAND,3DV3,128GBP,XXX,S4E,170G,T,ULGA110 | U8600,U8700,U8800,U8900 | CRITICAL | NAND:TO_512G     | 335S00327   | 4         | NAND,3DV3,256GBP,XXX,S4E,170G,SD,ULGA110 | U8600,U8700,U8800,U8900 | CRITICAL         | NAND:SD_1T | 335S00321 | 4            | NAND,3DV4,256GBM,S4E,256G,SS,ULGA110    | U8600,U8700,U8800,U8900 | CRITICAL  | NAND:SS_1T | 335S00332    | 4   | NAND,3DV3,256GBP,XXX,S4E,170G,T,ULGA110 | U8600,U8700,U8800,U8900 | CRITICAL  | NAND:TO_1T | 335S00322   | 4                         | NAND,3DV4,512GBM,S4E,256G,SS,ULGA110 | U8600,U8700,U8800,U8900 | CRITICAL        | NAND:SS_2T | 335S00244      | 4         | NAND,3DV3,512GBM,S4E,256G,SS,ULGA110 | U8600,U8700,U8800,U8900 | CRITICAL  | NAND:SS_2T_DOE   | 335S00302 | 4         | NAND,3DV3,85GBP,XXX,S4E,170G,SD,ULGA110 | U8600,U8700,U8800,U8900 | CRITICAL         | NAND:SD_256G_DOE | 998-11757 | 4             | NAND,3DV3,85GBP,XXX,S4E,170G,T,ULGA110 | U8600,U8700,U8800,U8900 | CRITICAL  | NAND:TO_256G_DOE | PART NUMBER   | QTY       | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION     | 335S00133 | 2         | IC,SPI SERIAL FLASH,8MBITS,3.0V,USON8 | U2890,UB090 | CRITICAL      | TBT_ROM:BLANK | 341S00931 | 1           | T29,TR1 (V1.1) PROTO-0,X941 | U2890     | CRITICAL   | TBT_LT_ROM:P0  | 341S00932 | 1                               | T29,TR2 (V1.1) PROTO-1,X941 | UB090     | CRITICAL    | TBT_RT_ROM:P0 | 341S00959 | 1                               | T29,TR1 (V5.1) DCB,X941 | U2890    | CRITICAL     | TBT_LT_ROM:DCB | 341S00960 | 1           | T29,TR2 (V5.1) DCB,X941 | UB090    | CRITICAL   | TBT_RT_ROM:DCB | 341S00967 | 1                                        | T29,TR1 (V6.52) PROTO-1,X941 | U2890    | CRITICAL  | TBT_LT_ROM:P1 | 341S00968 | 1                                        | T29,TR2 (V6.52) PROTO-1,X941 | UB090    | CRITICAL  | TBT_RT_ROM:P1 | 341S01028 | 1                                       | T29,TR1 (VXXXX) EVT,X941 | U2890    | CRITICAL  | TBT_LT_ROM:EVT | 341S01029 | 1           | T29,TR2 (VXXXX) EVT,X941 | UB090    | CRITICAL   | TBT_RT_ROM:EVT | PART NUMBER | QTY                                      | DESCRIPTION             | REFERENCE DES | CRITICAL   | BOM OPTION | 335S00256 | 1                                         | IC,SPI SERIAL FLASH,2MBIT,1.8V,DFN8 | U3750    | CRITICAL     | BT_ROM:BLANK | 341S0965 | 1           | IC,BT SFLASH ROM (V10.53.122) PROTO-0,X1190 | U3750    | CRITICAL   | BT_ROM:P0 | 341S0965 | 1                              | IC,BT SFLASH ROM (V21) PROTO-1,X1190 | U3750    | CRITICAL | BT_ROM:P1   | 341S01025 | 1           | IC,BT SFLASH ROM (VXX) EVT,X1190 | U3750    | CRITICAL   | BT_ROM:EVT | PART NUMBER | QTY                                      | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION  | 335S00214 | 1           | IC,EEPROM,SER,UWIRE,16K,1.8V,DFN8 | U3710    | CRITICAL   | WIFI_ROM:BLANK | 341S00725 | 1                                          | IC,WIFI ROM (V01) WW1,X665 | U3710    | CRITICAL | WIFI_ROM:P0 |     |             |               |          |            |           |   |                                        |              |          |  |             |     |             |               |          |            |           |   |                                           |                         |          |            |           |   |                                         |                         |          |            |           |   |                                         |                         |          |            |           |   |                                           |                         |          |             |           |   |                                         |                         |          |             |           |   |                                         |                         |          |             |  |  |  |  |  |  |  |  |  |
| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DESCRIPTION                                 | REFERENCE DES           | CRITICAL           | BOM OPTION       |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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|  |  |
| 335S00324   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NAND,3DV3,85GBP,XXX,S4E,170G,SD,ULGA110     | U8600,U8700,U8800,U8900 | CRITICAL           | NAND:SD_256G     |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |               |                                        |                         |           |                  |               |           |             |               |          |                |           |           |                                       |             |               |               |           |             |                             |           |            |                |           |                                 |                             |           |             |               |           |                                 |                         |          |              |                |           |             |                         |          |            |                |           |                                          |                              |          |           |               |           |                                          |                              |          |           |               |           |                                         |                          |          |           |                |           |             |                          |          |            |                |             |                                          |                         |               |            |            |           |                                           |                                     |          |              |              |          |             |                                             |          |            |           |          |                                |                                      |          |          |             |           |             |                                  |          |            |            |             |                                          |             |               |          |             |           |             |                                   |          |            |                |           |                                            |                            |          |          |             |     |             |               |          |            |           |   |                                        |              |          |  |             |     |             |               |          |            |           |   |                                           |                         |          |            |           |   |                                         |                         |          |            |           |   |                                         |                         |          |            |           |   |                                           |                         |          |             |           |   |                                         |                         |          |             |           |   |                                         |                         |          |             |  |  |  |  |  |  |  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| 335S00329   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |               |                                        |                         |           |                  |               |           |             |               |          |                |           |           |                                       |             |               |               |           |             |                             |           |            |                |           |                                 |                             |           |             |               |           |                                 |                         |          |              |                |           |             |                         |          |            |                |           |                                          |                              |          |           |               |           |                                          |                              |          |           |               |           |                 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                                        |                            |          |          |             |     |             |               |          |            |           |   |                                        |              |          |  |             |     |             |               |          |            |           |   |                                           |                         |          |            |           |   |                                         |                         |          |            |           |   |                                         |                         |          |            |           |   |                                           |                         |          |             |           |   |                                         |                         |          |             |           |   |                                         |                         |          |             |  |  |  |  |  |  |  |  |  |
| 335S00325   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |               |                                        |                         |           |                  |               |           |             |               |          |                |           |           |                                       |             |               |               |           |             |                             |           |            |                |           |                                 |                             |           |             |               |           |                                 |                         |          |              |                |           |             |                         |          |            |                |           |                                          |                              |          |           |               |           |                                          |                              |          |           |               |           |                                         |                          |          |           |                |           |             |                          |          |            |                |             |                                          |                         |               |            |            |           |                                           |                                     |          |              |              |          |             |                                             |          |            |           |          |                                |                                      |          |          |             |           |             |                                  |          |            |            |             |                                          |             |               |          |             |           |             |                                   |          |            |                |           |    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|  |  |
| 335S00330   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |               |                                        |                         |           |                  |               |           |             |               |          |                |           |           |                                       |             |               |               |           |             |                             |           |            |                |           |                                 |                             |           |             |               |           |                                 |                         |          |              |                |           |             |                         |          |            |                |           |                                          |                              |          |           |               |           |                                          |                              |          |           |               |           |                 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| 335S00327   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 335S00321   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 335S00332   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 335S00322   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 335S00244   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 335S00302   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 998-11757   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DESCRIPTION                                 | REFERENCE DES           | CRITICAL           | BOM OPTION       |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00133   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | IC,SPI SERIAL FLASH,8MBITS,3.0V,USON8       | U2890,UB090             | CRITICAL           | TBT_ROM:BLANK    |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S00931   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | T29,TR1 (V1.1) PROTO-0,X941                 | U2890                   | CRITICAL           | TBT_LT_ROM:P0    |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S00932   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S00959   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S00960   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S00967   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S00968   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S01028   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S01029   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00256   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | IC,SPI SERIAL FLASH,2MBIT,1.8V,DFN8         | U3750                   | CRITICAL           | BT_ROM:BLANK     |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S0965    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | IC,BT SFLASH ROM (V10.53.122) PROTO-0,X1190 | U3750                   | CRITICAL           | BT_ROM:P0        |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |       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| 341S0965    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 341S01025   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| PART NUMBER | QTY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DESCRIPTION                                 | REFERENCE DES           | CRITICAL           | BOM OPTION       |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00214   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 341S00725   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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|             | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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|             | <div>NAND Alternate Parts</div> <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>335S00302</td><td>335S00324</td><td>NAND:SD_256G</td><td>ALL</td><td>SanDisk 256G Old</td></tr><tr><td>335S00290</td><td>335S00329</td><td>NAND:TO_256G</td><td>ALL</td><td>Toshiba 256G Old</td></tr><tr><td>998-12418</td><td>335S00329</td><td>NAND:TO_256G</td><td>ALL</td><td>Toshiba 256G Sub 1</td></tr><tr><td>998-12419</td><td>335S00329</td><td>NAND:TO_256G</td><td>ALL</td><td>Toshiba 256G Sub 2</td></tr><tr><td>335S00303</td><td>335S00325</td><td>NAND:SD_512G</td><td>ALL</td><td>SanDisk 512G Old</td></tr><tr><td>335S00291</td><td>335S00330</td><td>NAND:TO_512G</td><td>ALL</td><td>Toshiba 512G Old</td></tr><tr><td>998-12420</td><td>335S00330</td><td>NAND:TO_512G</td><td>ALL</td><td>Toshiba 512G Sub 1</td></tr><tr><td>998-12421</td><td>335S00330</td><td>NAND:TO_512G</td><td>ALL</td><td>Toshiba 512G Sub 2</td></tr><tr><td>335S00305</td><td>335S00327</td><td>NAND:SD_1T</td><td>ALL</td><td>SanDisk 1T Old</td></tr><tr><td>335S00293</td><td>335S00332</td><td>NAND:TO_1T</td><td>ALL</td><td>Toshiba 1T Old</td></tr><tr><td>998-12424</td><td>335S00332</td><td>NAND:TO_1T</td><td>ALL</td><td>Toshiba 1T Sub 1</td></tr><tr><td>998-12426</td><td>335S00332</td><td>NAND:TO_1T</td><td>ALL</td><td>Toshiba 1T Sub 2</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                             |                         |                    |                  |   |   |   |  | PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION  | REF DES       | COMMENTS: | 335S00302  | 335S00324 | NAND:SD_256G | ALL                                      | SanDisk 256G Old        | 335S00290 | 335S00329    | NAND:TO_256G | ALL | Toshiba 256G Old                         | 998-12418               | 335S00329 | NAND:TO_256G | ALL       | Toshiba 256G Sub 1 | 998-12419                                | 335S00329               | NAND:TO_256G | ALL          | Toshiba 256G Sub 2 | 335S00303 | 335S00325                               | NAND:SD_512G            | ALL      | SanDisk 512G Old | 335S00291   | 335S00330 | NAND:TO_512G                             | ALL                     | Toshiba 512G Old | 998-12420  | 335S00330 | NAND:TO_512G | ALL                                     | Toshiba 512G Sub 1      | 998-12421 | 335S00330  | NAND:TO_512G | ALL | Toshiba 512G Sub 2                      | 335S00305               | 335S00327 | NAND:SD_1T | ALL         | SanDisk 1T Old            | 335S00293                            | 335S00332               | NAND:TO_1T      | ALL        | Toshiba 1T Old | 998-12424 | 335S00332                            | NAND:TO_1T              | ALL       | Toshiba 1T Sub 1 | 998-12426 | 335S00332 | NAND:TO_1T                              | ALL                     | Toshiba 1T Sub 2 |                  |           |               |                                        |                         |           |                  |               |           |             |               |          |                |           |           |                                       |             |               |               |           |             |                             |           |            |                |           |                                 |                             |           |             |               |           |                                 |                         |          |              |                |           |             |                         |          |            |                |           |                                          |                              |          |           |               |           |                                          |                              |          |           |               |           |       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| PART NUMBER | ALTERNATE FOR PART NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | BOM OPTION                                  | REF DES                 | COMMENTS:          |                  |   |   |   |  |             |                           |             |               |           |            |           |              |                                          |                         |           |              |              |     |                                          |                         |           |              |           |                    |                                          |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00302   | 335S00324                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00290   | 335S00329                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 998-12418   | 335S00329                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 998-12419   | 335S00329                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00303   | 335S00325                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00291   | 335S00330                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 998-12420   | 335S00330                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 998-12421   | 335S00330                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00305   | 335S00327                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 335S00293   | 335S00332                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                                   |                         |              |              |                    |           |                                         |                         |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |            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| 998-12424   | 335S00332                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| 998-12426   | 335S00332                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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|             | <div>Strategic Silicon</div> <table><tr><th>PART#</th><th>STRATEGIC VALUE</th><th>COMMENT</th></tr><tr><td>337S00456</td><td>08</td><td>CPU</td></tr><tr><td>333S00069</td><td>01</td><td>MEMORY</td></tr><tr><td>333S00070</td><td>01</td><td>MEMORY</td></tr><tr><td>333S00125</td><td>01</td><td>MEMORY</td></tr><tr><td>333S00126</td><td>01</td><td>MEMORY</td></tr><tr><td>333S00166</td><td>08</td><td>MEMORY</td></tr><tr><td>333S00167</td><td>08</td><td>MEMORY</td></tr><tr><td>339S00370</td><td>09</td><td>pSOC</td></tr><tr><td>339S00372</td><td>09</td><td>pSOC</td></tr><tr><td>338S00267</td><td>08</td><td>CALPE</td></tr><tr><td>335S00321</td><td>10</td><td>NAND</td></tr><tr><td>335S00322</td><td>10</td><td>NAND</td></tr><tr><td>335S00324</td><td>09</td><td>NAND</td></tr><tr><td>335S00325</td><td>09</td><td>NAND</td></tr><tr><td>335S00327</td><td>09</td><td>NAND</td></tr><tr><td>335S00329</td><td>09</td><td>NAND</td></tr></table> <table><tr><th>PART#</th><th>STRATEGIC 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  | MEMORY                                   | 333S00167               | 08           | MEMORY       | 339S00370          | 09        | pSOC                                    | 339S00372               | 09       | pSOC             | 338S00267   | 08        | CALPE                                    | 335S00321               | 10               | NAND       | 335S00322 | 10           | NAND                                    | 335S00324               | 09        | NAND       | 335S00325    | 09  | NAND                                    | 335S00327               | 09        | NAND       | 335S00329   | 09                        | NAND                                 | PART#                   | STRATEGIC VALUE | COMMENT    | 335S00330      | 09        | NAND                                 | 335S00332               | 09        | NAND             | 338S00410 | 08        | OCARINA                                 | 353S01442               | 05               | ACE              | 338S00408 | 08            | TITAN RIDGE                            | 353S01188               | 09        | DEBUG MUX2       | 353S01525     | 06        | SUONA       | 353S00928     | 06       | INVP8          | 353S00526 | 07        | TPS62180                              | 353S01077   | 02            | ISL95870HRUZ  | 353S01629 | 08          | TAS5770L                    | 335S00253 | 09         | SECURE ELEMENT | 353S4415  | 01                              | BACKLIGHT CONTROLLER        | 353S01320 | 08          | MIPI SWITCH   | 338S1163  | 01                              | ACCELEROMETER           |          |              |                |           |             |                         |          |            |                |           |                                          |                              |          |           |               |           |                                          |                              |          |           |               |           |      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| PART#       | STRATEGIC VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| 337S00456   | 08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 333S00069   | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 333S00070   | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 333S00125   | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 333S00126   | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 333S00166   | 08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 333S00167   | 08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 339S00370   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 339S00372   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 338S00267   | 08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 335S00321   | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 335S00322   | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 335S00324   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 335S00325   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 335S00329   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| PART#       | STRATEGIC VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| 335S00330   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 335S00332   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 338S00410   | 08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S01442   | 05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 338S00408   | 08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S01188   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S01525   | 06                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S00928   | 06                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S00526   | 07                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S01077   | 02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S01629   | 08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 335S00253   | 09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S4415    | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 353S01320   | 08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 338S1163    | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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|             | <div>BOM Configuration</div> <div>NOTICE OF PROPRIETARY PROPERTY:</div> <div>THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:</div> <div>I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE</div> <div>II NOT TO REPRODUCE OR COPY IT</div> <div>III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART</div> <div>IV ALL RIGHTS RESERVED</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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                        |          |                  |             |           |                                          |                         |                  |            |           |              |                                         |                         |           |            |              |     |                                         |                         |           |            |             |                           |                                      |                         |                 |            |                |           |                                      |                         |           |                  |           |           |                                         |                         |                  |                  |           |               |                                        |                         |           |                  |               |           |             |               |          |                |           |           |                                       |             |               |               |           |             |                             |           |            |                |           |                                 |                             |           |             |               |           |                                 |                         |          |              |                |           |             |                         |          |            |                |           |                                          |                              |          |           |               |           |                                          |                              |          |           |               |           |                                         |                          |          |           |                |           |             |                          |          |            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|             | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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|---|---|---|---|---|---|---|---|
| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|---|

## BOM Groups

| BOM GROUP      | BOM OPTIONS                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| X941_COMMON    | SCH,PCB,ALTERNATE,COMMON,X941_COMMON1,X941_COMMON2,X941_COMMON3,X941_PROGPARTS                                                        |
| X941_COMMON1   | TBT_TR:C1,ACE:C0_NFBCA,SE:DEV_2017,PBUS:3S,SYSDBT:FET,TBTTHERM_SNS,BOARD_ID,BOARD_REV:100,RF_DOE                                      |
| X941_COMMON2   | EDP_ENABLE,XDP:YES,SKIP_5V3V3:AUDIBLE                                                                                                 |
| X941_COMMON3   | LOADRC:YES                                                                                                                            |
| X941_PROGPARTS | TBT_LT_ROM:EVT,TBT_RT_ROM:EVT,WIFI_ROM:P0,BT_ROM:EVT                                                                                  |
| X941_DEVEL:ENG | ALTERNATE,ESPI_DBG,DBGLD,SOC_DBG,USBC_DBG,XDP_CONN:YES,WIFI_DBG,SSD_DBG,DBG_FAN,DEBUG_BUTTON,LOADISNS_AP_TEMP:YES,SENSOR:DEV,BOOTCFG0 |
| X941_DEVEL:DVT | ALTERNATE,SOC_DBG,USBC_DBG,XDP_CONN:YES,WIFI_DBG                                                                                      |
| X941_DEVEL:PVT | ALTERNATE                                                                                                                             |

## NAND Configs

| BOM GROUP           | BOM OPTIONS                              |
|---------------------|------------------------------------------|
| NANDCFG:SD_256G     | NAND:SD_256G,SOC:B0_1G,NAND_VCC:2.5V     |
| NANDCFG:TO_256G     | NAND:TO_256G,SOC:B0_1G,NAND_VCC:2.5V     |
| NANDCFG:SD_512G     | NAND:SD_512G,SOC:B0_1G,NAND_VCC:2.5V     |
| NANDCFG:TO_512G     | NAND:TO_512G,SOC:B0_1G,NAND_VCC:2.5V     |
| NANDCFG:SD_1T       | NAND:SD_1T,SOC:B0_2G,NAND_VCC:2.5V       |
| NANDCFG:SS_1T       | NAND:SS_1T,SOC:B0_2G,NAND_VCC:2.5V       |
| NANDCFG:TO_1T       | NAND:TO_1T,SOC:B0_2G,NAND_VCC:2.5V       |
| NANDCFG:SS_2T       | NAND:SS_2T,SOC:B0_2G,NAND_VCC:2.5V       |
| NANDCFG:SS_2T_DOE   | NAND:SS_2T_DOE,SOC:B0_2G,NAND_VCC:2.7V   |
| NANDCFG:SD_256G_DOE | NAND:SD_256G_DOE,SOC:B0_1G,NAND_VCC:2.5V |
| NANDCFG:TO_256G_DOE | NAND:TO_256G_DOE,SOC:B0_1G,NAND_VCC:2.5V |

## Main DRAM SPD Straps

| BOM GROUP      | BOM OPTIONS                                                       |
|----------------|-------------------------------------------------------------------|
| DRAMCFG:HY_8G  | DRAM:HY_8G, RAMCFG4_L, RAMCFG3_L, RAMCFG2_L, RAMCFG1_L, RAMCFG0_L |
| DRAMCFG:MI_8G  | DRAM:MI_8G, RAMCFG4_L, RAMCFG3_L, RAMCFG2_L, RAMCFG1_L            |
| DRAMCFG:SS_8G  | DRAM:SS_8G, RAMCFG3_L, RAMCFG2_L, RAMCFG0_L                       |
| DRAMCFG:HY_16G | DRAM:HY_16G, RAMCFG4_L, RAMCFG3_L, RAMCFG1_L, RAMCFG0_L           |
| DRAMCFG:MI_16G | DRAM:MI_16G, RAMCFG4_L, RAMCFG3_L, RAMCFG1_L                      |
| DRAMCFG:SS_16G | DRAM:SS_16G, RAMCFG3_L, RAMCFG0_L                                 |

## CPU DRAM CFG Chart

|         |       |
|---------|-------|
| DIE REV | CFG 4 |
| A       | 0     |
| B       | 1     |
|         |       |

| VENDOR  | CFG 1 | CFG 0 |
|---------|-------|-------|
| HYNIX   | 0     | 0     |
| MICRON  | 0     | 1     |
| SAMSUNG | 1     | 0     |
| N/A     | 1     | 1     |

| SPEED | CFG 3 |
|-------|-------|
| 2133  | 0     |
| 1866  | 1     |
|       |       |

| CAPACITY | CFG 2 |
|----------|-------|
| 8GB      | 0     |
| 16GB     | 1     |

## Variable BOM Groups

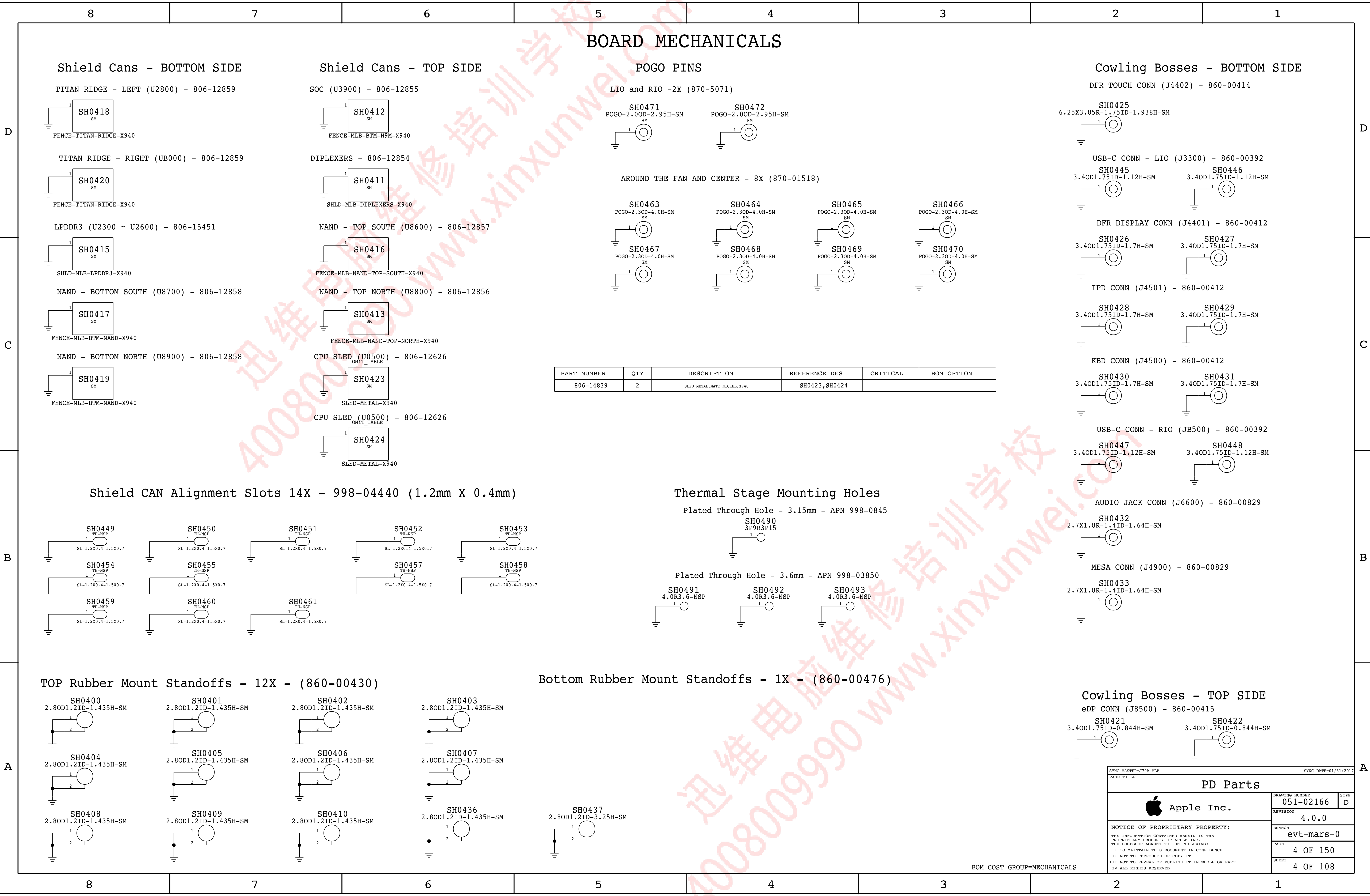
### Development/Base BOMs

| PART NUMBER | QTY | DESCRIPTION           | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-----------------------|---------------|----------|------------|
| 685-00187   | 1   | COMMON PARTS,MLB,X941 | BASE          | CRITICAL | BASE_BOM   |
| 985-00366   | 1   | DEV,MLB,X941          | DEVEL         | CRITICAL | DEVEL_BOM  |

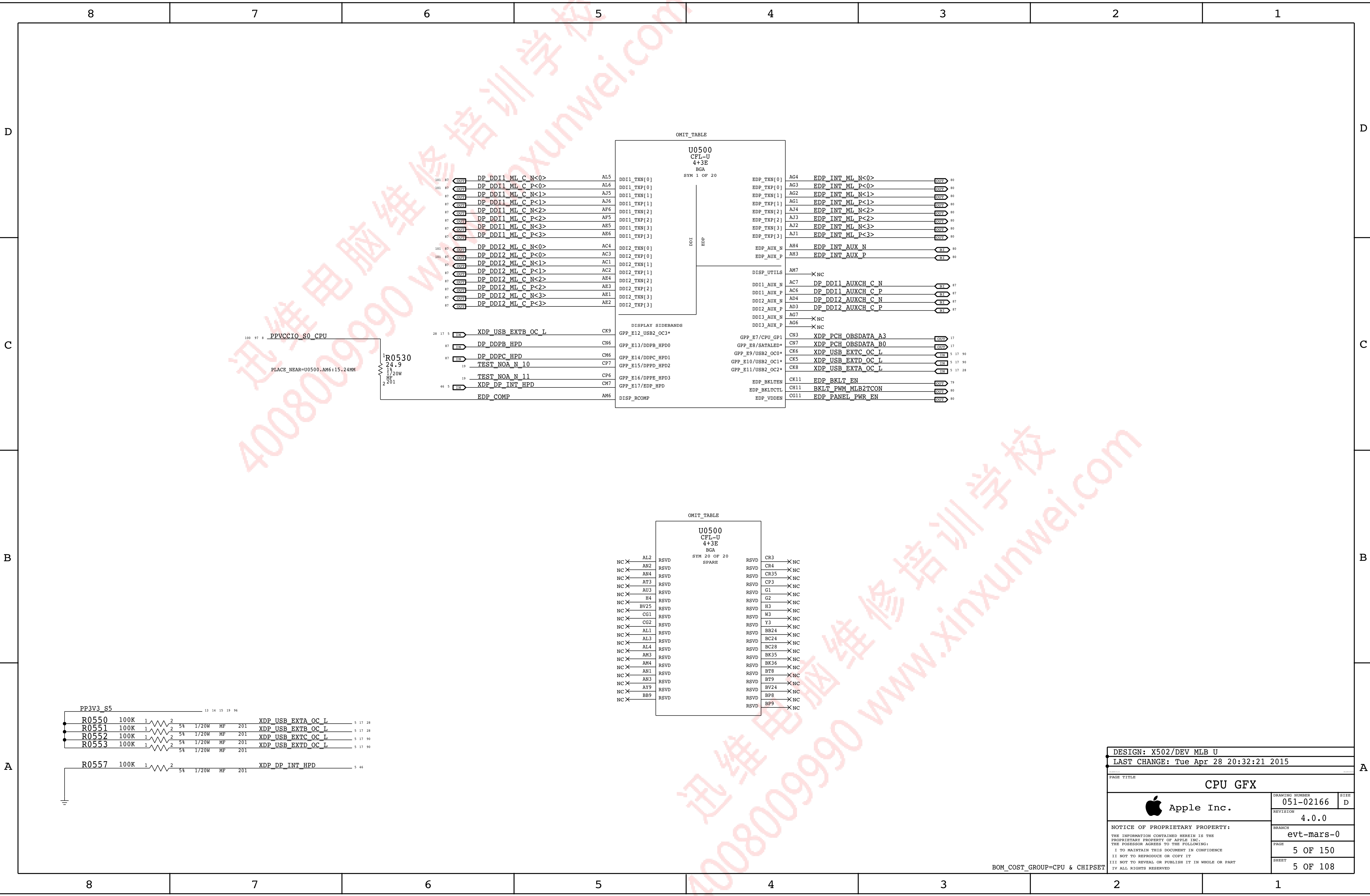
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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                           |            | BOM Configuration |            |
|                                                                                                                                                                                                 | Apple Inc. | DRAWING NUMBER    | 051-02166  |
|                                                                                                                                                                                                                                                                                      |            | REVISION          | 4.0.0      |
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| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
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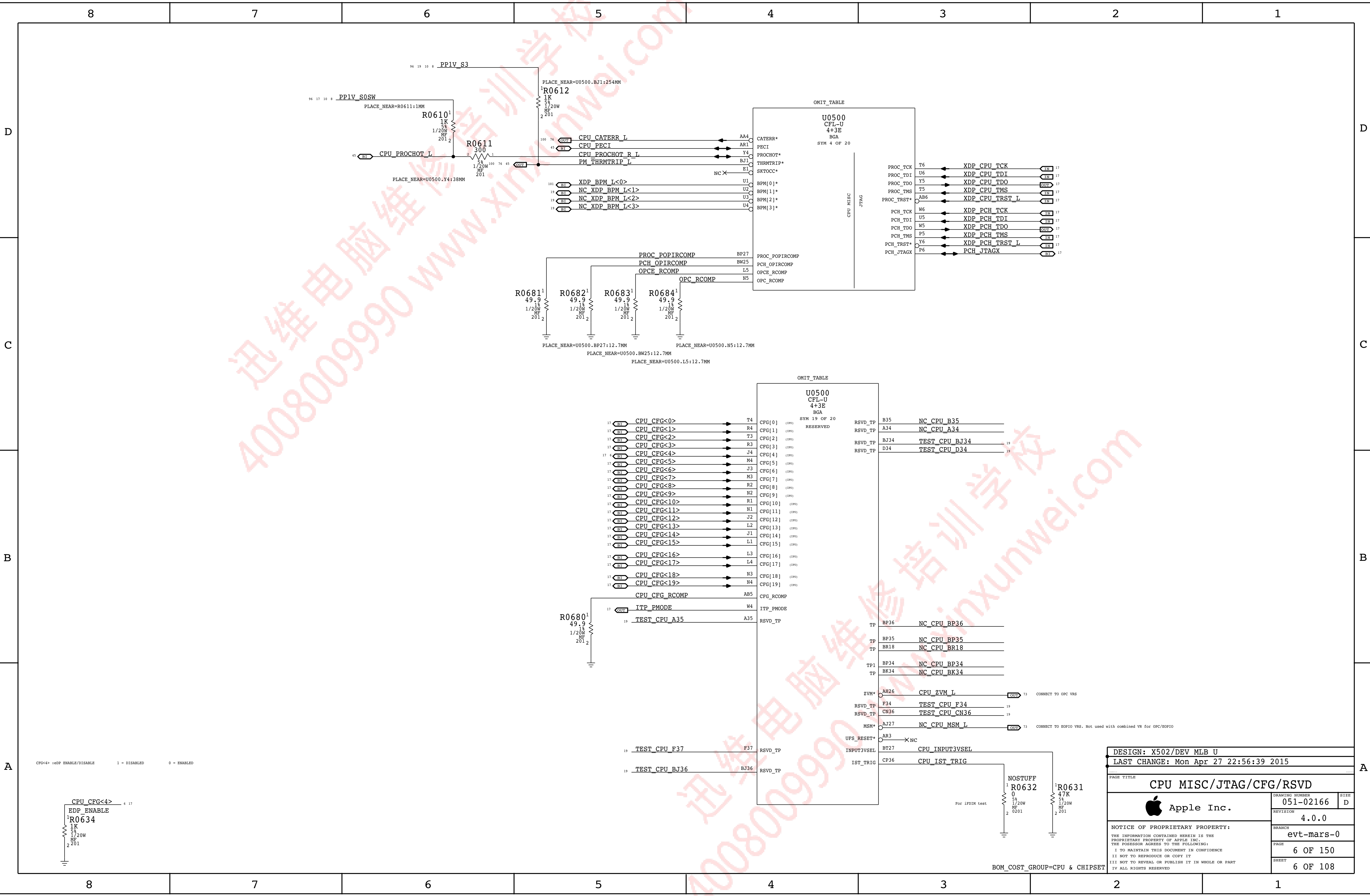










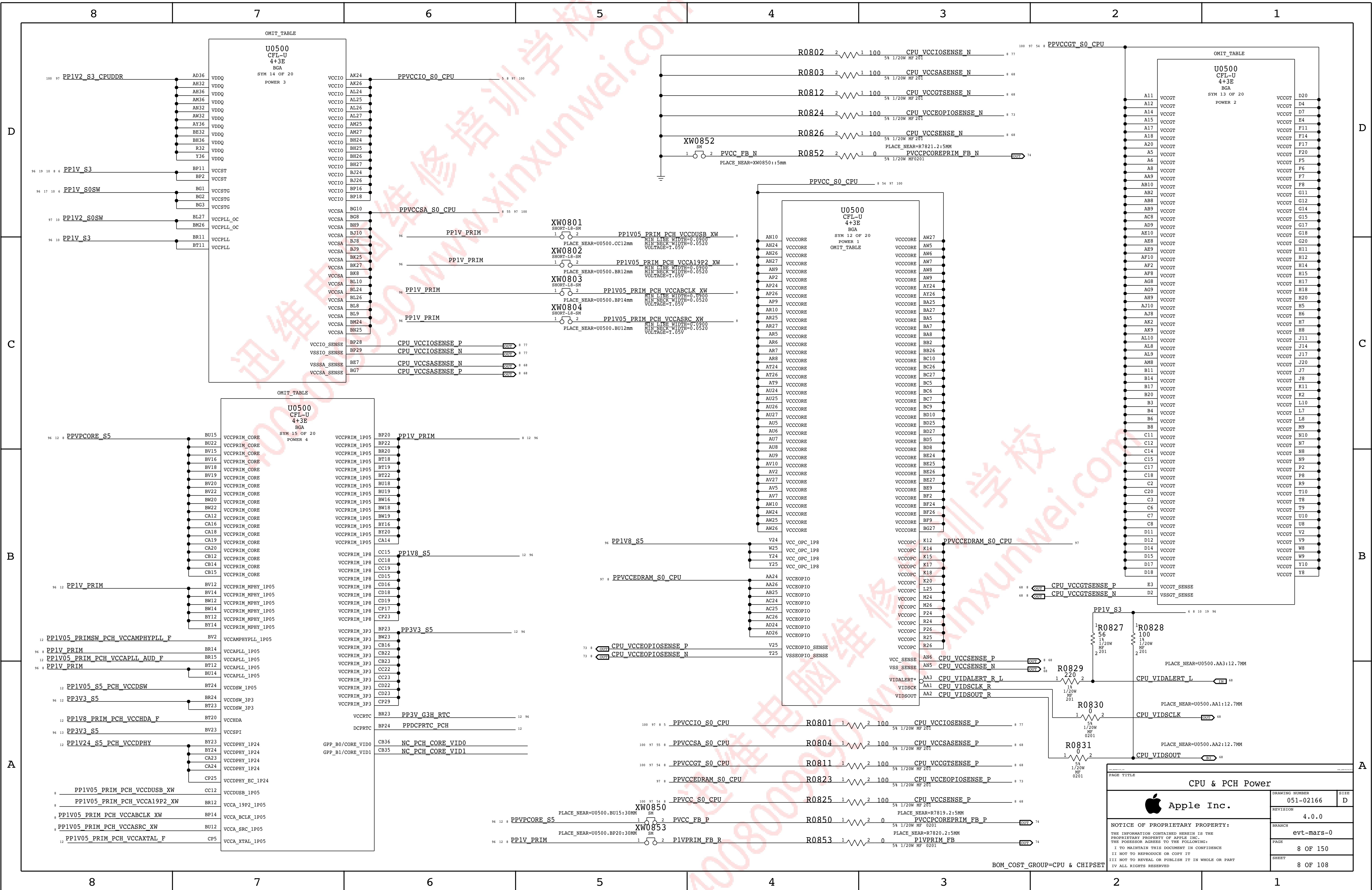


|                                                                                                                                                                                                                                                                                                                   |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| DESIGN: X502/DEV MLB U                                                                                                                                                                                                                                                                                            |                |            |
| LAST CHANGE: Mon Apr 27 22:56:39 2015                                                                                                                                                                                                                                                                             |                |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |            |
| CPU MISC/JTAG/CFG/RSVD                                                                                                                                                                                                                                                                                            |                |            |
|                                                                                                                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                   | REVISION       | 4.0.0      |
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|                                                                                                                                                                                                                                                                                                                   | PAGE           | 6 OF 150   |
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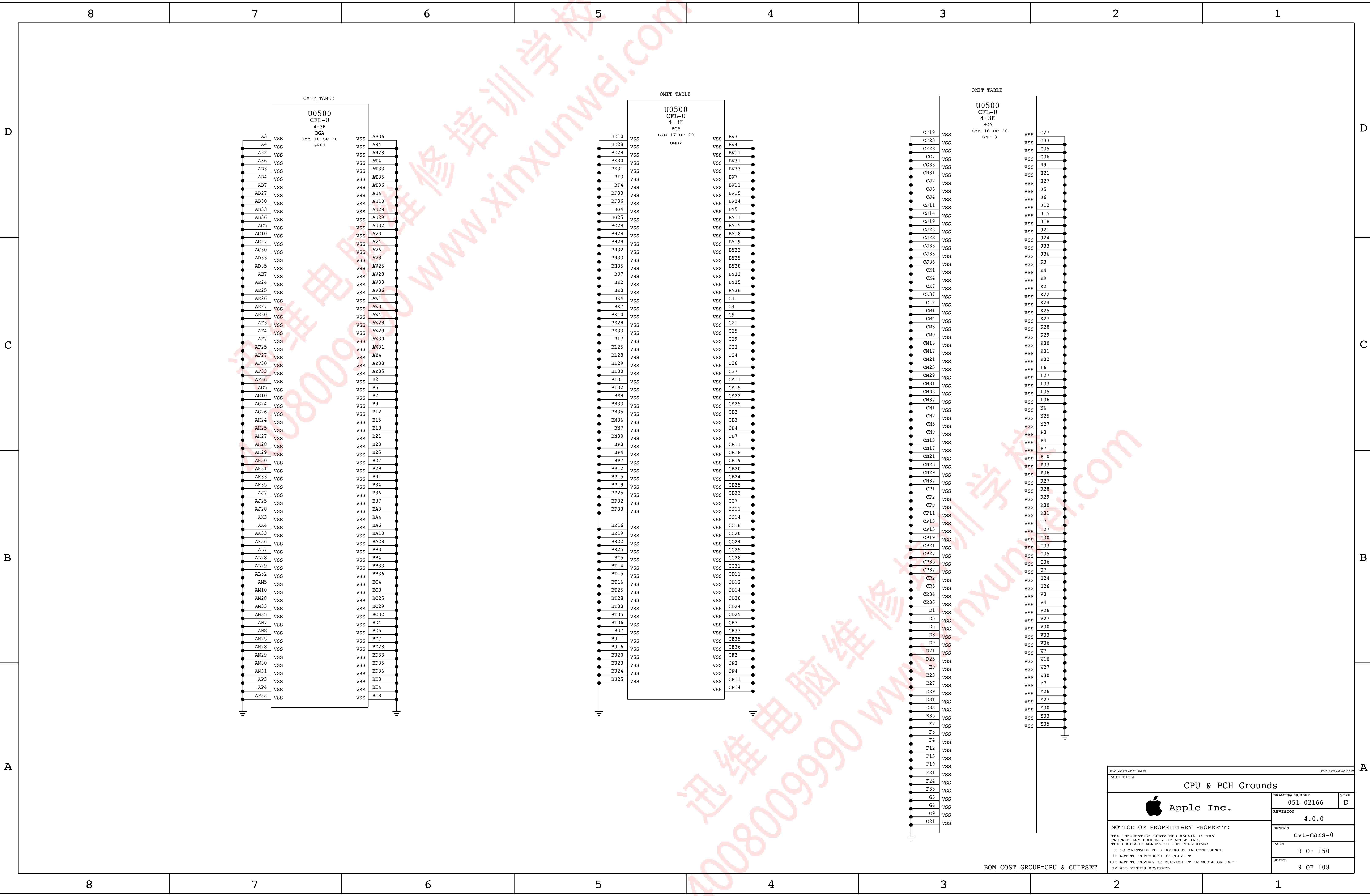












SYMC\_PARTNER-1132\_000001

SYMC\_DATE=02/03/2017

PAGE TITLE

CPU & PCH Grounds

 Apple Inc.

DRAWING NUMBER

051-02166

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BOM\_COST\_GROUP=CPU & CHIPSET

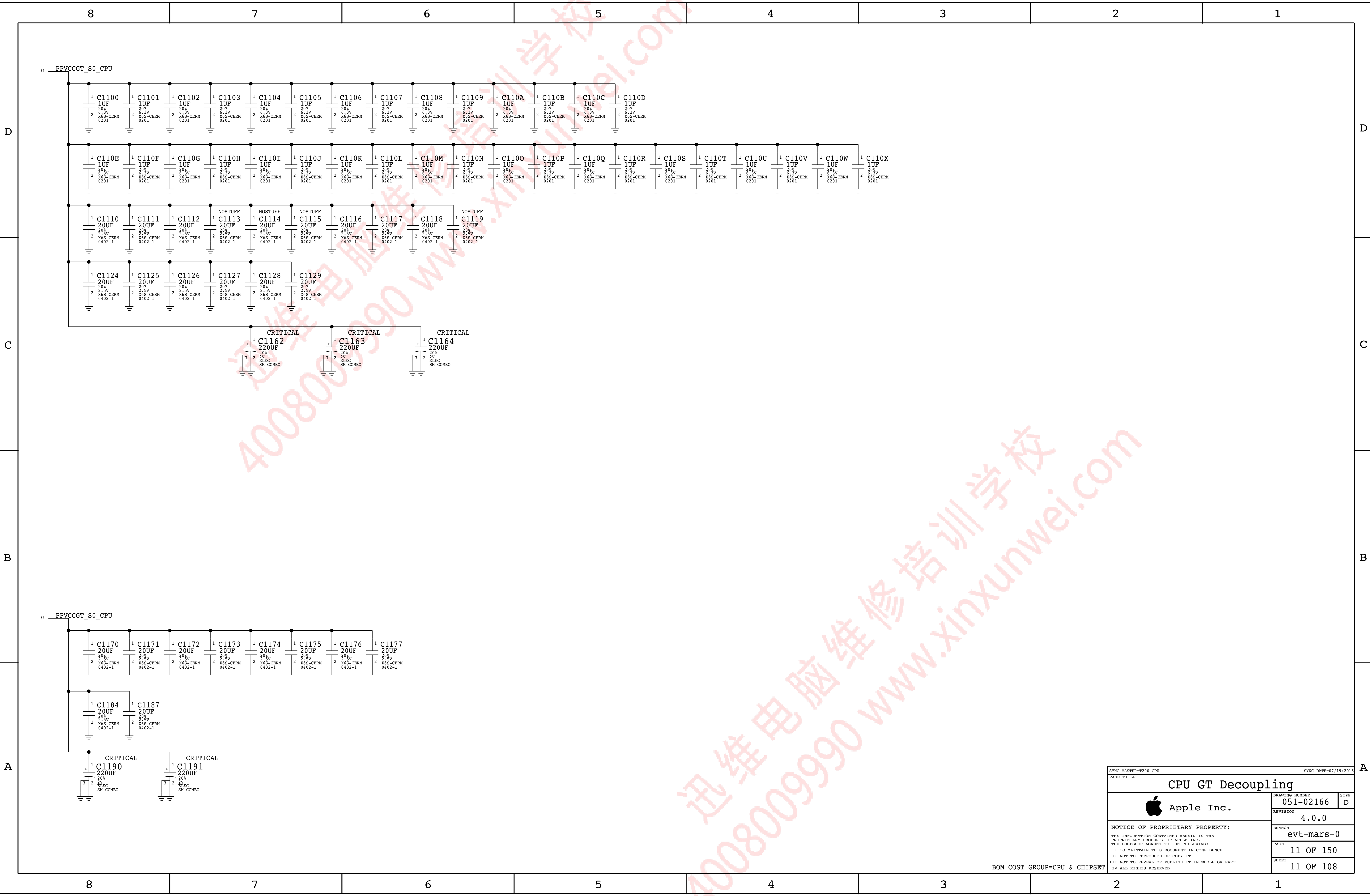




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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|---------------------|------------|--|
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |  |                | CPU Core Decoupling |            |  |
|                                                                                                                                                                                                                                                                                                                   |  | DRAWING NUMBER |                     | SIZE       |  |
|                                                                                                                                                                                                                                                                                                                   |  | 051-02166      |                     | D          |  |
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BOM\_COST\_GROUP=CPU & CHIPSET

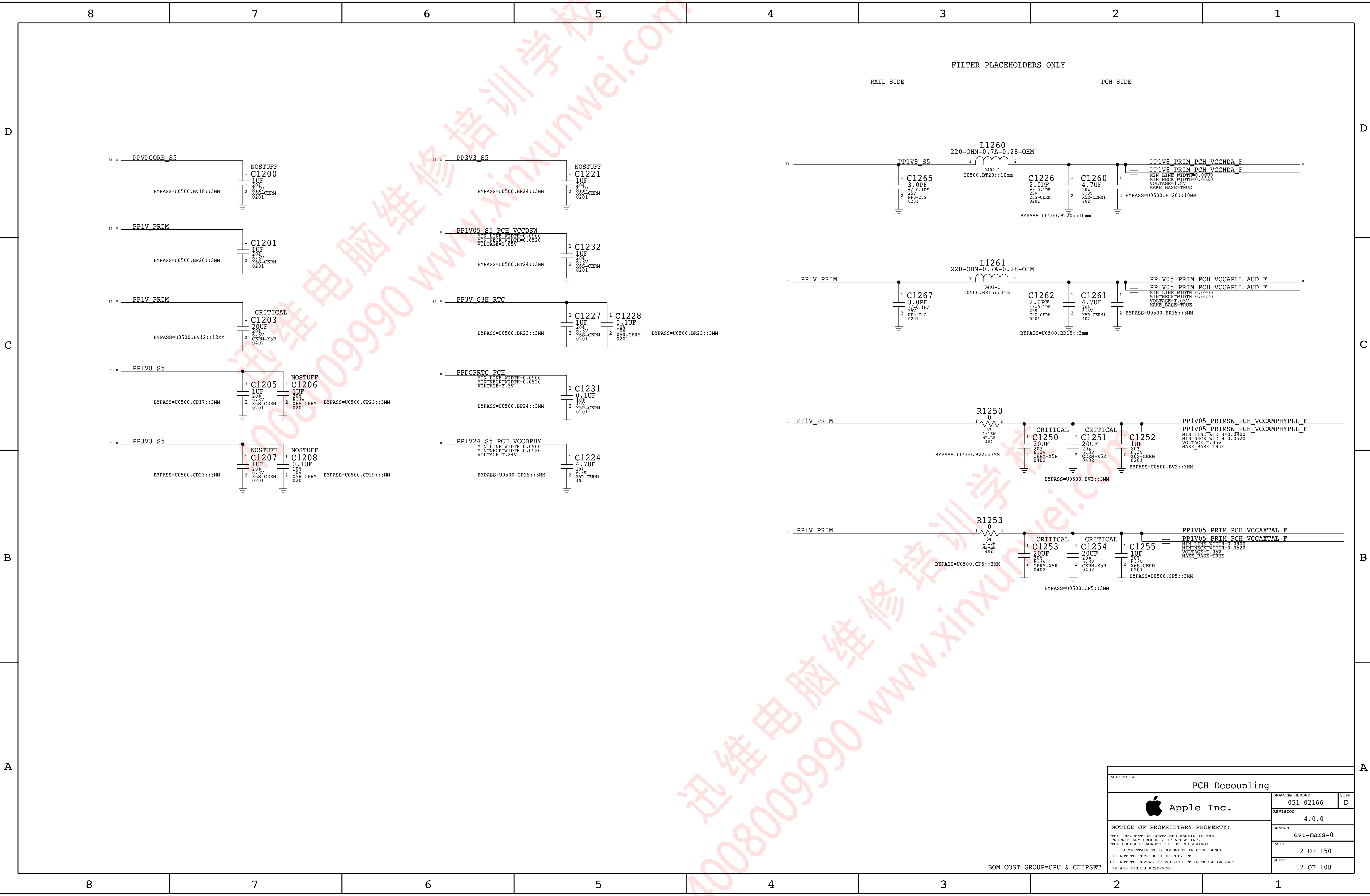




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| SYNC_MASTER=T290_CPU                                                                                                                                                                                                                                                                 |                | SYNC_DATE=07/19/2016 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                           |                |                      |           |
| CPU GT Decoupling                                                                                                                                                                                                                                                                    |                |                      |           |
|  Apple Inc.                                                                                                                                                                                     | DRAWING NUMBER | 051-02166            | SIZE<br>D |
|                                                                                                                                                                                                                                                                                      | REVISION       | 4.0.0                |           |
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BOM\_COST\_GROUP=CPU & CHIPSET

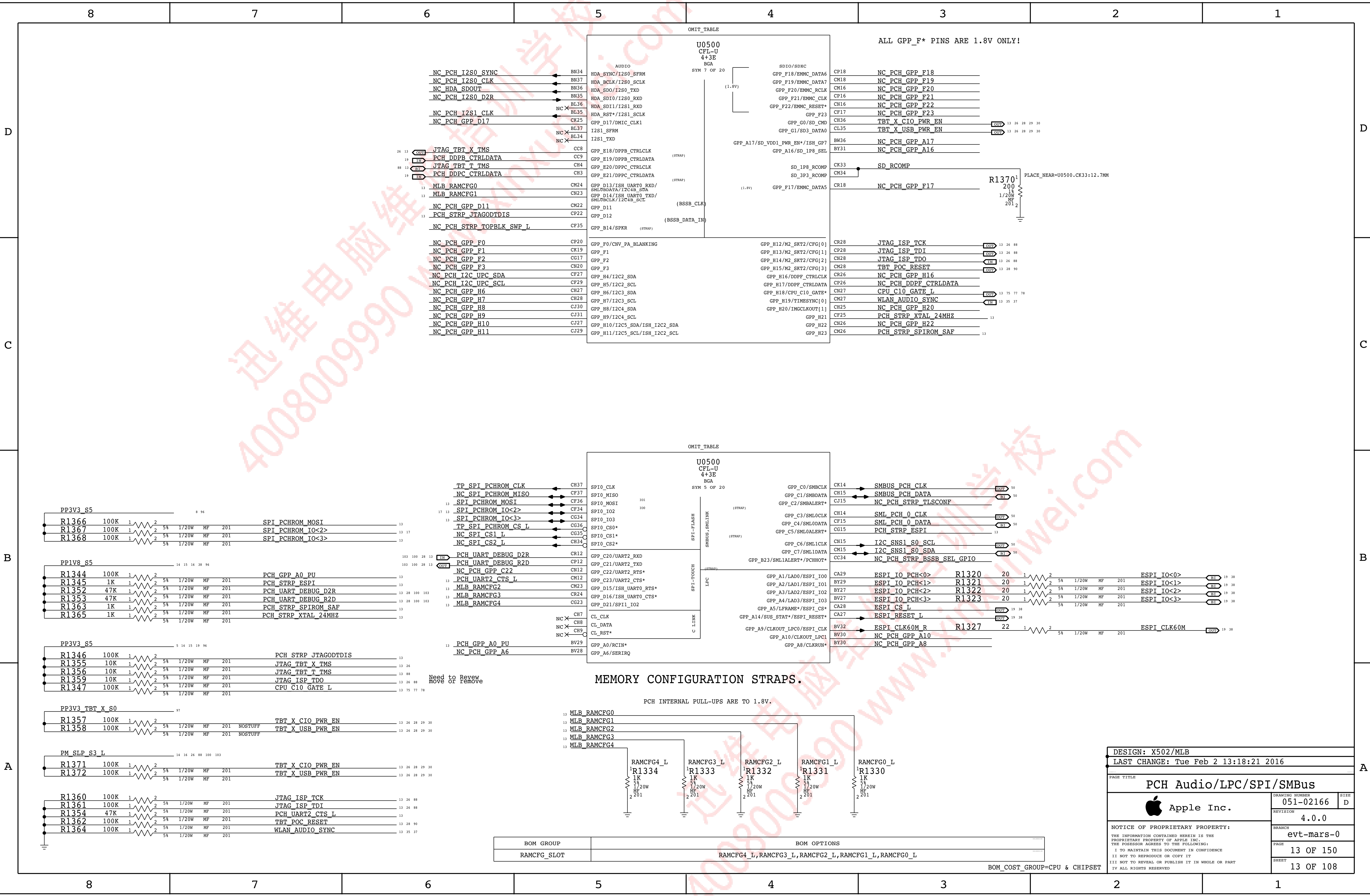




| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
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| PCH Decoupling                                                                                                                                                                                                                                                                                                          |                |            |
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                         | REVISION       | 4.0.0      |
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BOM\_COST\_GROUP=CPU & CHIPSET







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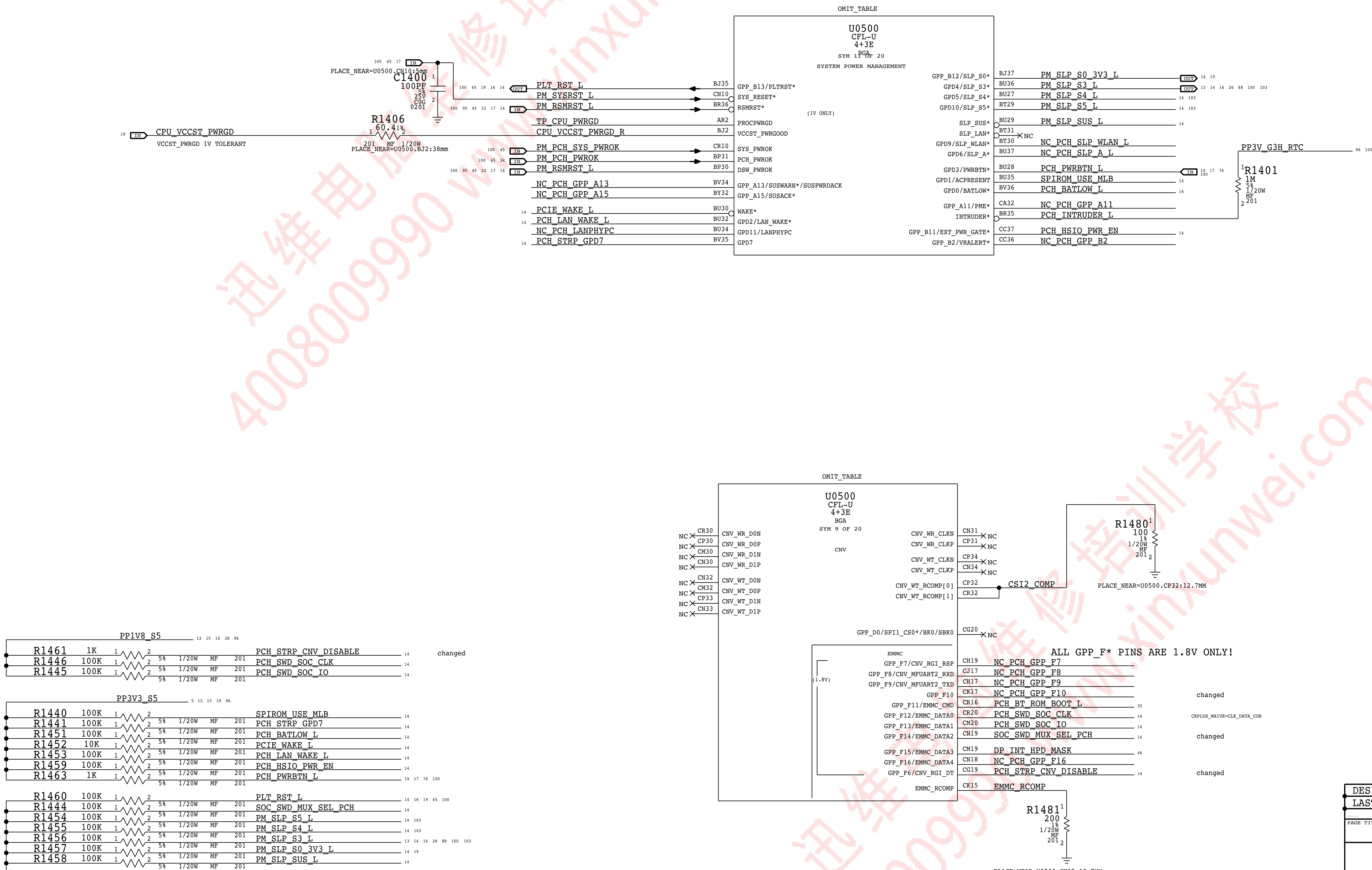
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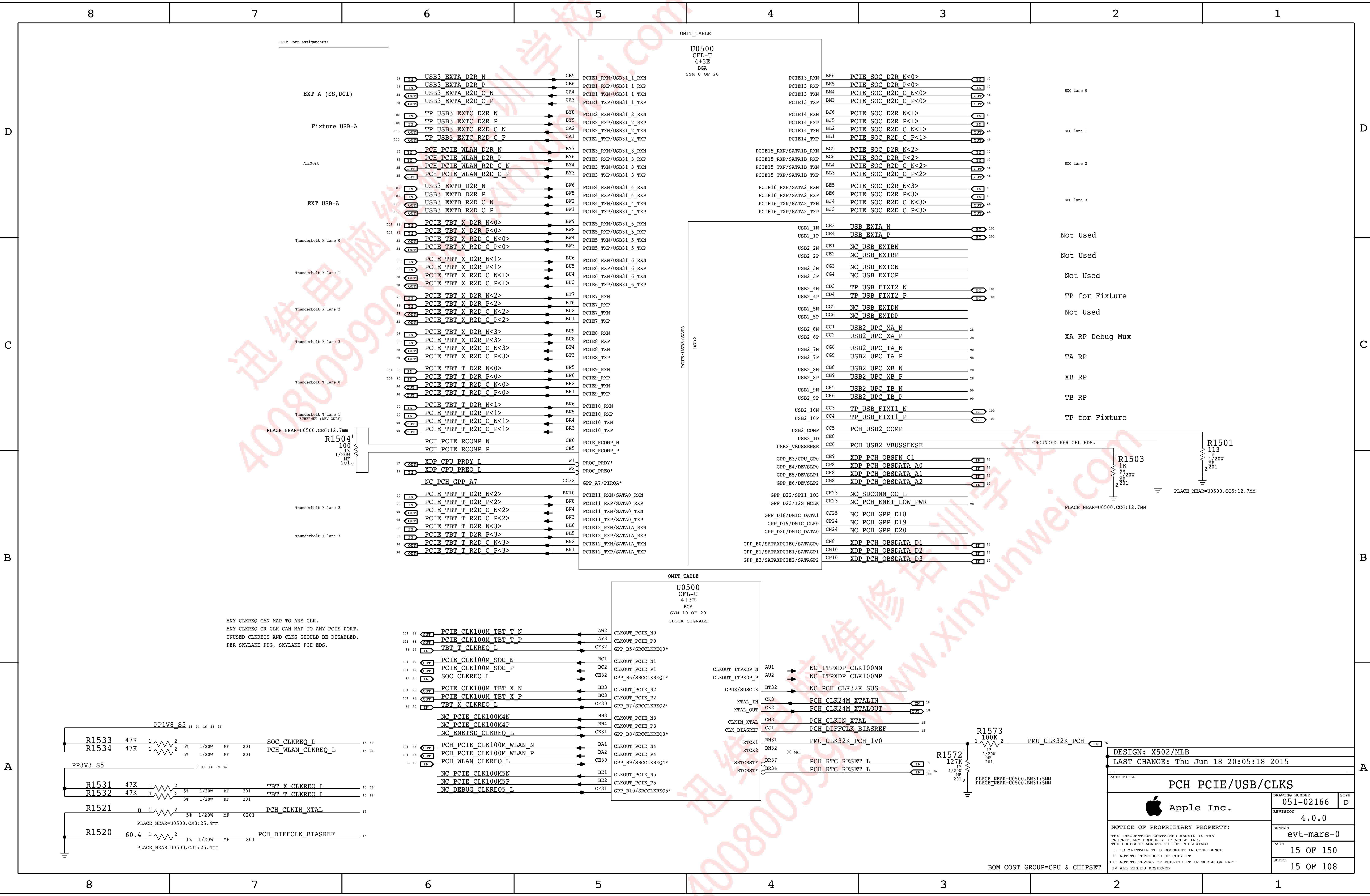


NOTE: =PM\_SLP\_S0\_L HAS INTERNAL PULL-UP BEFORE RSMRST\_L IS RELEASED.  
THIS CAUSES A VOLTAGE DIVIDER WITH THE PULL-DOWN HERE.  
THE SIGNAL IS DRIVEN HI AFTER RSMRST\_L IS RELEASED.

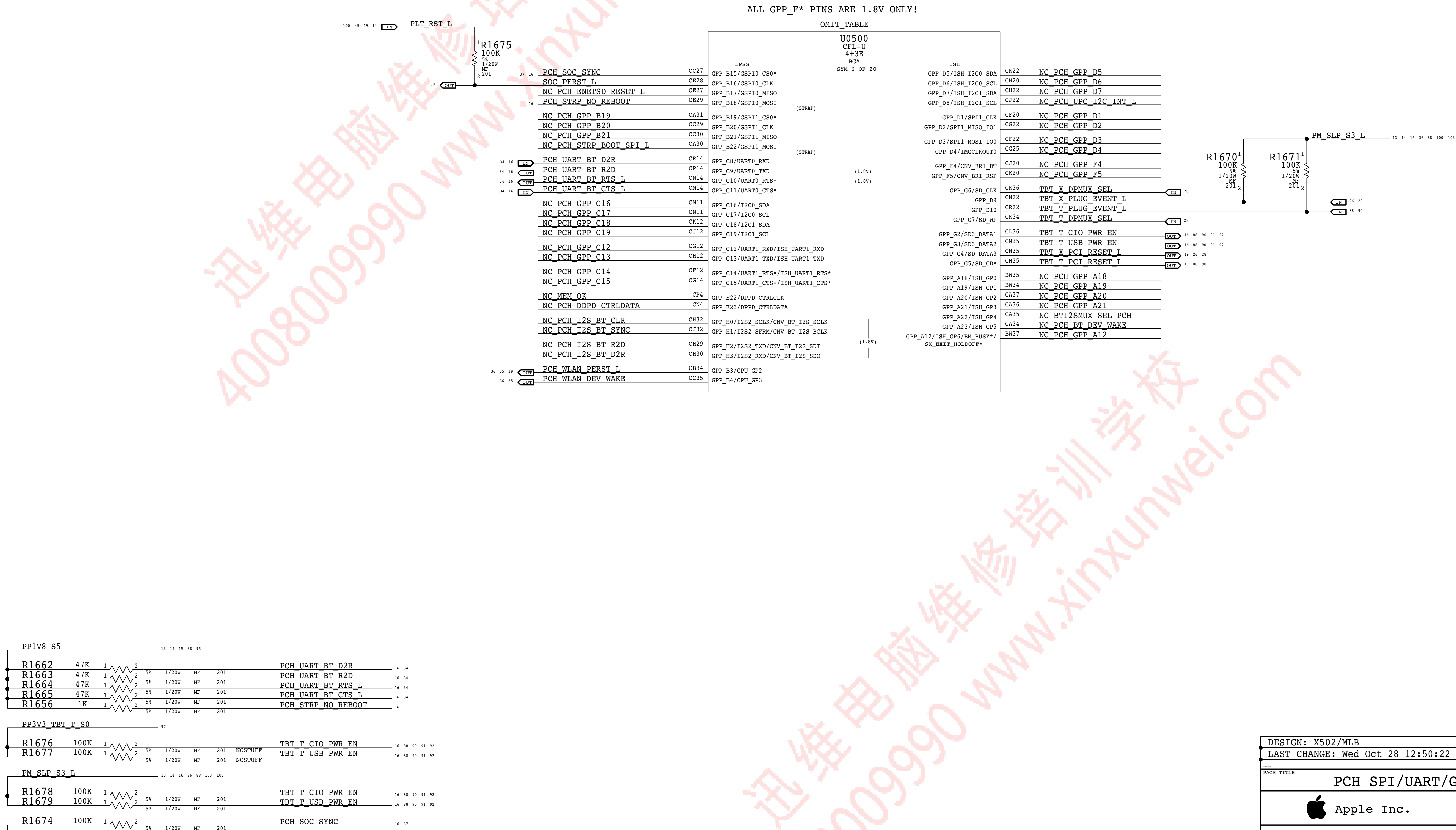
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|                                                                                                                                                                                                                                                                                                                         |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| DESIGN: X502/MLB                                                                                                                                                                                                                                                                                                        |                |            |
| LAST CHANGE: Tue Apr 5 13:08:54 2016                                                                                                                                                                                                                                                                                    |                |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
| PCH Power Management                                                                                                                                                                                                                                                                                                    |                |            |
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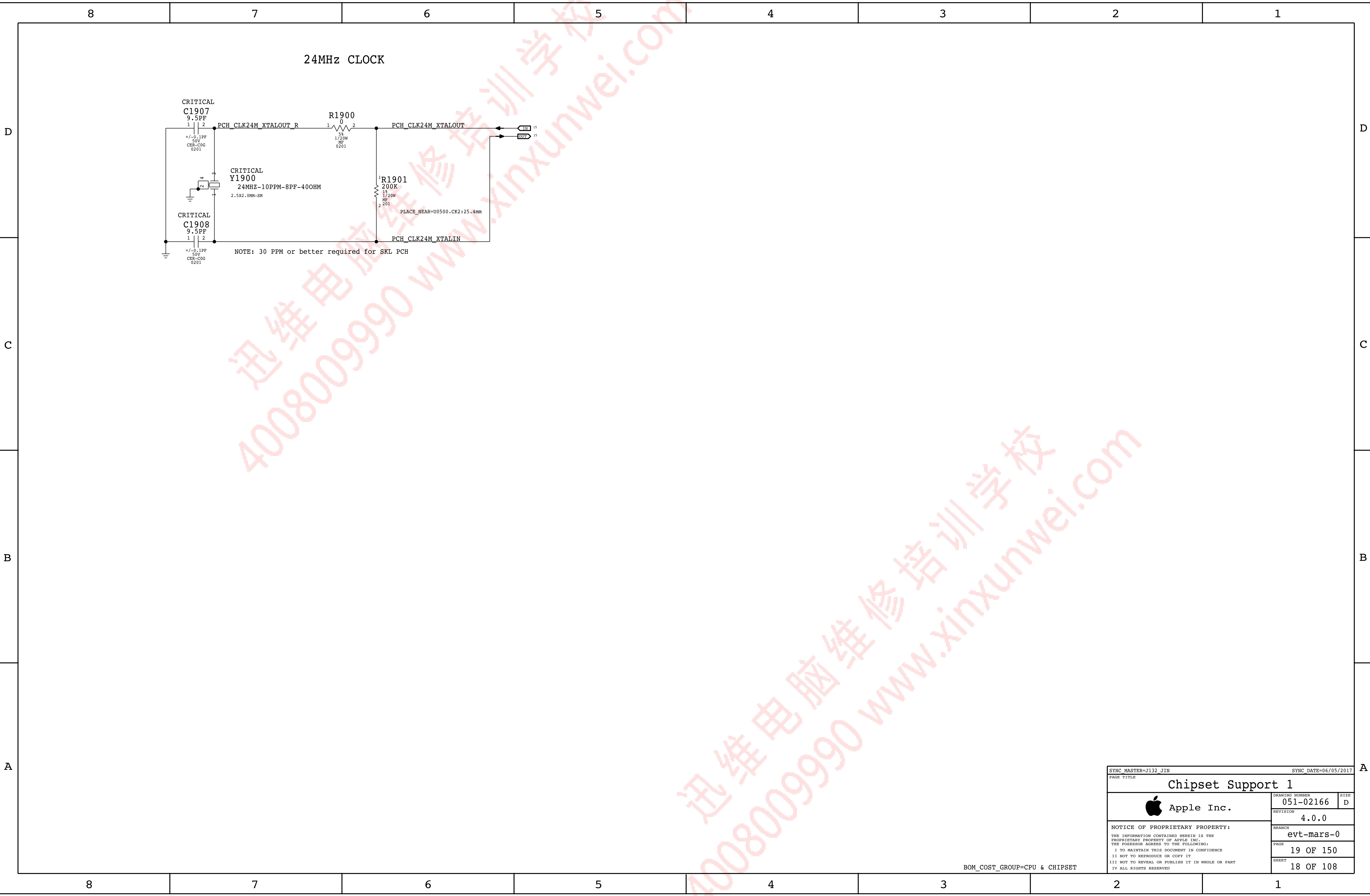


BOM COST GROUP=CPU & CHIPSET

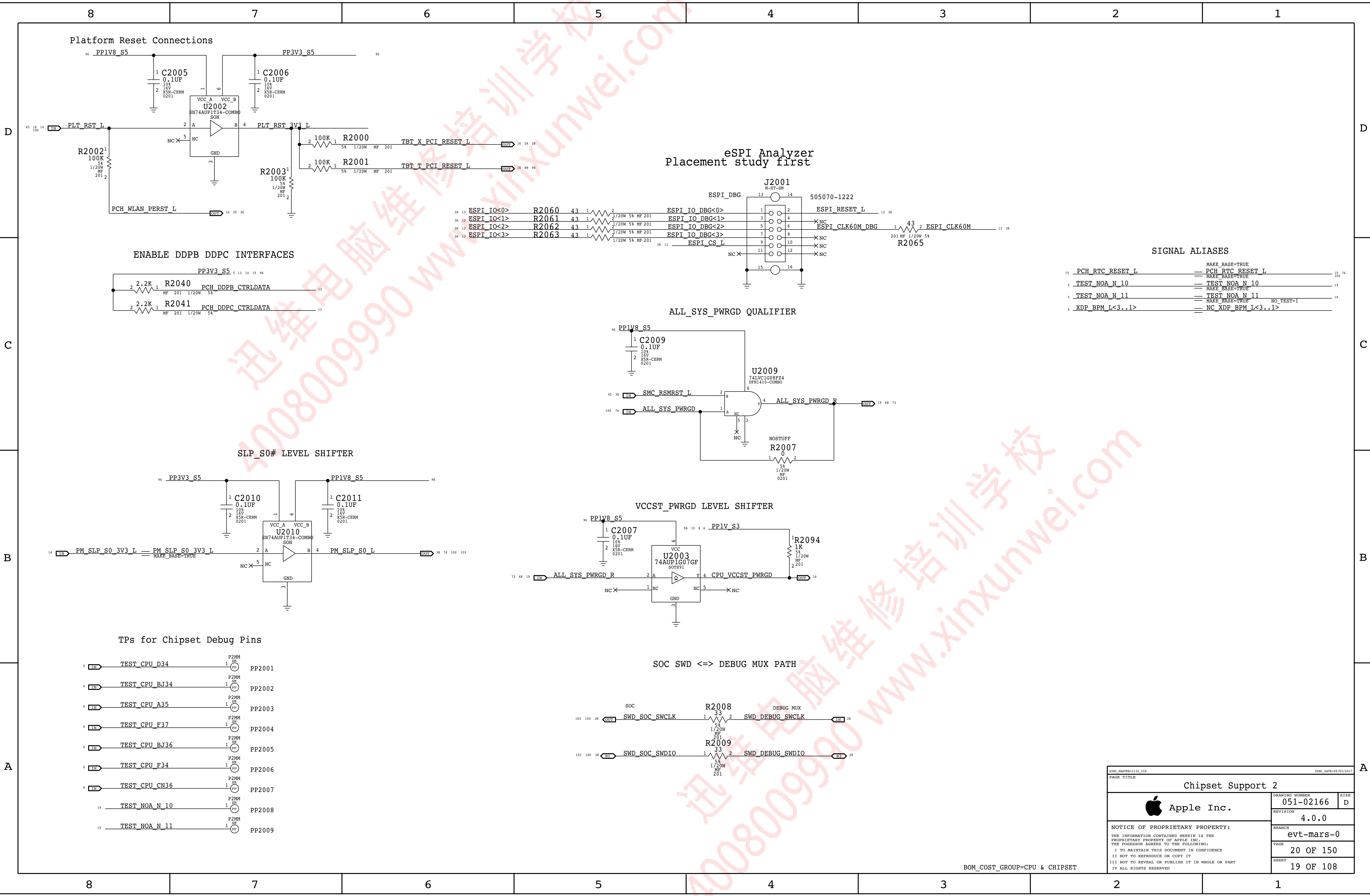












SYMC\_MASTER-2132\_218

SYMC\_DATE-09/05/2017

PAGE TITLE

Chipset Support 2



Apple Inc.

DRAWING NUMBER

051-02166

SIZE

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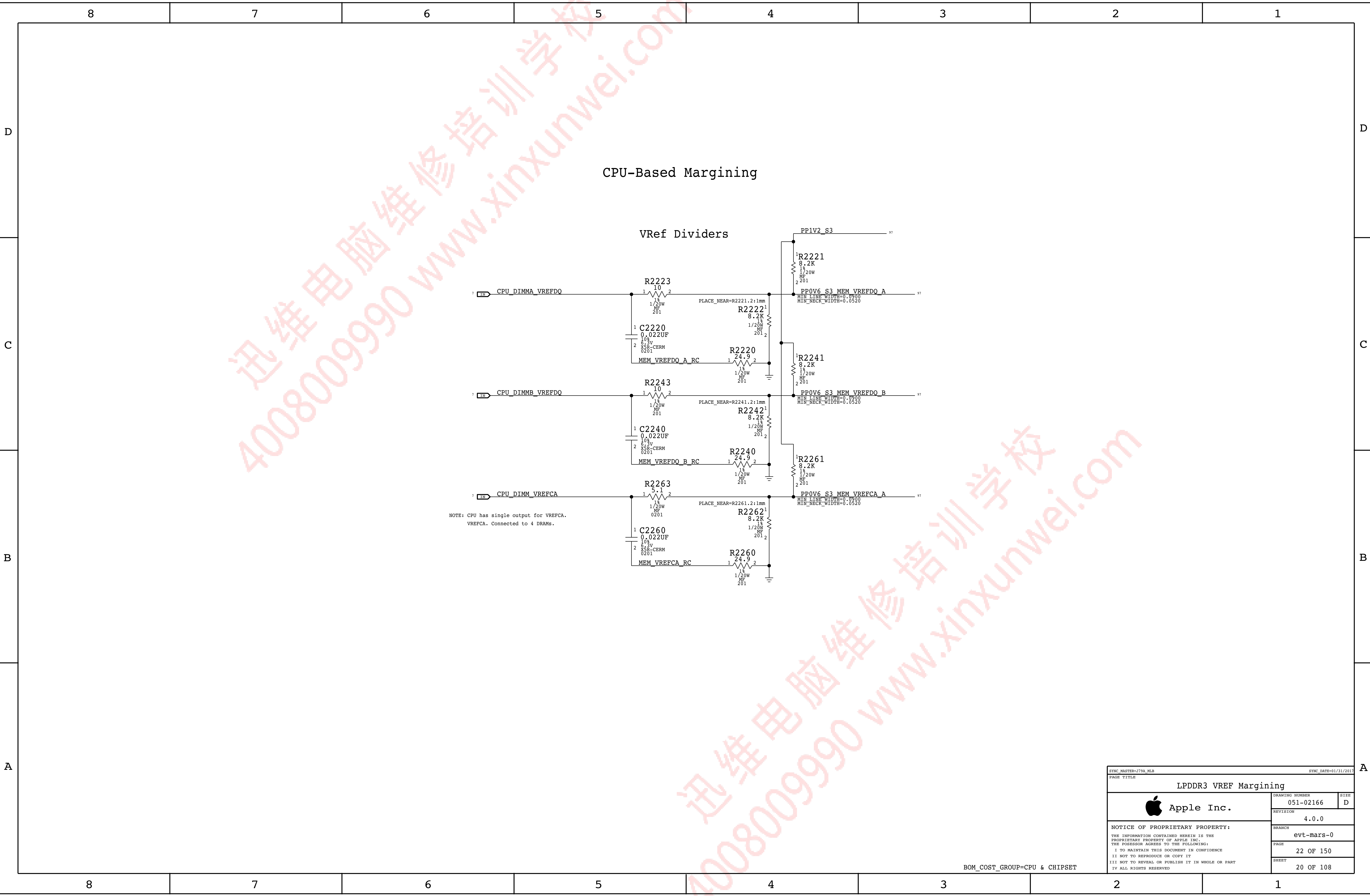
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NOTE: CPU has single output for VREFCA.  
VREFCA. Connected to 4 DRAMs.

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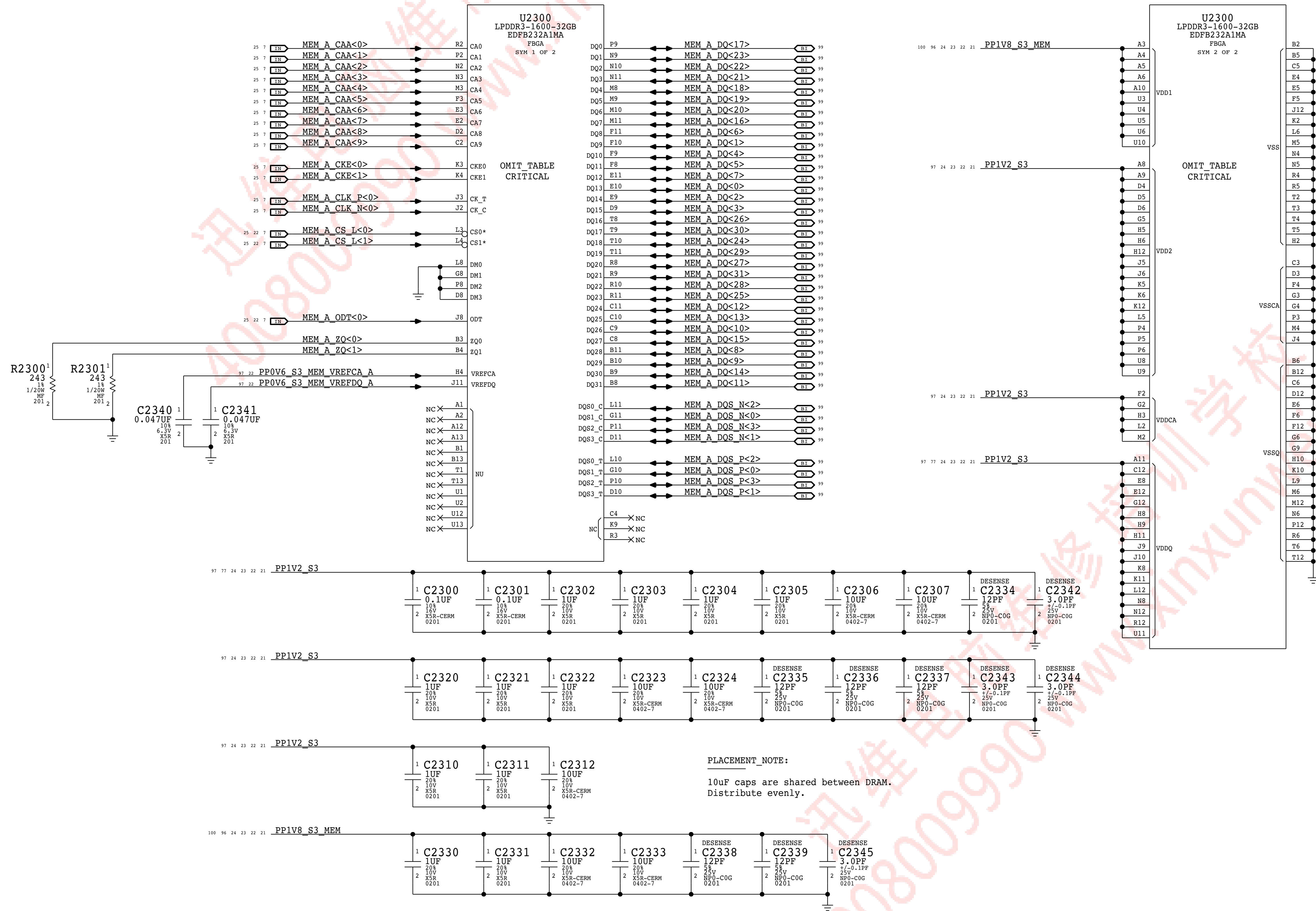
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BOM\_COST\_GROUP=CPU & CHIPSET

|                                                                                                                                                                                                                                                                                |                |                      |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|-----------|
| SYNC MASTER=J79A.MLB                                                                                                                                                                                                                                                           |                | SYNC DATE=01/31/2017 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                     |                |                      |           |
| LPDDR3 VREF Margining                                                                                                                                                                                                                                                          |                | DRAWING              |           |
|  Apple Inc.                                                                                                                                                                               | DRAWING NUMBER |                      | SIZE      |
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# LPDDR3 CHANNEL A (0-31)



|                                                                                                                                                                                                                                                                                       |  |                      |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|------------|
| SYMC MASTER-J79A MLB                                                                                                                                                                                                                                                                  |  | SYMC DATE=01/31/2017 |            |
| PAGE TITLE                                                                                                                                                                                                                                                                            |  |                      |            |
| LPDDR3 DRAM Channel A (00-31)                                                                                                                                                                                                                                                         |  |                      |            |
|  Apple Inc.                                                                                                                                                                                      |  | DRAWING NUMBER       | 051-02166  |
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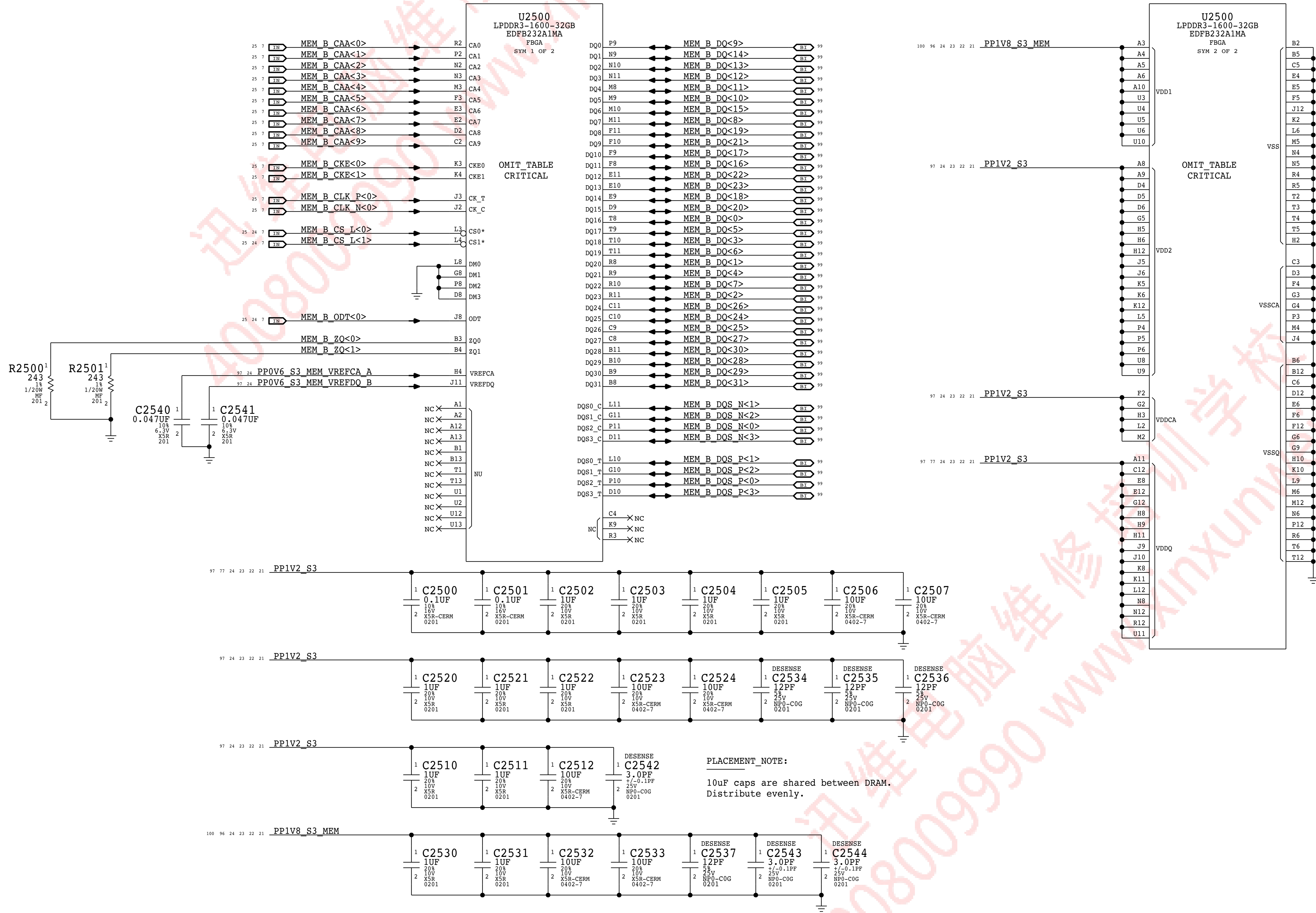
## D

A

BOM\_COST\_GROUP=DRAM



# LPDDR3 CHANNEL B (0-31)

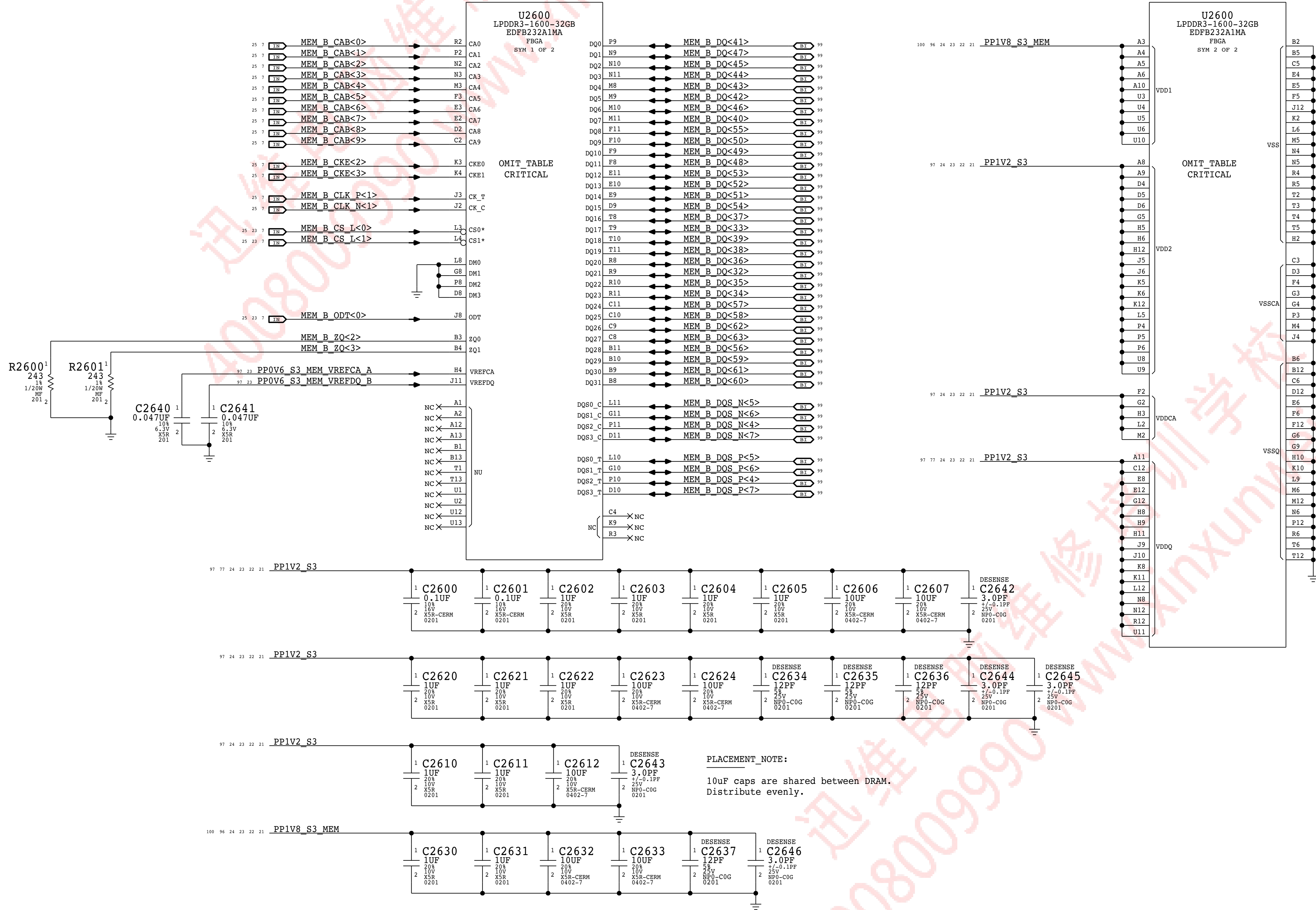


|                                                                                                                   |  |                               |            |
|-------------------------------------------------------------------------------------------------------------------|--|-------------------------------|------------|
| PAGE TITLE                                                                                                        |  | PAGE TITLE                    |            |
| LPDDR3 DRAM Channel B (00-31)                                                                                     |  | LPDDR3 DRAM Channel B (00-31) |            |
| Apple Inc.                                                                                                        |  | DRAWING NUMBER                | 051-02166  |
|                                                                                                                   |  | REVISION                      | 4.0.0      |
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BOM\_COST\_GROUP=DRAM



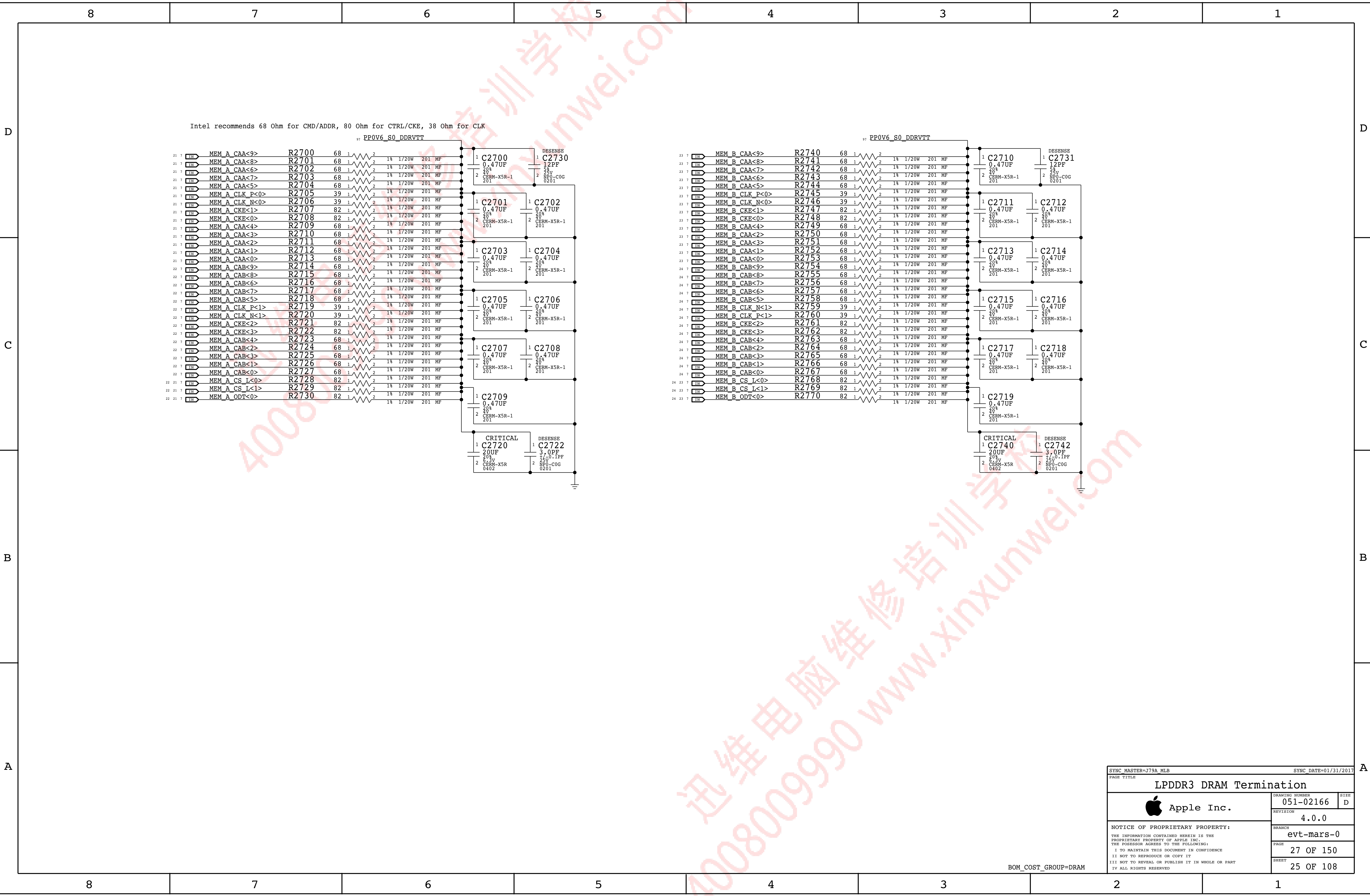
# LPDDR3 CHANNEL B (32-63)



BOM\_COST\_GROUP=DRAM

|                                                                                                                   |                |                      |           |
|-------------------------------------------------------------------------------------------------------------------|----------------|----------------------|-----------|
| SYNC MASTER=J79A MLB                                                                                              |                | SYNC DATE=01/31/2017 |           |
| PAGE TITLE                                                                                                        |                |                      |           |
| LPDDR3 DRAM Channel B (32-63)                                                                                     |                |                      |           |
|  Apple Inc.                  | DRAWING NUMBER | 051-02166            | SIZE<br>D |
|                                                                                                                   | REVISION       | 4.0.0                |           |
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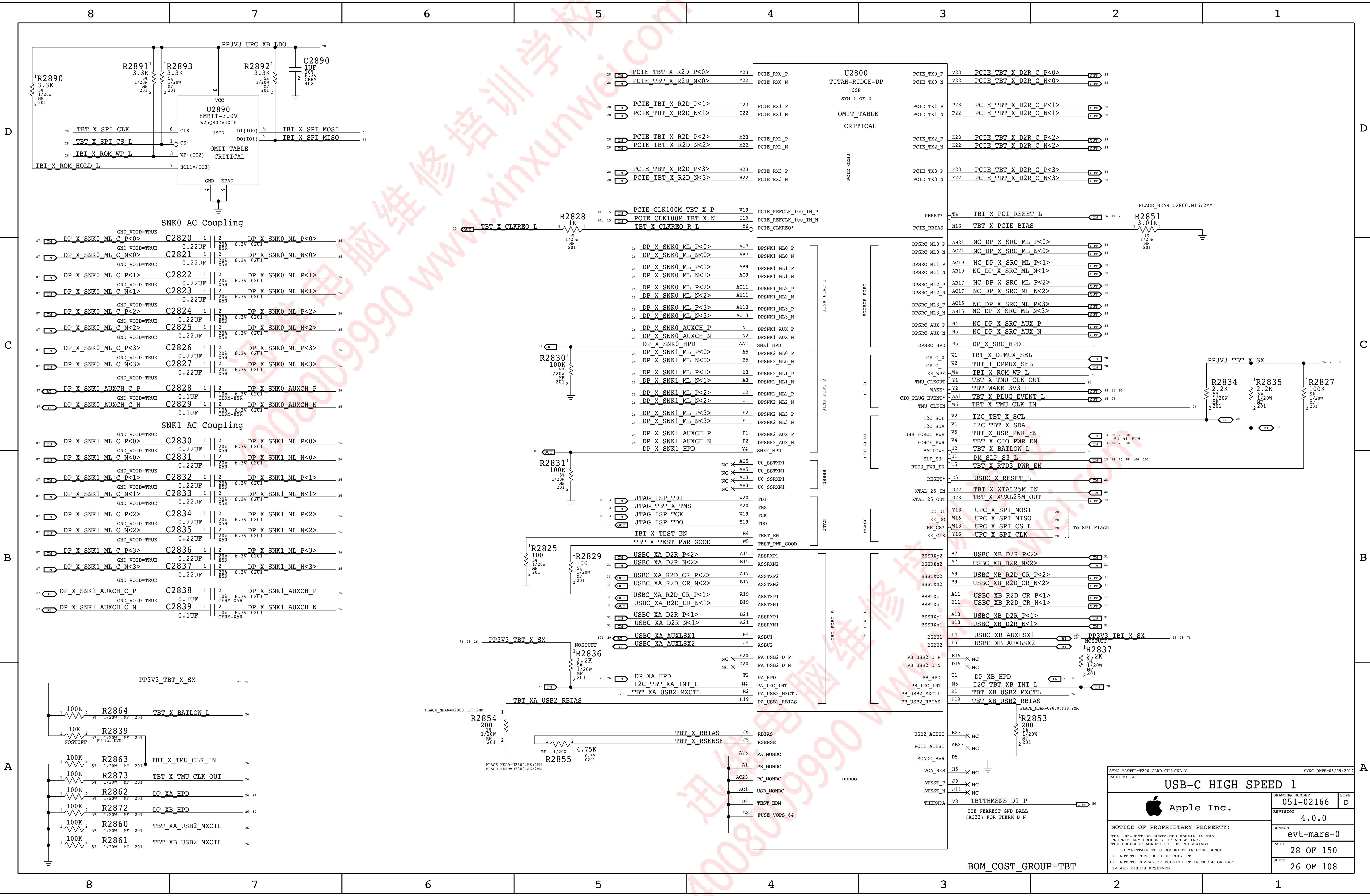




|                                                                                                                                                                                                                                                                                                                             |  |  |                      |  |            |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------|--|------------|--|
| SYNC_MASTER=J79A_MLB                                                                                                                                                                                                                                                                                                        |  |  | SYNC_DATE=01/31/2017 |  |            |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |  |                      |  |            |  |
| LPDDR3 DRAM Termination                                                                                                                                                                                                                                                                                                     |  |  |                      |  |            |  |
|  Apple Inc.                                                                                                                                                                                                                            |  |  | DRAWING NUMBER       |  | SIZE       |  |
|                                                                                                                                                                                                                                                                                                                             |  |  | 051-02166            |  | D          |  |
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|                                                                                                                                                                                                                                                                                                                             |  |  | SHEET                |  | 25 OF 108  |  |

BOM\_COST\_GROUP=DRAM

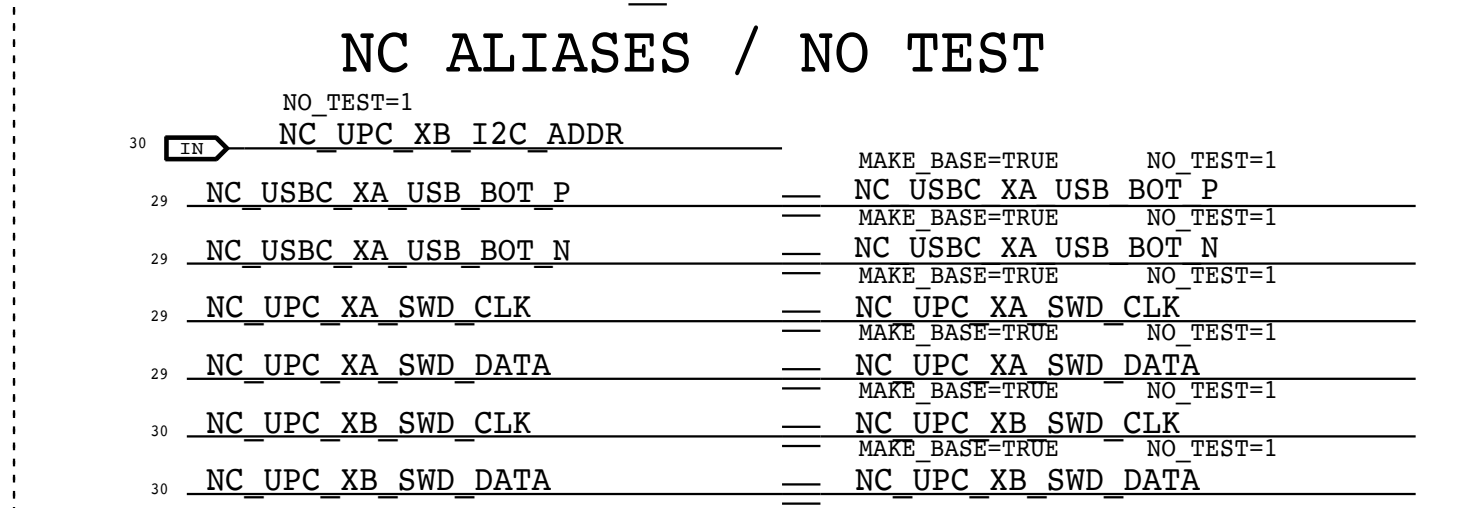
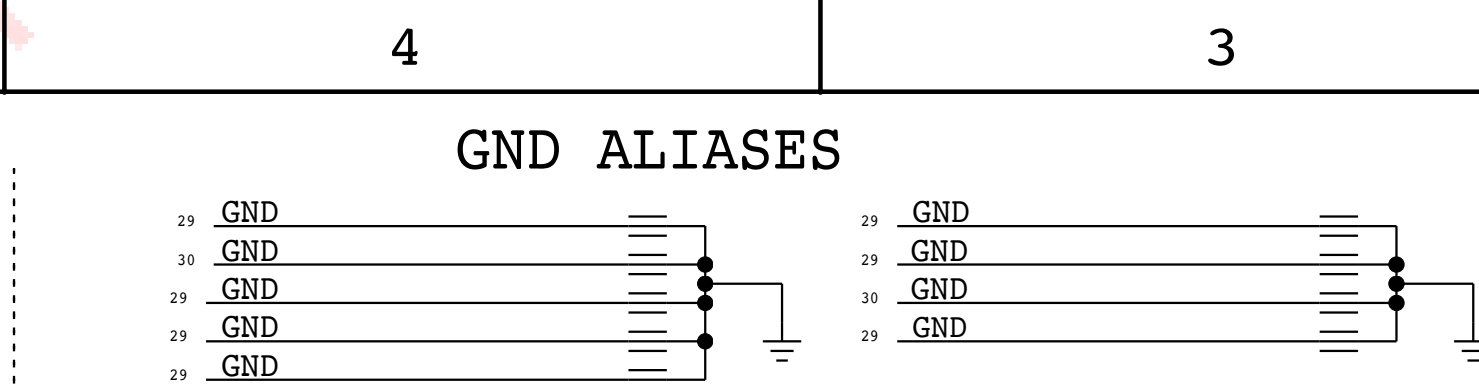
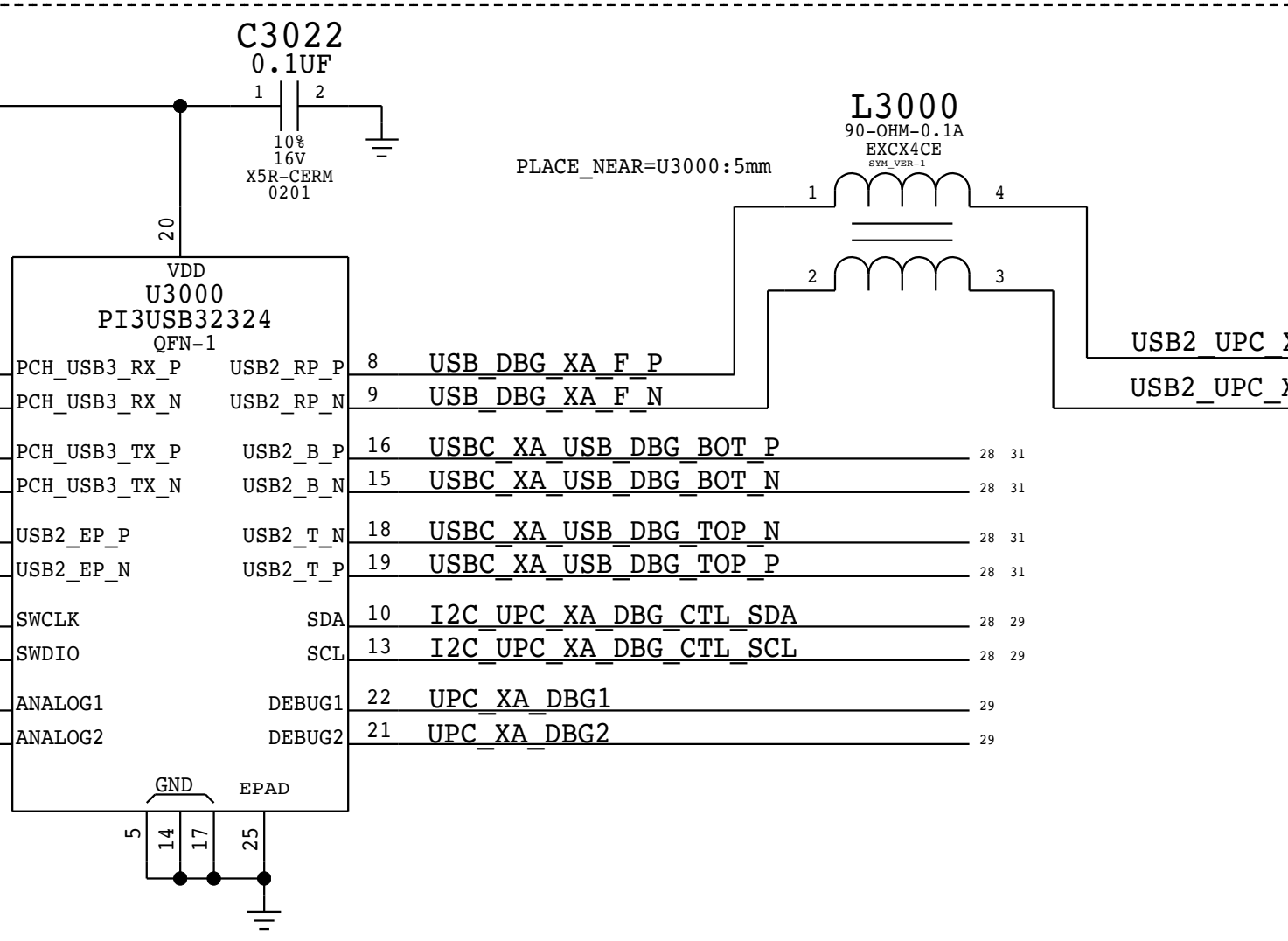
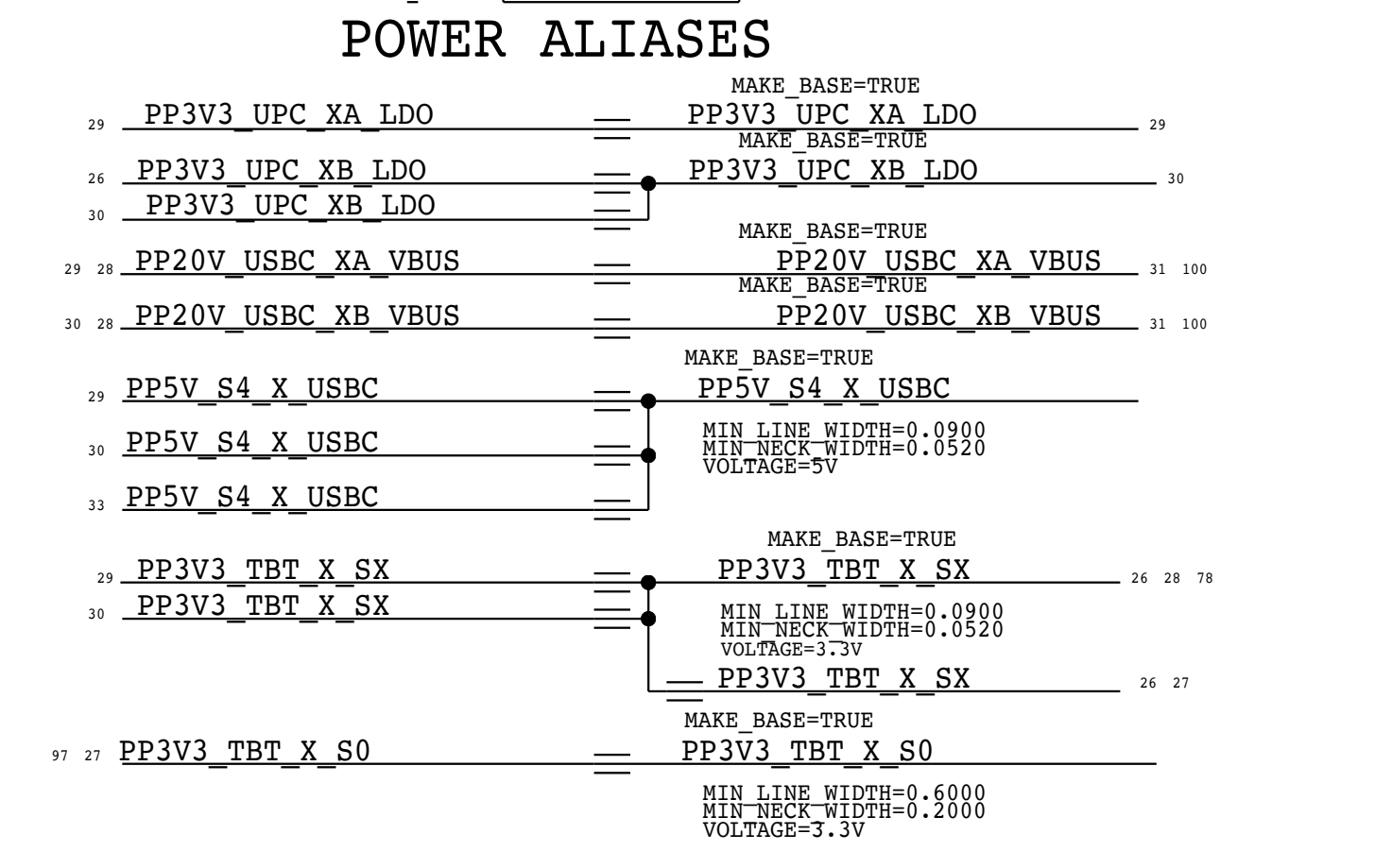
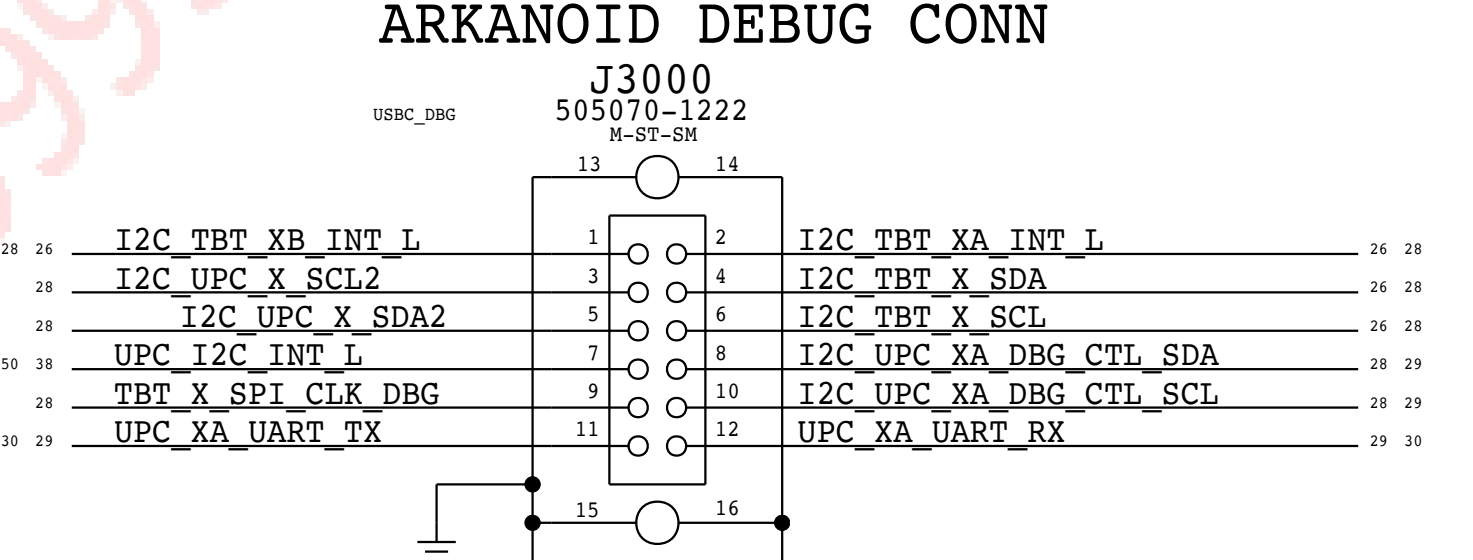
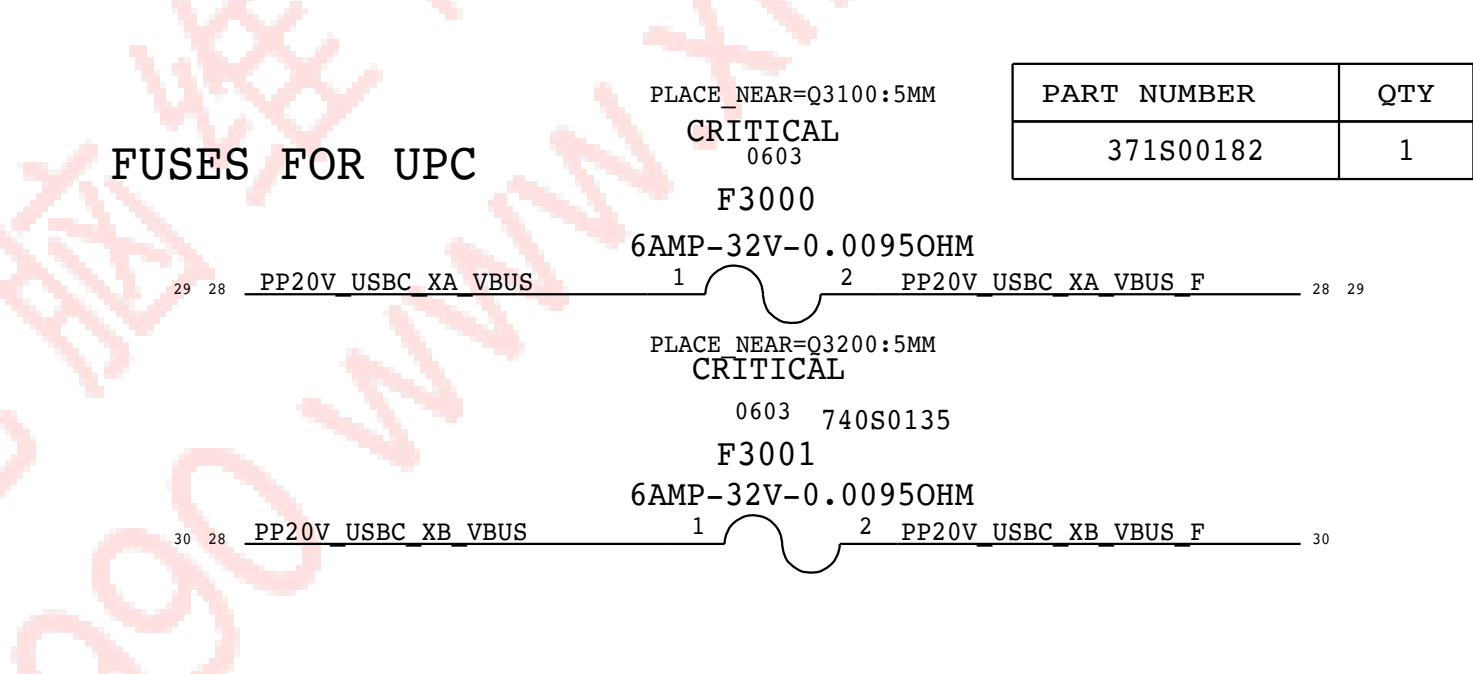
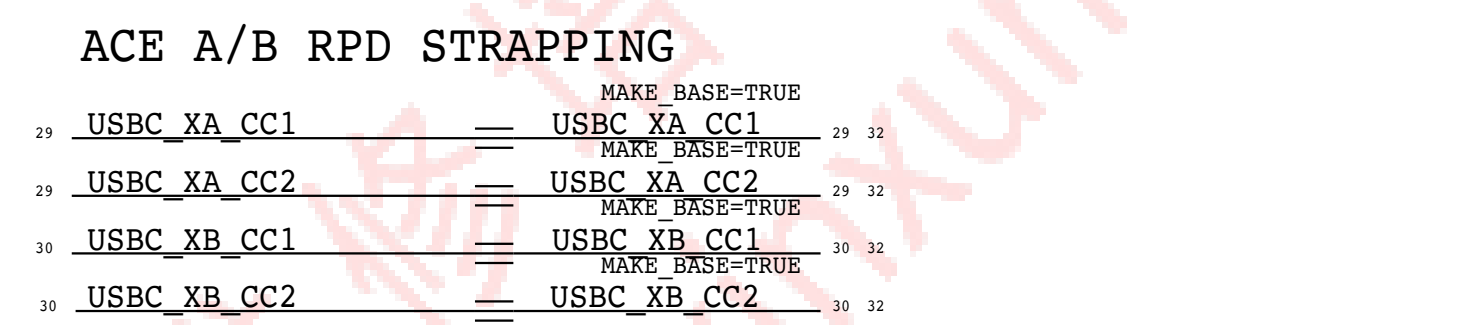
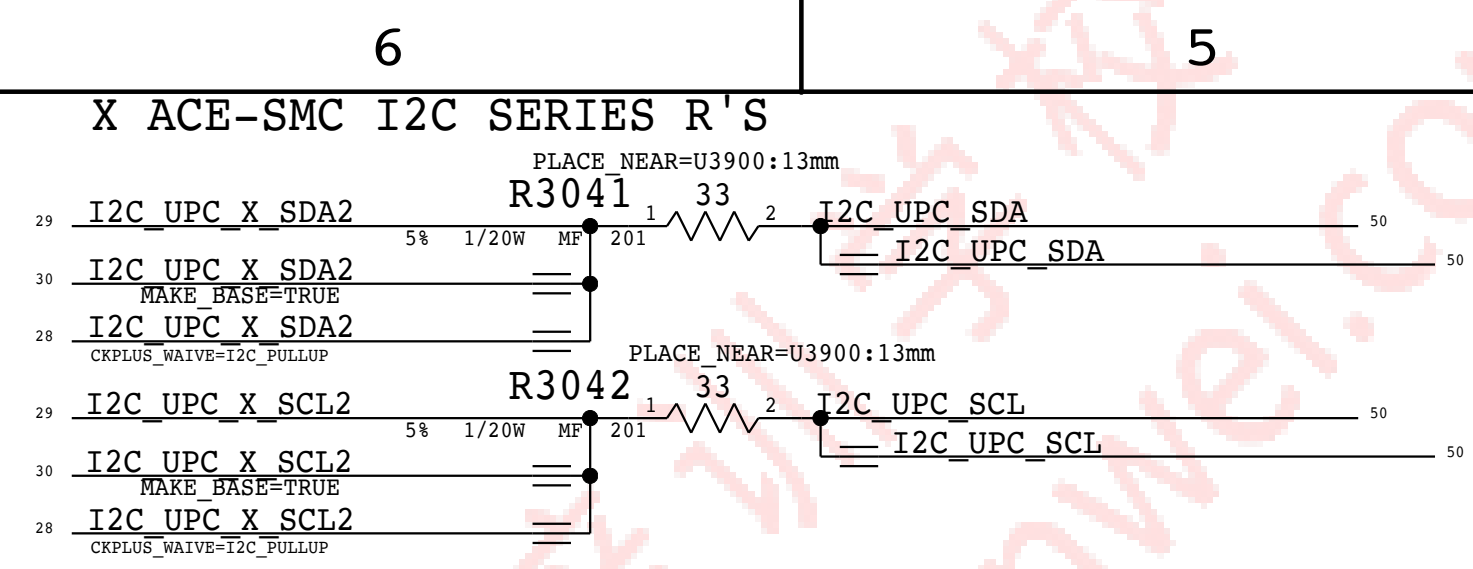
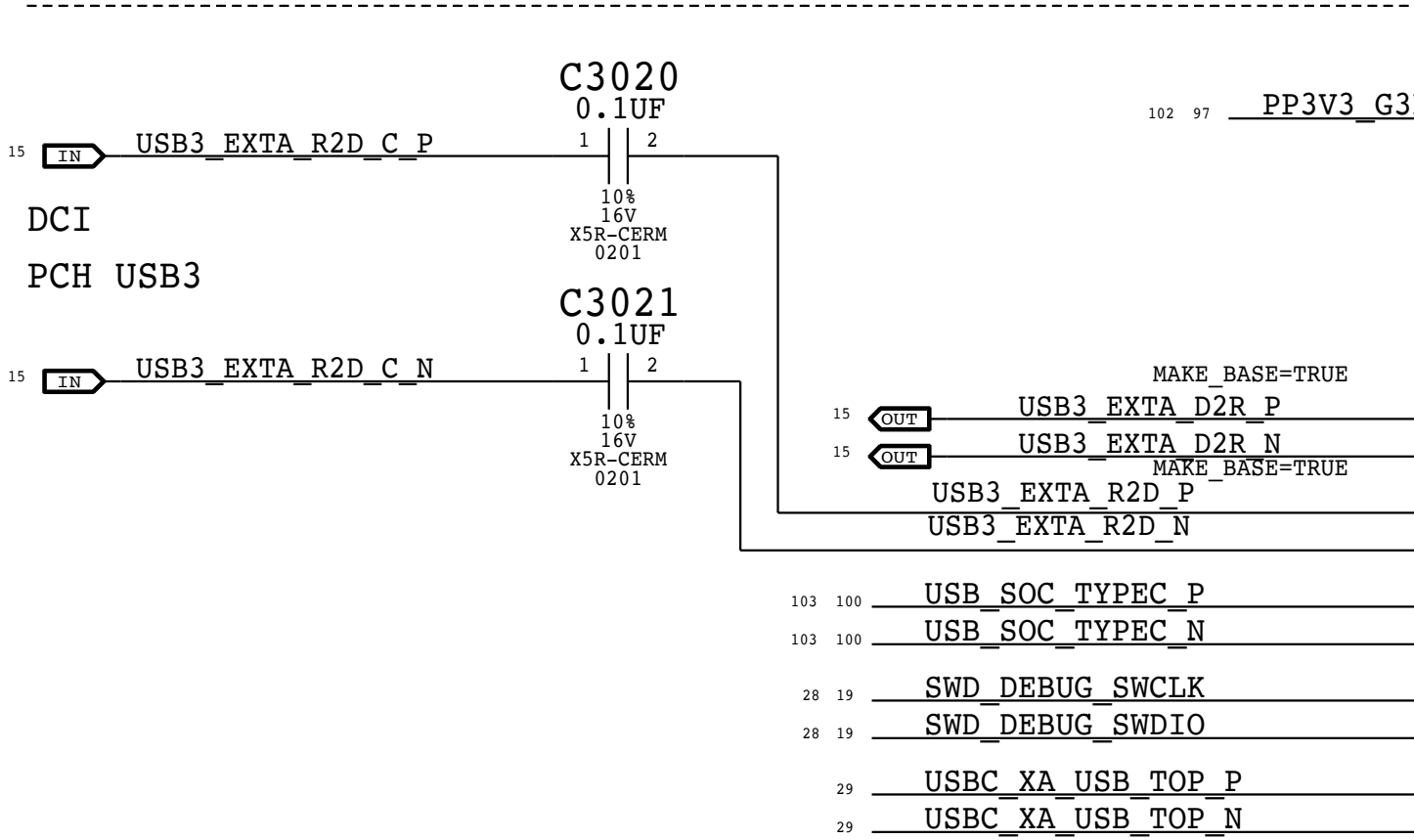
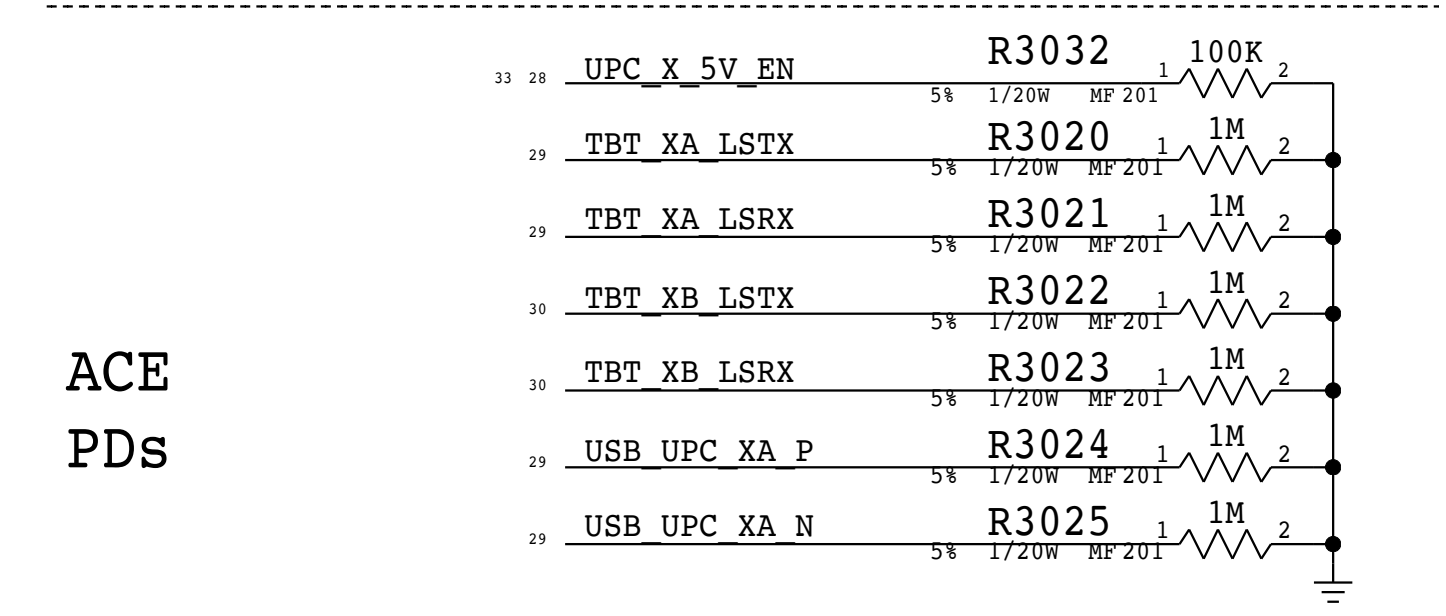
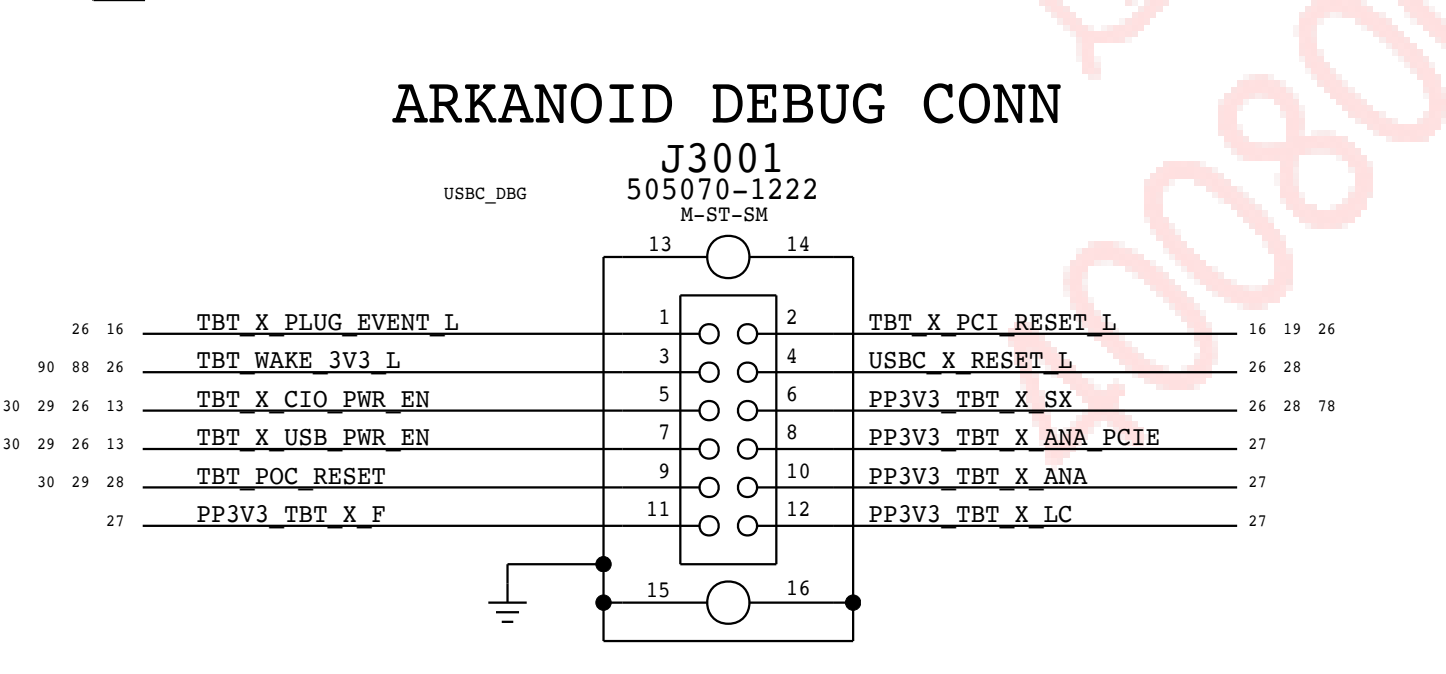
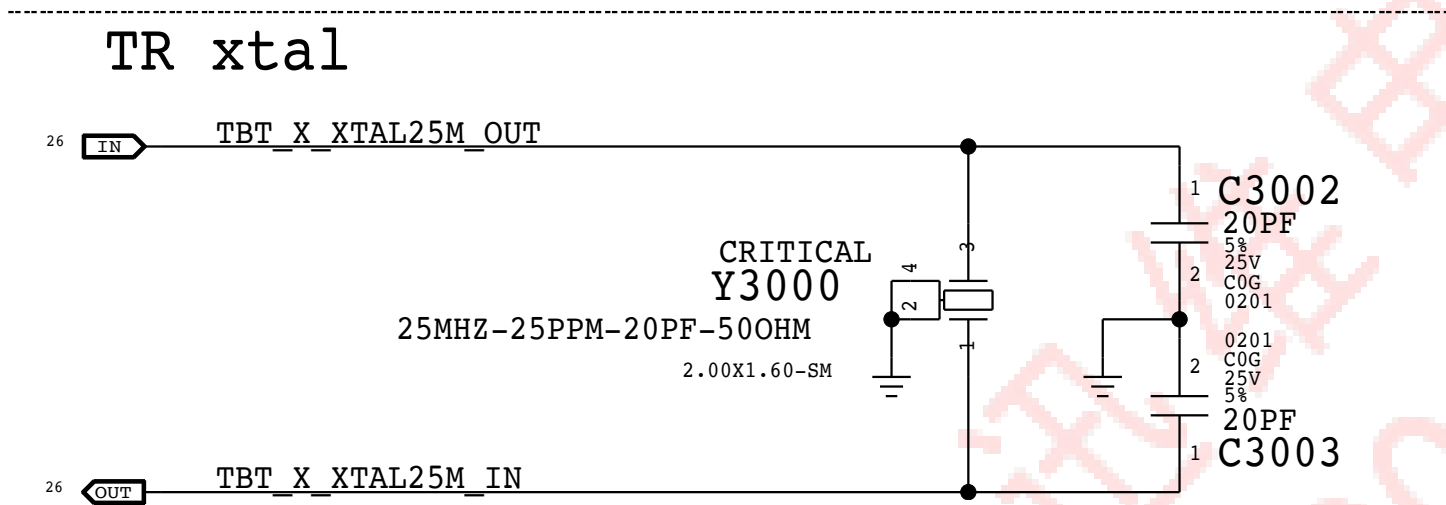
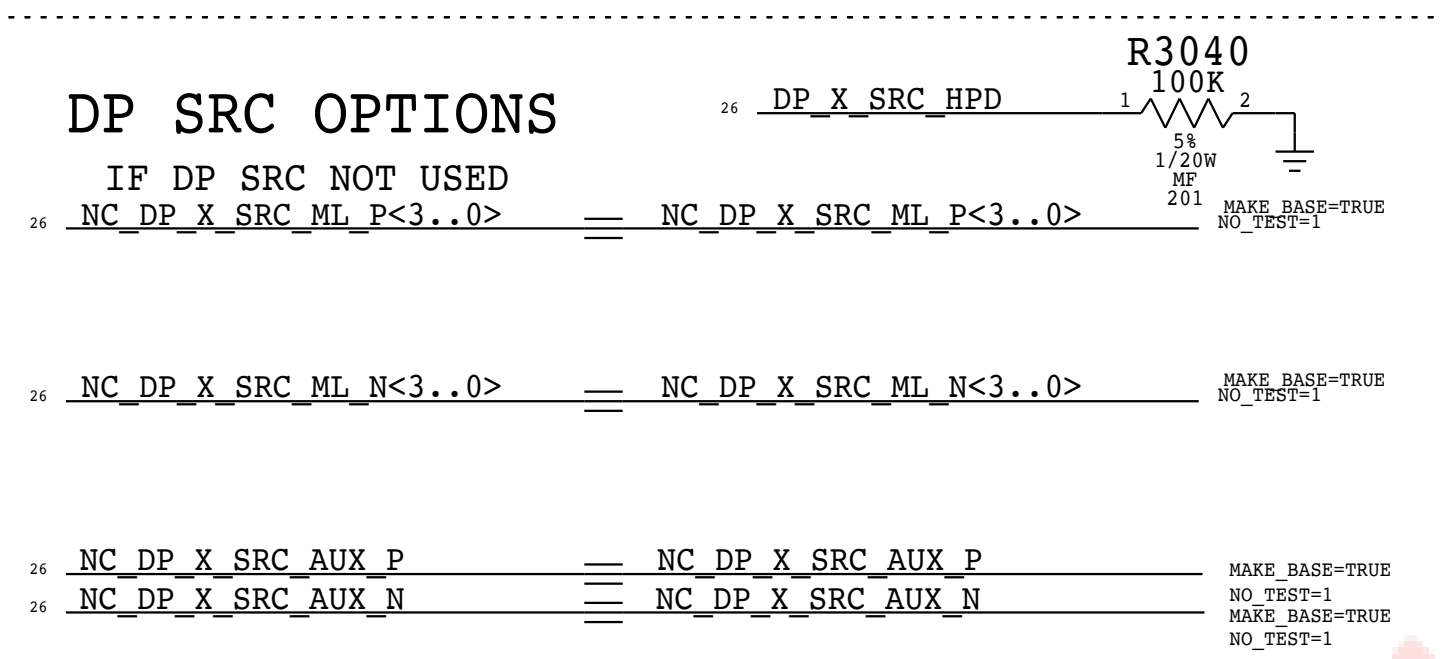
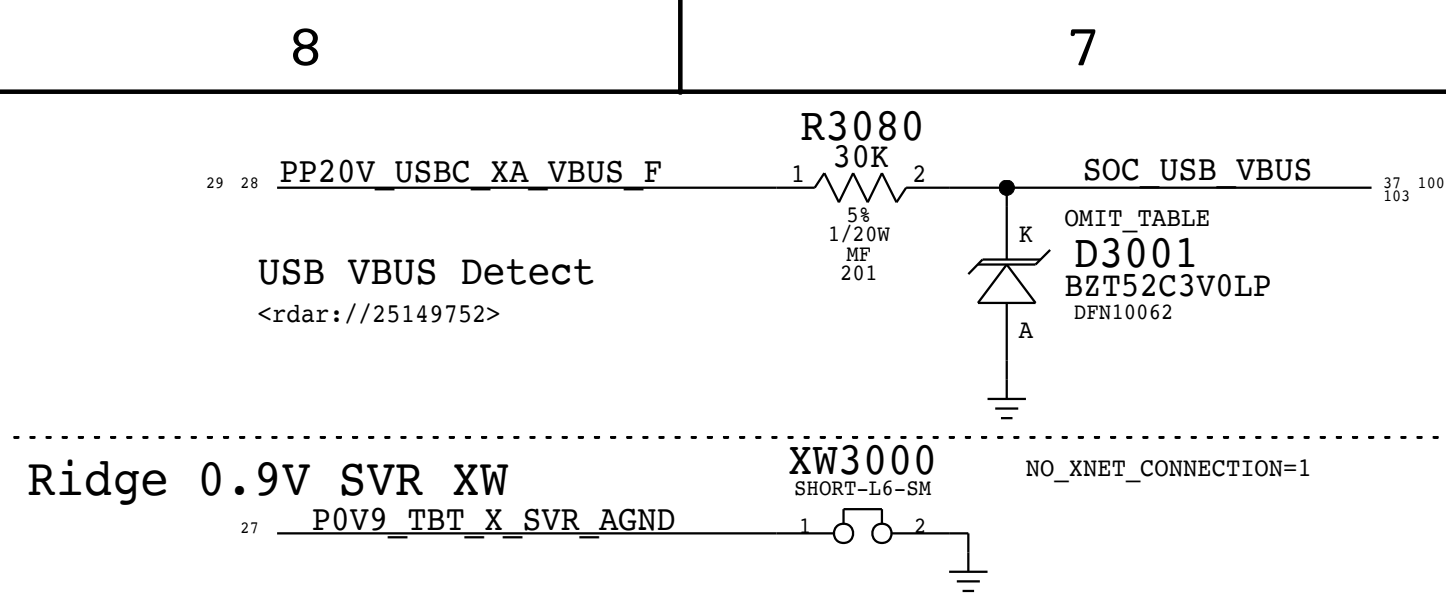




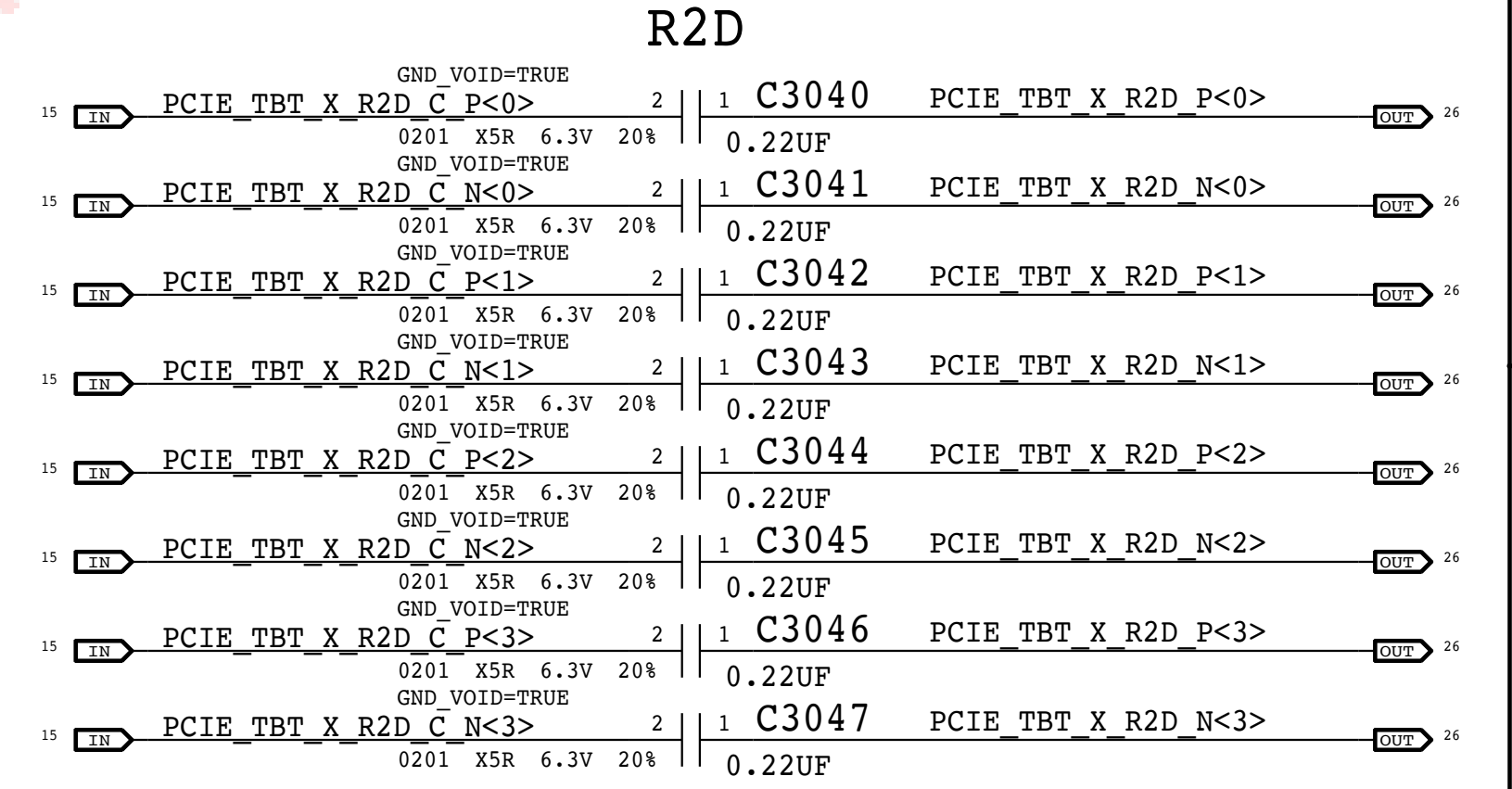
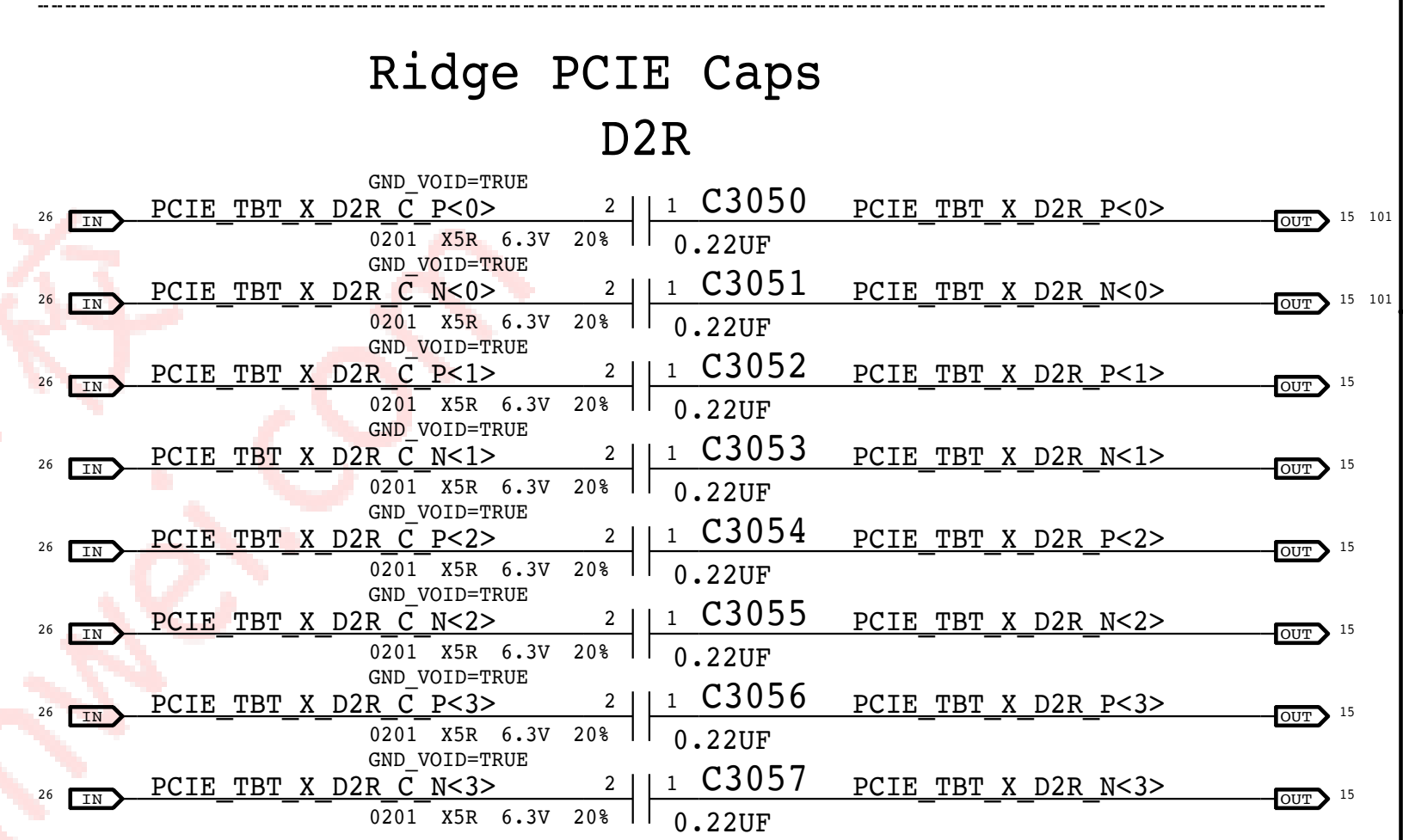
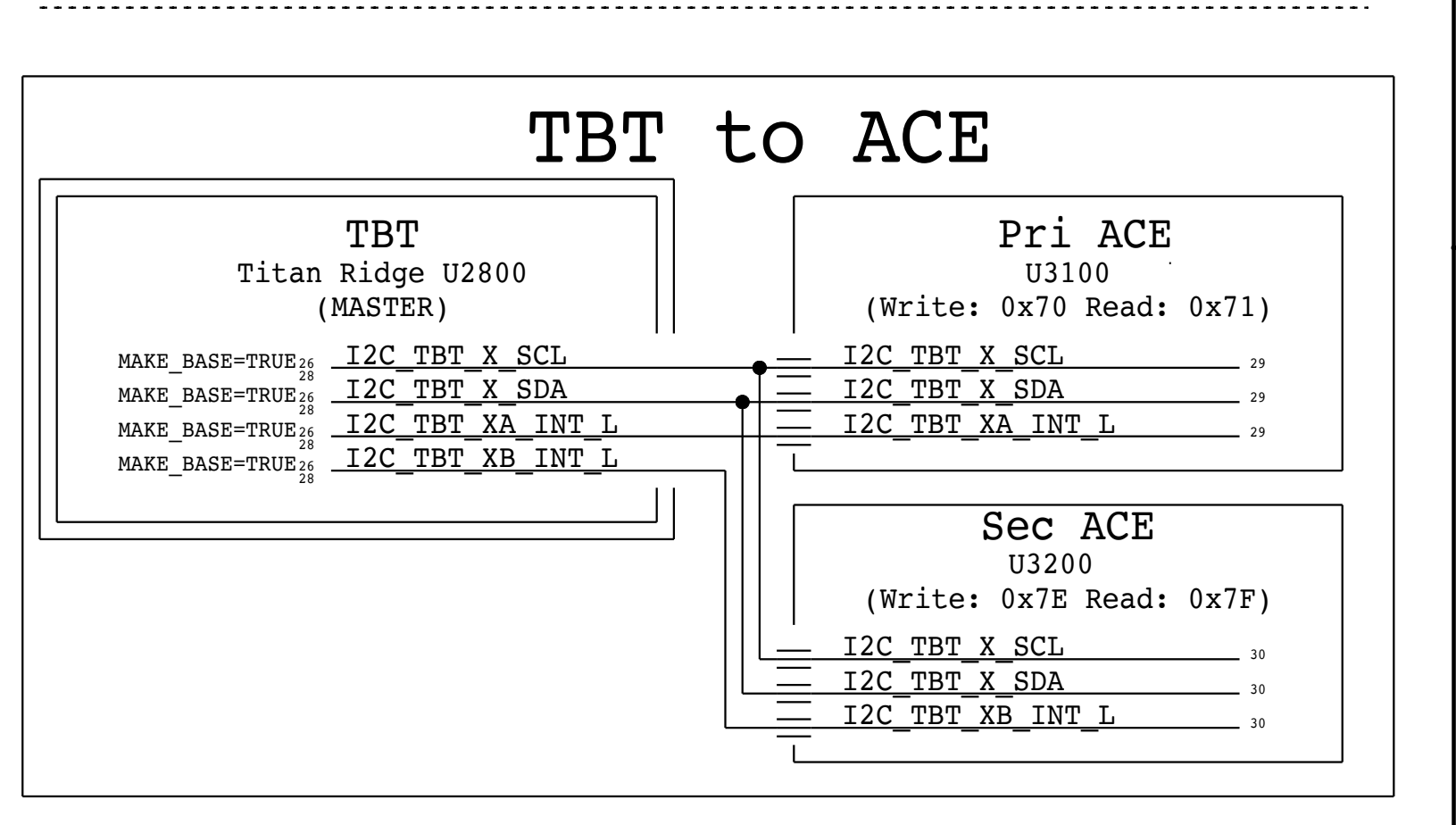
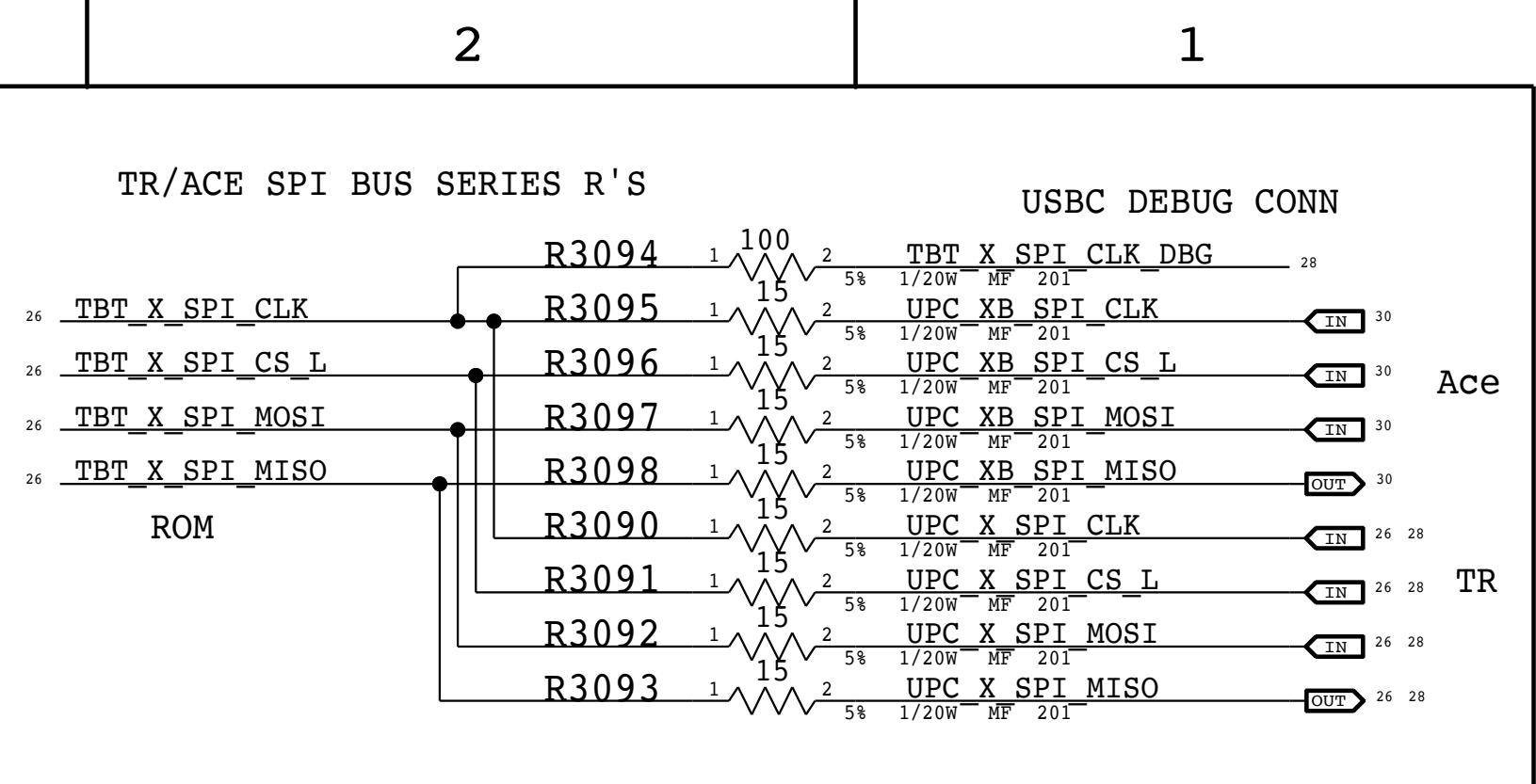
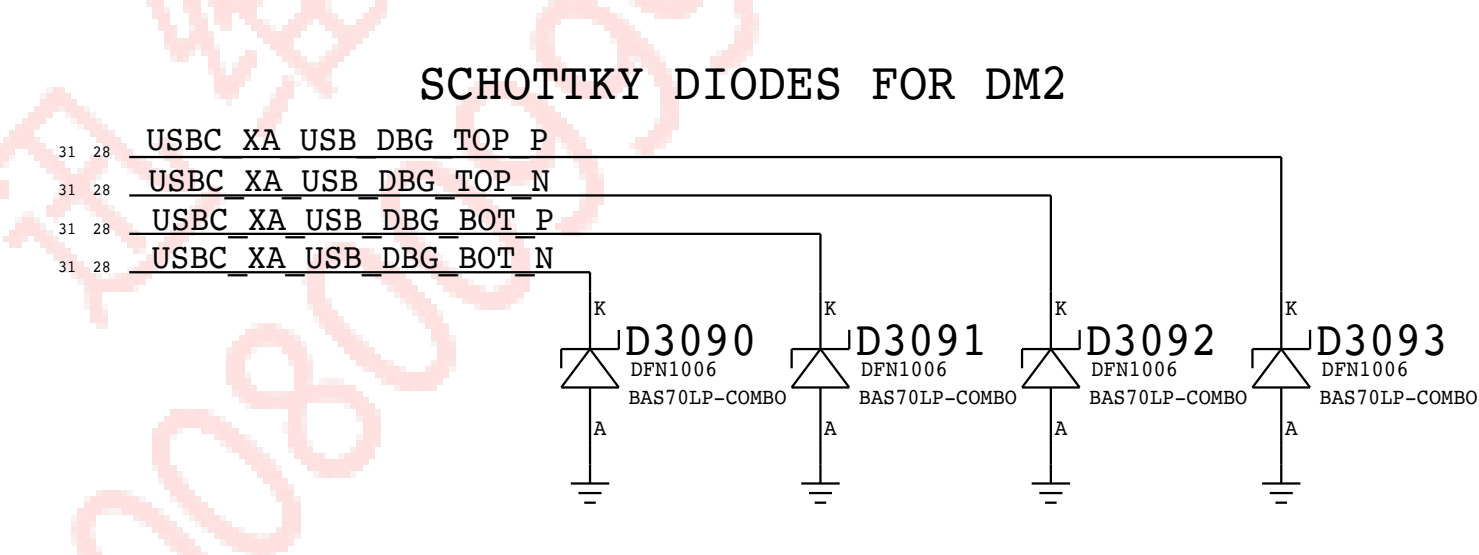
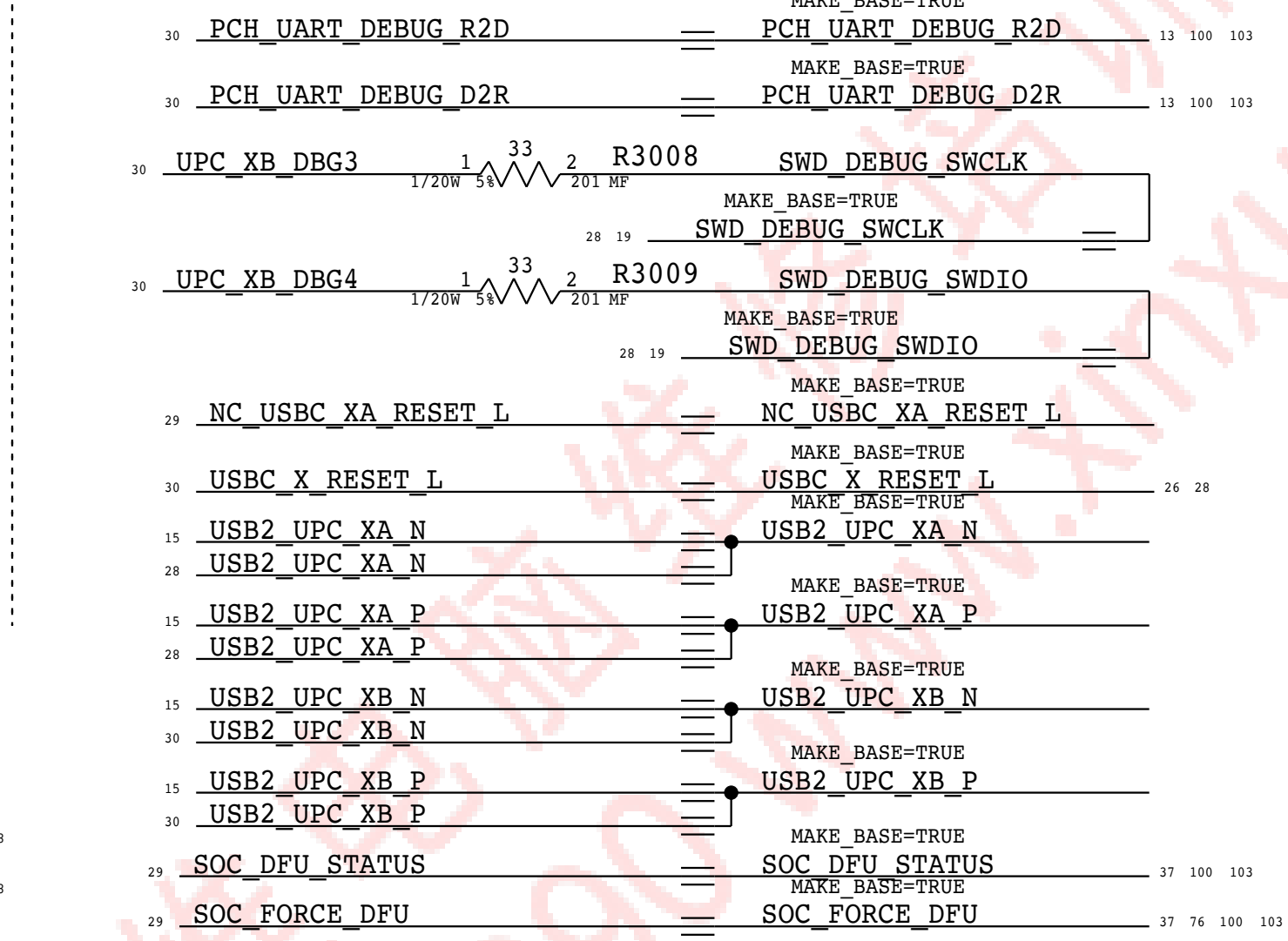
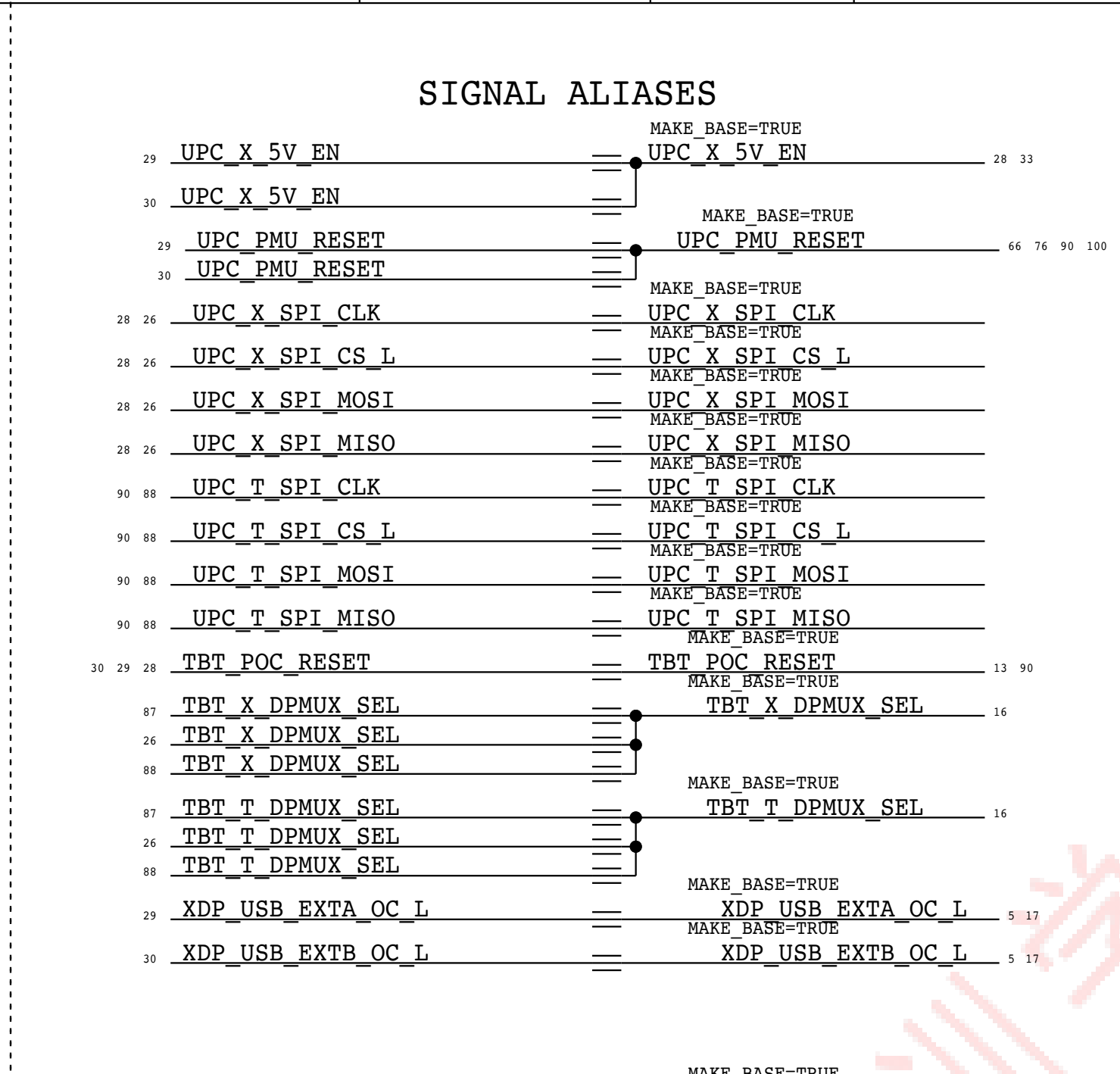








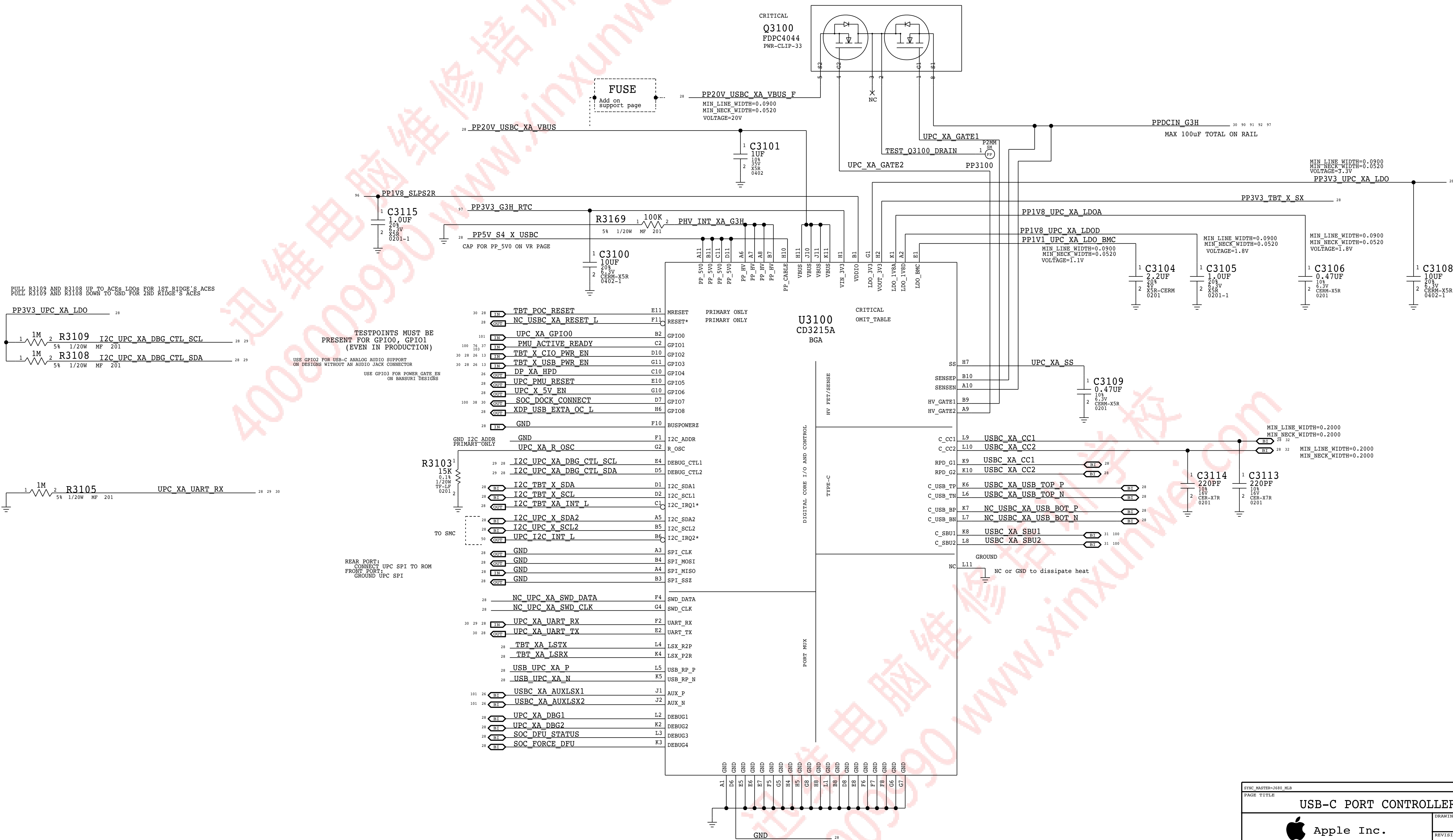
| DESCRIPTION                       | REFERENCE DES | CRITICAL | BOM OPTION |
|-----------------------------------|---------------|----------|------------|
| DIODE, ZENER, 3.0V, 74250NW, DFN2 | D3001         |          |            |



|                                                                                                                                                                                                                                                                                                                       |  |                |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                            |  |                |            |
|  Apple Inc.                                                                                                                                                                                                                      |  | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                       |  | REVISION       | 4.0.0      |
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PRIMARY ACE USB-C PORT CONTROLLER (UPC)

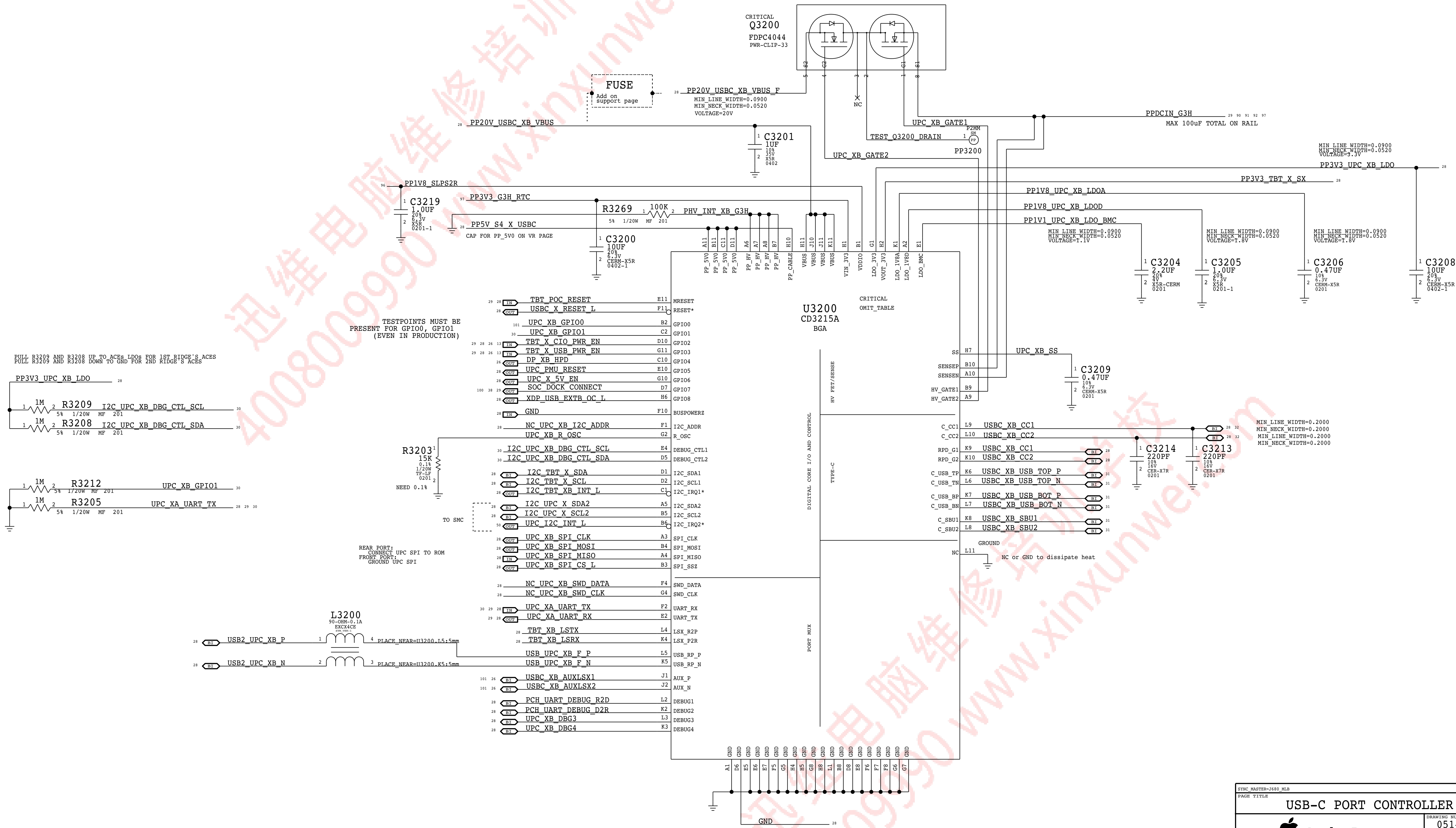


BOM\_COST\_GROUP=USB-C

|                                                                                                                                                                                                                                                                                                                   |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |            |
| USB-C PORT CONTROLLER A                                                                                                                                                                                                                                                                                           |                |            |
|                                                                                                                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02166  |
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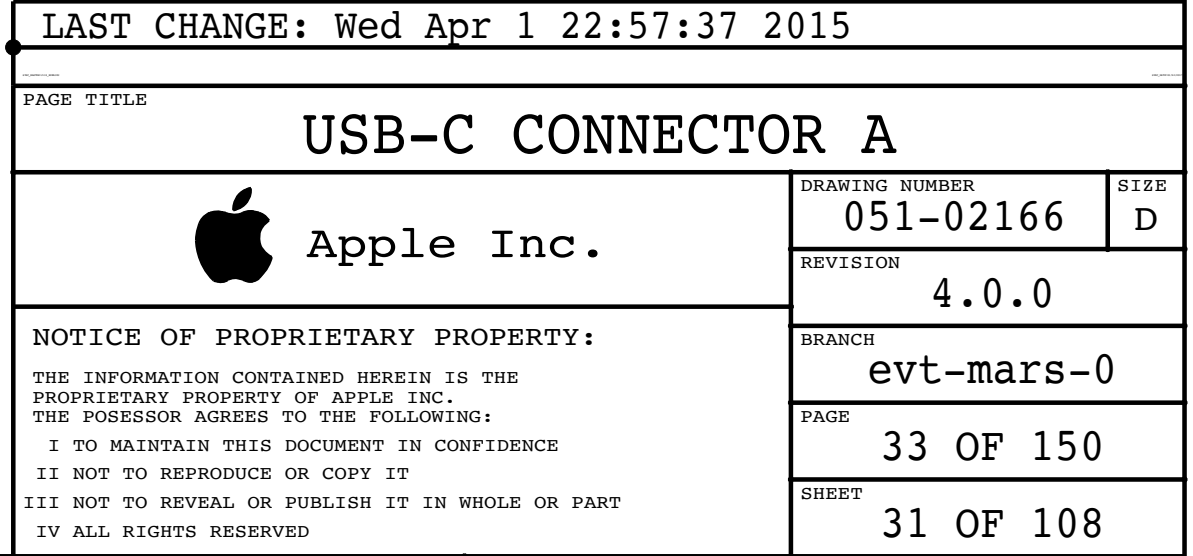
# SECONDARY ACE USB-C PORT CONTROLLER (UPC)



BOM COST GROUP=USB-C

|                                                                                                                                                                                                                                                                                                                             |                |                      |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|------|
| SYNC_MASTER=J680_MLB                                                                                                                                                                                                                                                                                                        |                | SYNC_DATE=03/30/2017 |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |                |                      |      |
| USB-C PORT CONTROLLER B                                                                                                                                                                                                                                                                                                     |                |                      |      |
|  Apple Inc.                                                                                                                                                                                                                            | DRAWING NUMBER |                      | SIZE |
|                                                                                                                                                                                                                                                                                                                             | 051-02166      |                      | D    |
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| SHEET                                                                                                                                                                                                                                                                                                                       |                |                      |      |
| 30 OF 108                                                                                                                                                                                                                                                                                                                   |                |                      |      |

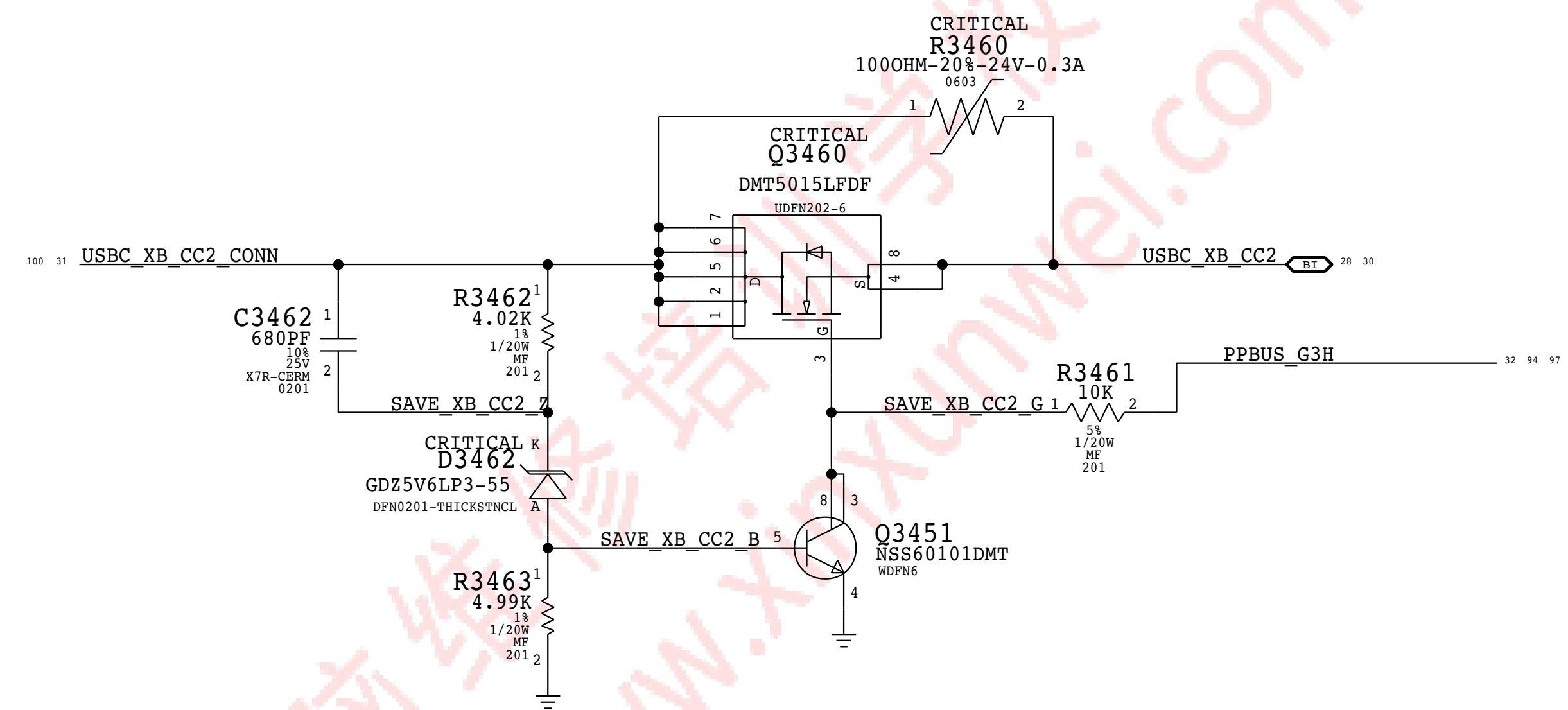
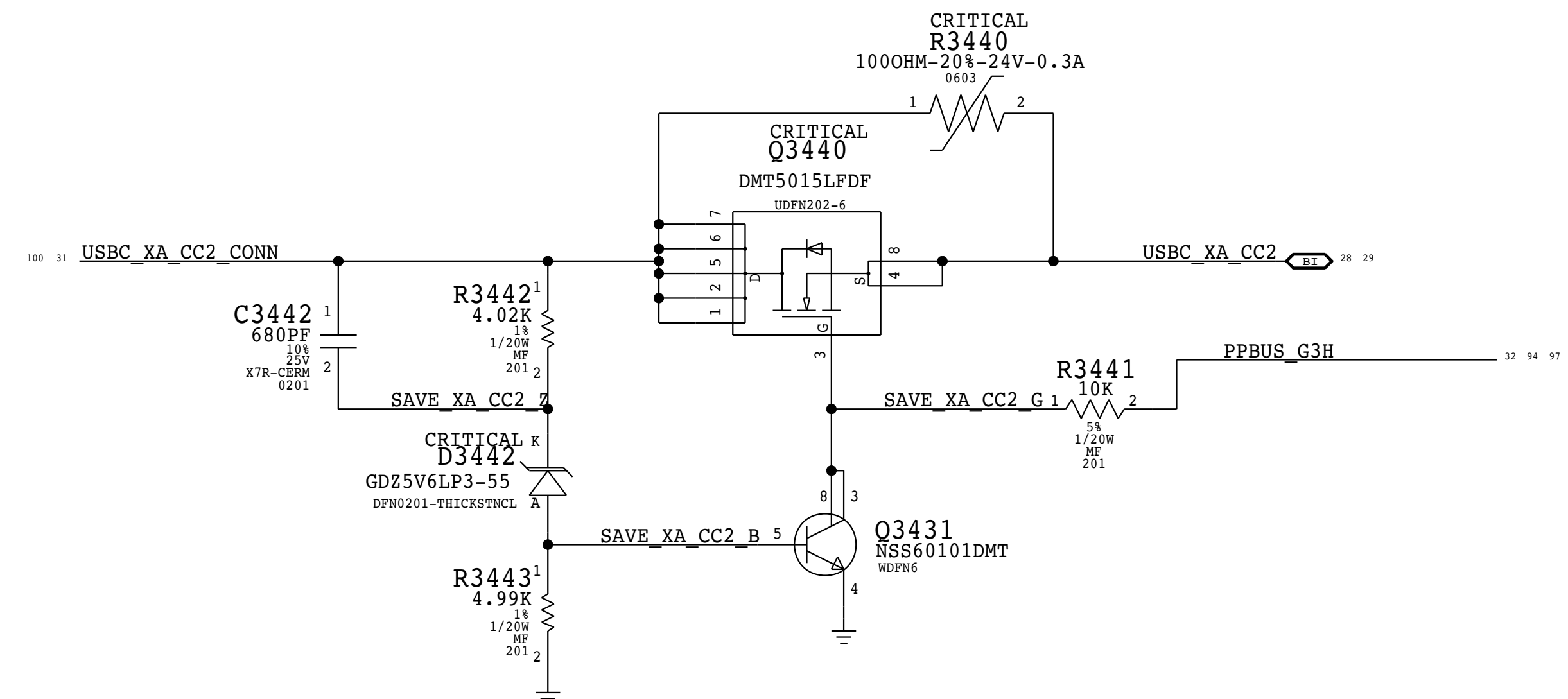






The schematic diagram illustrates the internal circuit of a USB to RS485 module. Key components and connections include:

- USB Connector:** USBC\_XA\_CC1\_CONN (pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100).
- Power Regulation:**
  - 5V Regulator (Q3431):** A 5V regulator (Q3431) powered by a 5V input (V5) and outputting 5V to the USB connector (USBC\_XA\_CC1).
  - 3.3V Regulator (Q3432):** A 3.3V regulator (Q3432) powered by a 3.3V input (V3.3) and outputting 3.3V to the USB connector (USBC\_XA\_CC1).
- USB to RS485 Converter:** The DMT5015LDFD chip converts USB signals to RS485 signals. It is powered by the 5V and 3.3V regulators and has its RS485 output connected to the 485 connector (485\_1).
- RS485 Connector:** The 485 connector (485\_1) provides the RS485 interface, with pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100.



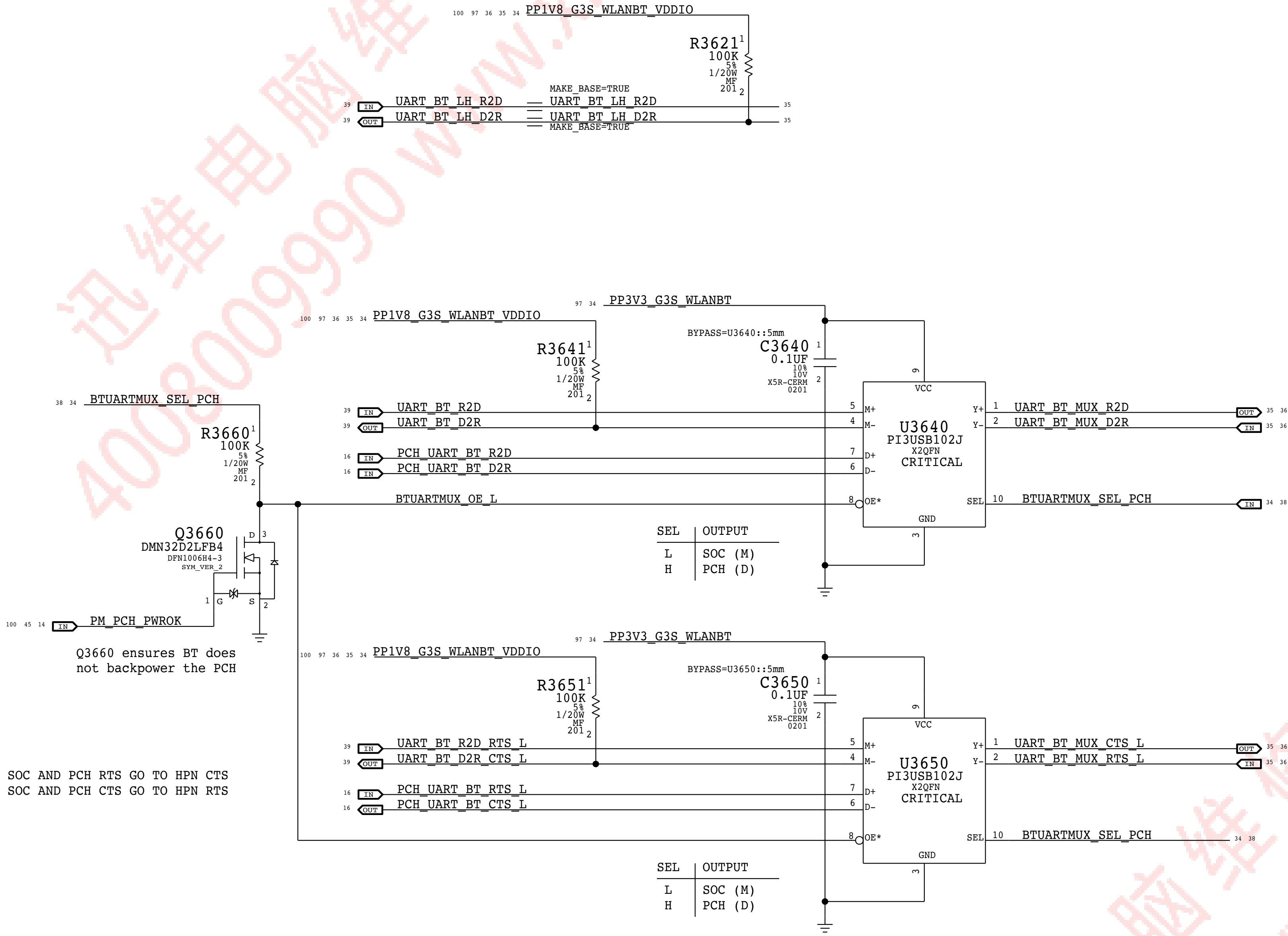
|                                                                                                                                                                                                                                                                                                                             |                |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------|
| DESIGN: X1032/MLB P4BP                                                                                                                                                                                                                                                                                                      |                |      |
| LAST CHANGE: Fri Jan 6 16:01:21 2017                                                                                                                                                                                                                                                                                        |                |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |                |      |
| USBC X Connector Support                                                                                                                                                                                                                                                                                                    |                |      |
|  Apple Inc.                                                                                                                                                                                                                            | DRAWING NUMBER | SIZE |
|                                                                                                                                                                                                                                                                                                                             | 051-02166      | D    |
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|                                                                                                                                                                                                                                                                                                                             | SHEET          |      |
| 32 OF 108                                                                                                                                                                                                                                                                                                                   |                |      |







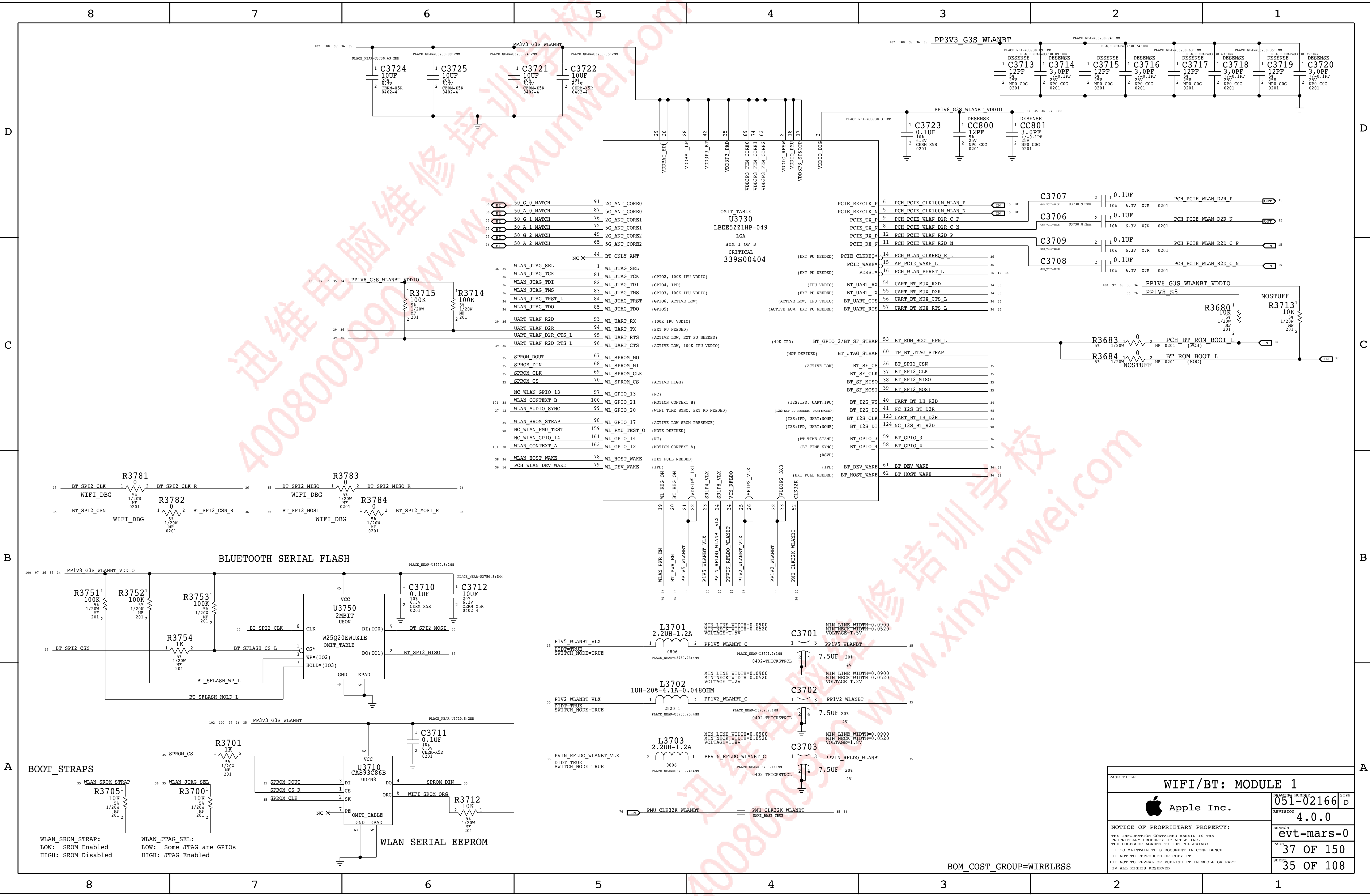
| Software        | I2S_SEL | UART_SEL | I2S I/F    | UART I/F   |
|-----------------|---------|----------|------------|------------|
| Gen1 (macOS)    | 0       | 1        | UART (PCH) | UART (PCH) |
| Gen1 (Bootcamp) | 1       | 1        | I2S (PCH)  | UART (PCH) |
| Gen2            | X       | 0        | X          | UART (SOC) |



|                                                                                                                                                                                                                                                                                                                               |  |                      |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|------------|
| SYNC_MASTER=J132 JACK                                                                                                                                                                                                                                                                                                         |  | SYNC_DATE=05/30/2017 |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                    |  |                      |            |
| WIFI/BT: Support                                                                                                                                                                                                                                                                                                              |  |                      |            |
|  Apple Inc.                                                                                                                                                                                                                              |  | DRAWING NUMBER       | 051-02166  |
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BOM\_COST\_GROUP=WIRELESS

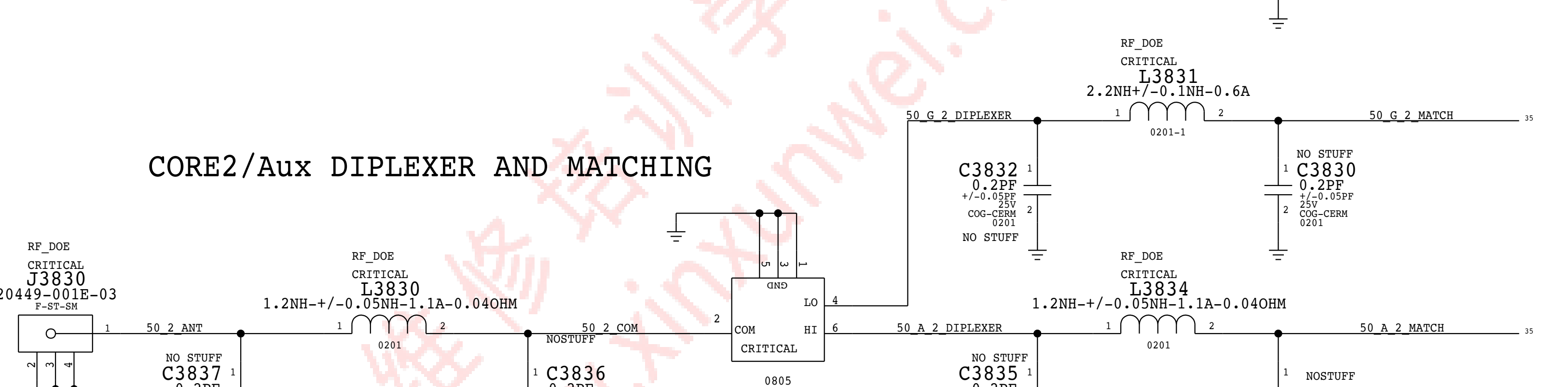
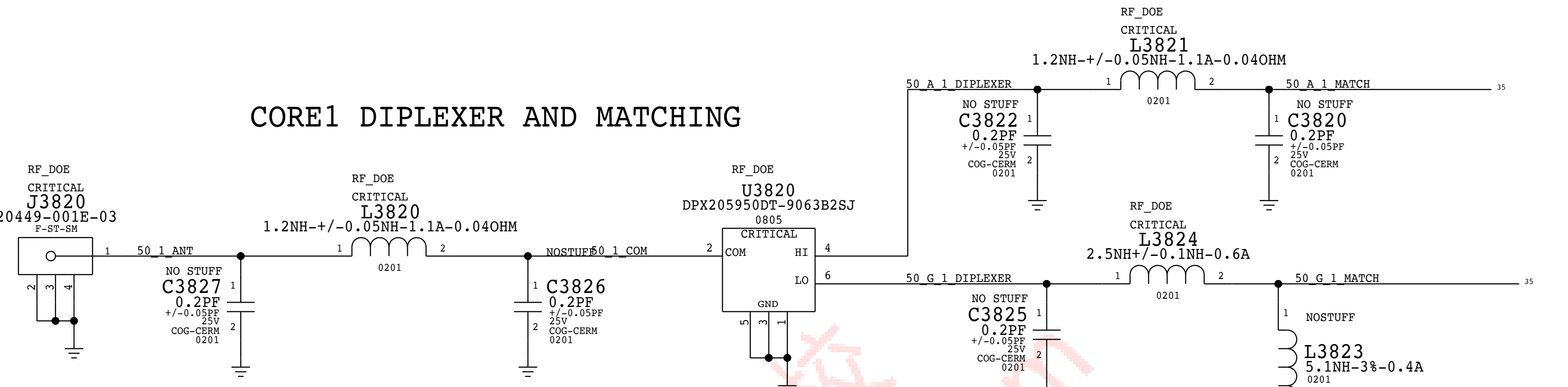
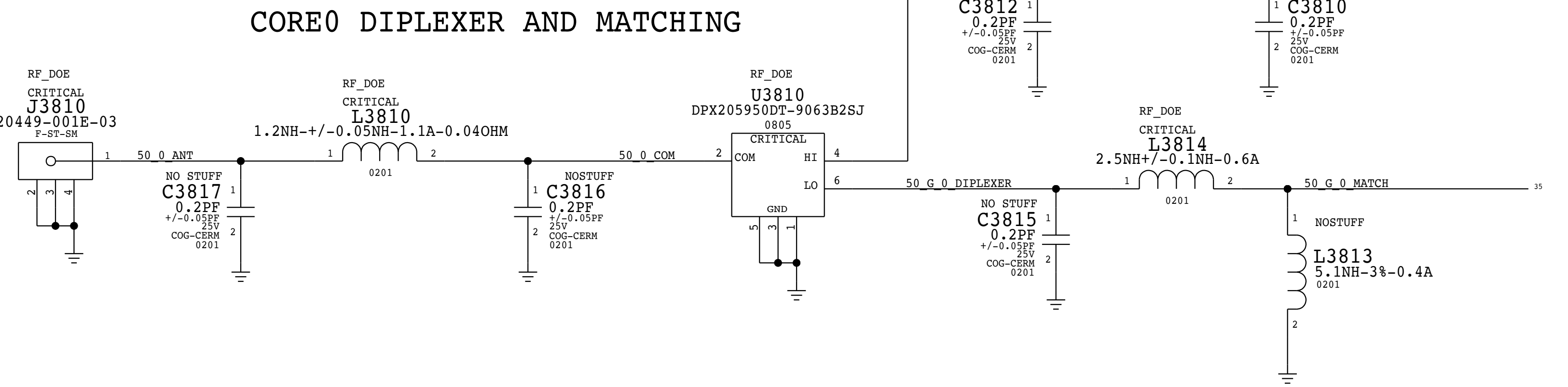
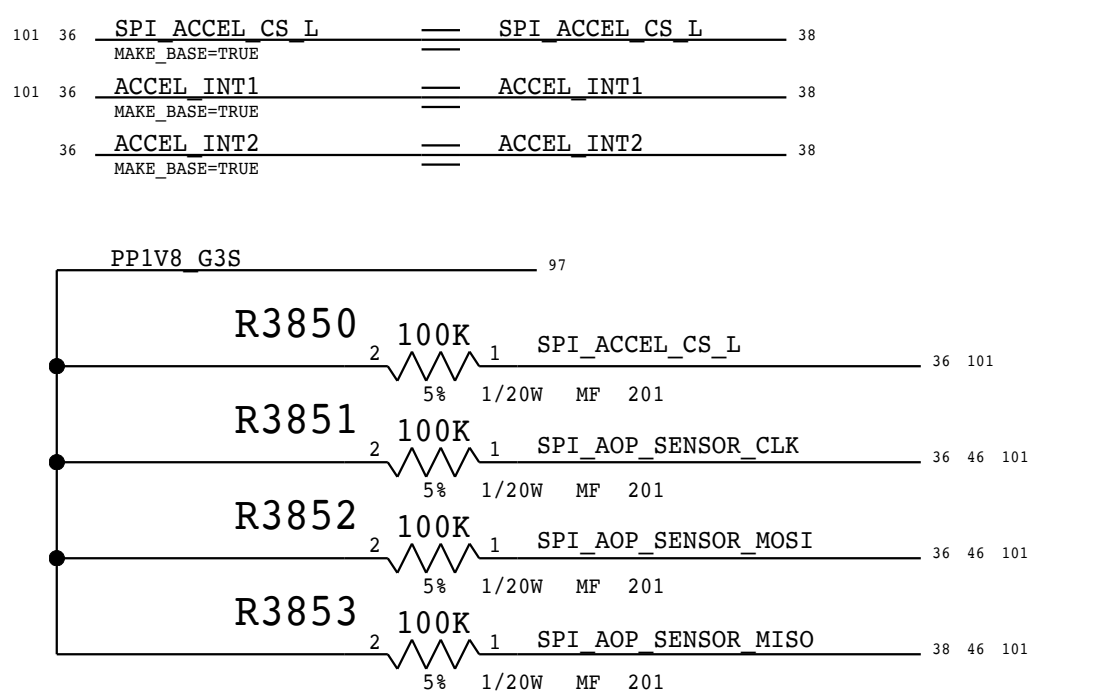
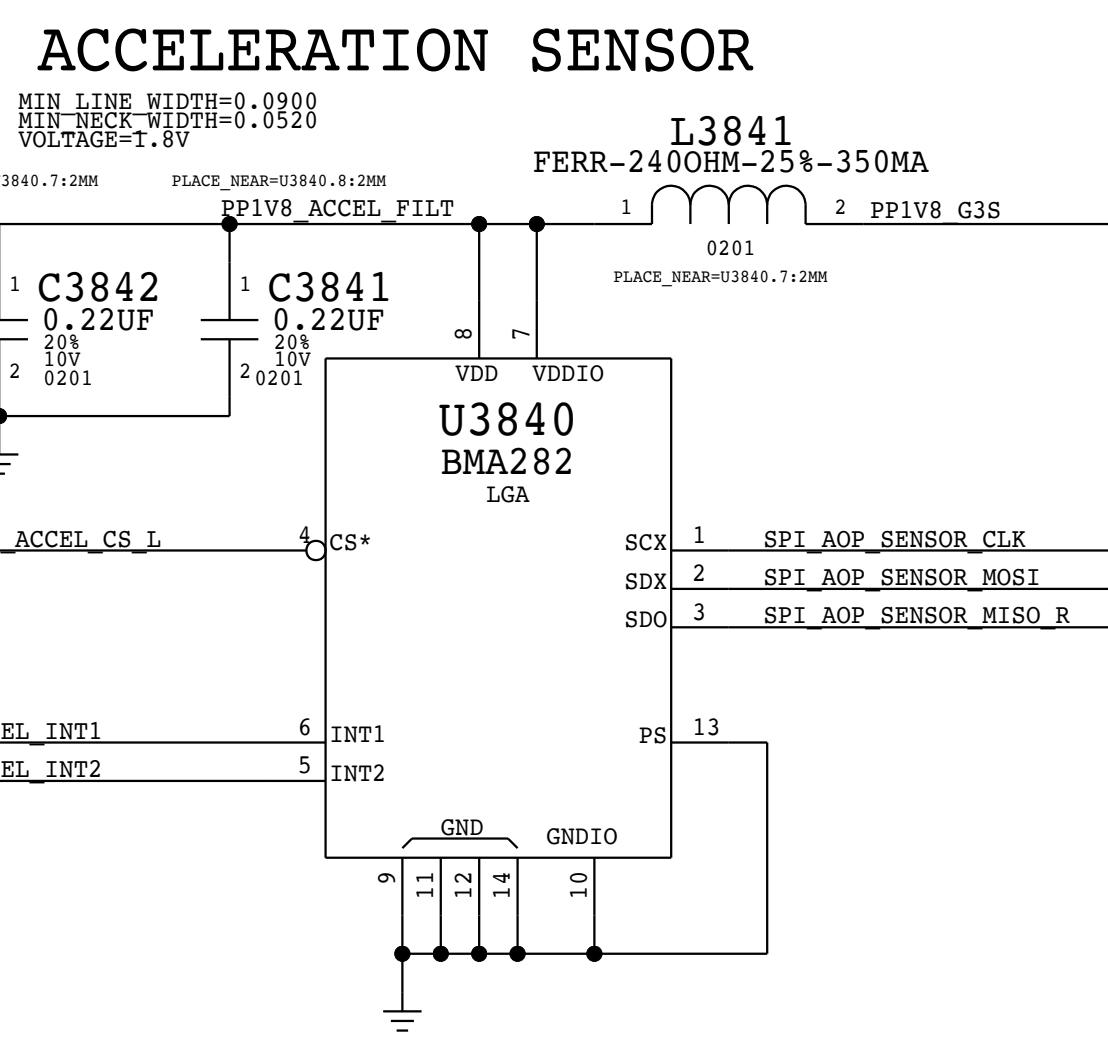
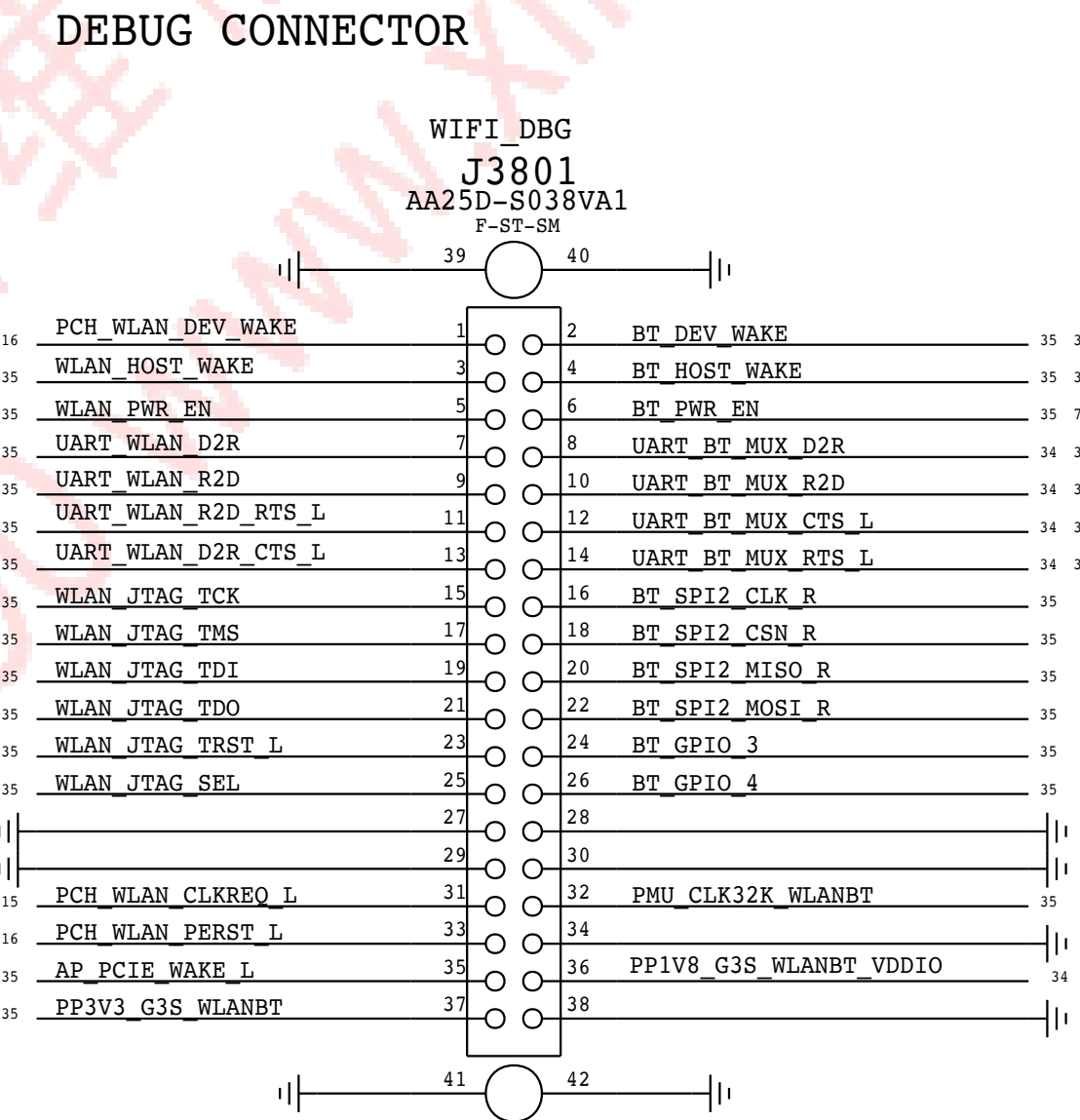
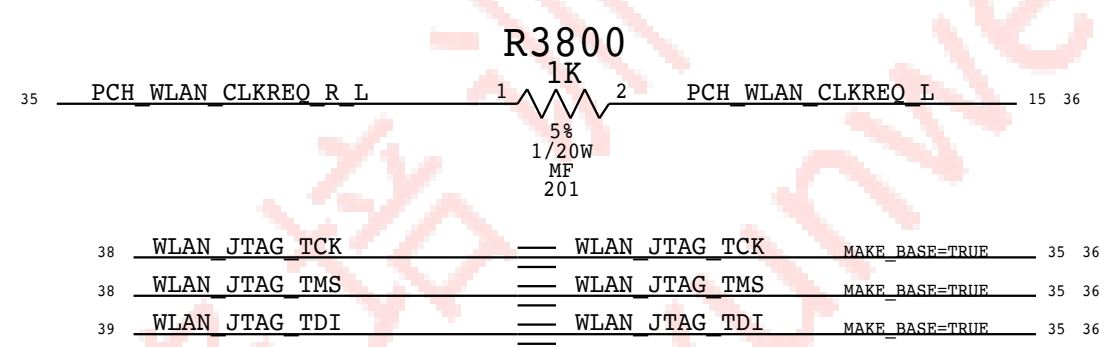
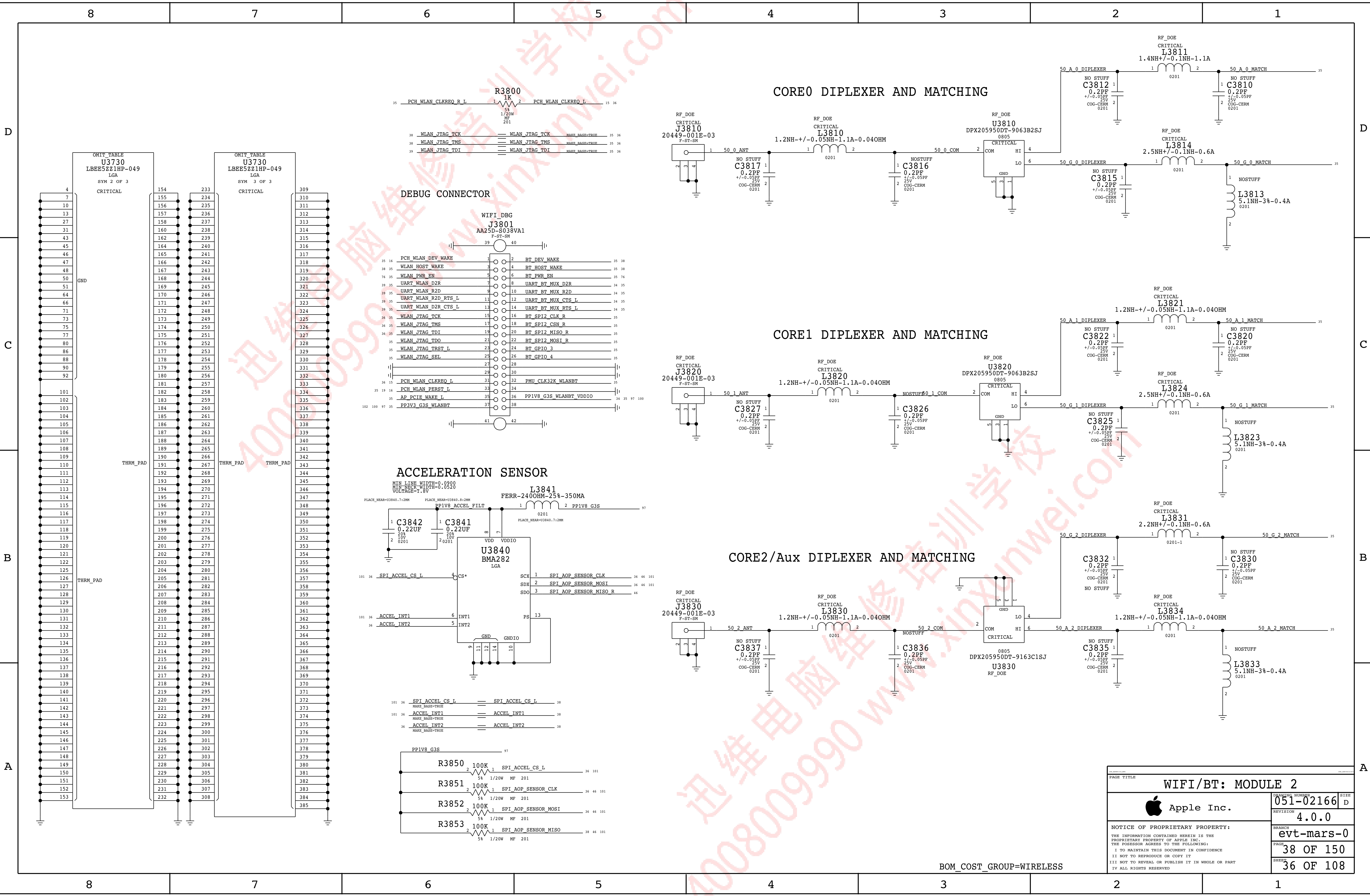




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| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |                    |
| WIFI/BT: MODULE 1                                                                                                                                                                                                                                                                                                 |                |                    |
|                                                                                                                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02166          |
|                                                                                                                                                                                                                                                                                                                   | REVISION       | 4.0.0              |
|                                                                                                                                                                                                                                                                                                                   | BRANCH         | evt-mars-0         |
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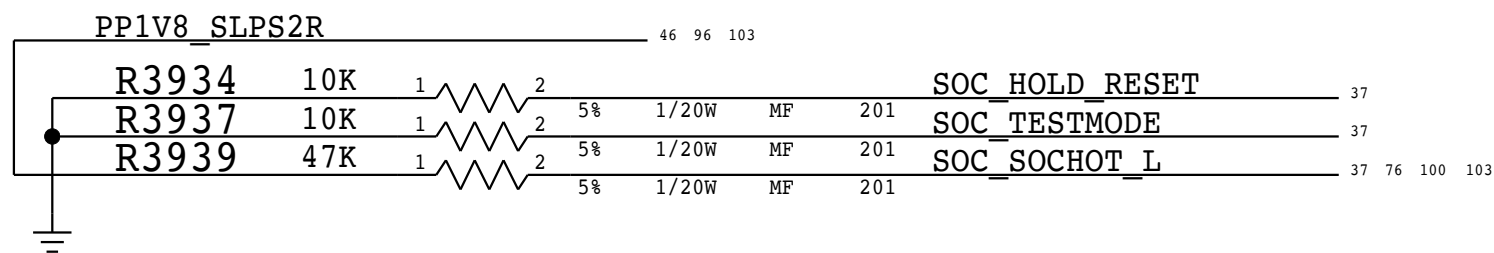
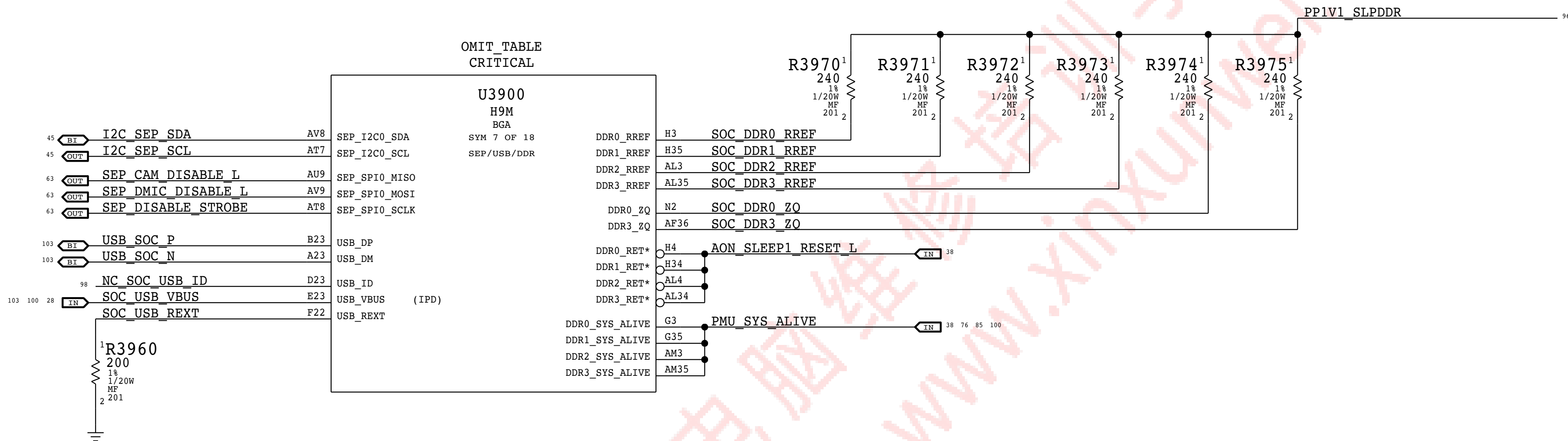
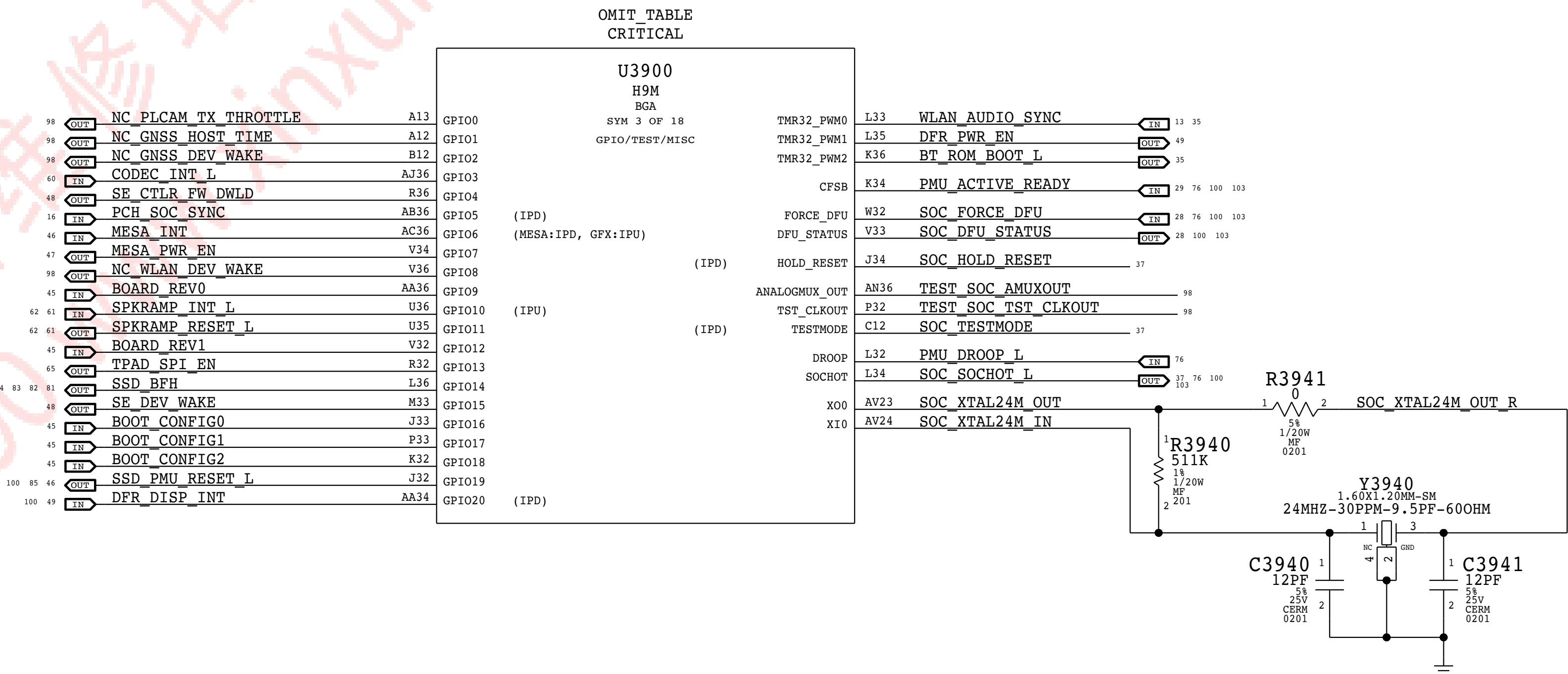


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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |           |            |
| WIFI/BT: MODULE 2                                                                                                                                                                                                                                                                                                 |                |           |            |
|                                                                                                                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02166 | SIZE       |
|                                                                                                                                                                                                                                                                                                                   | REVISION       | 4.0.0     | D          |
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|                                                                                                                                                                                                                                                                                                                   |                |           |            |

BOM\_COST\_GROUP=WIRELESS



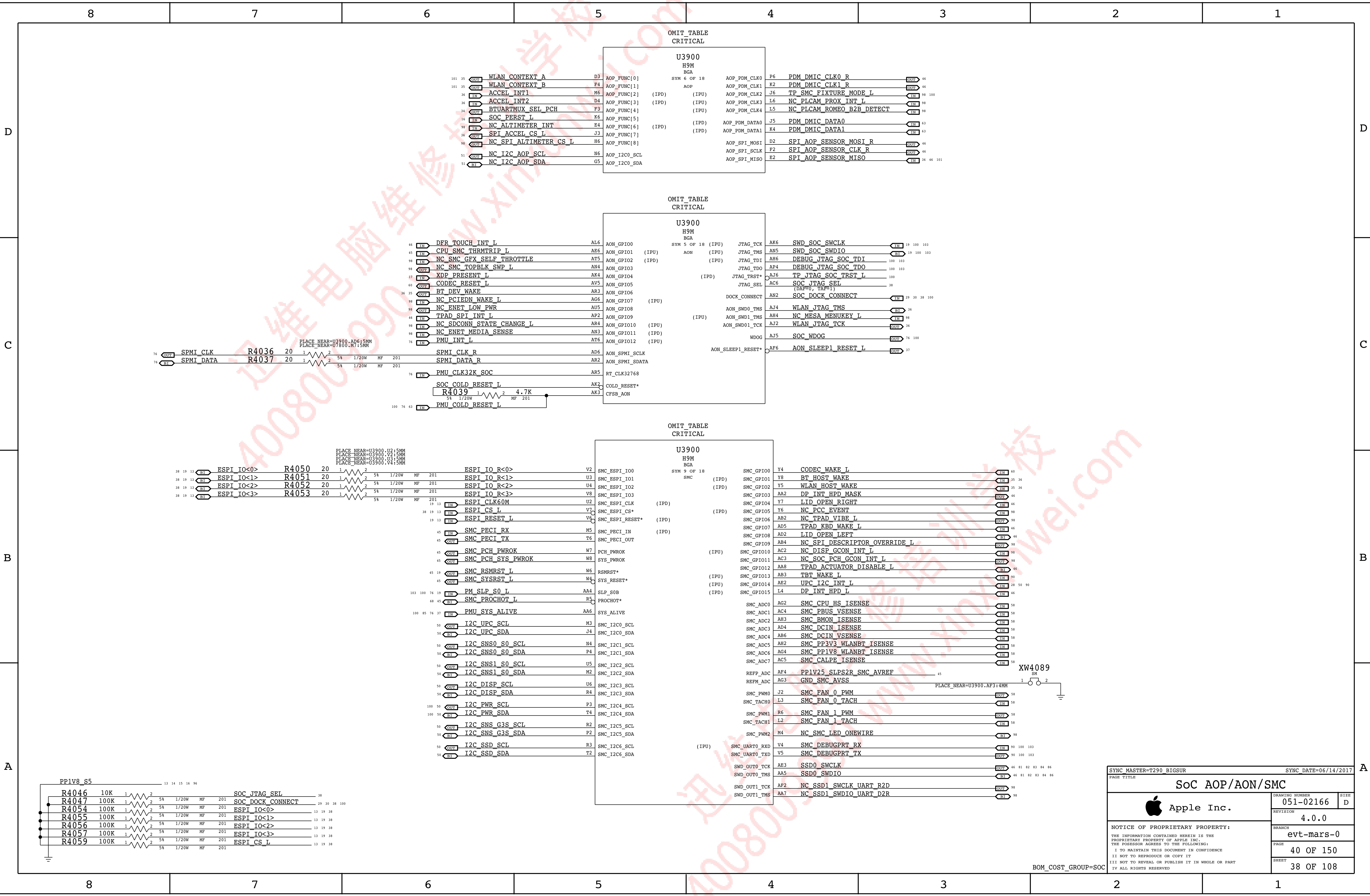
Note 1) IPU represents SW configured state, not HW default



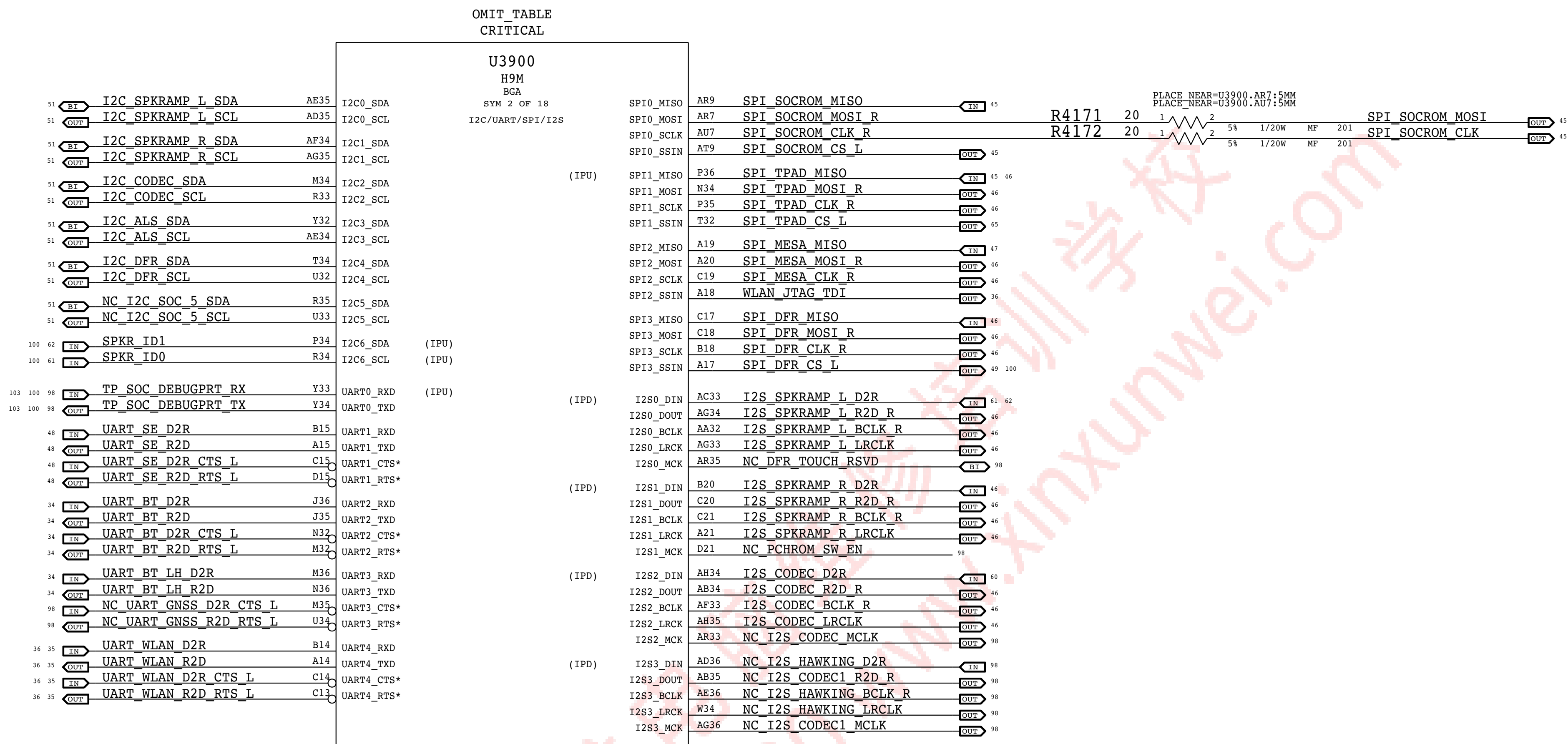
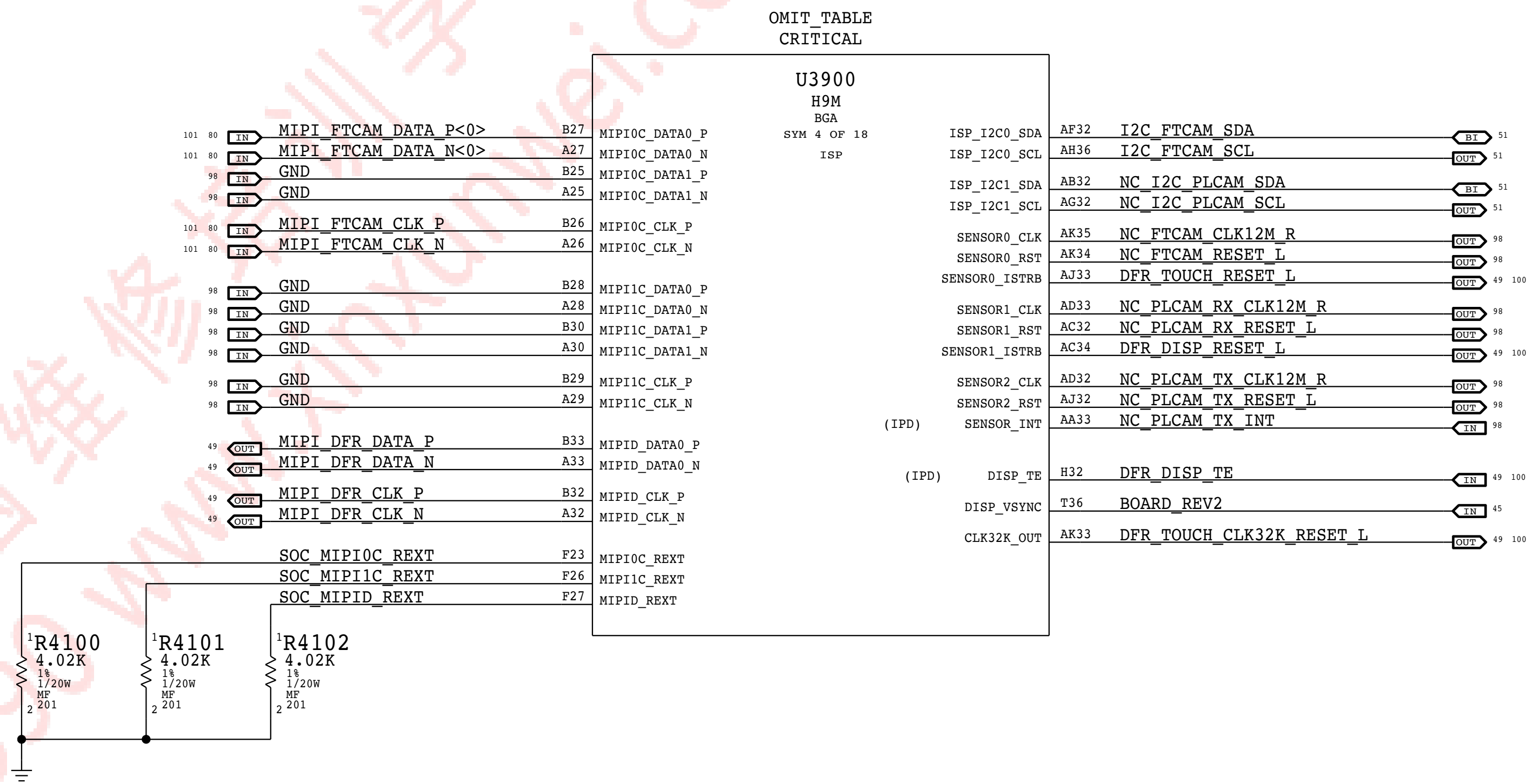
BOM\_COST\_GROUP=SOC

|                                                                                                                                                                                                                                                                                                                             |                |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|-----------|
| SYNC MASTER=T290 BIGSUR                                                                                                                                                                                                                                                                                                     |                | SYNC DATE=06/14/2017 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |                |                      |           |
| SoC GPIO/SEP/USB/DDR/Test                                                                                                                                                                                                                                                                                                   |                |                      |           |
|  Apple Inc.                                                                                                                                                                                                                            | DRAWING NUMBER |                      | SITE      |
|                                                                                                                                                                                                                                                                                                                             | 051-02166      |                      | D         |
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|                                                                                                                                                                                                                                                                                                                             | PAGE           |                      | 39 OF 150 |
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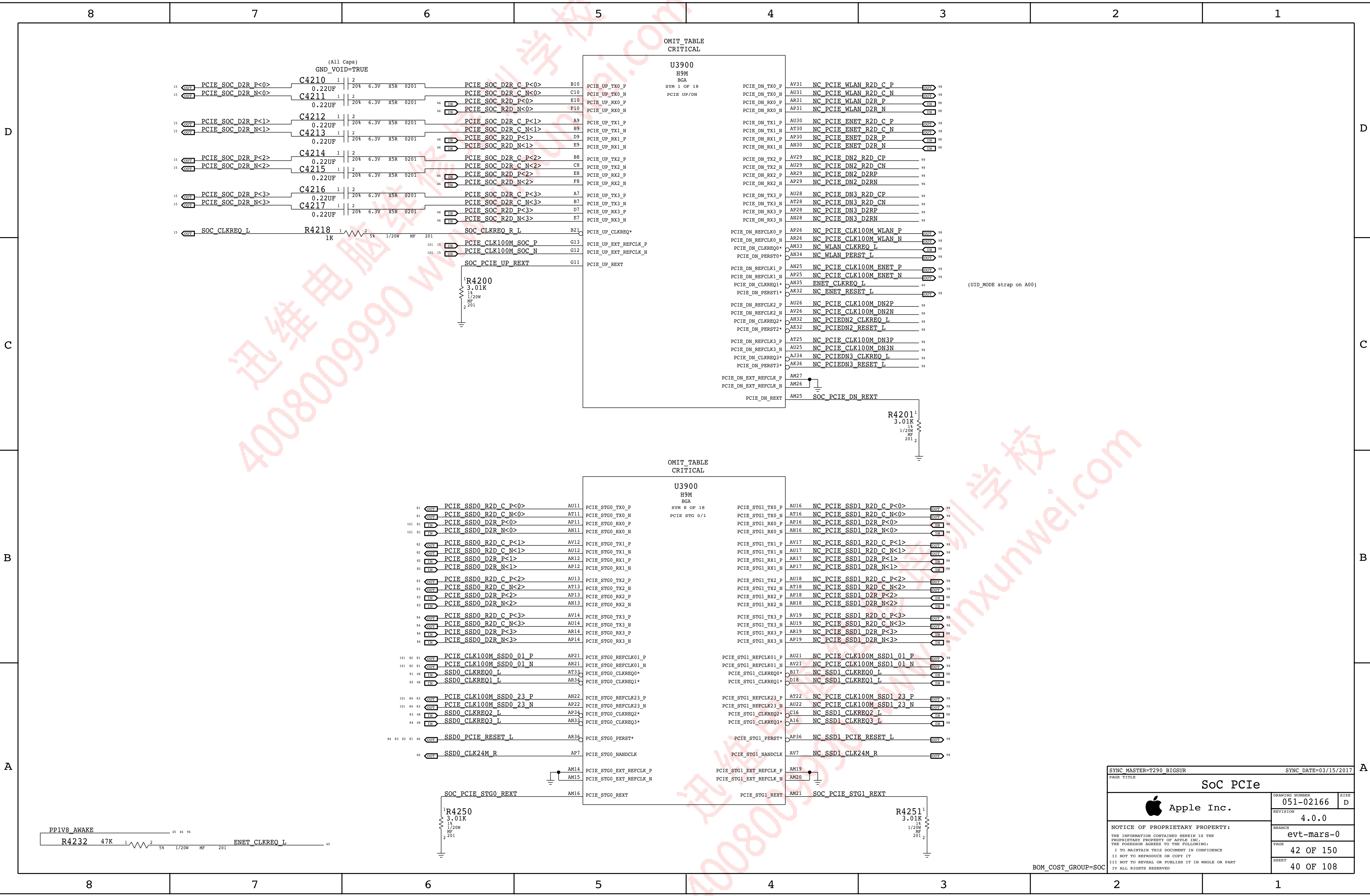




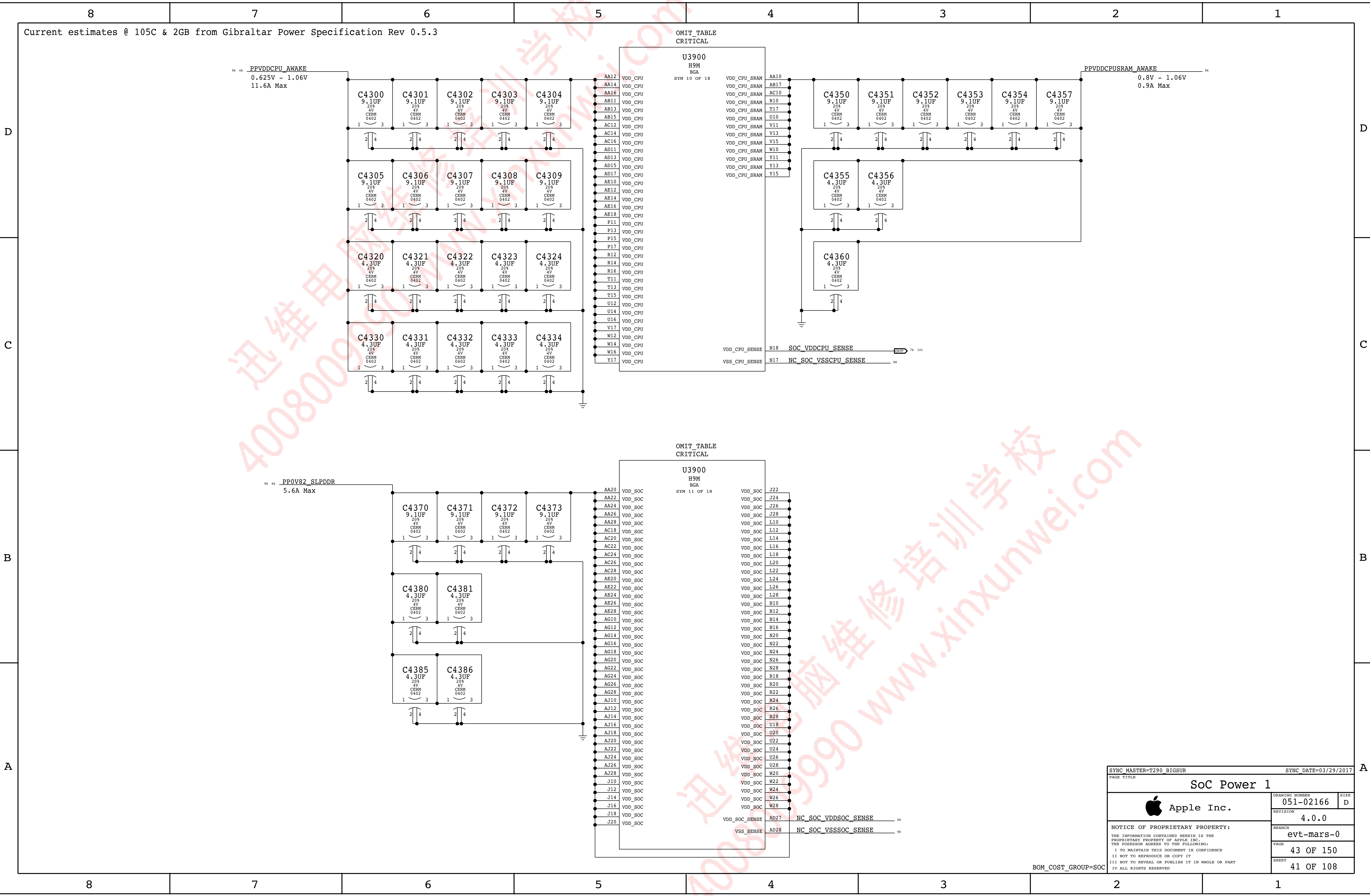
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| SYNC MASTER=T290 BIGSUR                                                                                                                                                                                                                                                                                                     |  | SYNC DATE=05/26/2017 |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |                      |            |
| SoC ISP/I2C/UART/SPI/I2S                                                                                                                                                                                                                                                                                                    |  |                      |            |
|  Apple Inc.                                                                                                                                                                                                                            |  | DRAWING NUMBER       | SIZE       |
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|                                                                                                                                                                                                                                                                                                                             |  |                      | evt-mars-0 |
|                                                                                                                                                                                                                                                                                                                             |  | PAGE                 | 41 OF 150  |
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|                                                                                                                                                                                                                                                                                                                             |  |                      | 39 OF 108  |

BOM\_COST\_GROUP=SOC









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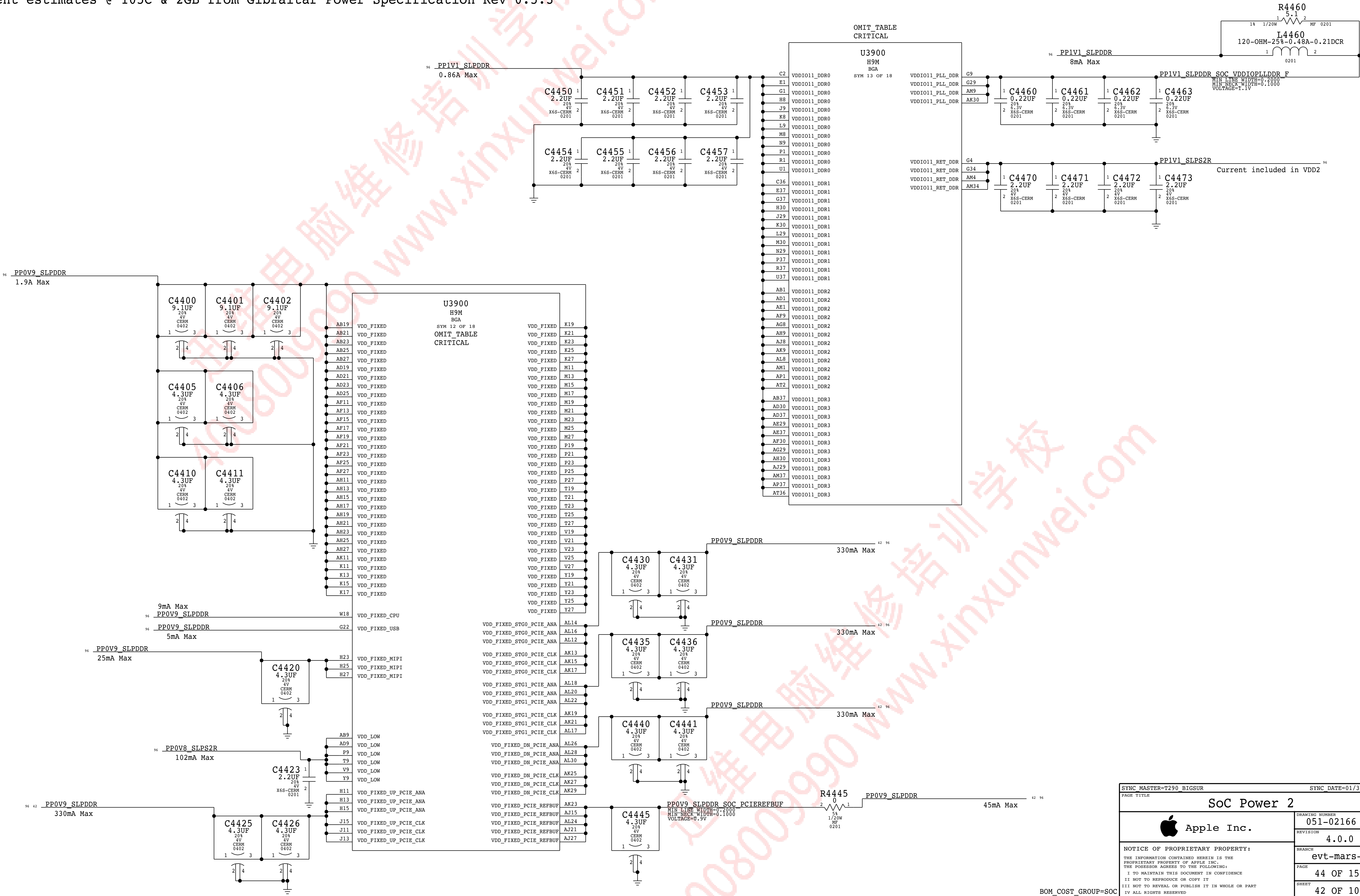
C

B

A



Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3

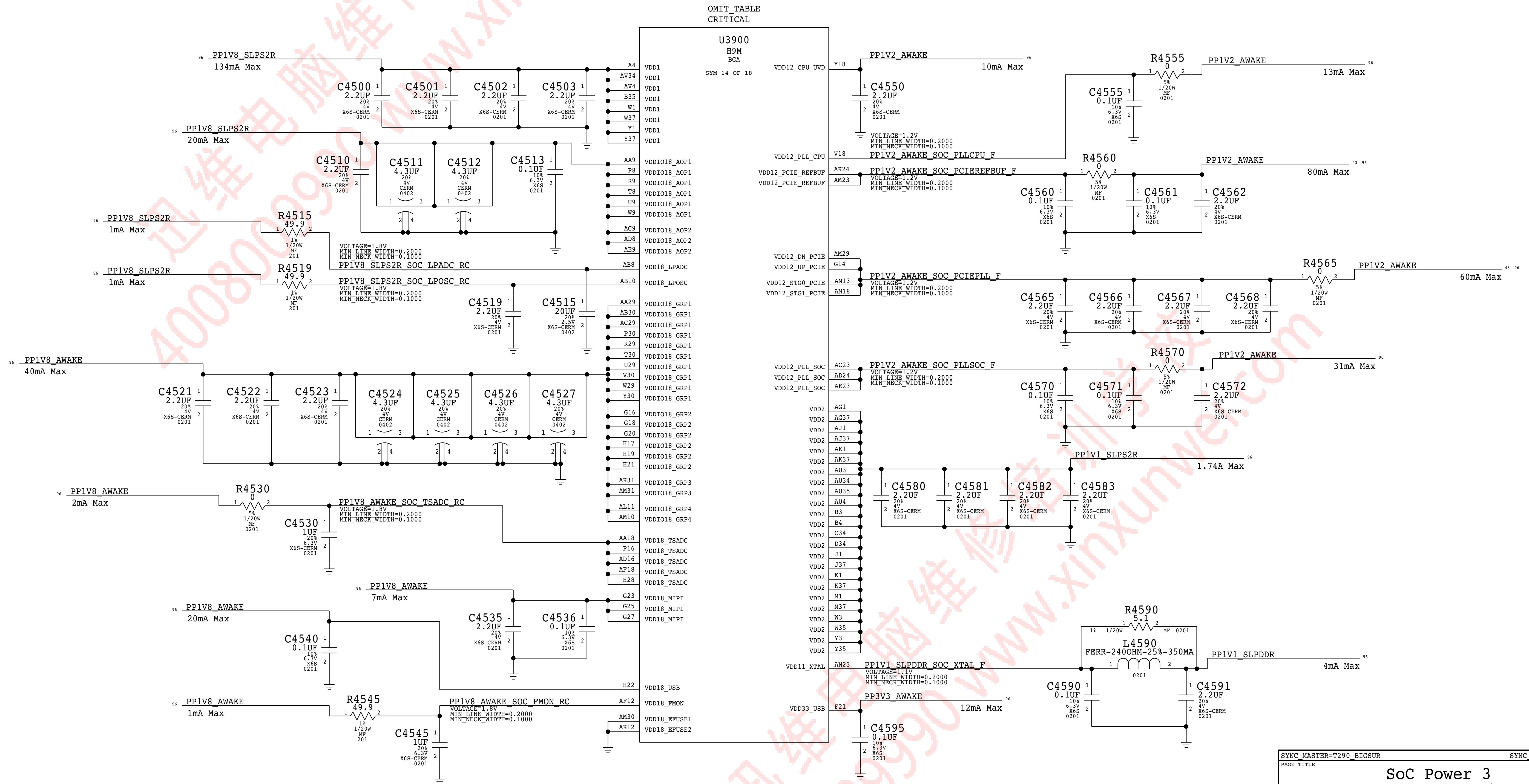


|                                                                                                                   |                |                      |            |
|-------------------------------------------------------------------------------------------------------------------|----------------|----------------------|------------|
| SYNC MASTER=T290 BIGSUR                                                                                           |                | SYNC DATE=01/31/2017 |            |
| PAGE TITLE                                                                                                        |                |                      |            |
| SoC Power 2                                                                                                       |                |                      |            |
|  Apple Inc.                  | DRAWING NUMBER |                      | SITE       |
|                                                                                                                   | 051-02166      |                      | D          |
|                                                                                                                   |                | REVISION             | 4.0.0      |
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BOM\_COST\_GROUP=SOC

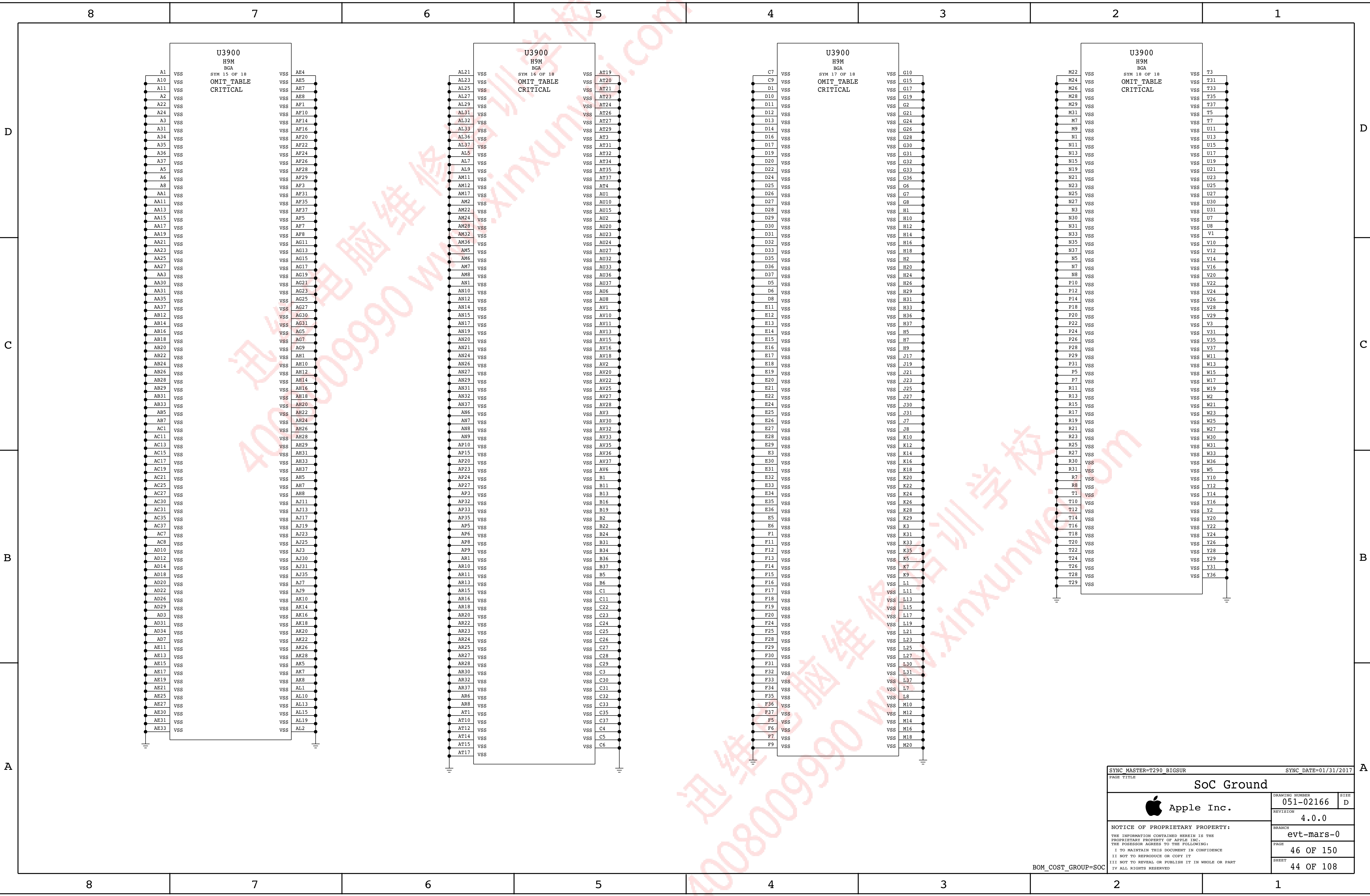


Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3



|                                                                                                                                                                                                                                                                                                                                  |  |                      |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|------------|
| SYNC MASTER=T290 BIGSUR                                                                                                                                                                                                                                                                                                          |  | SYNC DATE=08/15/2017 |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                       |  |                      |            |
| SoC Power 3                                                                                                                                                                                                                                                                                                                      |  |                      |            |
|  Apple Inc.                                                                                                                                                                                                                                 |  | DRAWING NUMBER       | 051-02166  |
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|                                                                                                                                                                                                                                                                                      |                |                      |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|-----------|
| SYNC_MASTER=T290_BIGSUR                                                                                                                                                                                                                                                              |                | SYNC_DATE=01/31/2017 |           |
| PAGE_TITLE                                                                                                                                                                                                                                                                           |                |                      |           |
| SoC Ground                                                                                                                                                                                                                                                                           |                |                      |           |
|                                                                                                                                                                                                 | DRAWING_NUMBER |                      | SIZE      |
|                                                                                                                                                                                                                                                                                      | 051-02166      |                      | D         |
| Apple Inc.                                                                                                                                                                                                                                                                           |                | REVISION             |           |
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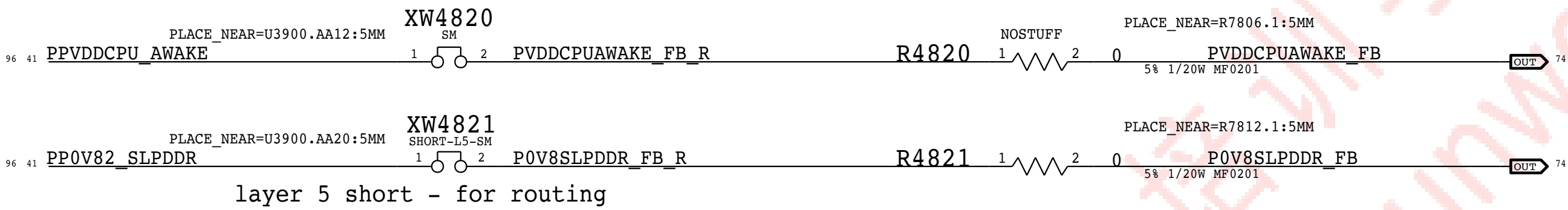
BOM\_COST\_GROUP=SOC



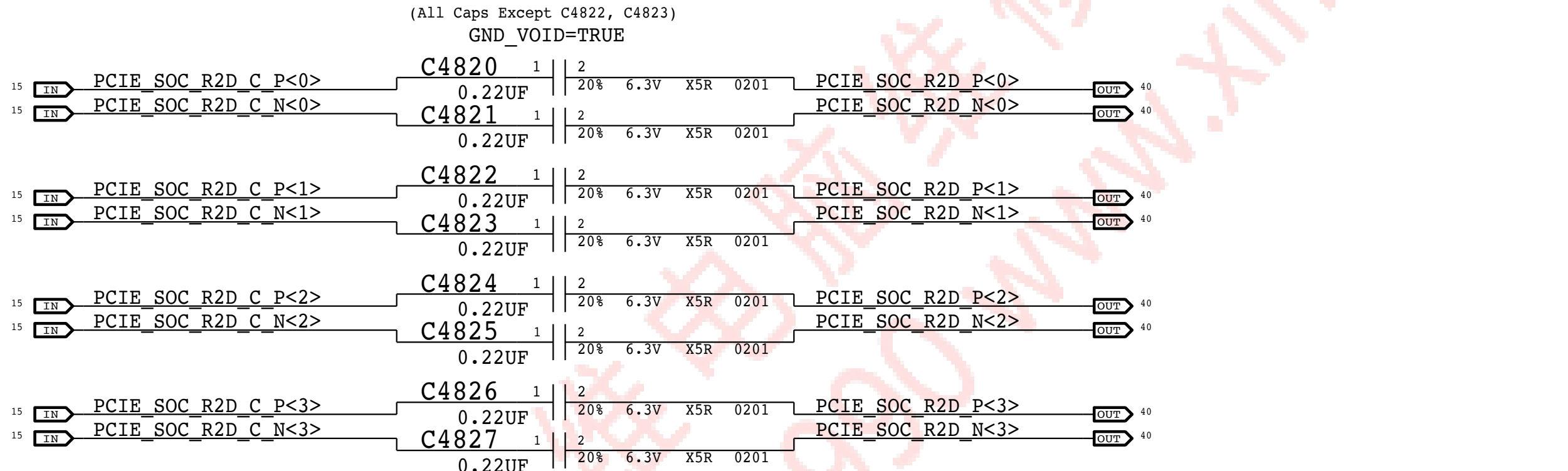




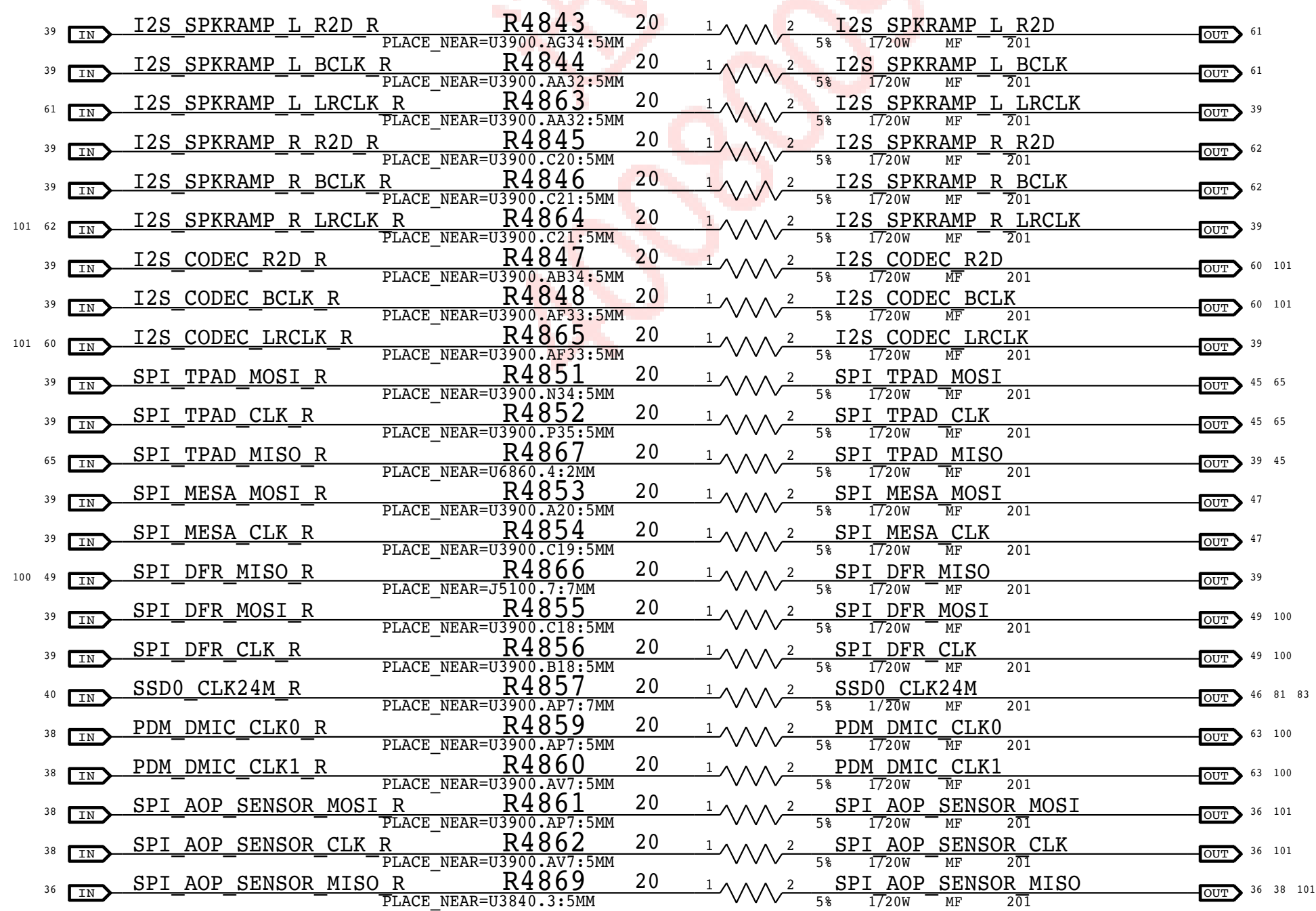
Alternate Feedback Sense



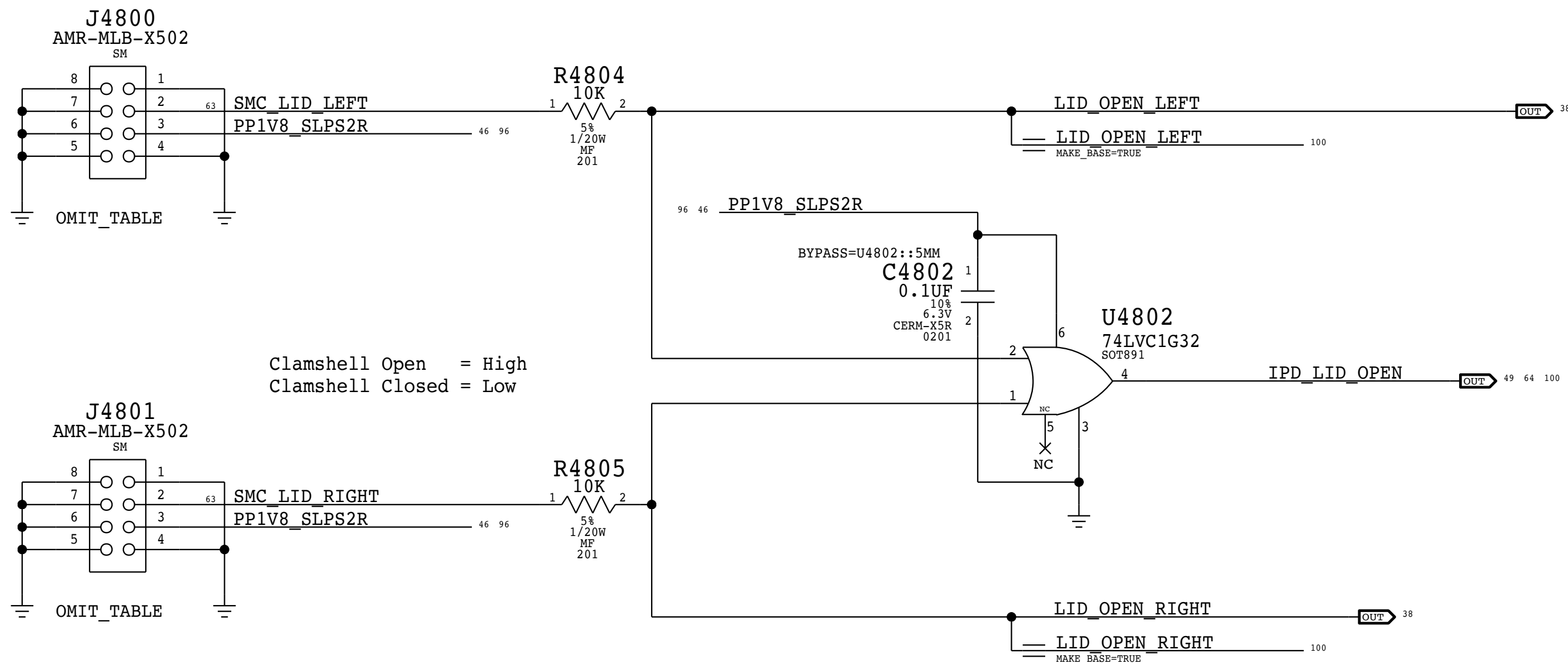
PCIe Up R2D AC Caps



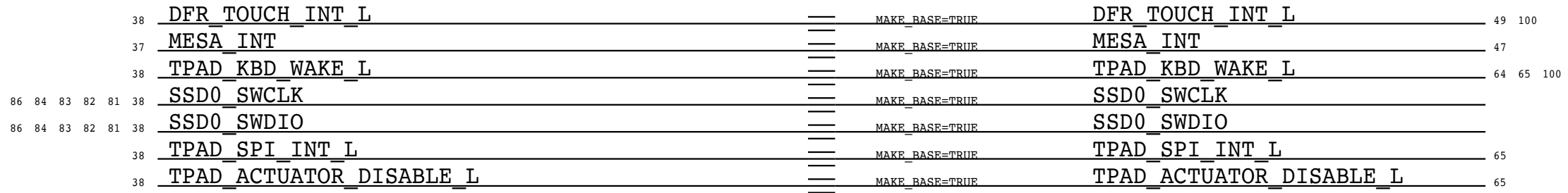
GPIO Source Termination



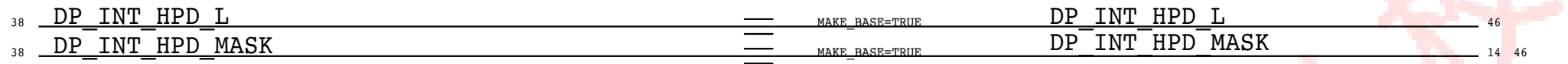
Lid Detect Sensors



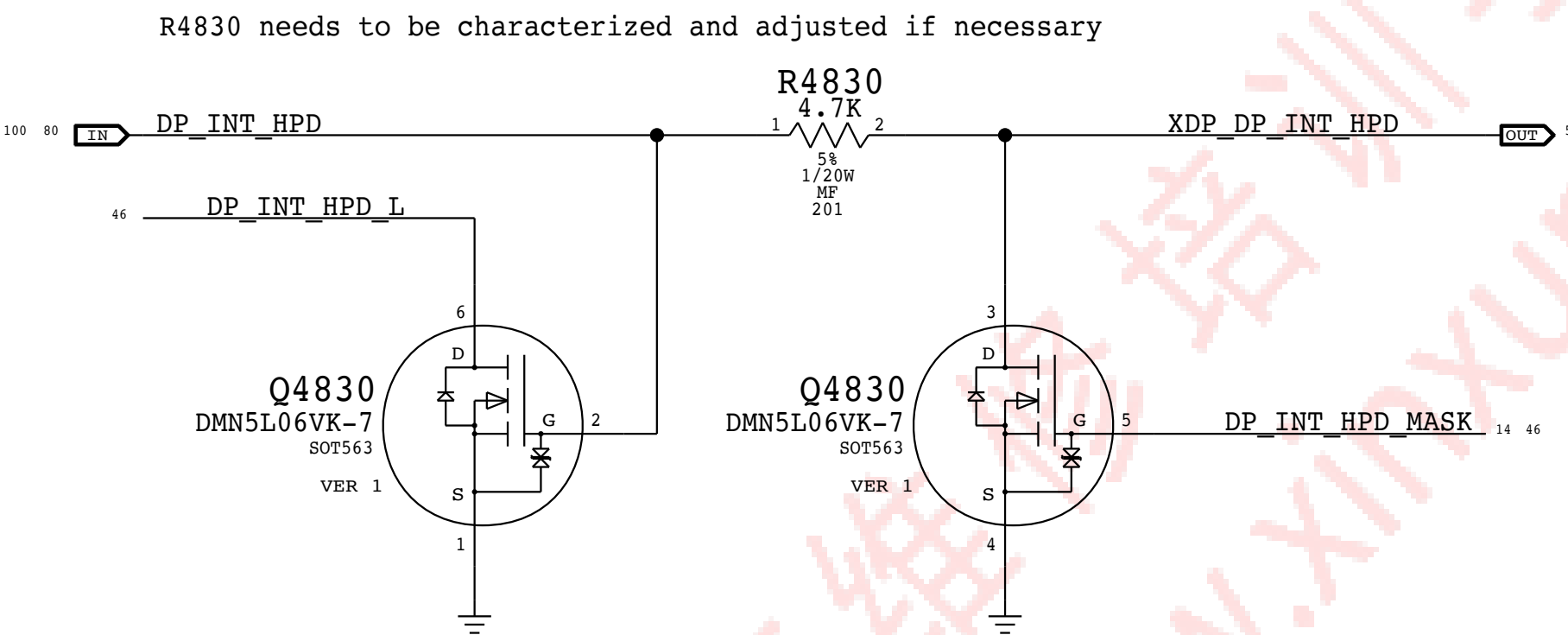
SOC Overloaded GPIOs



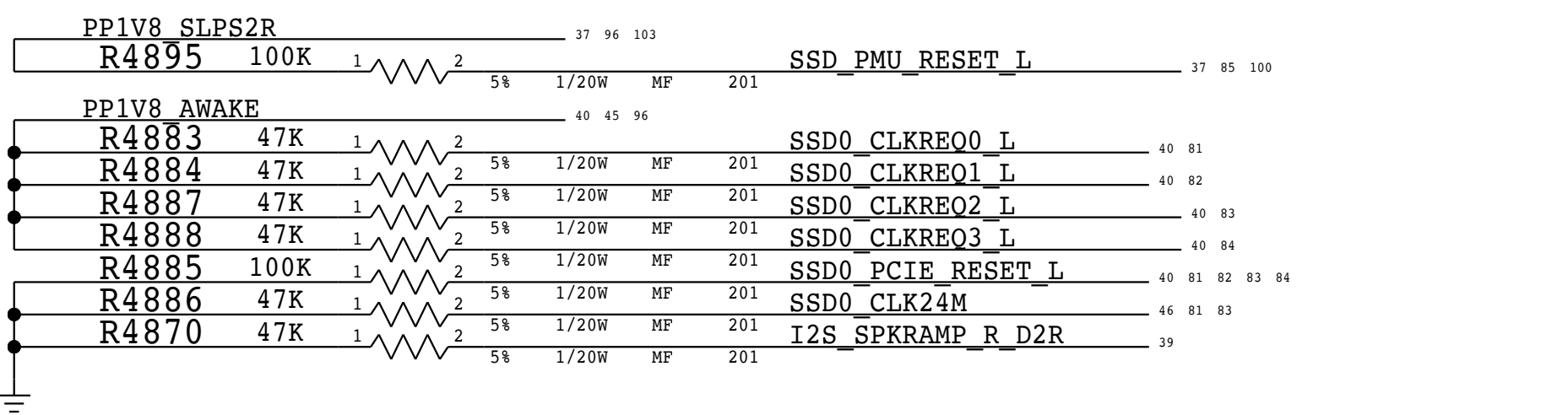
Platform-Specific GPIOs



HPD KSF Comp Circuit



Project Specific Pull-Ups



| BOM GROUP     | BOM OPTIONS                   |
|---------------|-------------------------------|
| BOARD_REV:111 | BOARDREV2,BOARDREV1,BOARDREV0 |
| BOARD_REV:110 | BOARDREV2,BOARDREV1           |
| BOARD_REV:101 | BOARDREV2,BOARDREV0           |
| BOARD_REV:100 | BOARDREV2                     |
| BOARD_REV:011 | BOARDREV1,BOARDREV0           |

| BOM GROUP | BOM OPTIONS       |
|-----------|-------------------|
| BOARD_ID  | BOARDID2,BOARDID3 |

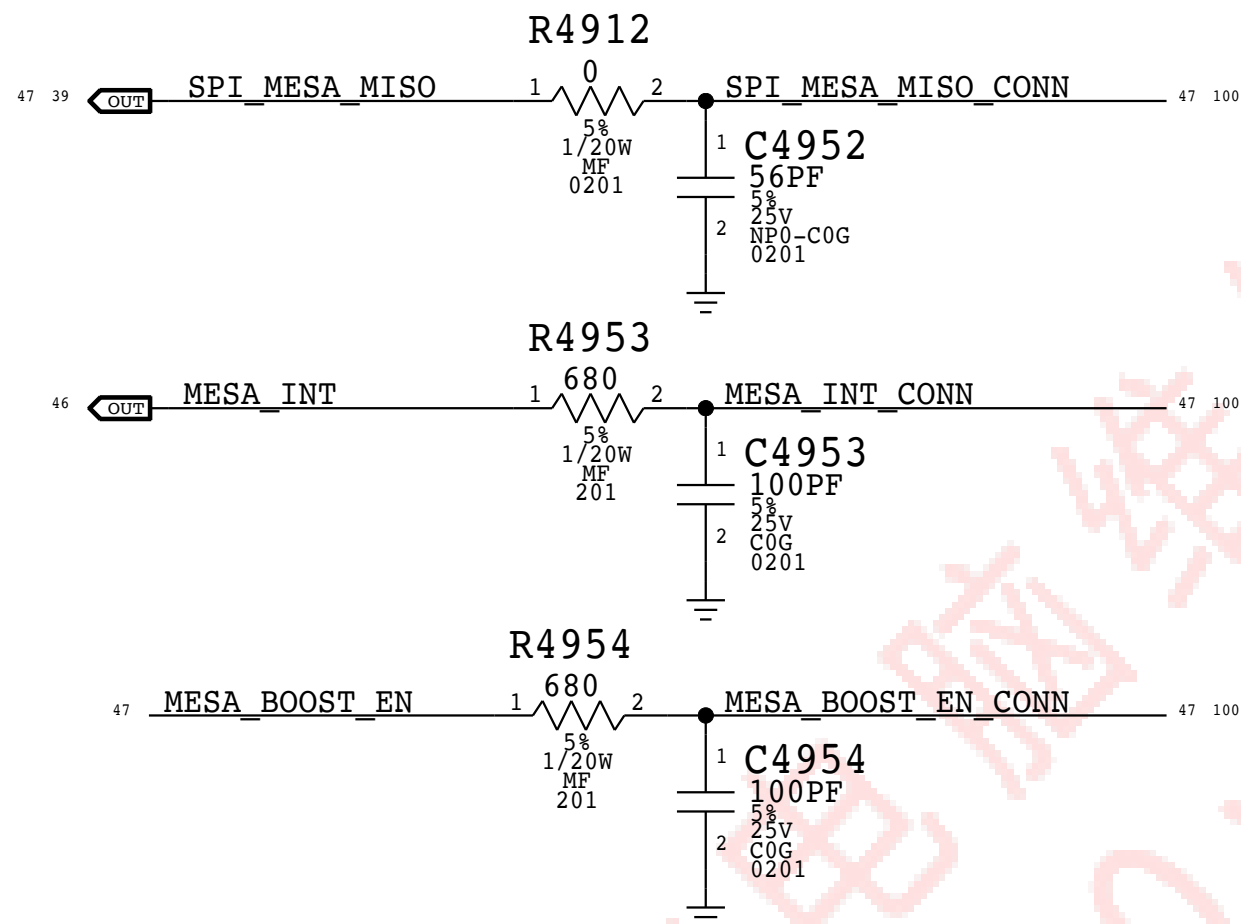
BOM\_COST\_GROUP=SOC

|                                                                                                                                                                                                                                                                                                                                              |                |                      |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|------------|
| SYNC_MASTER=J112 JACK                                                                                                                                                                                                                                                                                                                        |                | SYNC_DATE=11/03/2017 |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                                   |                |                      |            |
| SoC Project Support                                                                                                                                                                                                                                                                                                                          |                |                      |            |
|  Apple Inc.                                                                                                                                                                                                                                             | DRAWING NUMBER |                      | SHEET      |
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ISOLATE FROM OTHER COMPONENTS/NETS AS MUCH AS POSSIBLE

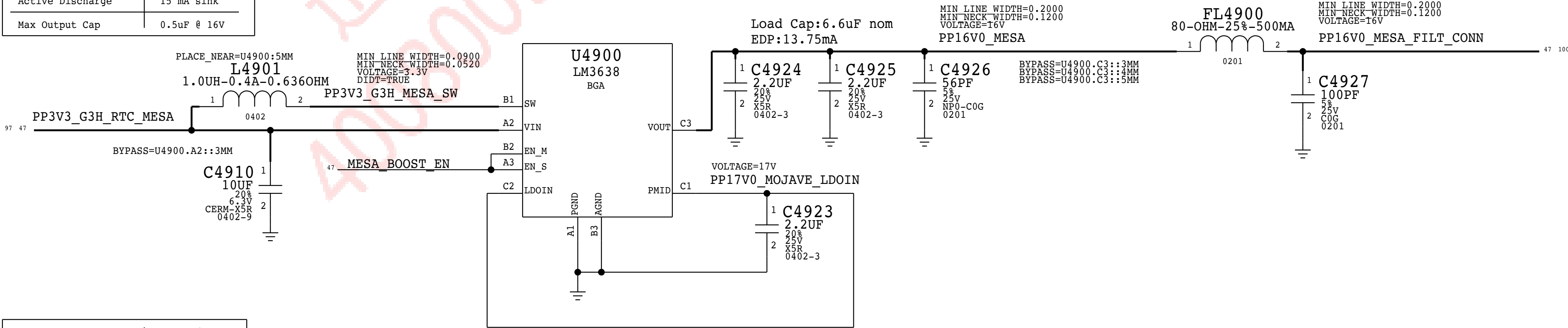
ESD Filters



Mesa Power Sequencing Requirements

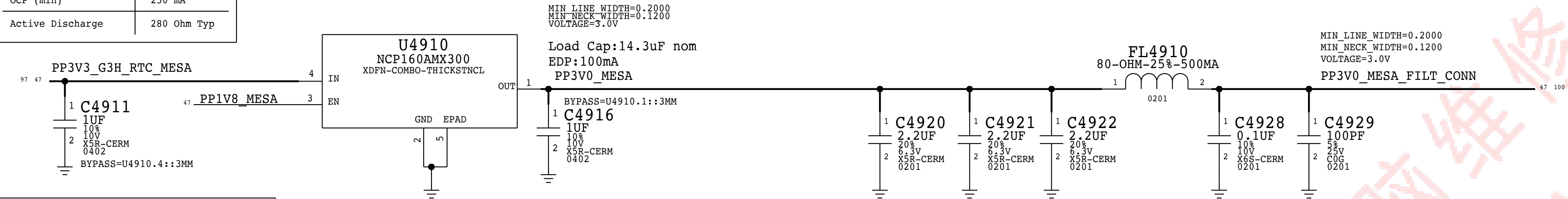
Power On: 1V8 -> 3V3 -> 16V0

MOJAVE 16V BOOST



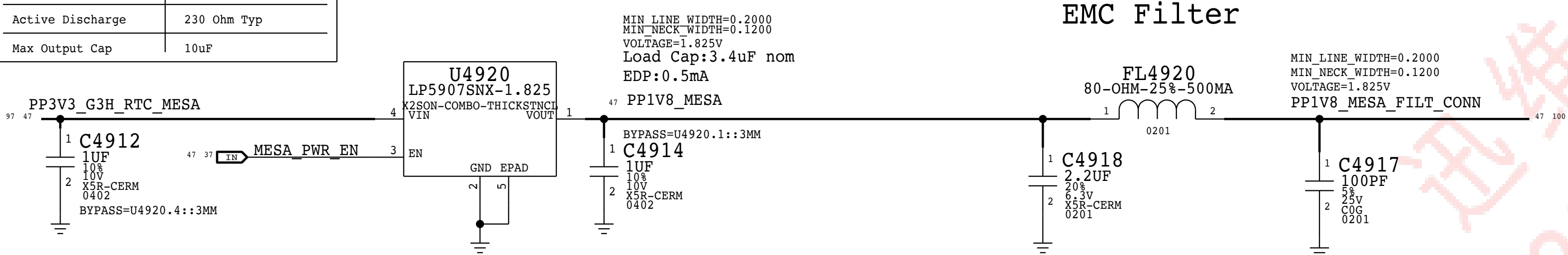
|                  |             |
|------------------|-------------|
| Output Voltage   | 3.0V +/- 2% |
| Iout (max avg)   | 250mA       |
| Dropout Voltage  | 155mV       |
| OCP (min)        | 250 mA      |
| Active Discharge | 280 Ohm Typ |

3.0V MESA



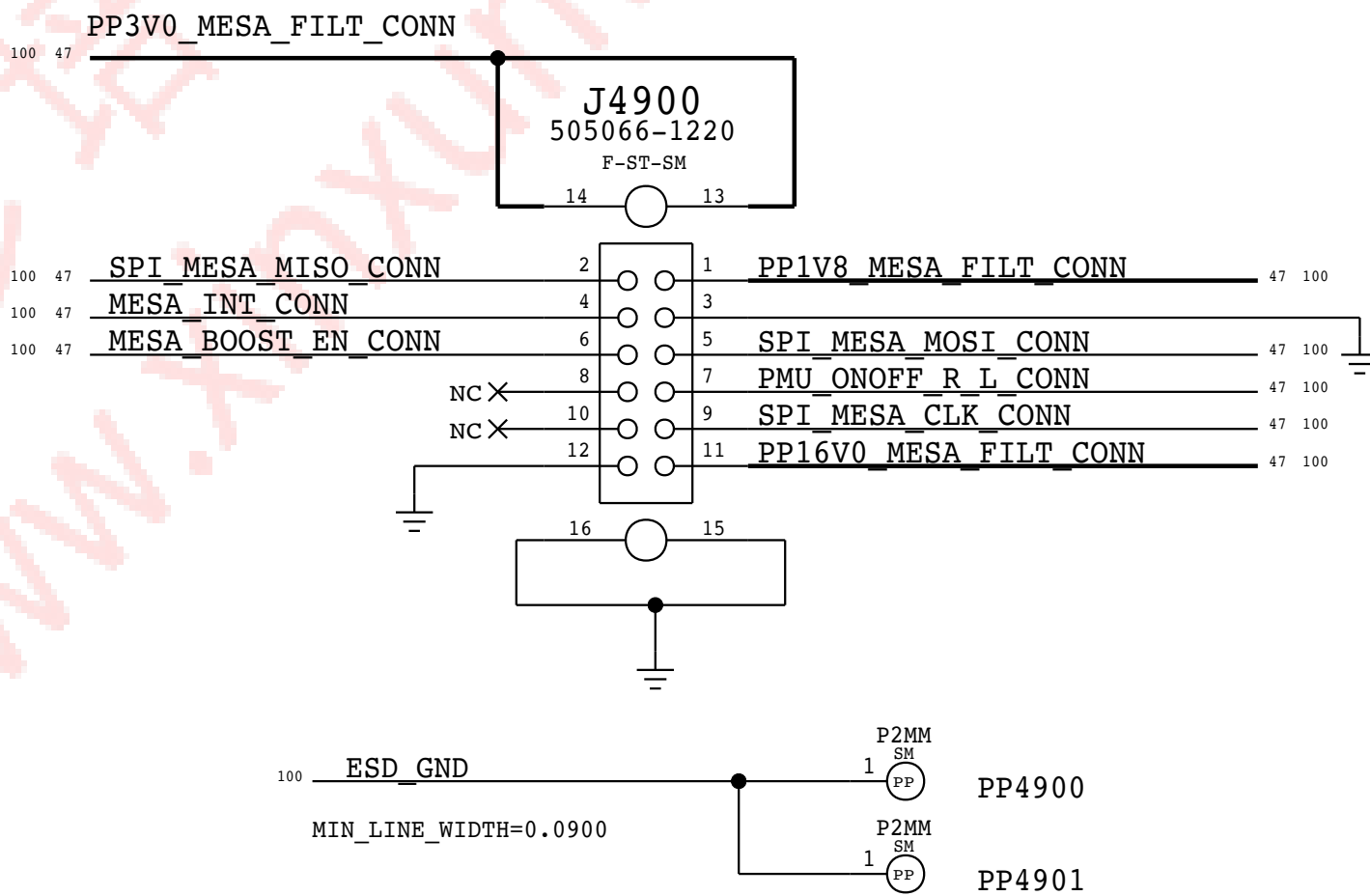
|                  |                  |
|------------------|------------------|
| Output Voltage   | 1.825V +/- 2%    |
| Iout (max avg)   | 250mA            |
| Dropout Voltage  | 50mV Typ @ 100mA |
| OCP (min)        | 250 mA           |
| Active Discharge | 230 Ohm Typ      |
| Max Output Cap   | 10uF             |

1.8V MESA



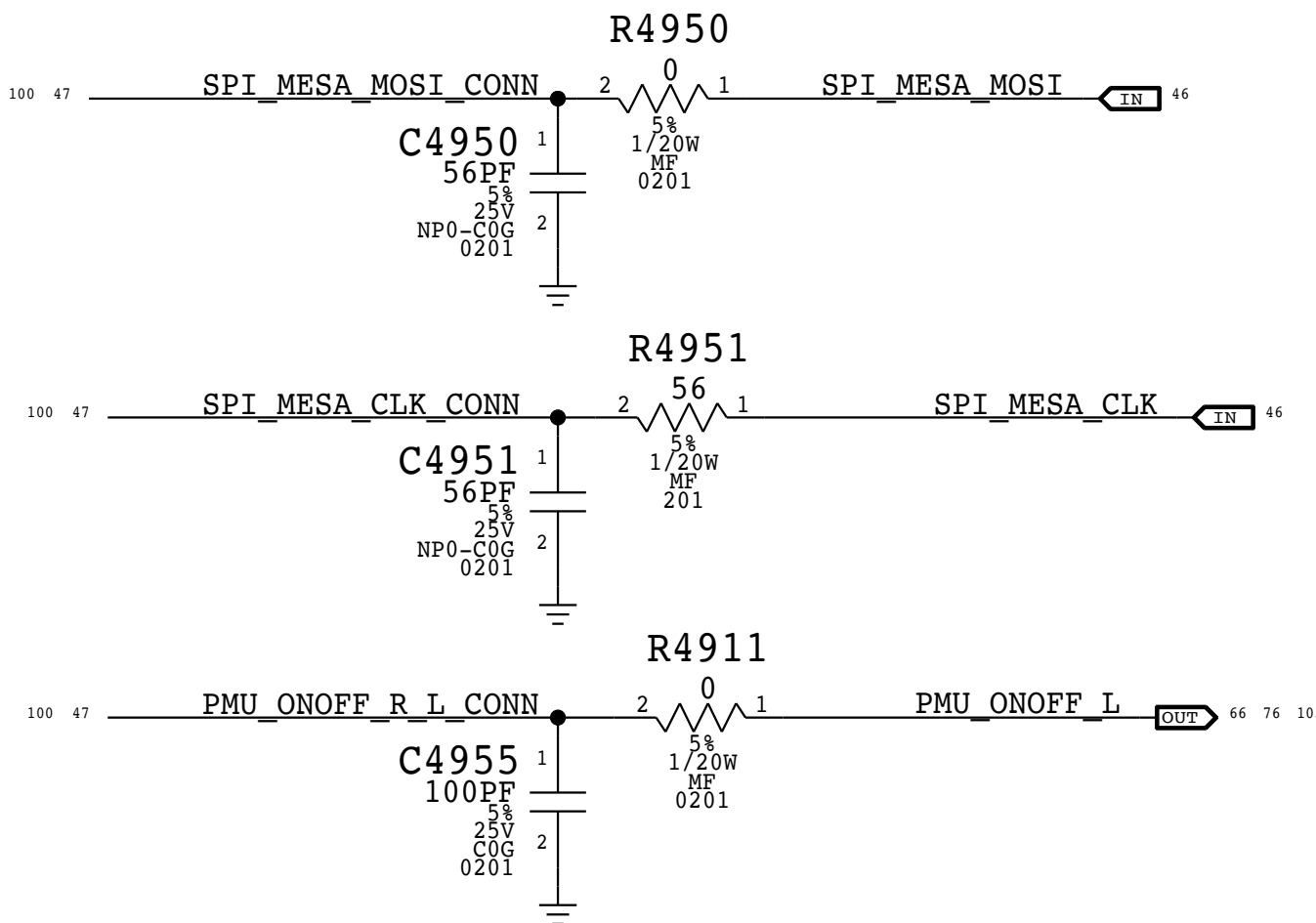
T151 FLEX CONNECTOR

PLUG (516S00115) - X434/ X435 Jumper  
RECEPTACLE (516S00203) - MLB



ISOLATE FROM OTHER COMPONENTS/NETS AS MUCH AS POSSIBLE

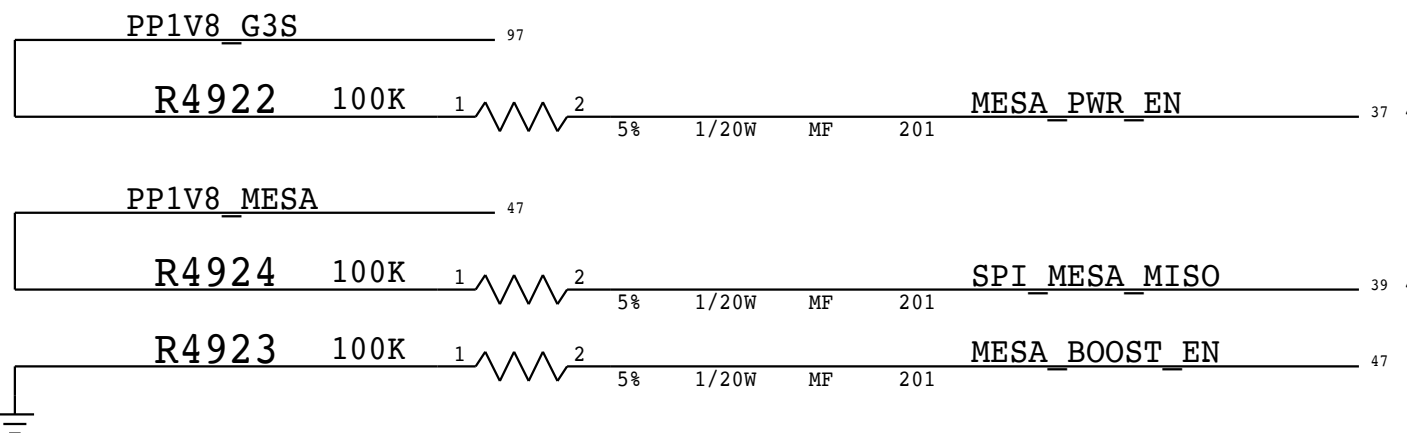
ESD Filters



EMC Filter

EMC Filter

EMC Filter



|                                                                                                                                                                                                                                                                                                                     |                |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                          |                |            |
| T151                                                                                                                                                                                                                                                                                                                |                |            |
|                                                                                                                                                                                                                                                                                                                     | DRAWING NUMBER | 051-02166  |
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|                                                                                                                                                                                                                                                                                                                     | PAGE           | 49 OF 150  |
|                                                                                                                                                                                                                                                                                                                     | SHEET          | 47 OF 108  |

BOM\_COST\_GROUP=T151



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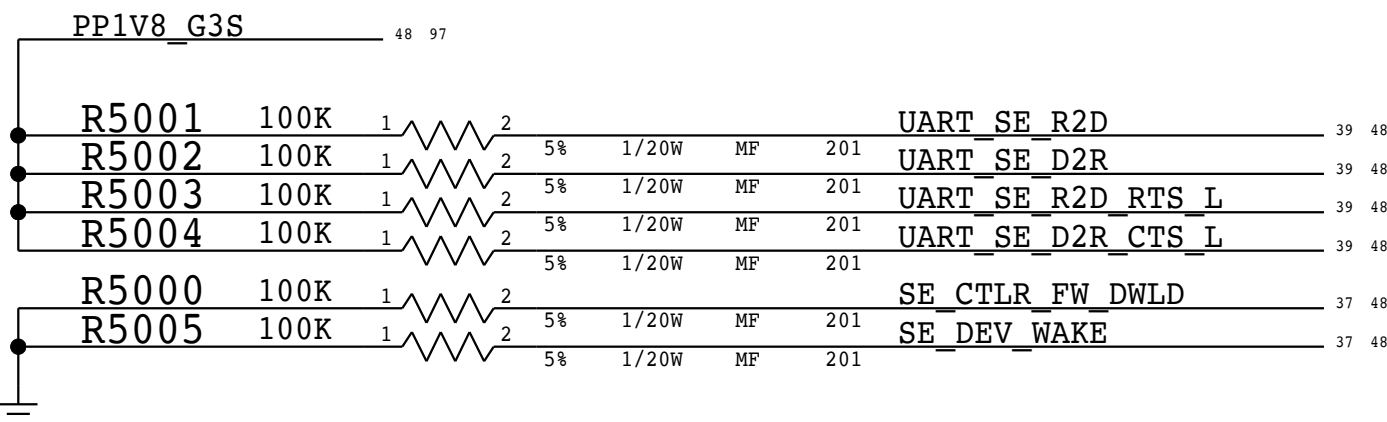
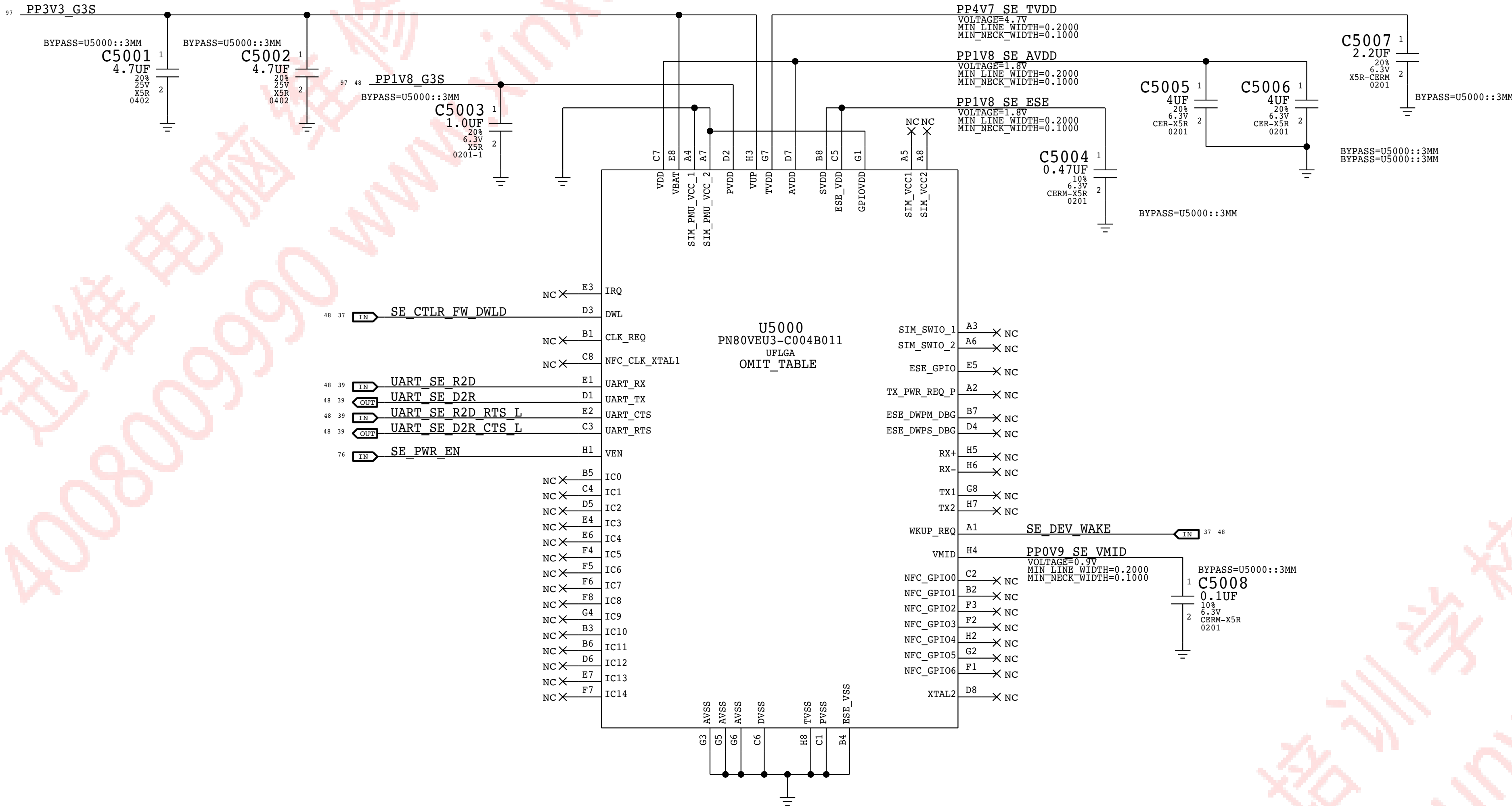
D

C

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A

STOCKHOLM

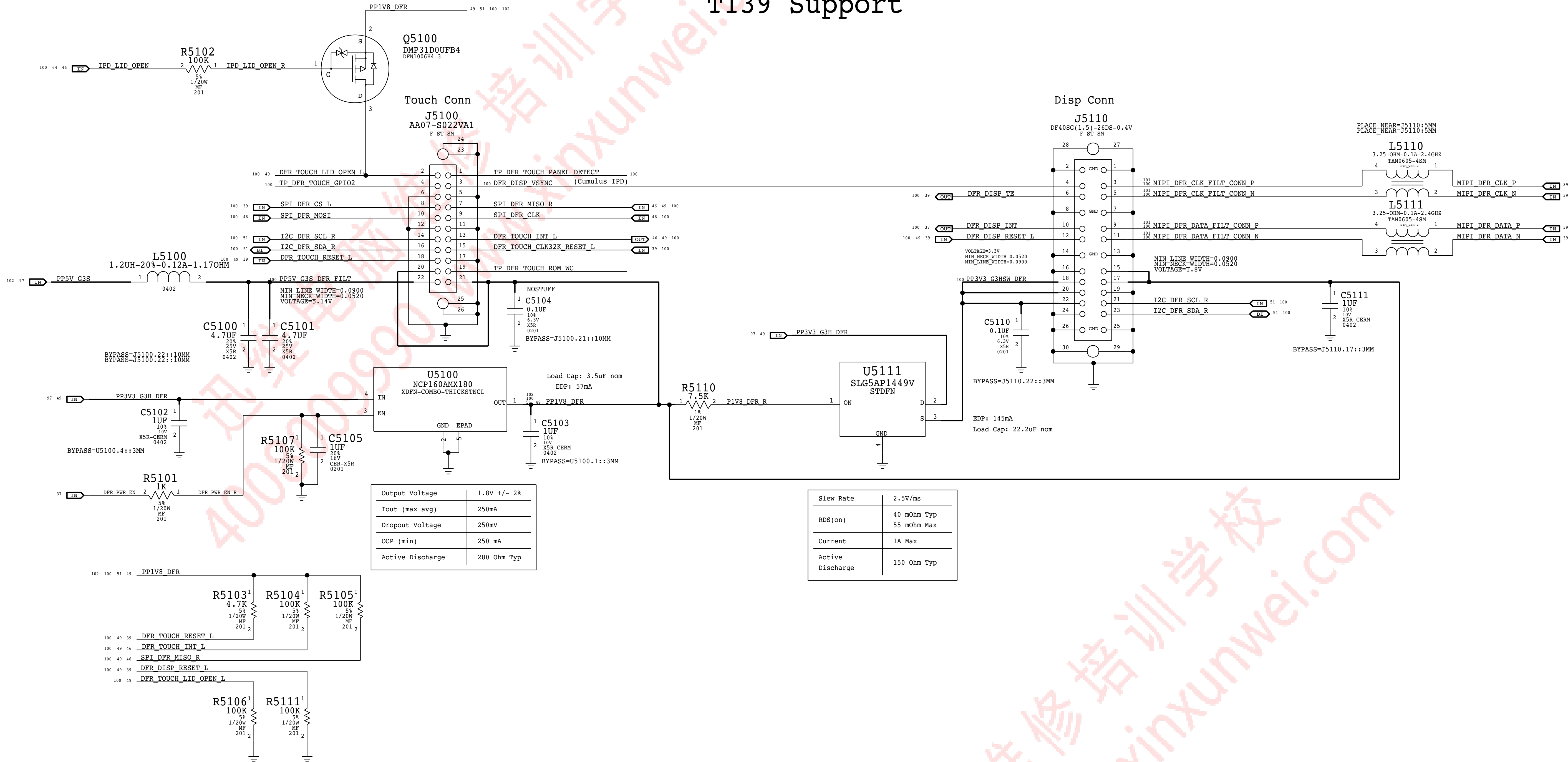


BOM\_COST\_GROUP=SOC

|                                                                                                                                                                                                                                                                                                                         |  |  |                |            |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------|------------|------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  |  | PAGE TITLE     |            |      |
| Secure Element                                                                                                                                                                                                                                                                                                          |  |  | Secure Element |            |      |
|                                                                                                                                                                                                                                                                                                                         |  |  | DRAWING NUMBER | 051-02166  | SIZE |
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T139 Support



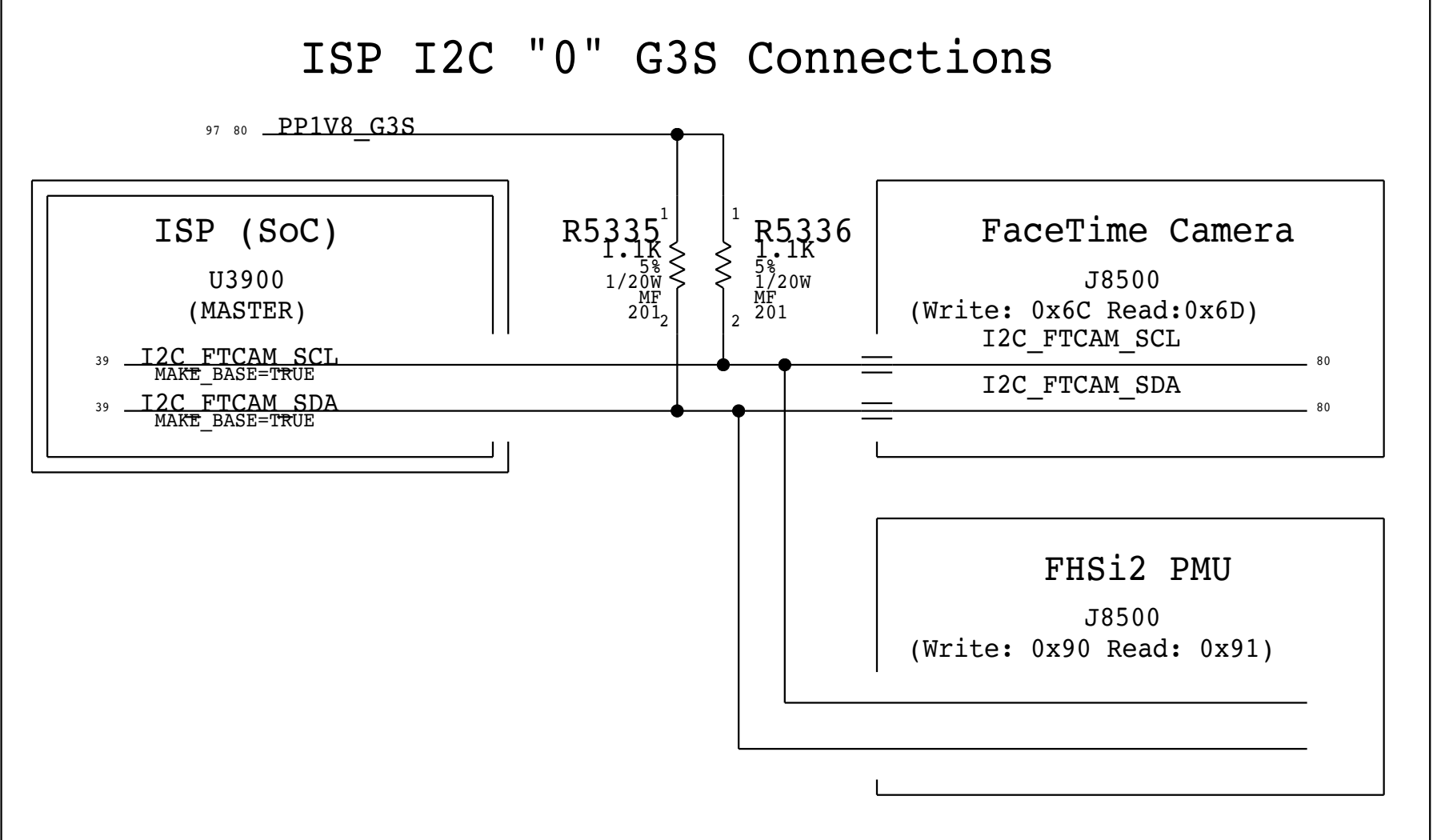
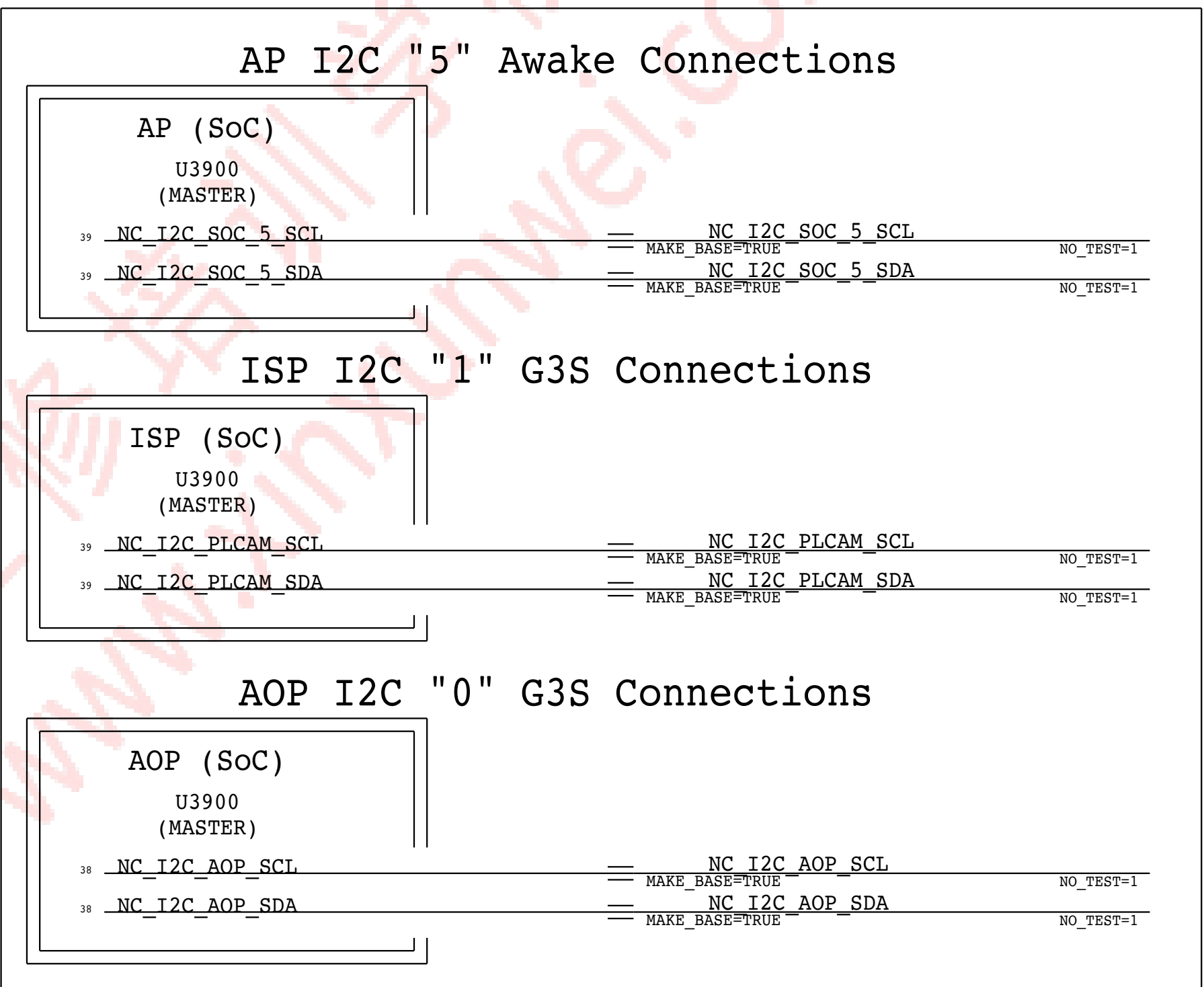
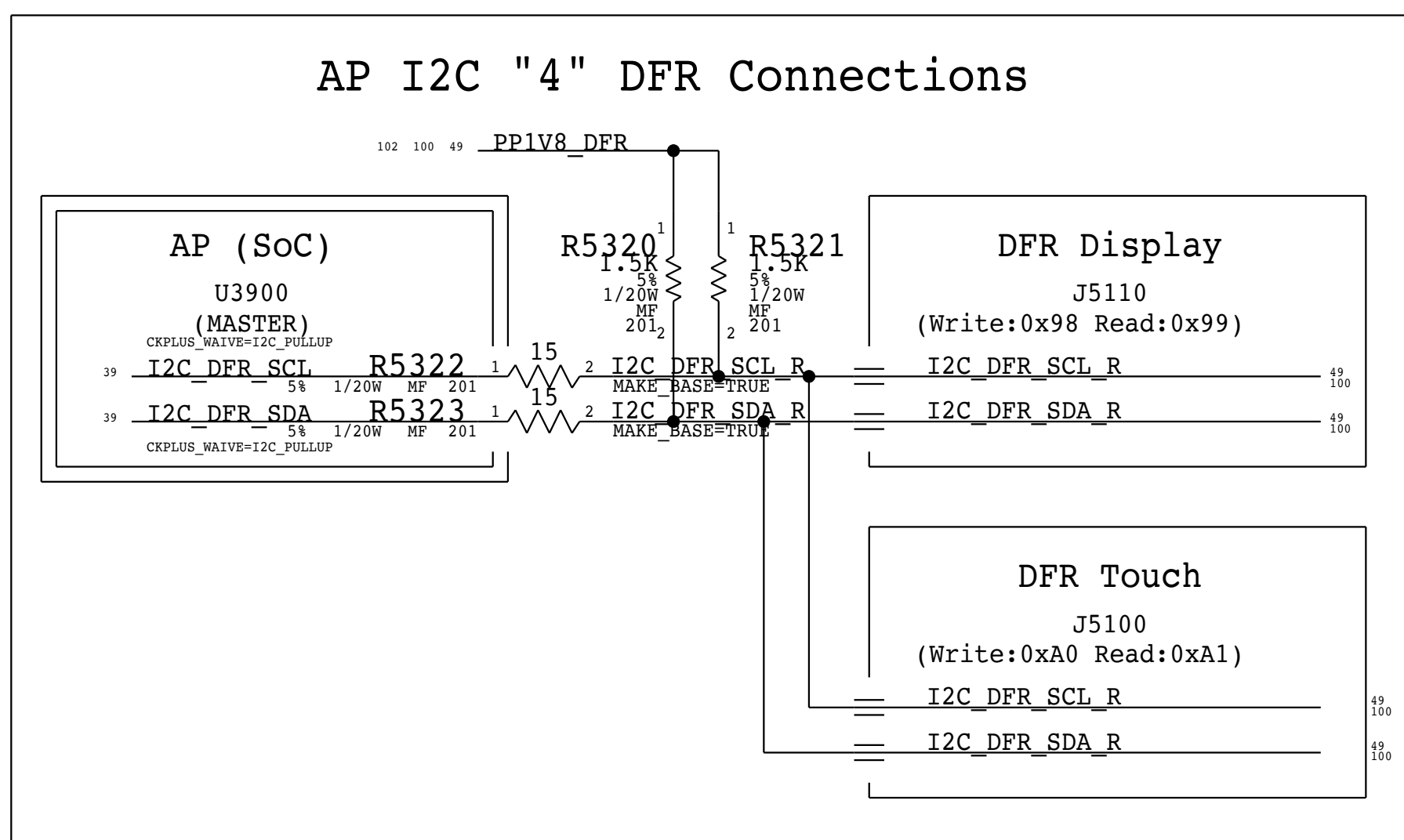
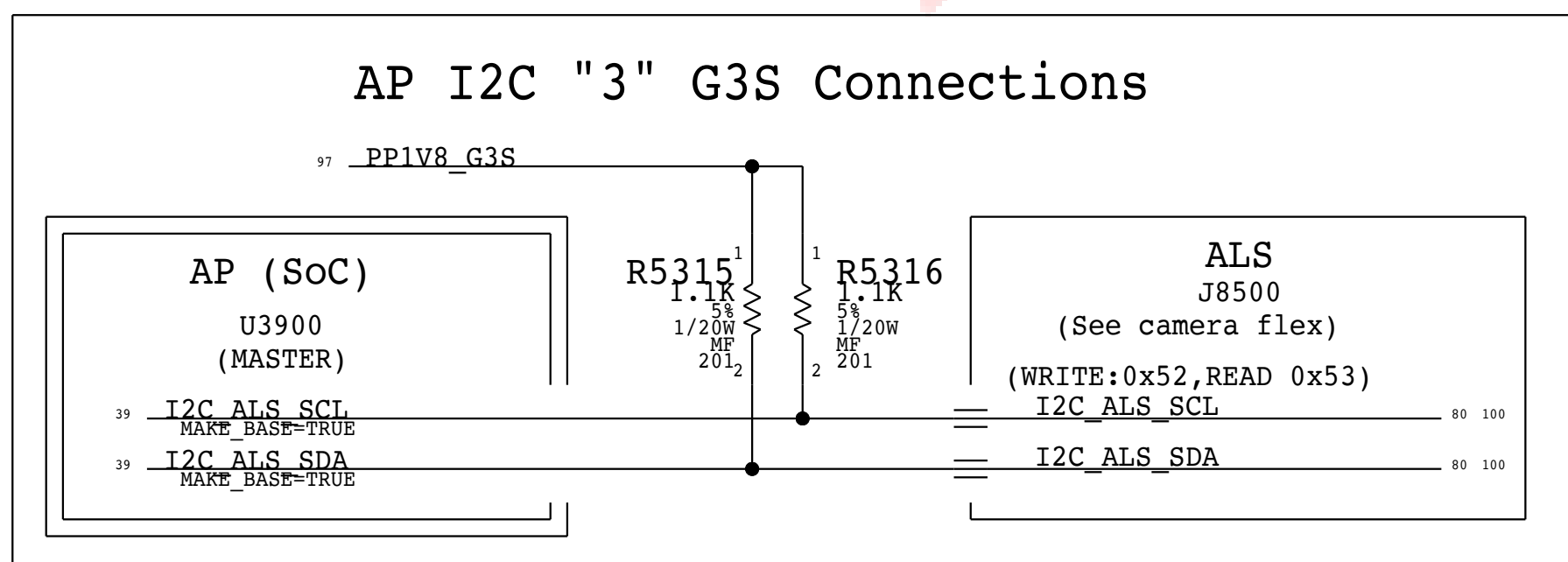
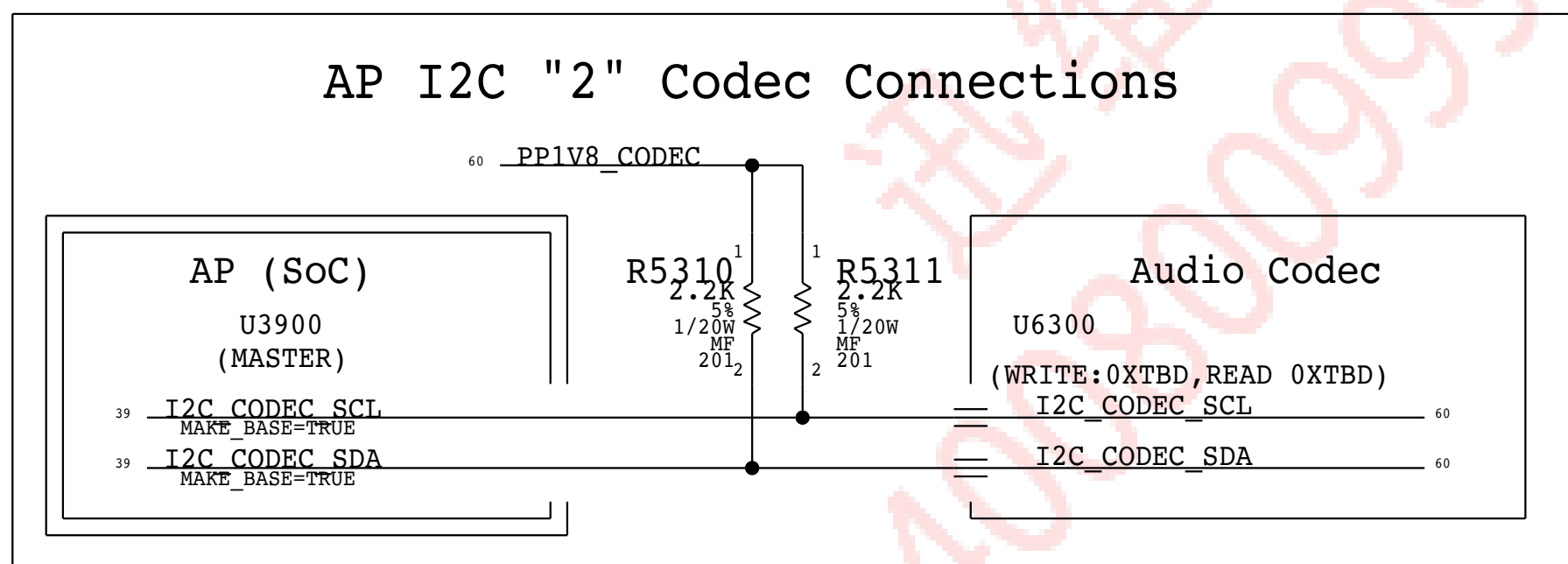
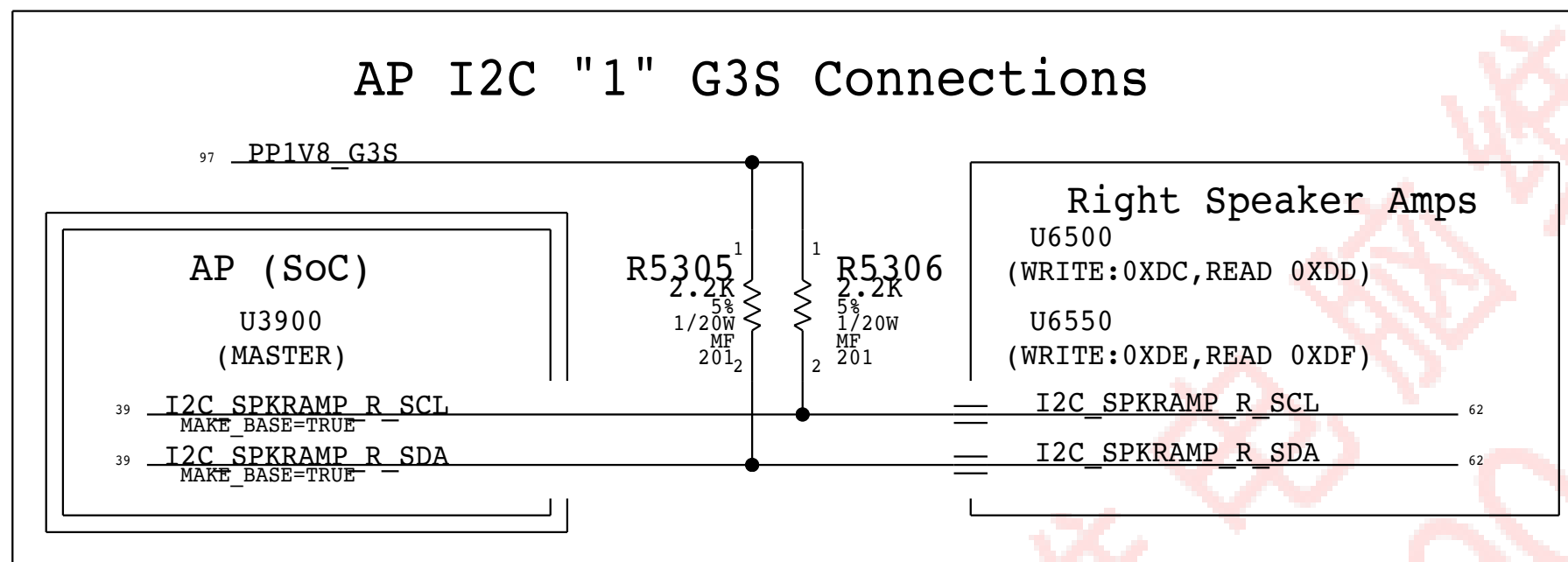
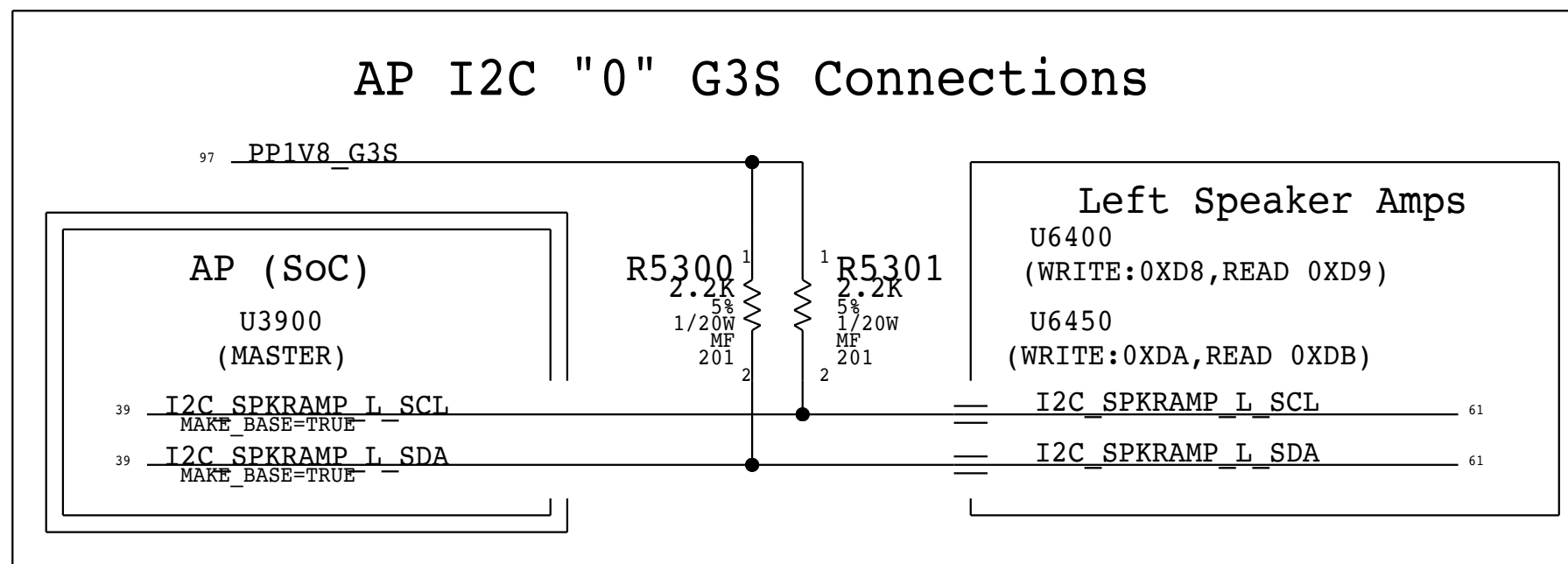
BOM\_COST\_GROUP=T139

| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| T139 Support                                                                                                                                                                                                                                                                                                      |                |            |
|                                                                                                                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02166  |
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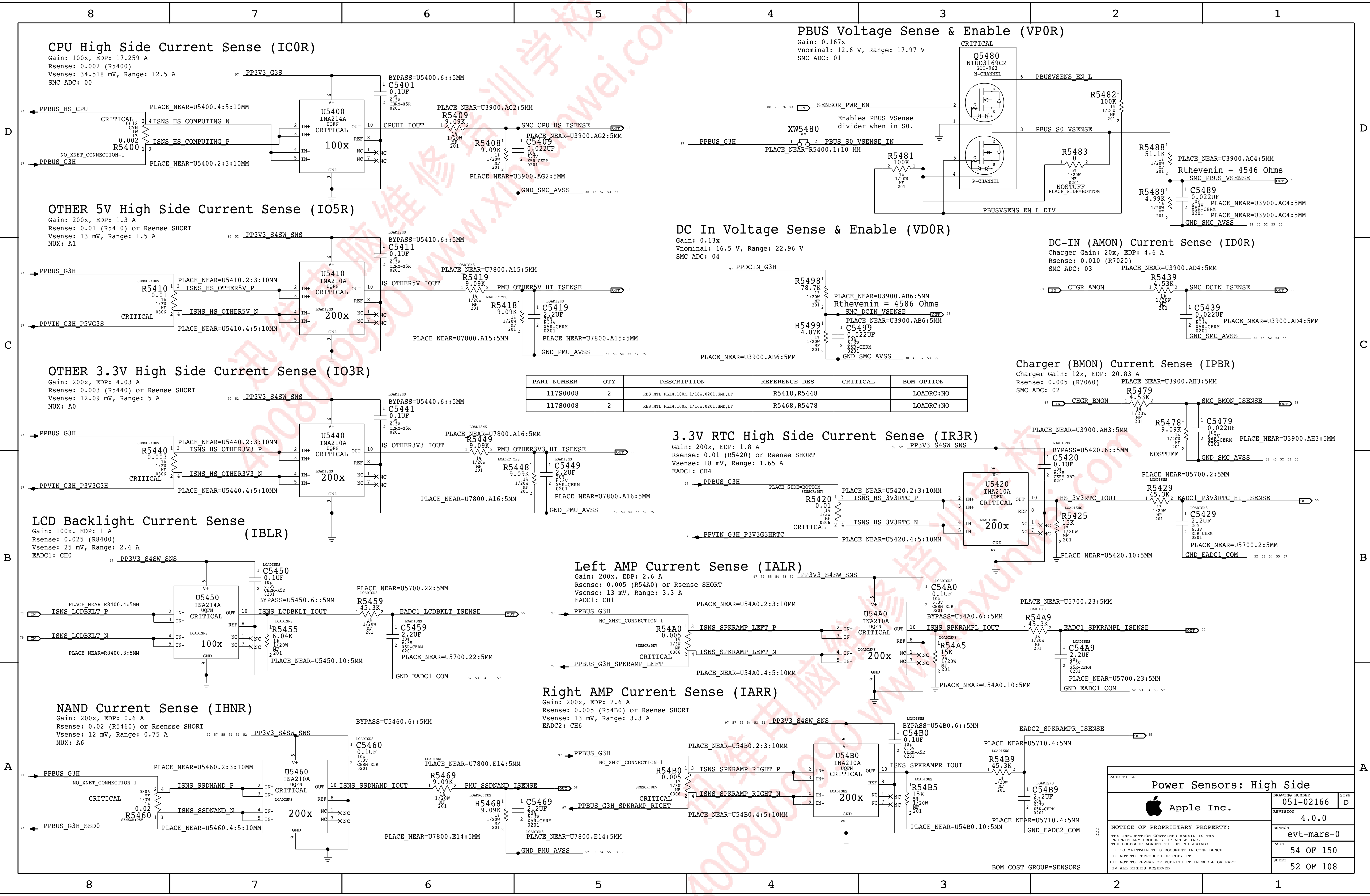




| Device                  | SMC IF | ADDR. (8b) |
|-------------------------|--------|------------|
| ACE XA                  | I2C0   | 0X70/1     |
| ACE XB                  | I2C0   | 0X7E/F     |
| ACE TA                  | I2C0   | 0X40/1     |
| ACE TB                  | I2C0   | 0X4E/F     |
| NC.                     | I2C1   |            |
| Temp. Sensor Left       | I2C2   | 0X90/1     |
| Temp. Sensor Right      | I2C2   | 0X96/7     |
| Platform Thermal Sensor | I2C2   | 0X92/3     |
| PCH                     | I2C2   | 0X88/9     |
| TCON                    | I2C3   | 0X10-1F    |
| Charger                 | I2C4   | 0X12/3     |
| Battery                 | I2C4   | 0X16/7     |
| Calpe                   | I2C4   | 0XE8/9     |
| Trackpad                | I2C5   | 0X98/9     |
| EADC1                   | I2C5   | 0X10/1     |
| EADC2                   | I2C5   | 0X12/3     |
| SSD                     | I2C6   | 0XF2/3     |
| SoC IF                  |        |            |
| Left Spkr Amp.(U6400)   | I2C0   | 0XD8/9     |
| Left Spkr Amp.(U6450)   | I2C0   | 0XDA/B     |
| Right Spkr Amp.(U6500)  | I2C1   | 0XDC/D     |
| Right Spkr Amp.(U6550)  | I2C1   | 0XDE/F     |
| Audio Codec             | I2C2   | 0X90/1     |
| ALS                     | I2C3   | 0X52/3     |
| DFR Display             | I2C4   | 0X98/9     |
| DFR Touch               | I2C4   | 0XA0/1     |
| NC.                     | I2C5   |            |
| Spkr ID1                | I2C6   |            |
| Spkr ID0                | I2C6   |            |
| SIP IF                  |        |            |
| FT Camera               | I2C0   | 0X6C/D     |
| FHSi2                   | I2C0   | 0X90/1     |
| NC.                     | I2C1   |            |
| AOP IF                  |        |            |
| NC.                     | I2C0   |            |

|                                                                                                                   |                |                              |      |
|-------------------------------------------------------------------------------------------------------------------|----------------|------------------------------|------|
| FORM NO. 100 (REVISED 11-79)                                                                                      |                | FORM NO. 100 (REVISED 11-79) |      |
| PAGE TITLE                                                                                                        |                |                              |      |
| I2PC Connections 2                                                                                                |                |                              |      |
|                              | DRAWING NUMBER |                              | SIZE |
|                                                                                                                   | 051-02166      |                              | D    |
| Apple Inc.                                                                                                        | REVISION       |                              |      |
|                                                                                                                   | 4.0.0          |                              |      |
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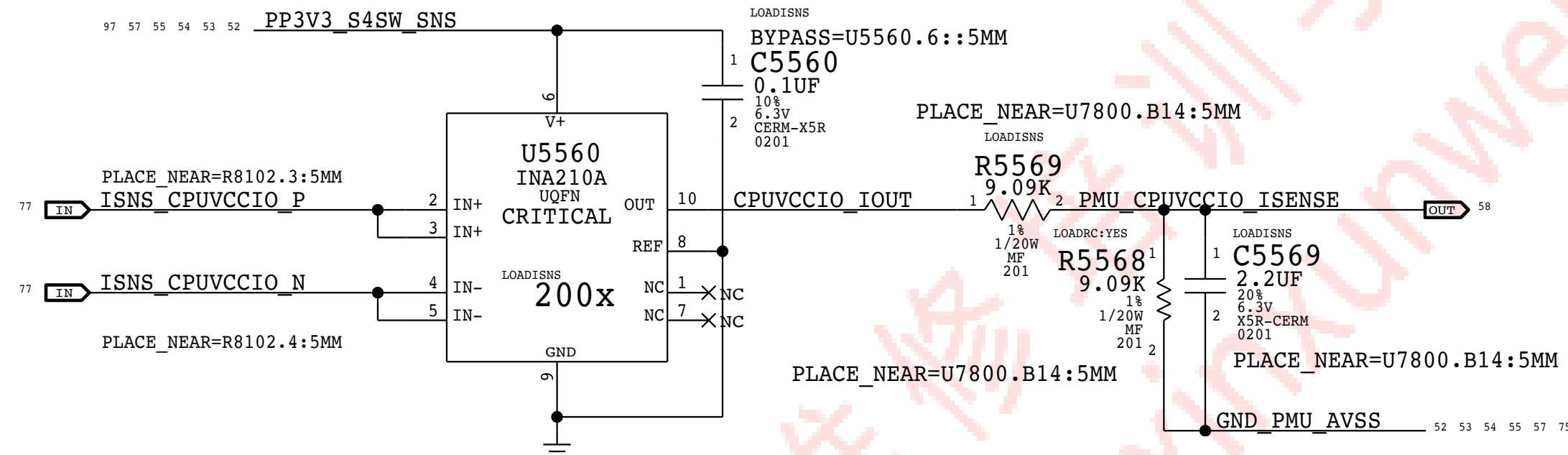






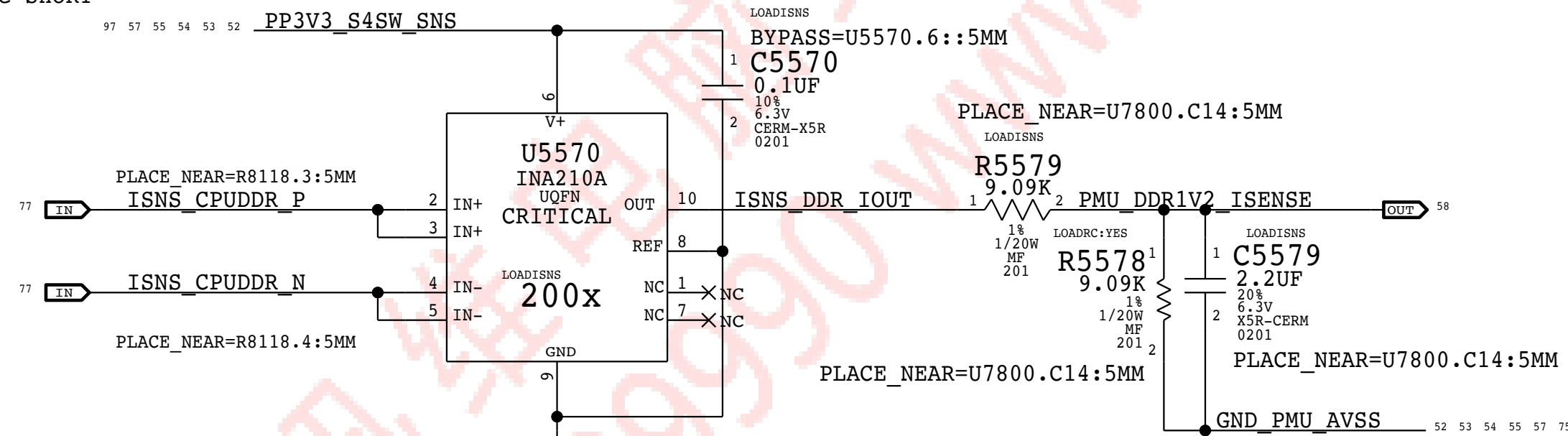
### CPU VCCIO Current Sense (IC1C)

Gain: 200x, EDP: 3.6 A  
Rsense: 0.003 (R8102) or Rsense SHORT  
Vsense: 10.8 mV, Range: 5 A  
MUX: A3



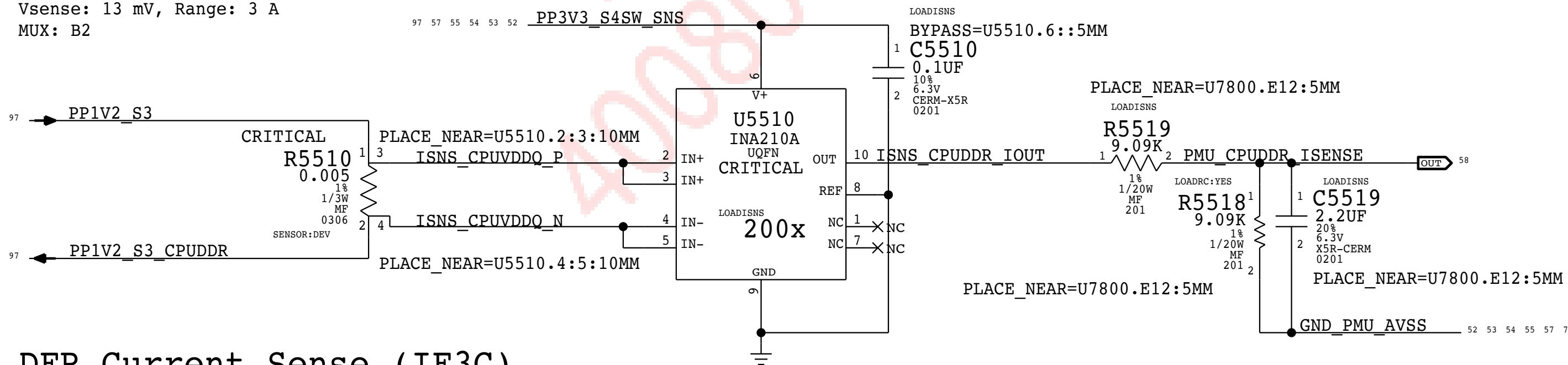
### DDR 1.2V S3 (CPU & Memory) Current Sense (IM0C)

Gain: 200x, EDP: 6.9 A  
Rsense: 0.002 (R8118) or Rsense SHORT  
Vsense: 13.8 mV, Range: 7.5 A  
MUX: A4



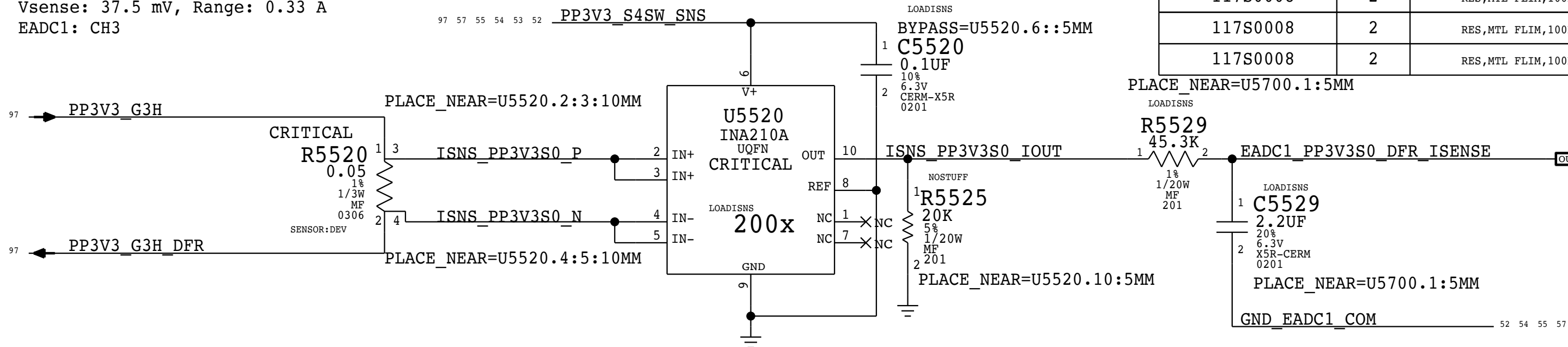
### CPU DDR 1.2V S3 (CPU Only) Current Sense (IMCC)

Gain: 200x, EDP: 2.6 A  
Rsense: 0.005 (R5510) or Rsense SHORT  
Vsense: 13 mV, Range: 3 A  
MUX: B2



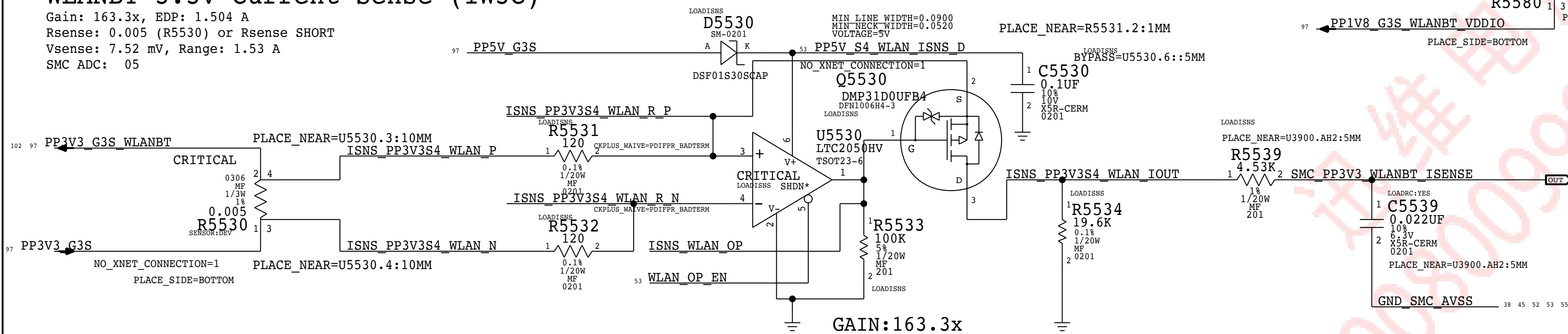
### DFR Current Sense (IF3C)

Gain: 200x, EDP: 0.75 A  
Rsense: 0.05 (R5520) or Rsense SHORT  
Vsense: 37.5 mV, Range: 0.33 A  
EADC1: CH3



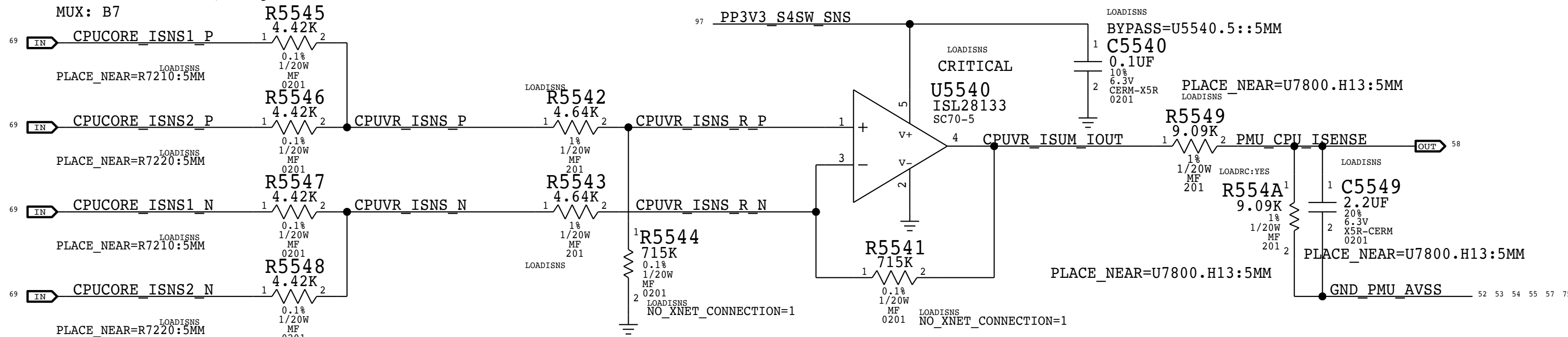
### WLANBT 3.3V Current Sense (IW3C)

Gain: 163.3x, EDP: 1.504 A  
Rsense: 0.005 (R5530) or Rsense SHORT  
Vsense: 7.52 mV, Range: 1.53 A  
SMC ADC: 05



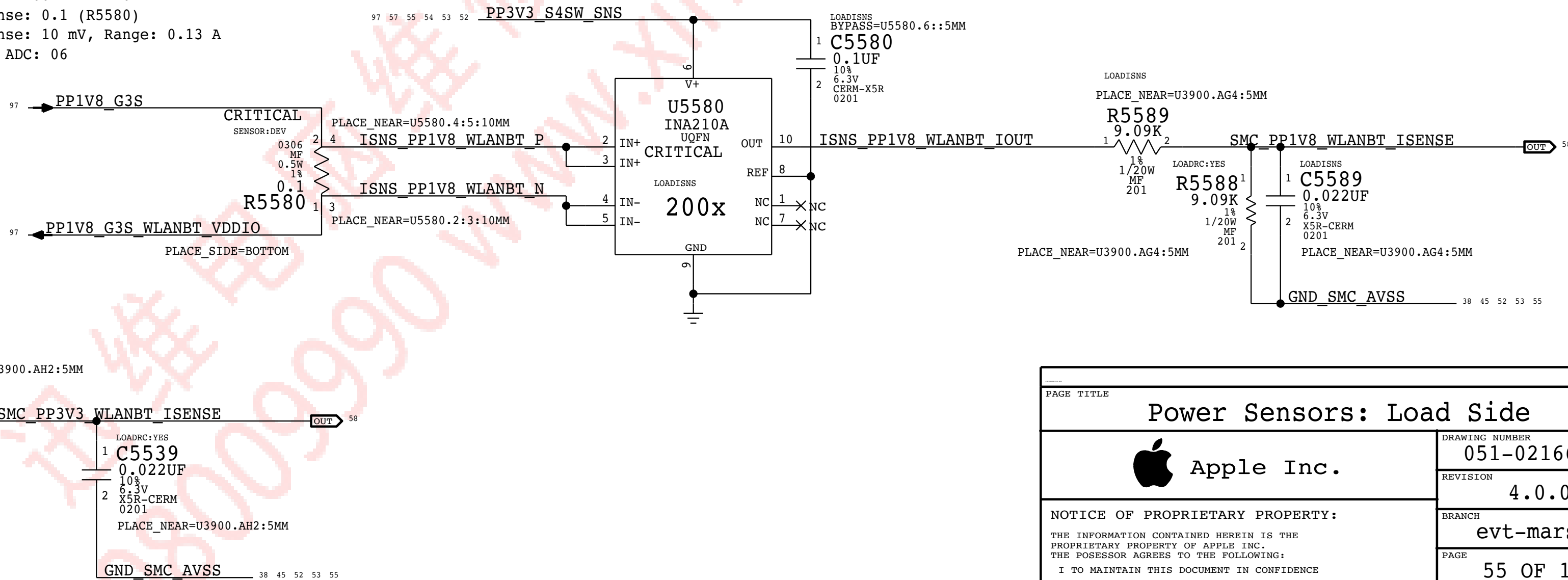
### CPU Fixed Current Sense (ICAC)

Gain: 275.74x, EDP: 29 A  
Rsense: 2x of 0.00075 (R7310, R7320), Rsum: 0.000375  
Vsense: 10.875 mV, Range: 29.01 A  
MUX: B7



### WLANBT 1.8V Current Sense (IW2C)

Gain: 200x, EDP: 0.1 A  
Rsense: 0.1 (R5580)  
Vsense: 10 mV, Range: 0.13 A  
SMC ADC: 06



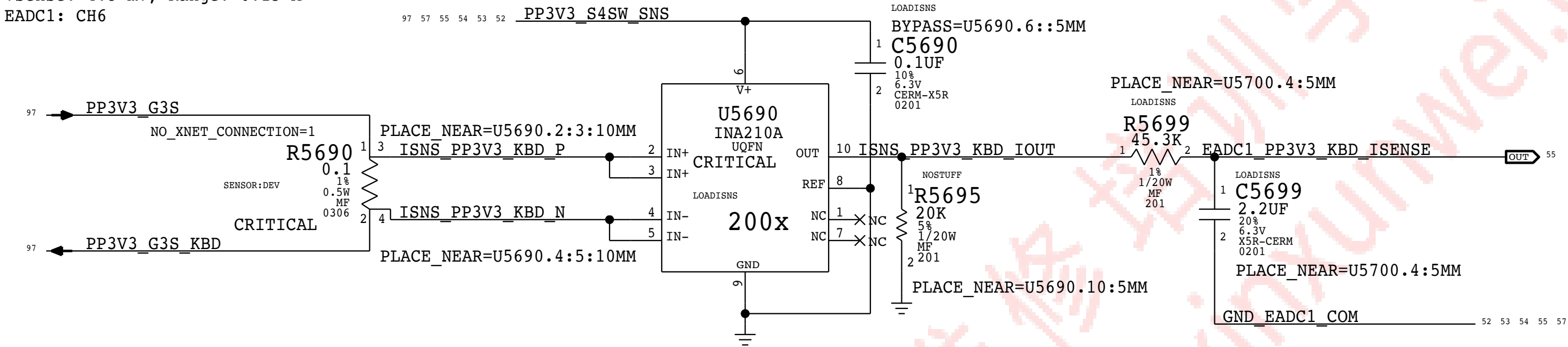
| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------------------|---------------|----------|------------|
| 117S0008    | 2   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R5568,R5518   |          | LOADRC:NO  |
| 117S0008    | 2   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R554A,R5578   |          | LOADRC:NO  |
| 117S0008    | 2   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5539,R5588   |          | LOADRC:NO  |

|                                                                                                                                                                                                                                                                                                                         |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
| Power Sensors: Load Side                                                                                                                                                                                                                                                                                                |                |            |
| Apple Inc.                                                                                                                                                                                                                                                                                                              | DRAWING NUMBER | 051-02166  |
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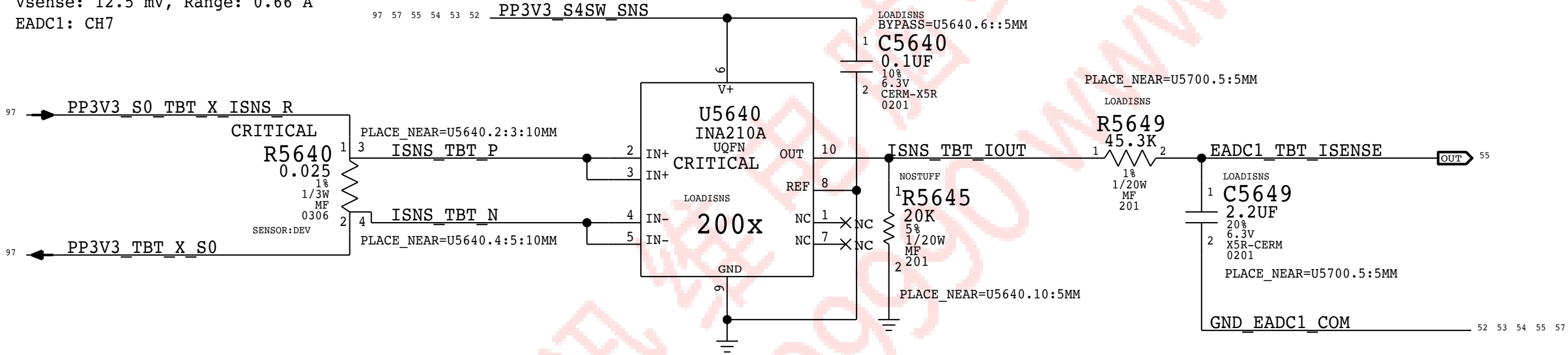
### Keyboard 3V Current Sense (IK3C)

Gain: 200x, EDP: 0.043 A  
Rsense: 0.1 (R5690) or Rsense SHORT  
Vsense: 4.3 mV, Range: 0.15 A  
EADC1: CH6



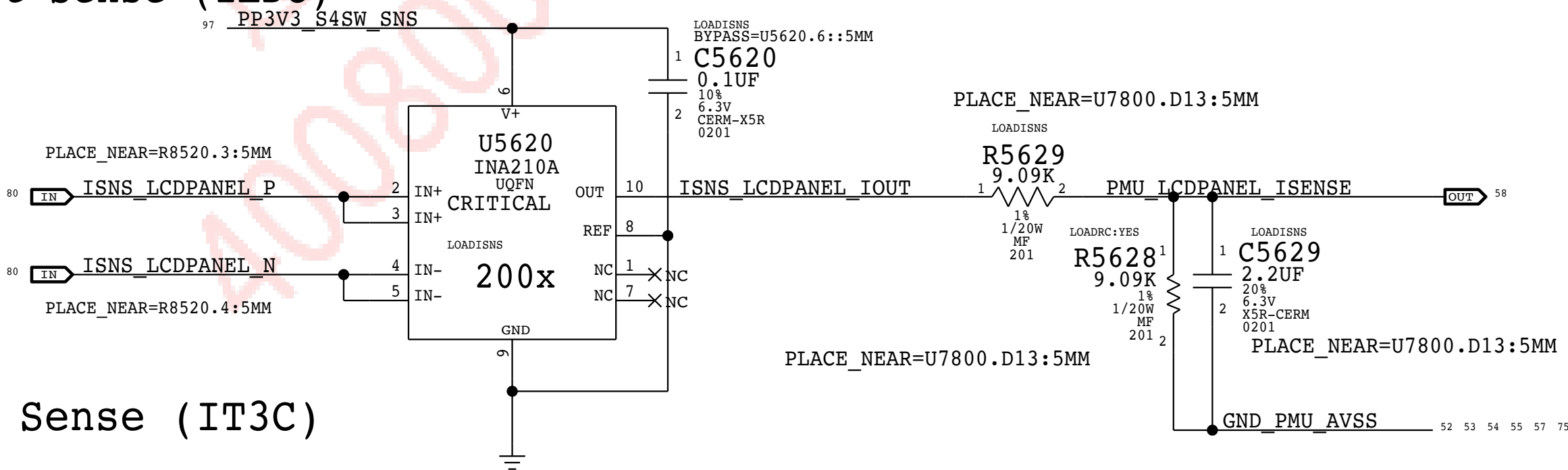
### Thunderbolt TBT Current Left (IULC)

Gain: 200x, EDP: 0.8 A  
Rsense: 0.025 (R5640) or Rsense SHORT  
Vsense: 12.5 mV, Range: 0.66 A  
EADC1: CH7



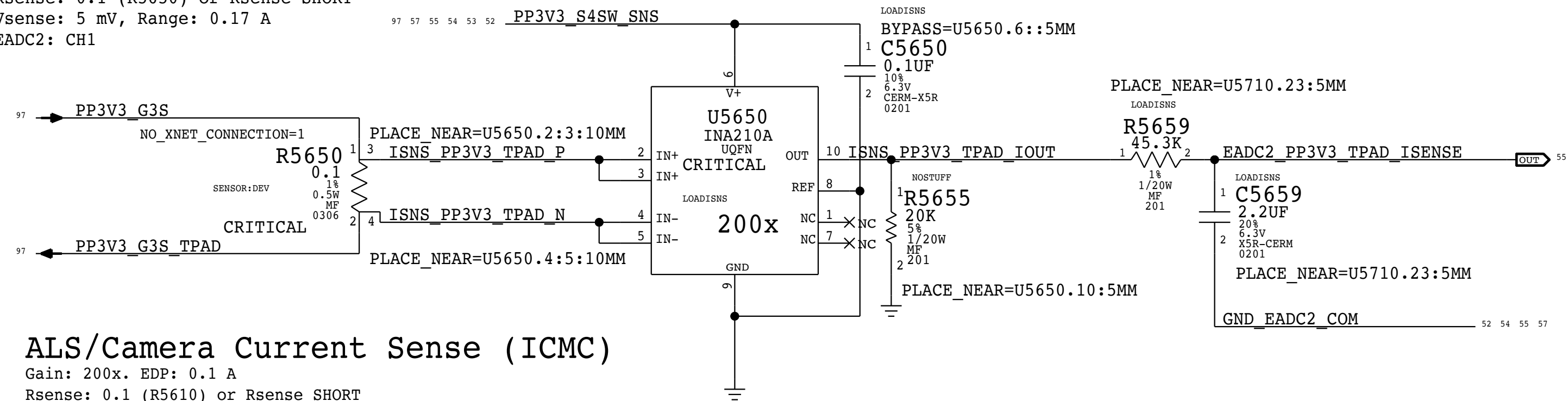
### LCD Panel 3V Current Sense (ILDC)

Gain: 200x, EDP: 1 A  
RSENSE: 0.01 (R8520) or Rsense SHORT  
Vsense: 5 mV, Range: 1.65 A  
EADC2: CH0



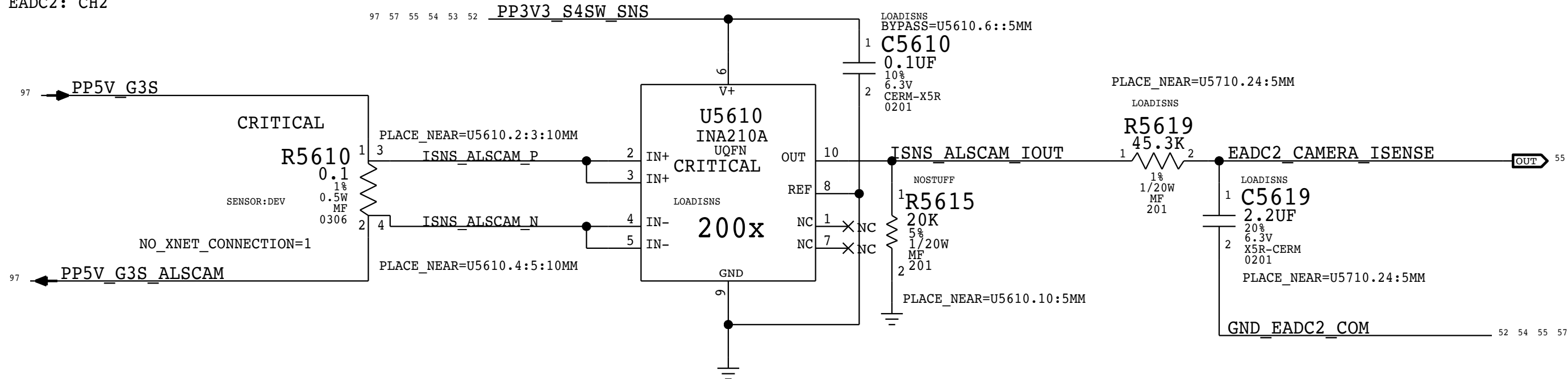
### Trackpad 3V Current Sense (IT3C)

Gain: 200x, EDP: 0.05 A  
Rsense: 0.1 (R5650) or Rsense SHORT  
Vsense: 5 mV, Range: 0.17 A  
EADC2: CH1



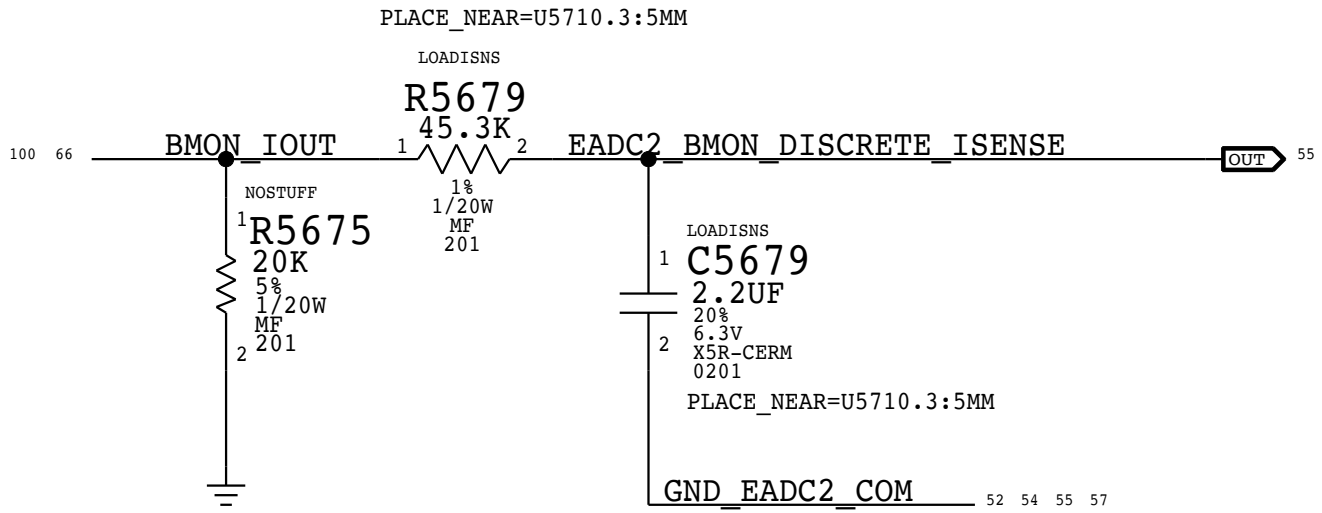
### ALS/Camera Current Sense (ICMC)

Gain: 200x, EDP: 0.1 A  
Rsense: 0.1 (R5610) or Rsense SHORT  
Vsense: 10 mV, Range: 0.17 A  
EADC2: CH2



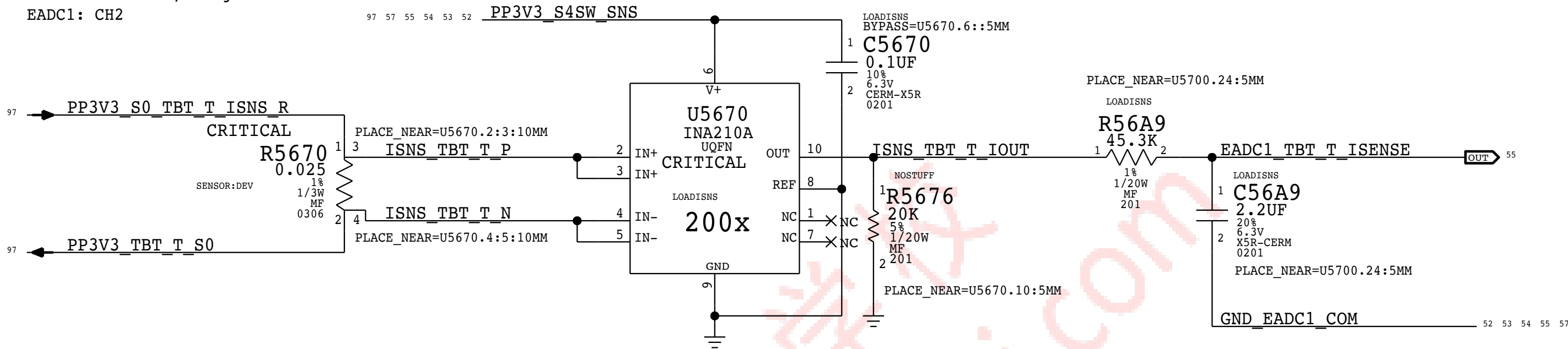
### Battery Discrete Current Sense (IB0L)

Gain: 2940x, EDP: 8 A  
Rsense: 0.003 (R501//R502)  
Vsense: 24 mV, Range: 0.28 A  
EADC2: CH5



### Thunderbolt TBT Current Right (IURC)

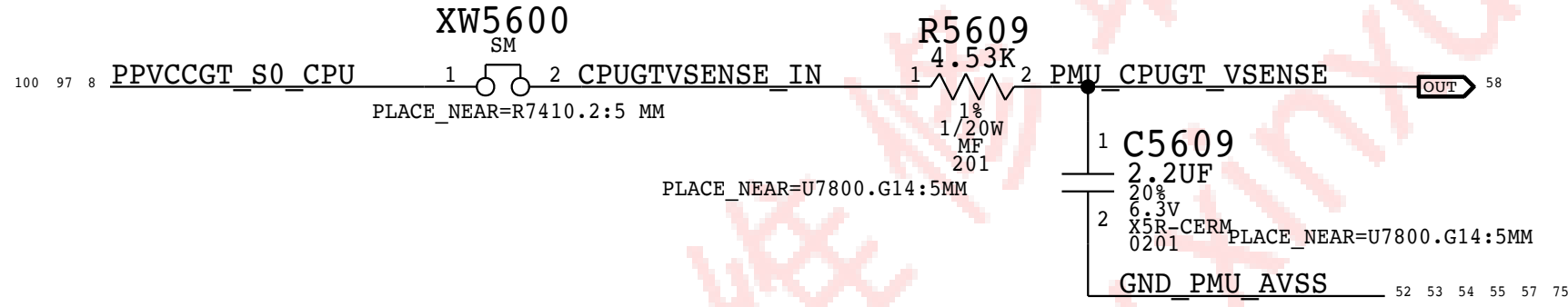
Gain: 200x, EDP: 0.8 A  
Rsense: 0.025 (R5670) or Rsense SHORT  
Vsense: 12.5 mV, Range: 0.66 A  
EADC1: CH2



| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------------------|---------------|----------|------------|
| 117S0008    | 2   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R5698,R5628   |          | LOADRC:NO  |

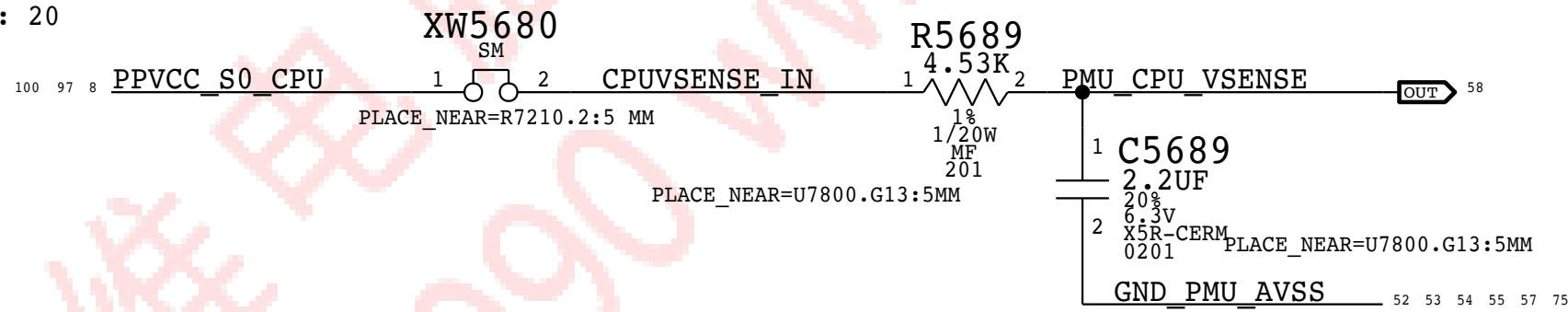
### CPU GT Voltage Sense (VCGC)

SMC ADC: 21



### CPU Core Voltage Sense (VCAC)

SMC ADC: 20



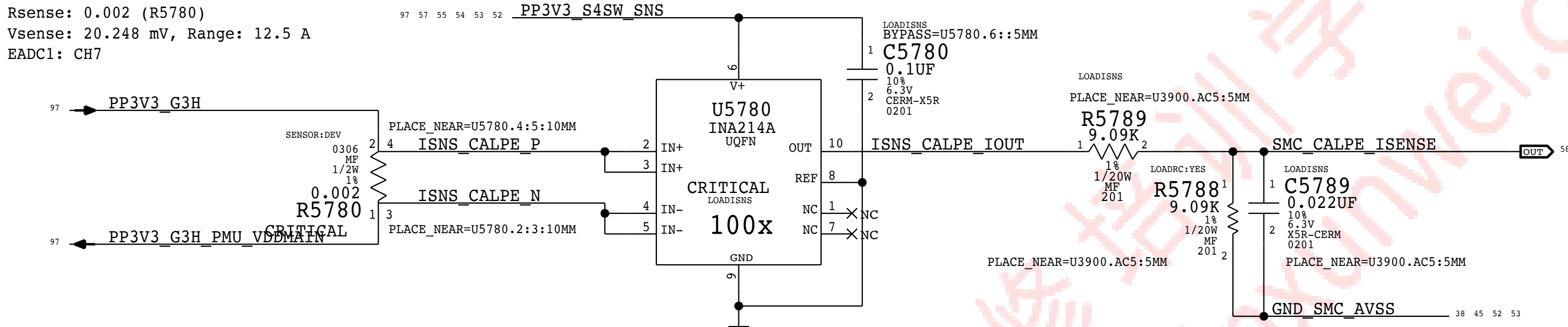
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| Power Sensors: Extended                                                                                                                                                                                                                                                                                                 |                |            |
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02166  |
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BOM\_COST\_GROUP=SENSORS



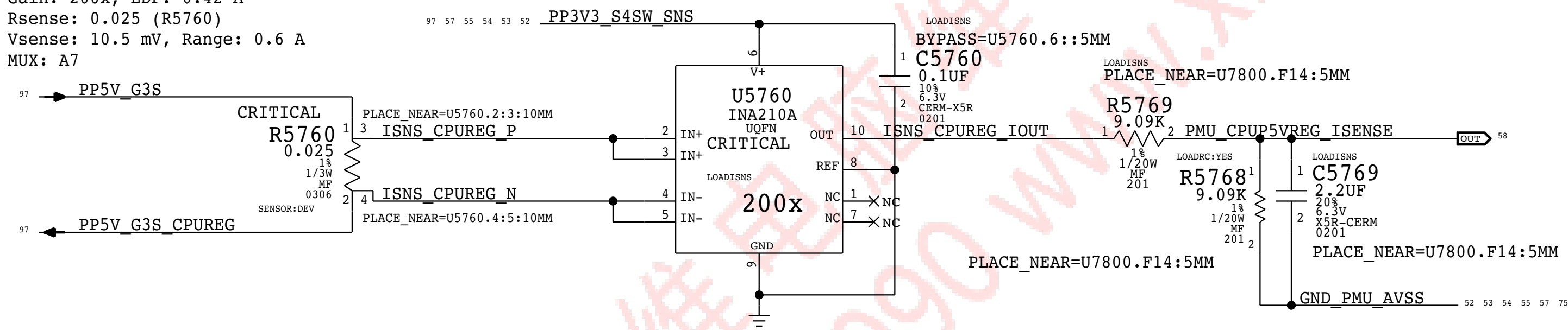
### CALPE 3.3V INPUT Current Left (IP3C)

Gain: 200x, EDP: 10.124 A  
Rsense: 0.002 (R5780)  
Vsense: 20.248 mV, Range: 12.5 A  
EADC1: CH7



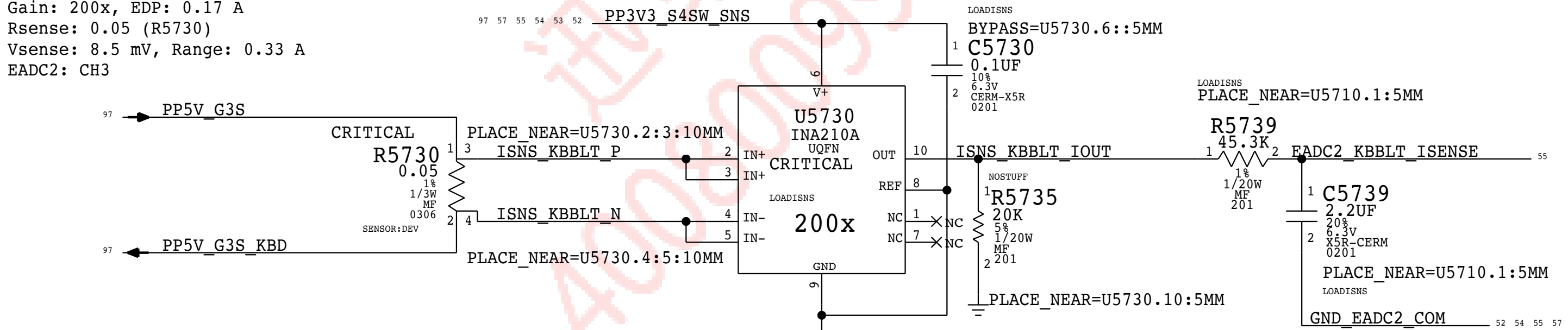
### CPU REG.5V Current Sense (IC5C)

Gain: 200x, EDP: 0.42 A  
Rsense: 0.025 (R5760)  
Vsense: 10.5 mV, Range: 0.6 A  
MUX: A7



### KB backlite Current Sense (IKBC)

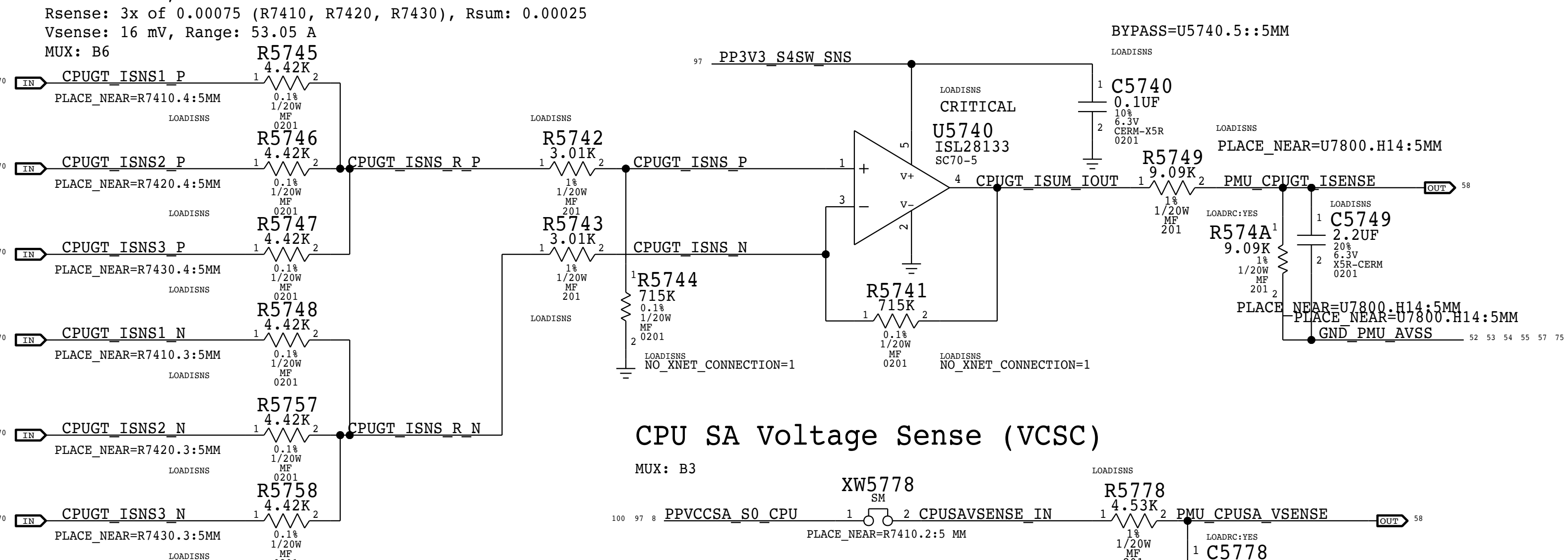
Gain: 200x, EDP: 0.17 A  
Rsense: 0.05 (R5730)  
Vsense: 8.5 mV, Range: 0.33 A  
EADC2: CH3



| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------------------|---------------|----------|------------|
| 117S0008    | 2   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R5768,R5777   |          | LOADRC:NO  |
| 117S0008    | 2   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R574A,R5788   |          | LOADRC:NO  |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | R5798         |          | LOADRC:NO  |

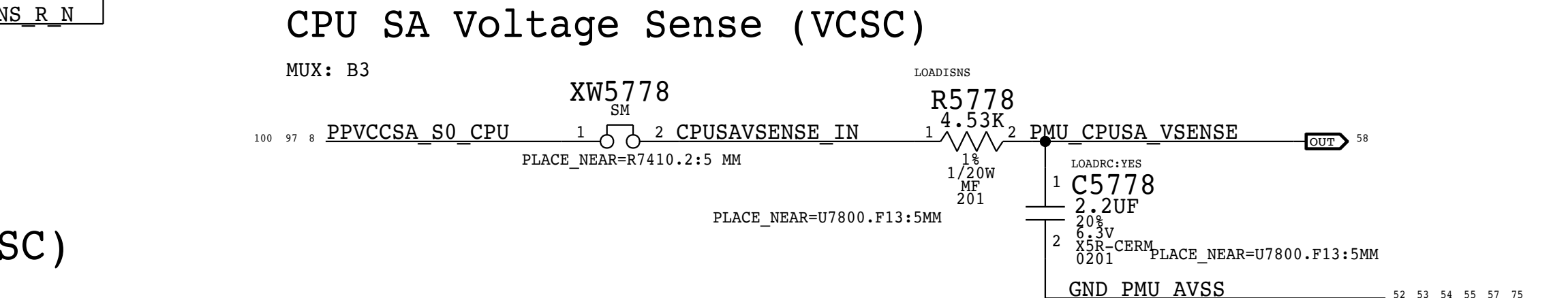
### CPU GT+GTx Current Sense (ICGC)

Gain: 188.49x, EDP: 64 A  
Rsense: 3x of 0.00075 (R7410, R7420, R7430), Rsum: 0.00025  
MUX: B6



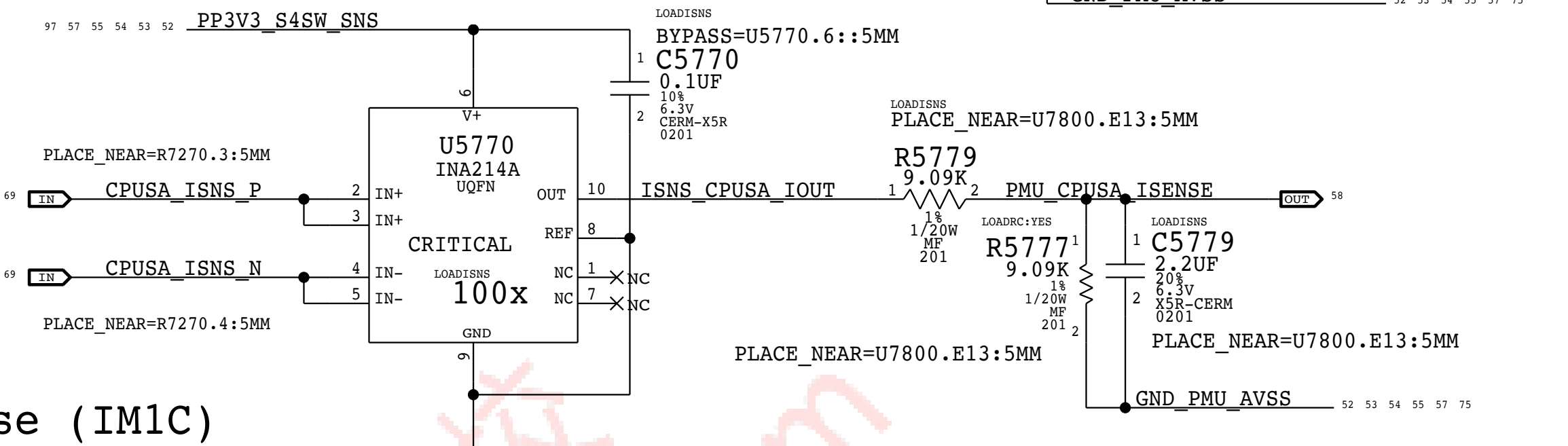
### CPU SA Voltage Sense (VCSC)

MUX: B3



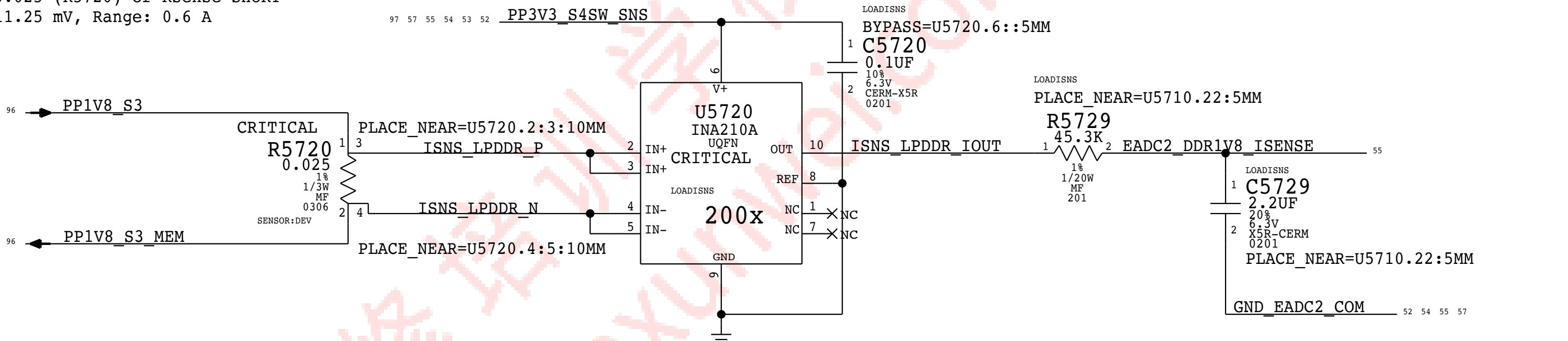
### CPU SA Current Sense (ICSC)

Gain: 100x, EDP: 9.5 A  
Rsense: 0.002 (R7270)  
Vsense: 19 mV, Range: 15 A  
MUX: B1



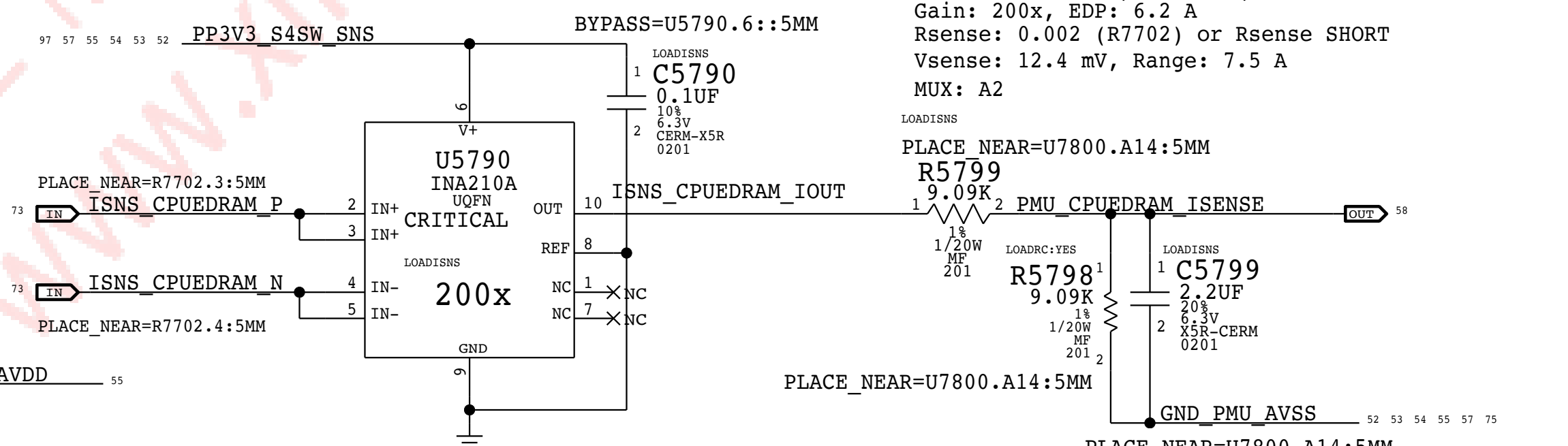
### LPDDR 1.8V Current Sense (IM1C)

Gain: 200x, EDP: 0.45 A  
Rsense: 0.025 (R5720) or Rsense SHORT  
Vsense: 11.25 mV, Range: 0.6 A  
MUX: B0

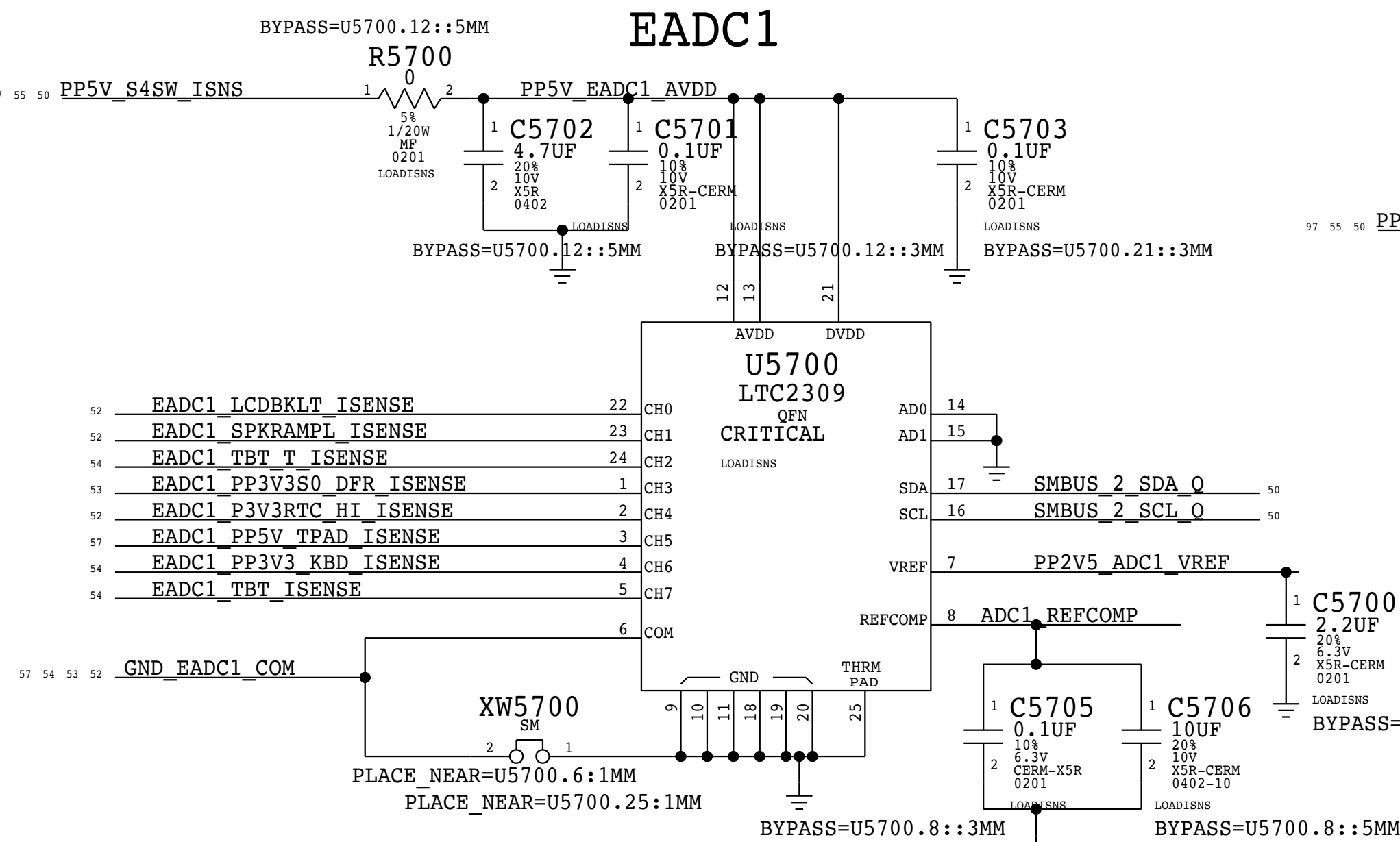


### CPU EDRAM Current Sense (ICEC)

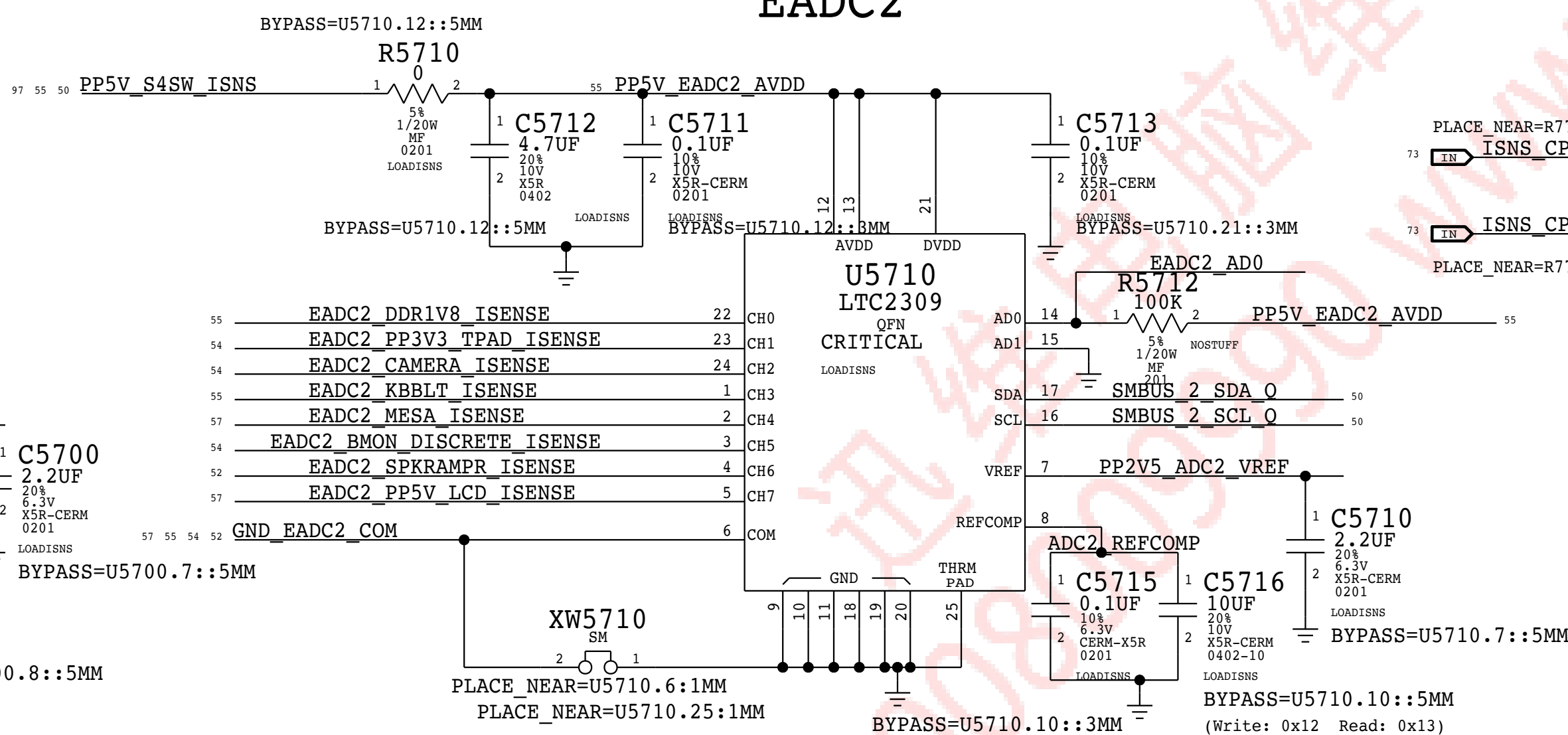
Gain: 200x, EDP: 6.2 A  
Rsense: 0.002 (R7702) or Rsense SHORT  
Vsense: 12.4 mV, Range: 7.5 A  
MUX: A2



### EADC1



### EADC2



| Power Sensors: Extended 2                                                                                                                                                                                                                                                                                         |                |            |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|--------|
|                                                                                                                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02166  | SIZE D |
|                                                                                                                                                                                                                                                                                                                   | REVISION       | 4.0.0      |        |
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BOM\_COST\_GROUP=SENSORS



Thermal Sensor A:  
Thunderbolt Die, Airflow Left

I2C Write: 0xD8, I2C Read: 0xD9

Thermal Diode: TBT Die (TTLD)

Placement Note:

The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AC22.

Thermal Sensor C:  
Thunderbolt Die, Air Flow Right

I2C Write: 0xB8, I2C Read: 0xB9

Thermal Diode: TBT Die (TRD)

Placement Note:

The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AC22.

Thermal Sensor B & CPU High Peak Detection:  
CPU Proximity, Memory Proximity, Fin Stack Left, Fin Stack Right

I2C Write: 0xB8, I2C Read: 0xB9

Thermal Diode: Fin Stack Left (Th2H)

Placement Note:

Place Q5871, Airflow thermal indicator, above the X100, on the TOP side.

Thermal Diode: CPU Proximity (TCOP)

Placement Note:

Place Q5872 under the CPU, on the TOP side.

Thermal Diode: Memory Proximity (TMOP)

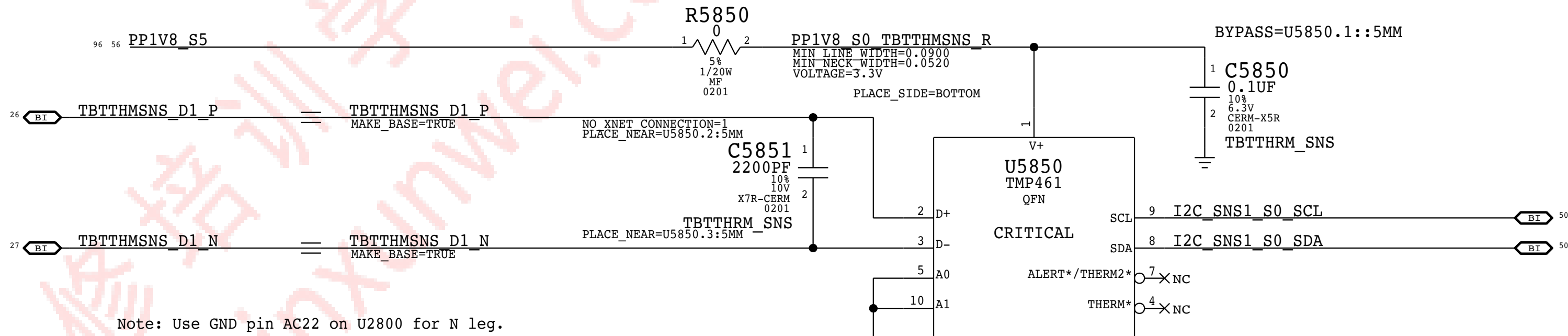
Placement Note:

Place Q5873 between two rows of Memory devices, between channel A and B, on the BOTTOM side.

HARPOON PROXIMITY (TWOP)

Placement note:

PLACE Q5874 ON BOTTOM NEAR HARPOON



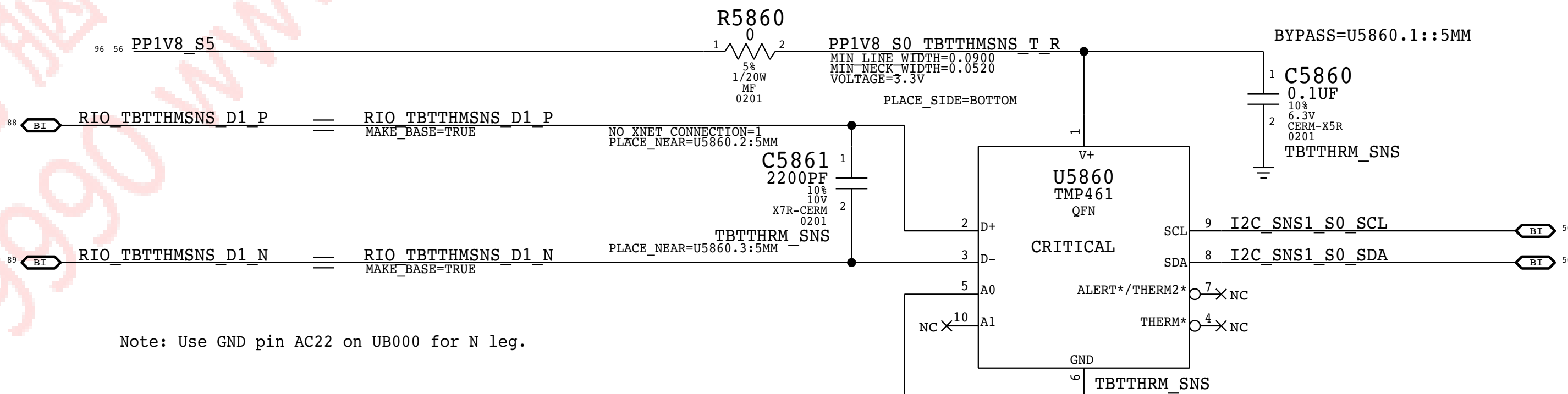
Note: Use GND pin AC22 on U2800 for N leg.

U5850 I2C Address: TMP461 is 0x90/0x91.

Thermal Sensor: Airflow Left (TaLC)

Placement Note:

Place U5860 on the BOTTOM side, at the lower corner near left fan.



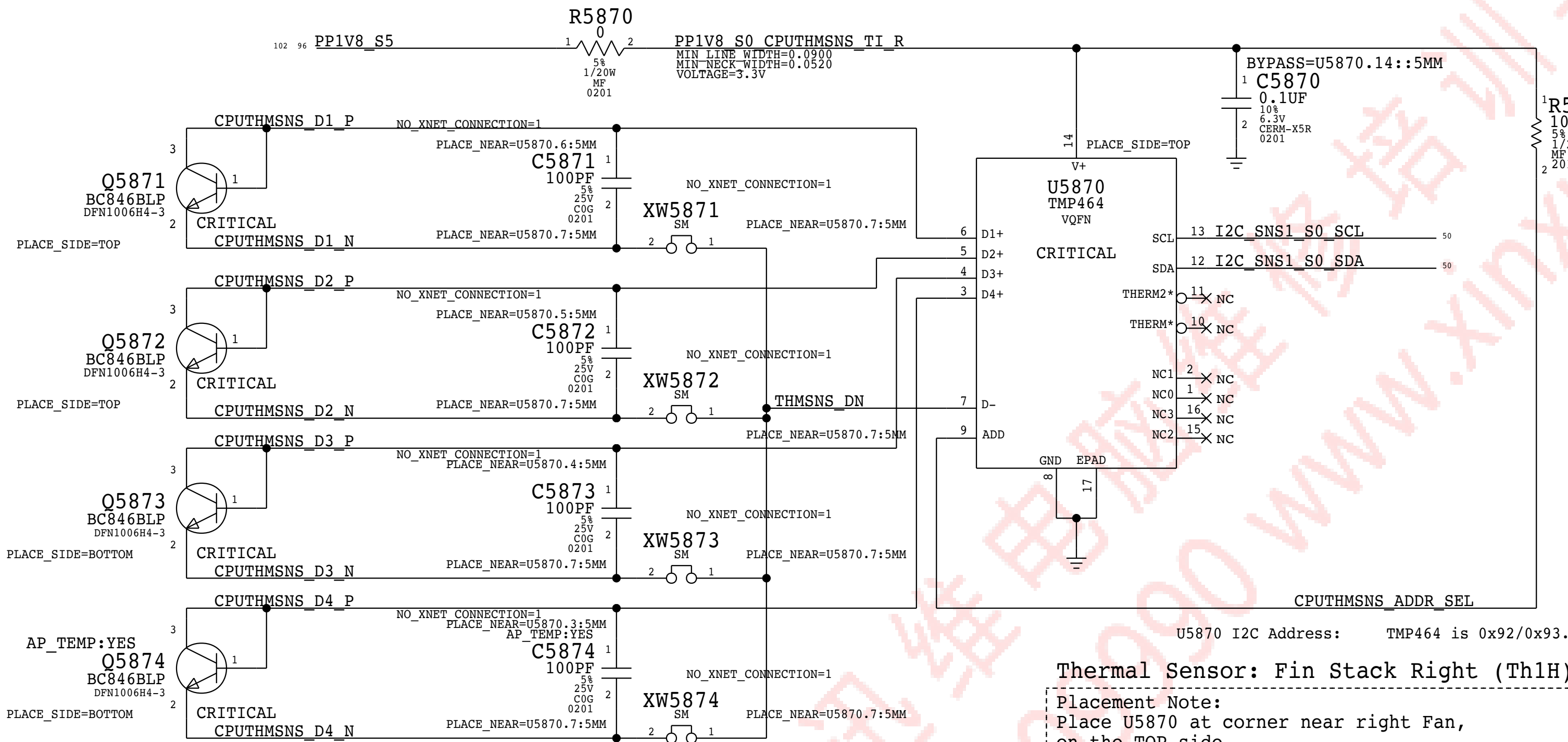
Note: Use GND pin AC22 on UB000 for N leg.

U5860 I2C Address: TMP461 is 0x96/0x97.

Thermal Sensor: Airflow Right (TaRC)

Placement Note:

Place U5860 on the BOTTOM side, at the lower corner near right fan.



U5870 I2C Address: TMP464 is 0x92/0x93.

Thermal Sensor: Fin Stack Right (Th1H)

Placement Note:

Place U5870 at corner near right Fan, on the TOP side.

| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------------------|---------------|----------|------------|
| 117S0002    | 1   | RES,WTL,FLIM,0ohm,1/20W,0201,SMD,LF | C5874         |          | AP_TEMP:NO |

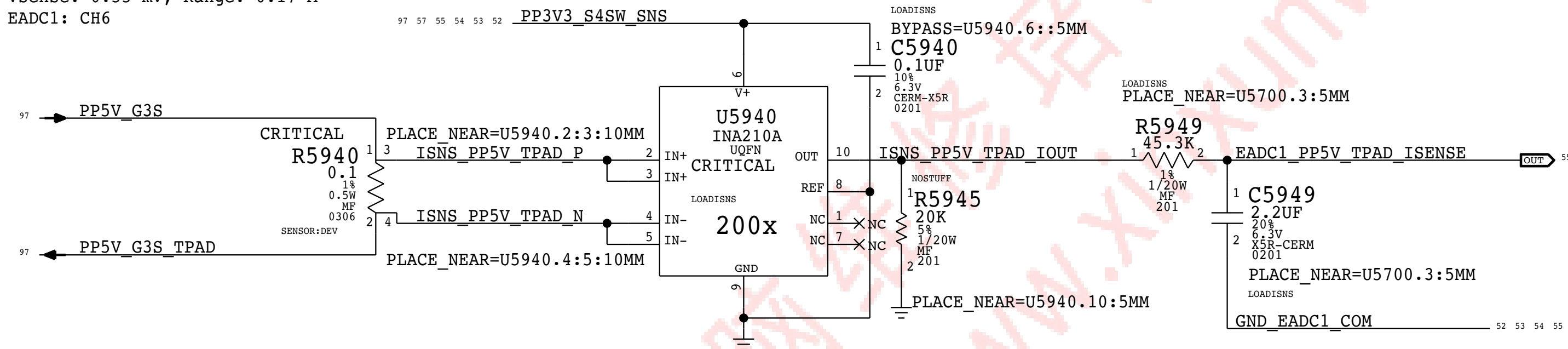
BOM\_COST\_GROUP=SENSORS

| PAGE TITLE                                                                                                        |                |            |
|-------------------------------------------------------------------------------------------------------------------|----------------|------------|
| Thermal Sensors                                                                                                   |                |            |
|                                                                                                                   | DRAWING NUMBER | 051-02166  |
|                                                                                                                   | REVISION       | 4.0.0      |
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| BRANCH                                                                                                            |                | evt-mars-0 |
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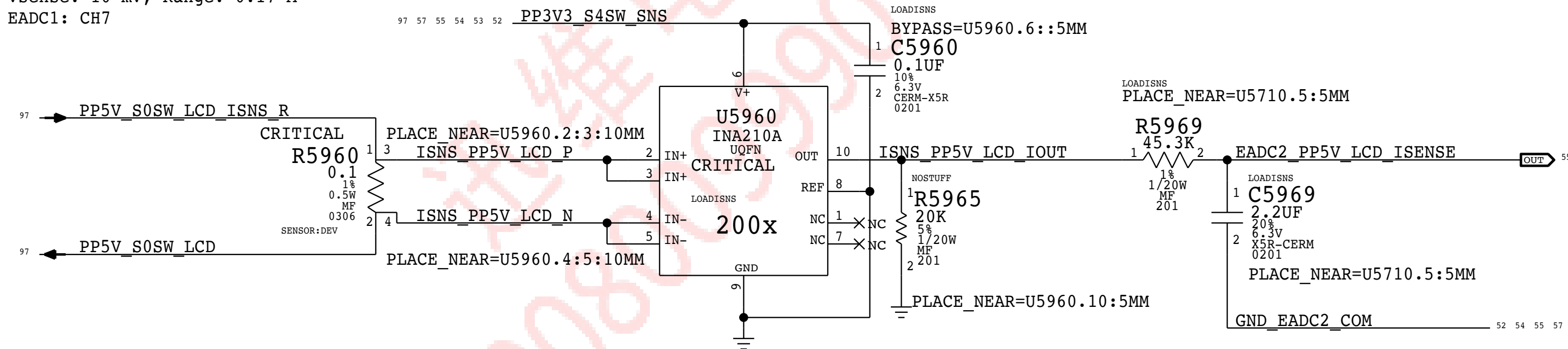
### Trackpad 5V Current Sense (IT5C)

Gain: 200x, EDP: 0.0055 A  
Rsense: 0.1 (R5940) or Rsense SHORT  
Vsense: 0.55 mV, Range: 0.17 A  
EADC1: CH6



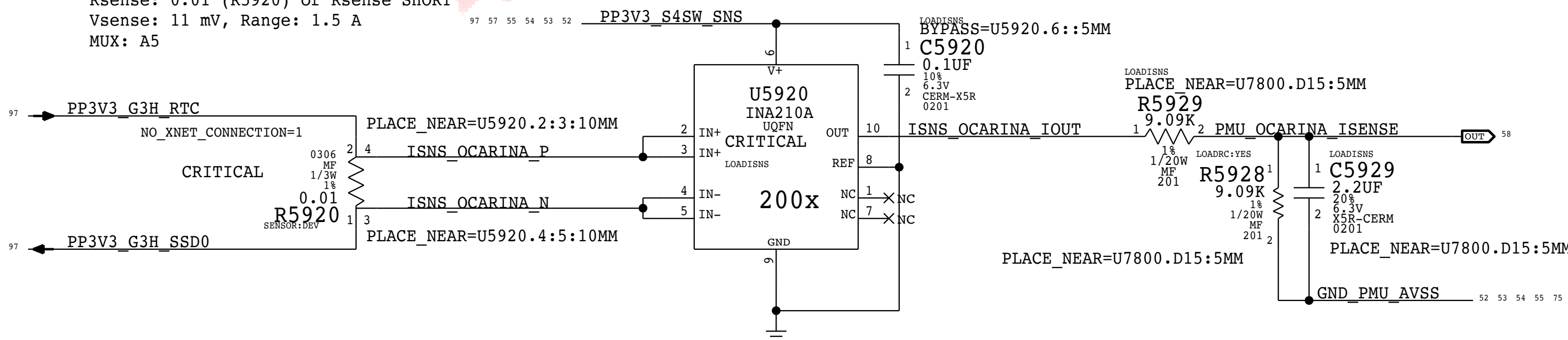
### LCD Panel 5V Current Sense (IL5C)

Gain: 200x, EDP: 0.1 A  
Rsense: 0.1 (R5960) or Rsense SHORT  
Vsense: 10 mV, Range: 0.17 A  
EADC1: CH7



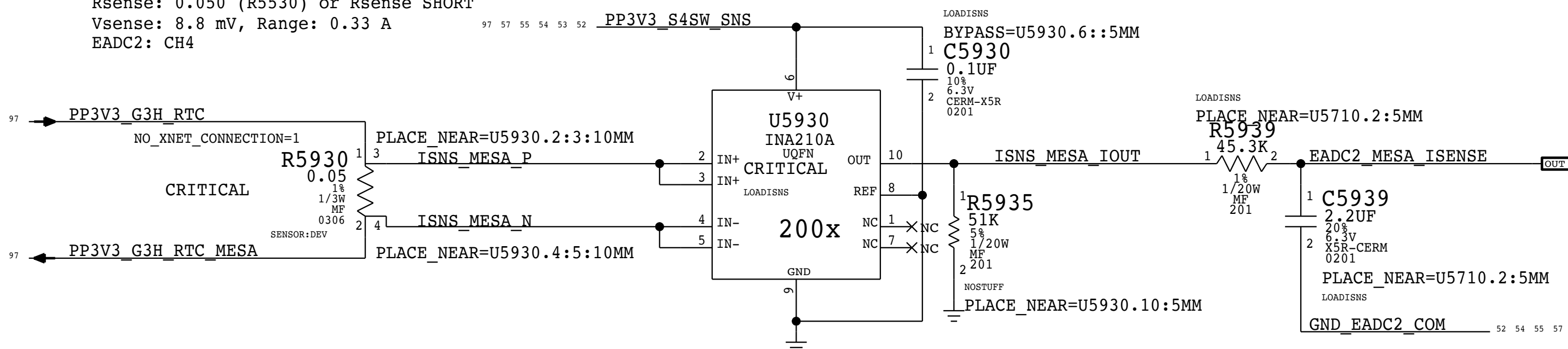
### Ocarina Current Sense (IHCC)

Gain: 200x, EDP: 1.1 A  
Rsense: 0.01 (R5920) or Rsense SHORT  
Vsense: 11 mV, Range: 1.5 A  
MUX: A5



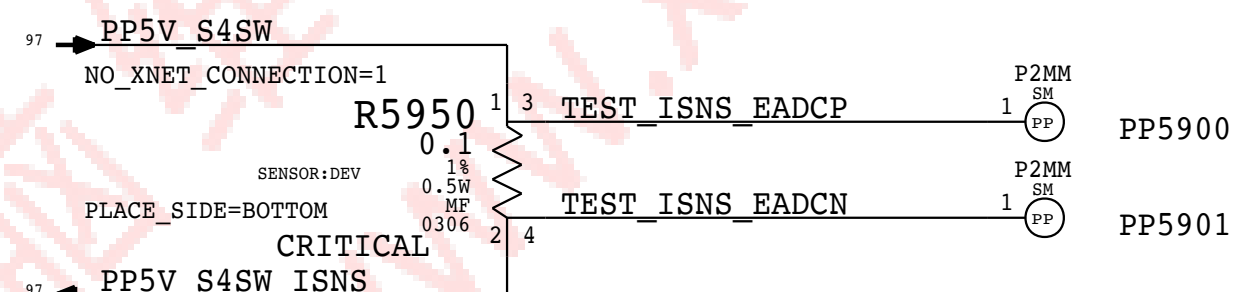
### MESA Current Sense (IIDC)

Gain: 200x, EDP: 0.176 A  
Rsense: 0.050 (R5930) or Rsense SHORT  
Vsense: 8.8 mV, Range: 0.33 A  
EADC2: CH4



### EADC Current Sense

EDP: 6m A  
Rsense: 0.1 (R5950) or Rsense SHORT  
Vsense: 0.6 mV



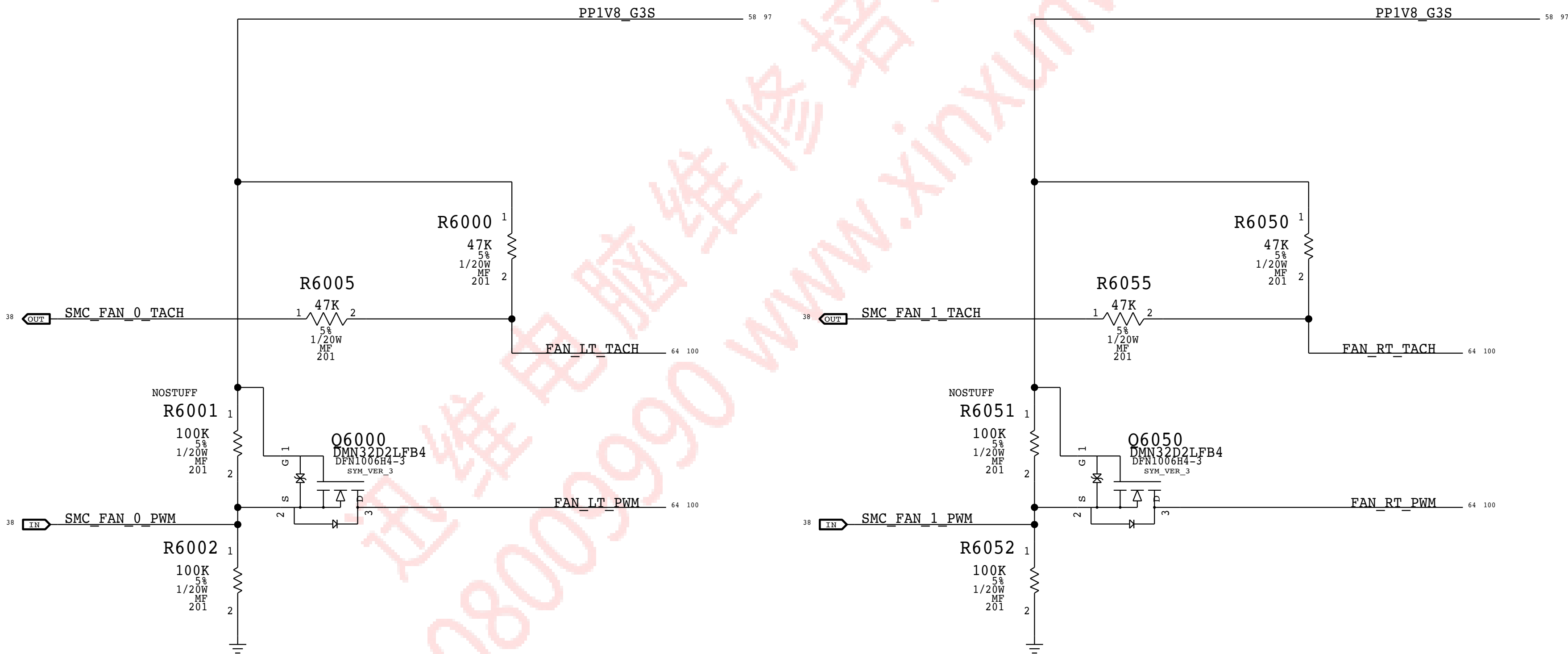
| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------------------|---------------|----------|------------|
| 117S0008    | 2   | RES,MTL FLTM,100K,1/16W,0201,SMD,LF | R5928,R5958   |          | LOADRC:NO  |

BOM\_COST\_GROUP=SENSORS

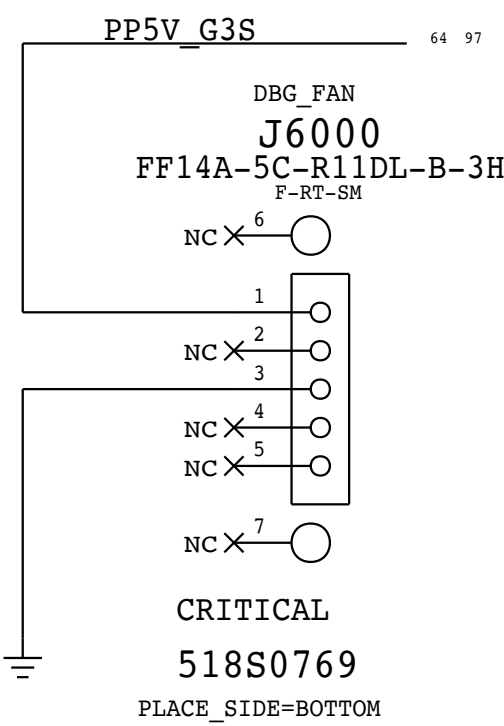
| Power Sensors:Extended 3                                                                                                                                                                                                                                                                                                |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02166  |
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# FAN CONTROL



# FOR DEBUG FAN POWER



## SMC ADC Assignments

|    |    |                         |    |                         |     |    |
|----|----|-------------------------|----|-------------------------|-----|----|
| 52 | IN | SMC_CPU_HS_ISENSE       | == | SMC_CPU_HS_ISENSE       | OUT | 38 |
| 52 | IN | SMC_PBUS_VSENSE         | == | SMC_PBUS_VSENSE         | OUT | 38 |
| 52 | IN | SMC_BMON_ISENSE         | == | SMC_BMON_ISENSE         | OUT | 38 |
| 52 | IN | SMC_DCIN_ISENSE         | == | SMC_DCIN_ISENSE         | OUT | 38 |
| 52 | IN | SMC_DCIN_VSENSE         | == | SMC_DCIN_VSENSE         | OUT | 38 |
| 53 | IN | SMC_PP3V3_WLANBT_ISENSE | == | SMC_PP3V3_WLANBT_ISENSE | OUT | 38 |
| 53 | IN | SMC_PP1V8_WLANBT_ISENSE | == | SMC_PP1V8_WLANBT_ISENSE | OUT | 38 |
| 55 | IN | SMC_CALPE_ISENSE        | == | SMC_CALPE_ISENSE        | OUT | 38 |

KBL-U NB Assignments

CPUGT ISNS IACORE ISNS

ADC5 ADC6

Desktop Assignments

GPU\_AUX GPU\_CORE 12V\_VSNS 12V\_ISNS Same

ADC0 ADC1 ADC2 ADC3 ADC4-7

## PMU ADC Assignments

|    |    |                        |    |                        |     |    |
|----|----|------------------------|----|------------------------|-----|----|
| 52 | IN | PMU_OTHER3V3_HI_ISENSE | == | PMU_OTHER3V3_HI_ISENSE | OUT | 76 |
| 52 | IN | PMU_OTHER5V_HI_ISENSE  | == | PMU_OTHER5V_HI_ISENSE  | OUT | 76 |
| 55 | IN | PMU_CPUEDRAM_ISENSE    | == | PMU_CPUEDRAM_ISENSE    | OUT | 76 |
| 53 | IN | PMU_CPUVCCIO_ISENSE    | == | PMU_CPUVCCIO_ISENSE    | OUT | 76 |
| 53 | IN | PMU_DDR1V2_ISENSE      | == | PMU_DDR1V2_ISENSE      | OUT | 76 |
| 57 | IN | PMU_OCARINA_ISENSE     | == | PMU_OCARINA_ISENSE     | OUT | 76 |
| 57 | IN | PMU_SSDNAND_ISENSE     | == | PMU_SSDNAND_ISENSE     | OUT | 76 |
| 55 | IN | PMU_CPUP5VREG_ISENSE   | == | PMU_CPUP5VREG_ISENSE   | OUT | 76 |
|    | IN | NC_PMU_AMUX_AY         | == | NC_PMU_AMUX_AY         | OUT | 76 |
| 54 | IN | PMU_LCDPANEL_ISENSE    | == | PMU_LCDPANEL_ISENSE    | OUT | 76 |
| 55 | IN | PMU_CPUSA_ISENSE       | == | PMU_CPUSA_ISENSE       | OUT | 76 |
| 53 | IN | PMU_CPUDDR_ISENSE      | == | PMU_CPUDDR_ISENSE      | OUT | 76 |
| 55 | IN | PMU_CPUSA_VSENSE       | == | PMU_CPUSA_VSENSE       | OUT | 76 |
| 54 | IN | PMU_CPU_VSENSE         | == | PMU_CPU_VSENSE         | OUT | 76 |
| 54 | IN | PMU_CPUGT_VSENSE       | == | PMU_CPUGT_VSENSE       | OUT | 76 |
| 55 | IN | PMU_CPUGT_ISENSE       | == | PMU_CPUGT_ISENSE       | OUT | 76 |
| 53 | IN | PMU_CPU_ISENSE         | == | PMU_CPU_ISENSE         | OUT | 76 |
|    | IN | NC_PMU_AMUX_BY         | == | NC_PMU_AMUX_BY         | OUT | 76 |

BOM\_COST\_GROUP=FAN

|                                                                                                                                                                                                                                                                                                                   |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |            |
| Fans/SMC/AMUX Support                                                                                                                                                                                                                                                                                             |                |            |
| Apple Inc.                                                                                                                                                                                                                                                                                                        | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                   | REVISION       | 4.0.0      |
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|                                                                                                                                                                                                                                                                                                                   | PAGE           | 60 OF 150  |
|                                                                                                                                                                                                                                                                                                                   | SHEET          | 58 OF 108  |



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| D |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| A |   |   |   |   |   |   |   |   |
|   | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

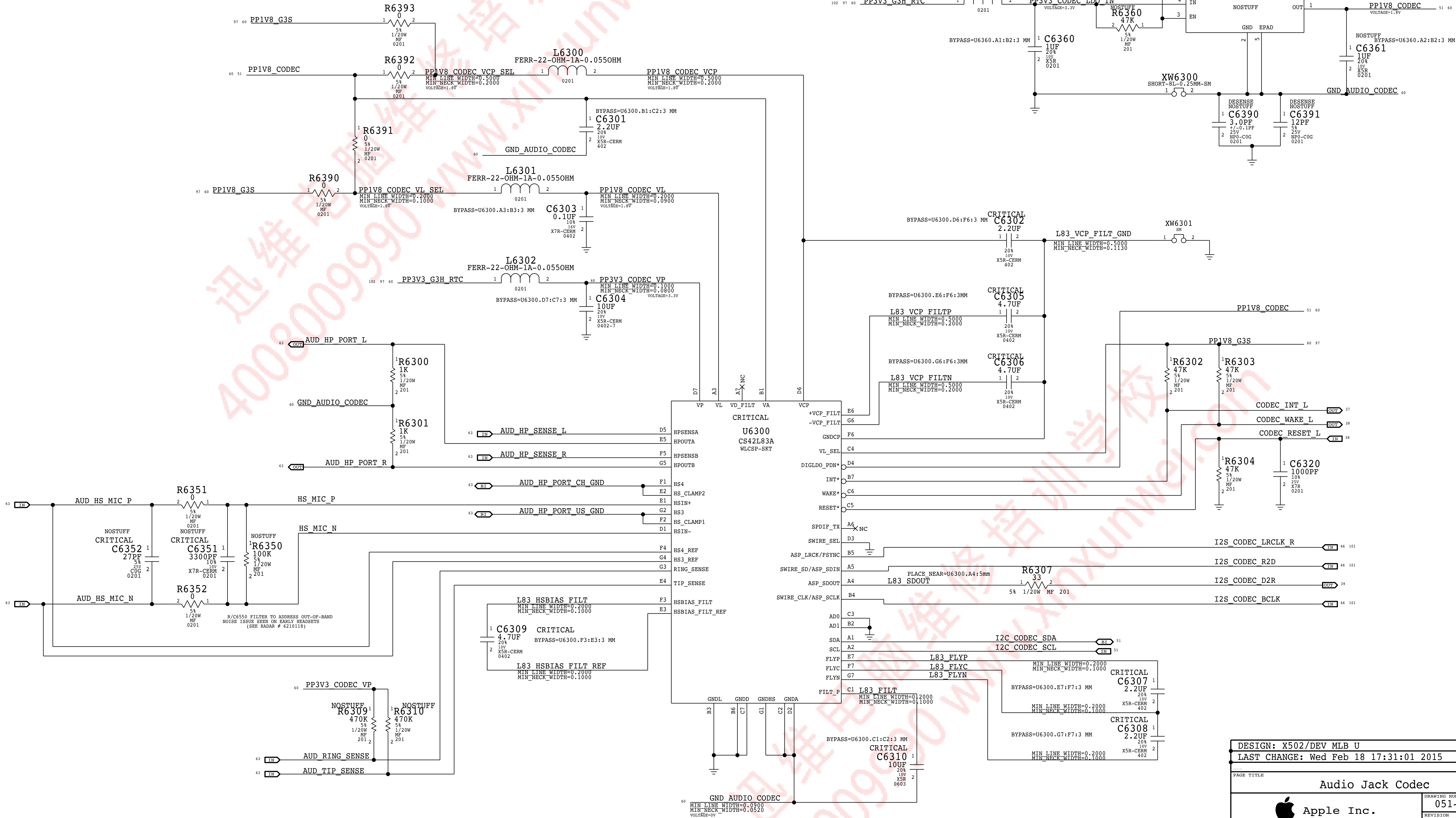
迅维电脑维修培训学校  
4008009990 www.xinxunwei.com

|                                                                                                                                                                                                                                                                                                                         |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| DESIGN: X502/DEV MLB U                                                                                                                                                                                                                                                                                                  |                |            |
| LAST CHANGE: Wed Feb 18 17:12:24 2015                                                                                                                                                                                                                                                                                   |                |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
| Audio Placeholder                                                                                                                                                                                                                                                                                                       |                |            |
|  Apple Inc.                                                                                                                                                                                                                        | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                         | REVISION       | 4.0.0      |
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|                                                                                                                                                                                                                                                                                                                         | SHEET          | 59 OF 108  |

BOM\_COST\_GROUP=AUDIO



| AUDIO JACK CODEC I2C ADDRESS |      |          |
|------------------------------|------|----------|
| AD1                          | AD0  | ADDRESS  |
| GND                          | GND  | 0x48 <-- |
| GND                          | 1.8V | 0x49     |
| 1.8V                         | GND  | 0x4A     |
| 1.8V                         | 1.8V | 0x4B     |



BOM\_COST\_GROUP=AUDIO

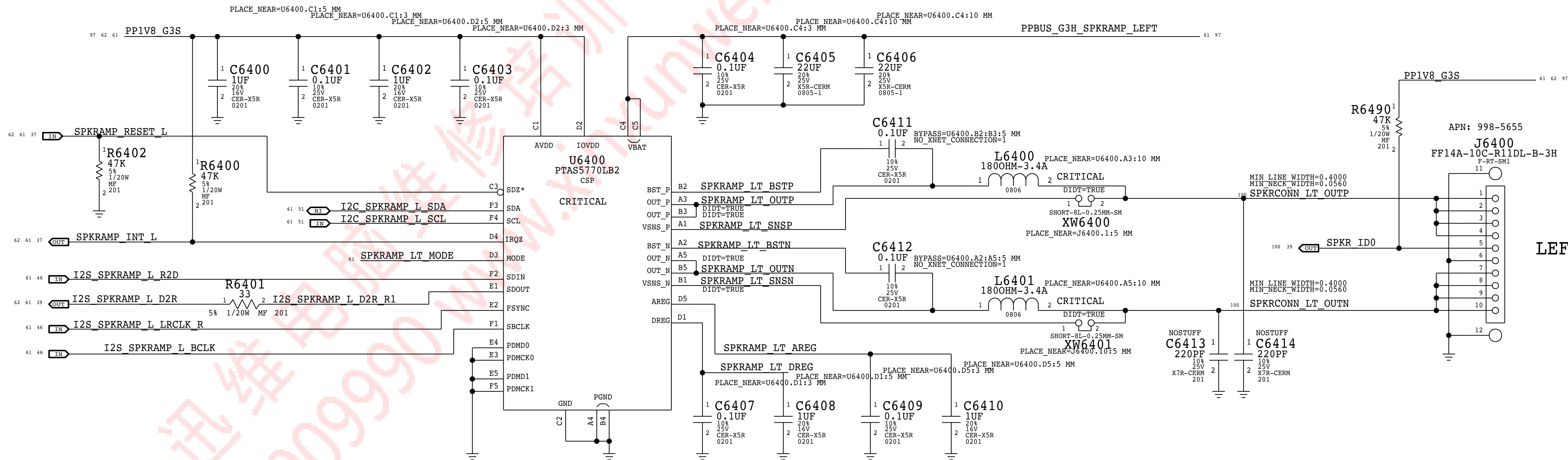
|                                                                                                                   |                |           |
|-------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| DESIGN: X502/DEV MLB U                                                                                            |                |           |
| LAST CHANGE: Wed Feb 18 17:31:01 2015                                                                             |                |           |
| PAGE TITLE                                                                                                        |                |           |
| Audio Jack Codec                                                                                                  |                |           |
|                                                                                                                   | DRAWING NUMBER | 051-02166 |
|                                                                                                                   | REVISION       | 4.0.0     |
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| BRANCH                                                                                                            | evt-mars-0     |           |
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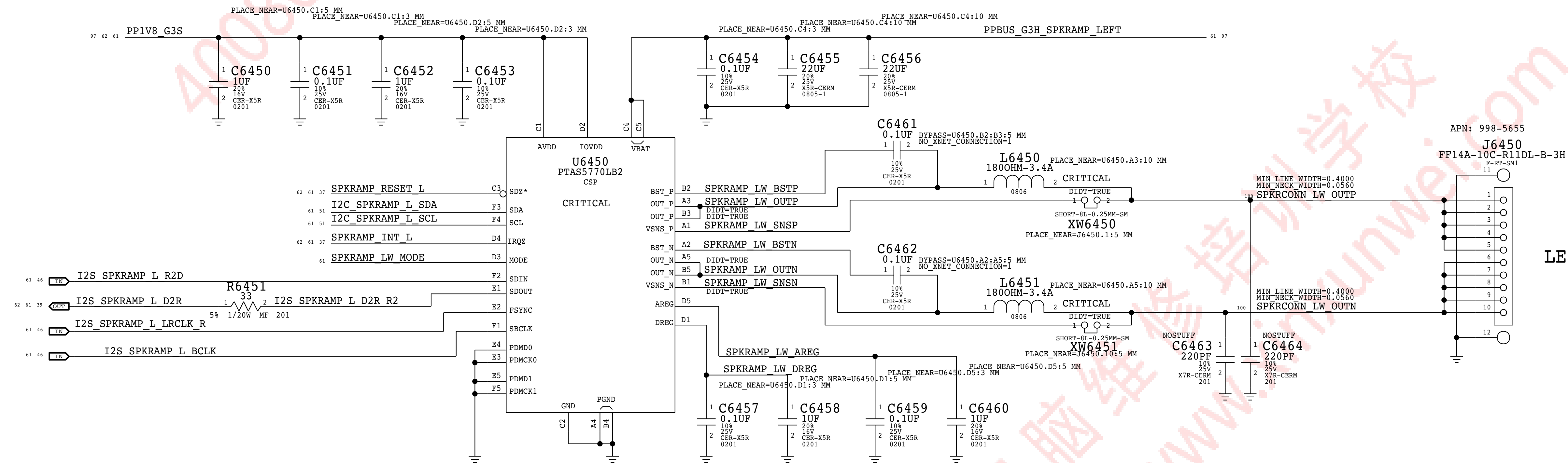
## 2X MONO SPEAKER LEFT AMPLIFIERS

APN: 353801252

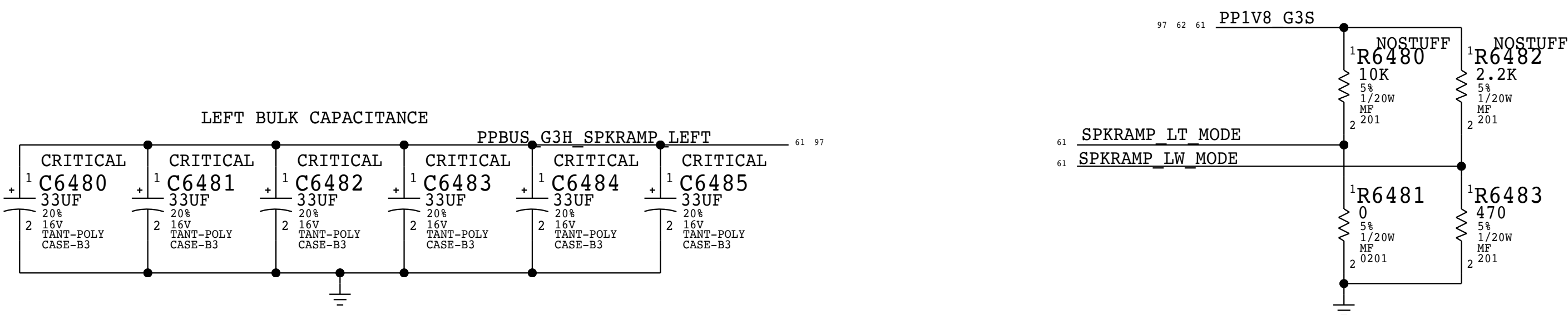
GAIN: 0DBFS = xxVRMS



LEFT TWEETER



LEFT WOOFER



| MODE         | PIN  | I2C ADDR | CHANNEL |
|--------------|------|----------|---------|
| GND          | 0x31 |          | L TW    |
| 470 to GND   | 0x32 |          | L WF    |
| 470 to IOVDD | 0x33 |          | R TW    |
| 2k2 to GND   | 0x34 |          | R WF    |
| 2k2 to IOVDD | 0x35 |          |         |
| 10k to GND   | 0x36 |          |         |
| 10k to IOVDD | 0x37 |          |         |
| 47k to IOVDD | 0x38 |          |         |

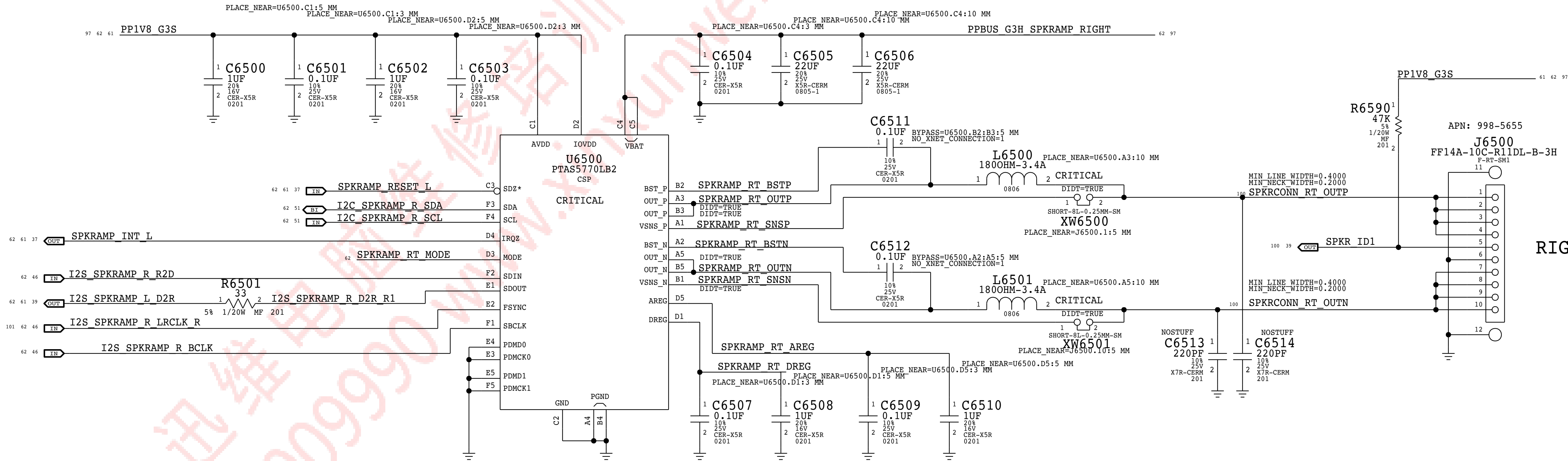
BOM\_COST\_GROUP=AUDIO

|                                                                                                                   |                |            |      |
|-------------------------------------------------------------------------------------------------------------------|----------------|------------|------|
| PAGE TITLE                                                                                                        |                |            |      |
| Audio Left Amplifiers                                                                                             |                |            |      |
|                                                                                                                   | DRAWING NUMBER | 051-02166  | SIZE |
|                                                                                                                   | REVISION       | 4.0.0      | D    |
|                                                                                                                   | BRANCH         | evt-mars-0 |      |
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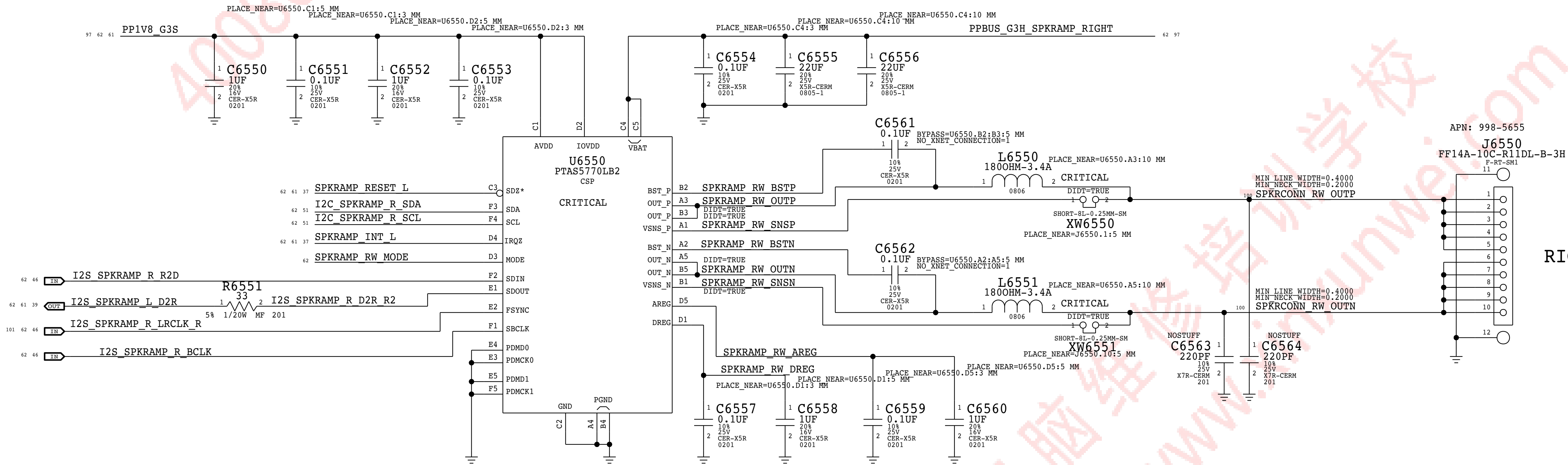


2X MONO SPEAKER RIGHT AMPLIFIERS

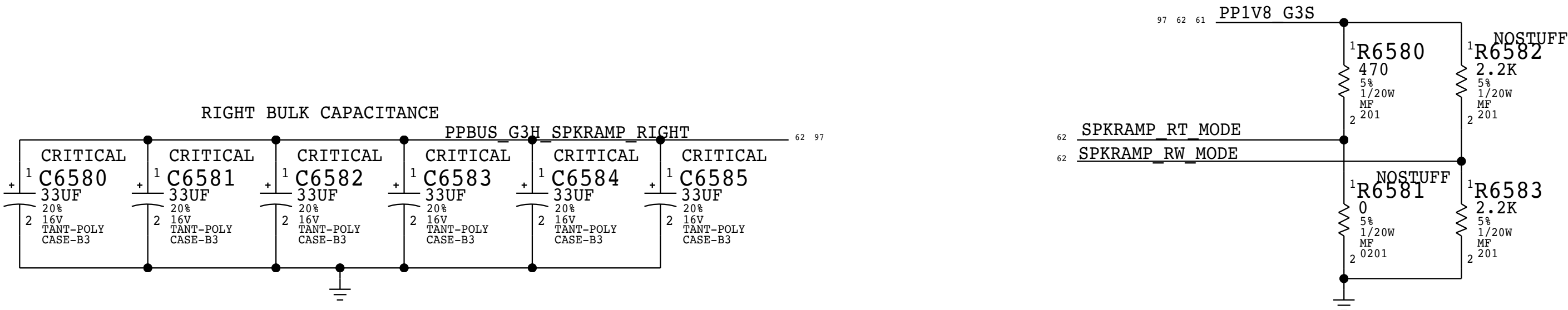
APN: 353801252  
GAIN: 0DBFS = xxVRMS



RIGHT TWEETER



RIGHT WOOFER

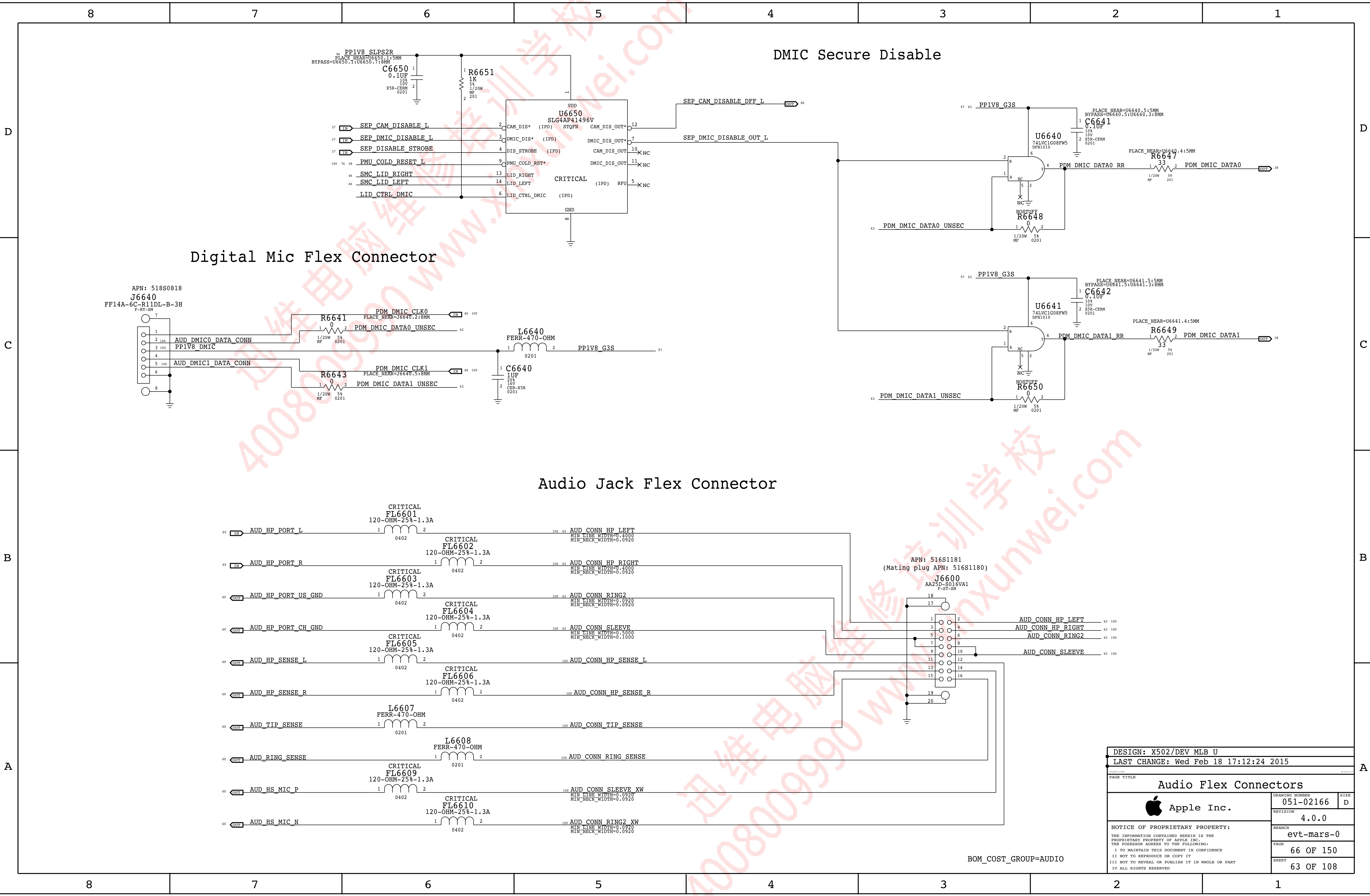


| MODE         | PIN | I2C ADDR | CHANNEL |
|--------------|-----|----------|---------|
| GND          |     | 0x31     | L TW    |
| 470 to GND   |     | 0x32     | L WF    |
| 470 to IOVDD |     | 0x33     | R TW    |
| 2K2 to GND   |     | 0x34     | R WF    |
| 2k2 to IOVDD |     | 0x35     |         |
| 10k to GND   |     | 0x36     |         |
| 10k to IOVDD |     | 0x37     |         |
| 47k to IOVDD |     | 0x38     |         |

BOM\_COST\_GROUP=AUDIO

|                                                                                                                   |                |            |           |
|-------------------------------------------------------------------------------------------------------------------|----------------|------------|-----------|
| PAGE TITLE                                                                                                        |                |            |           |
| Audio Right Amplifiers                                                                                            |                |            |           |
|                                                                                                                   | DRAWING NUMBER | 051-02166  | SIZE      |
|                                                                                                                   | REVISION       | 4.0.0      | D         |
|                                                                                                                   | BRANCH         | evt-mars-0 |           |
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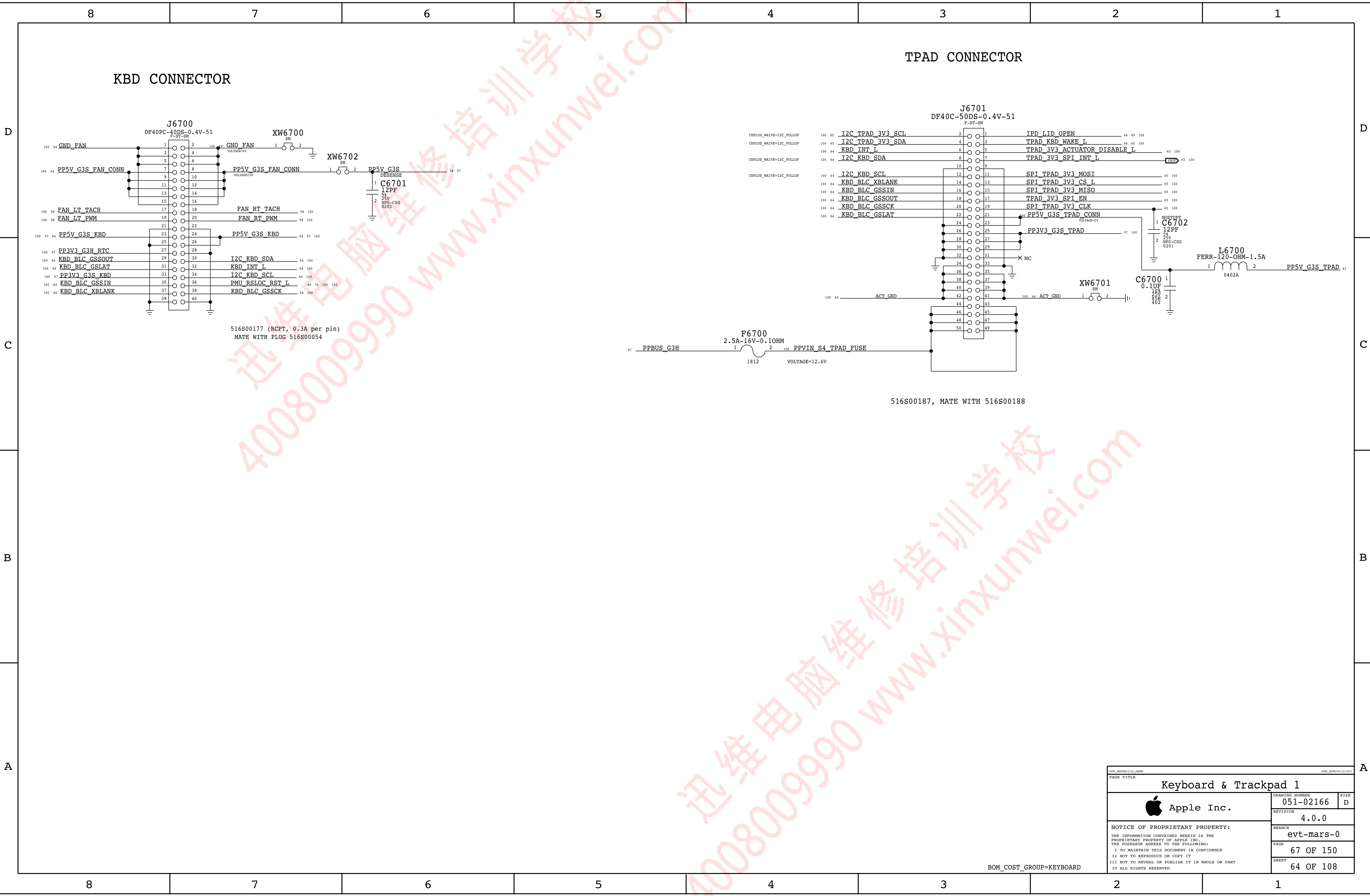


Digital Mic Flex Connector

Audio Jack Flex Connector

|                                                                                                                                                                                                                                                                                                                         |                |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| DESIGN: X502/DEV MLB U                                                                                                                                                                                                                                                                                                  |                |                    |
| LAST CHANGE: Wed Feb 18 17:12:24 2015                                                                                                                                                                                                                                                                                   |                |                    |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |                    |
| Audio Flex Connectors                                                                                                                                                                                                                                                                                                   |                |                    |
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02166          |
|                                                                                                                                                                                                                                                                                                                         | REVISION       | 4.0.0              |
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|                                                                                                                                                                                                                                                                                                                         |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
| Keyboard & Trackpad 1                                                                                                                                                                                                                                                                                                   |                |            |
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                         | REVISION       | 4.0.0      |
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BOM\_COST\_GROUP=KEYBOARD

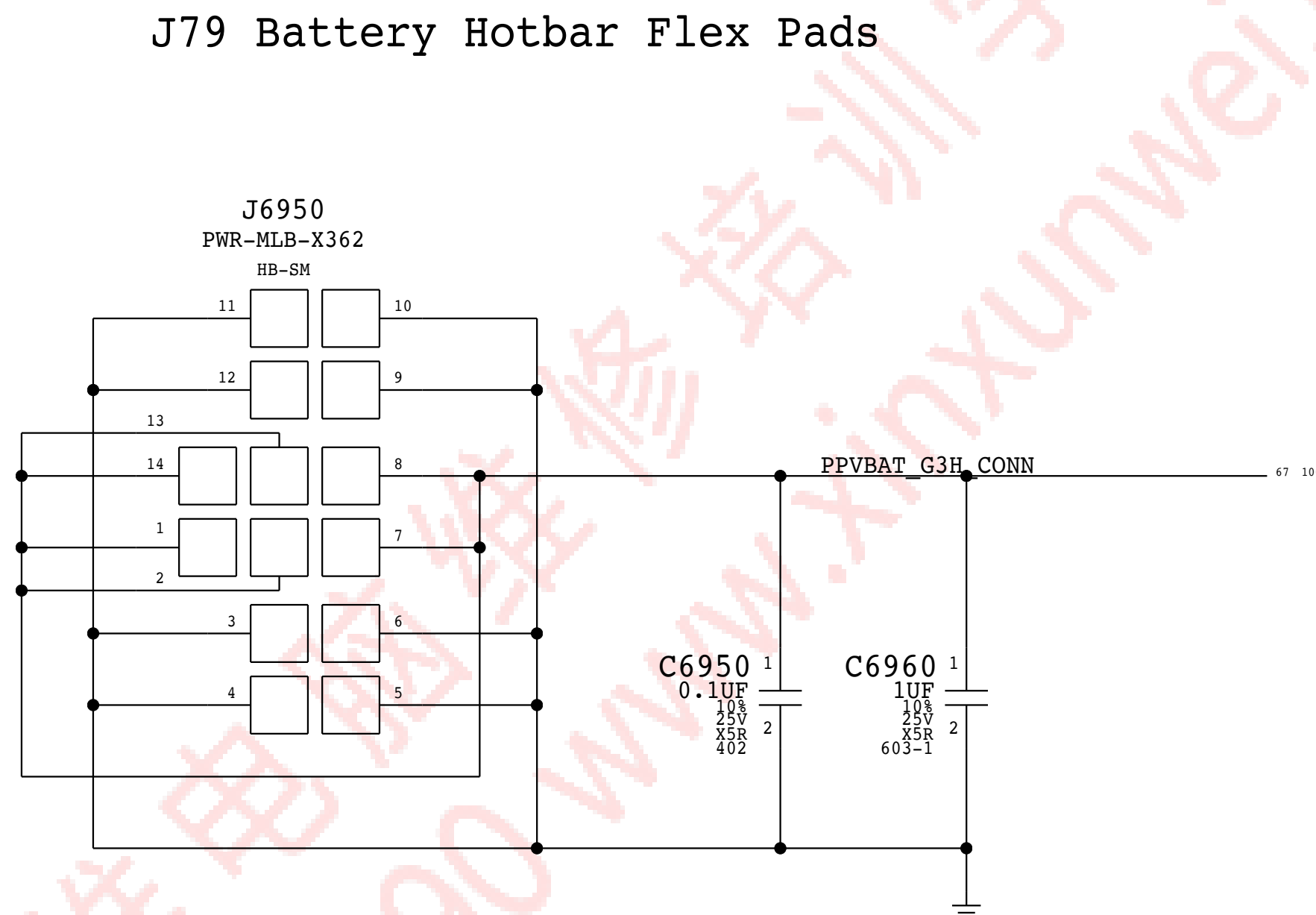


|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|---|

A

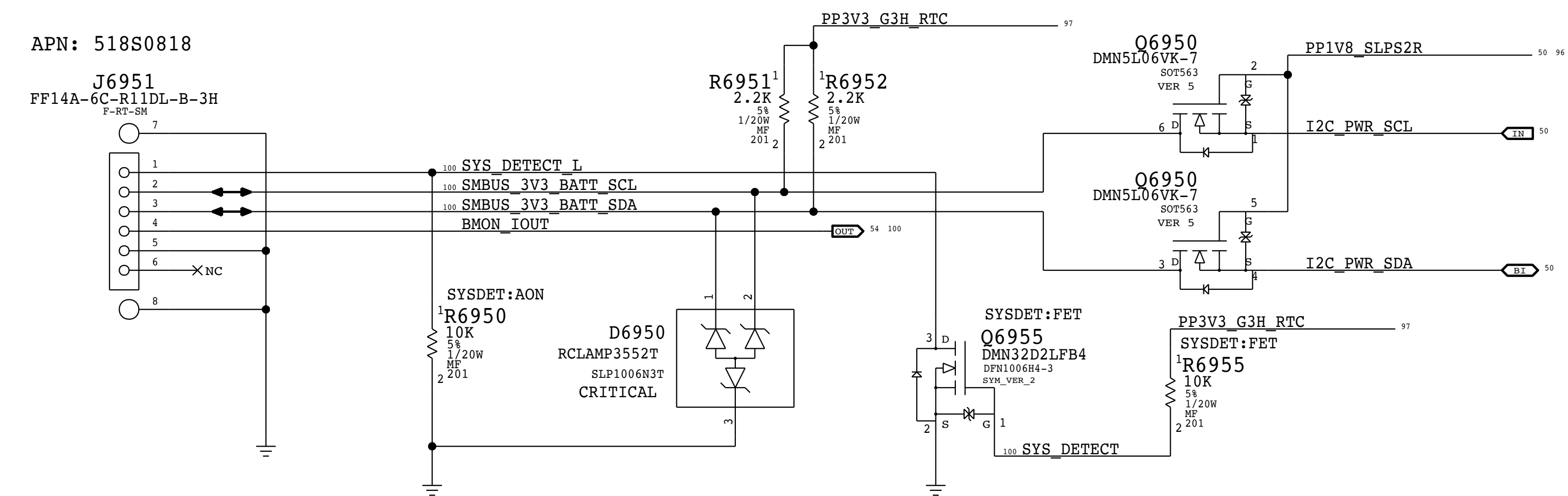
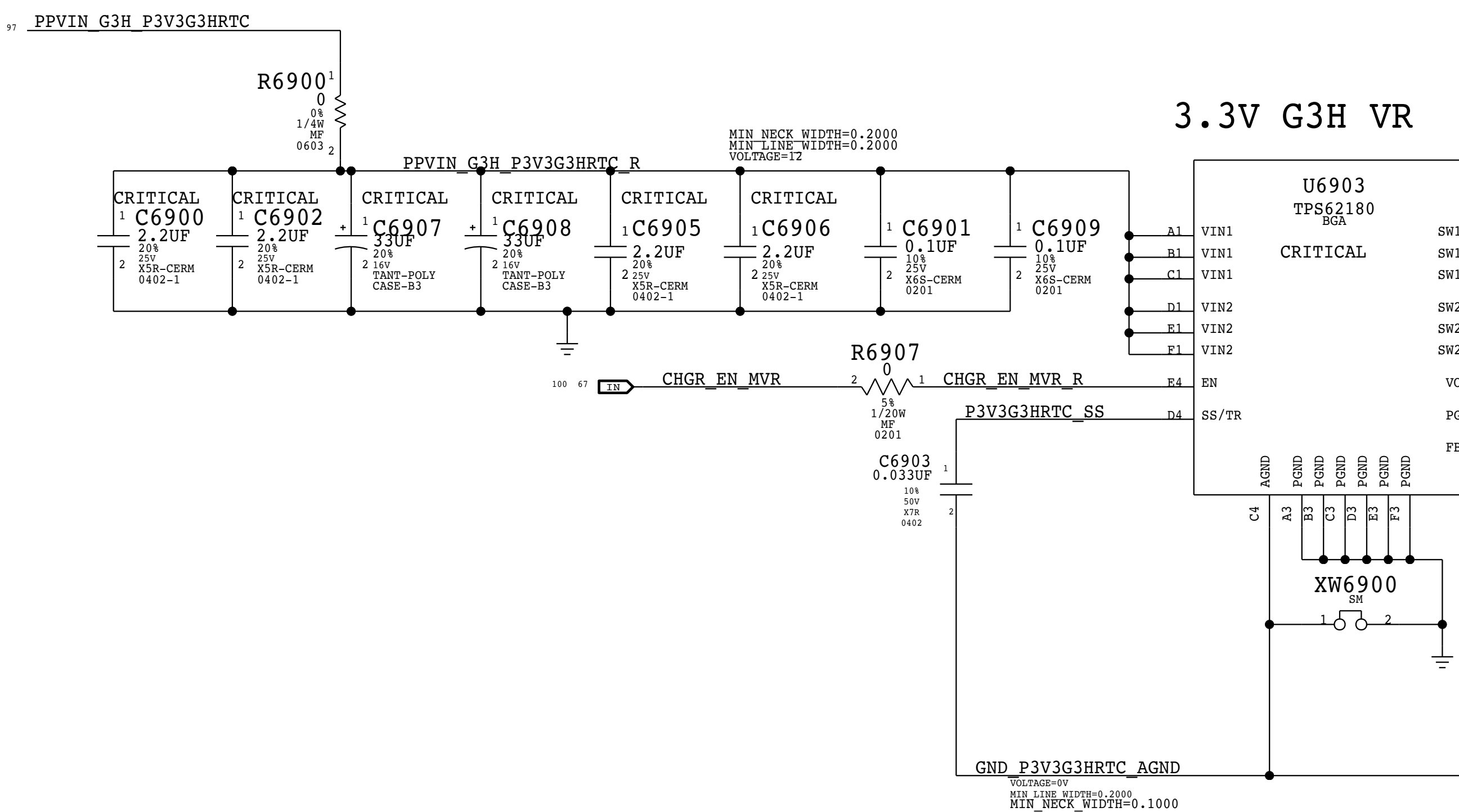
|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|---|





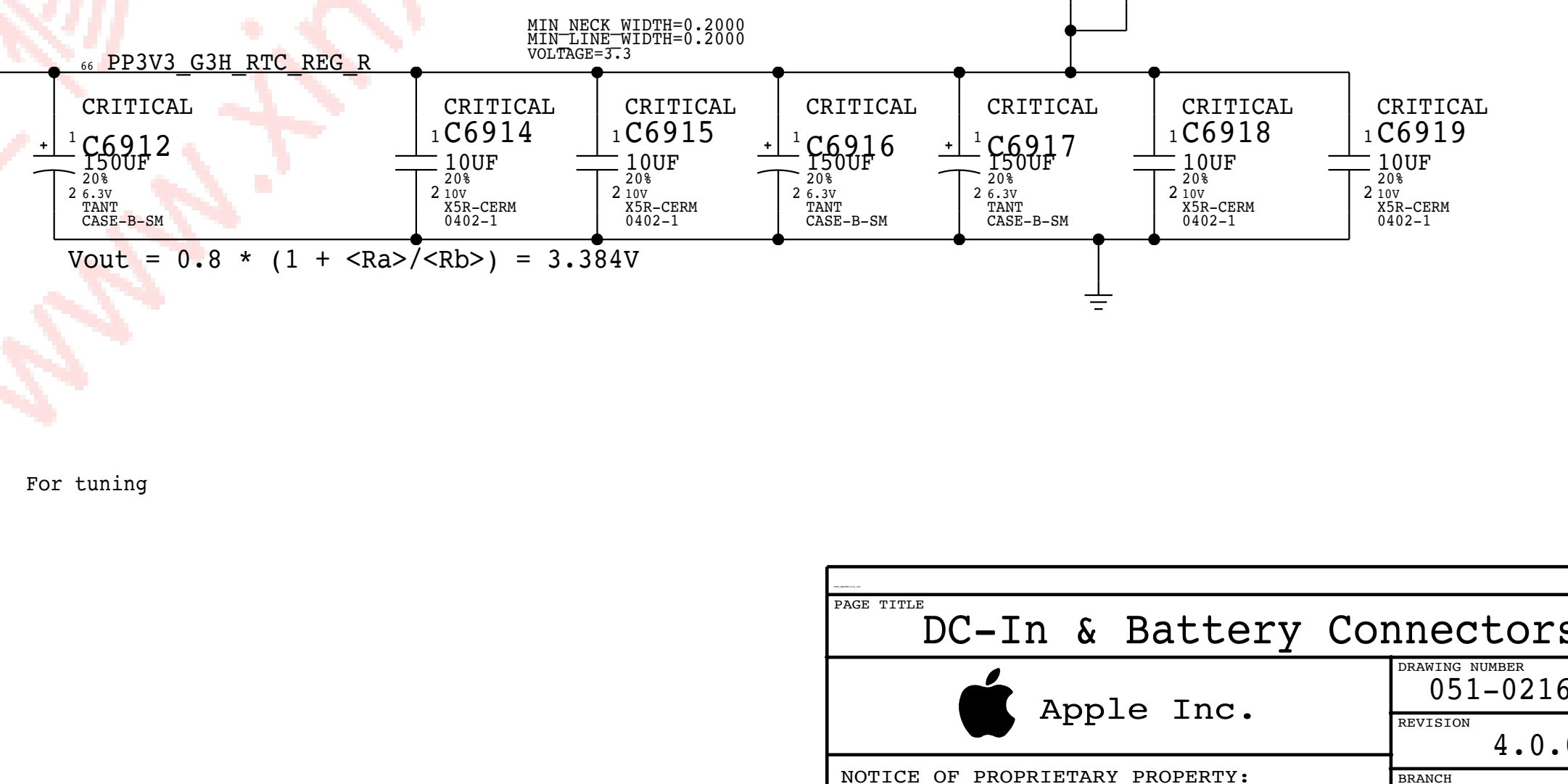
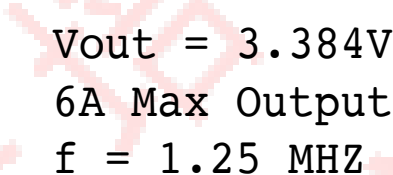
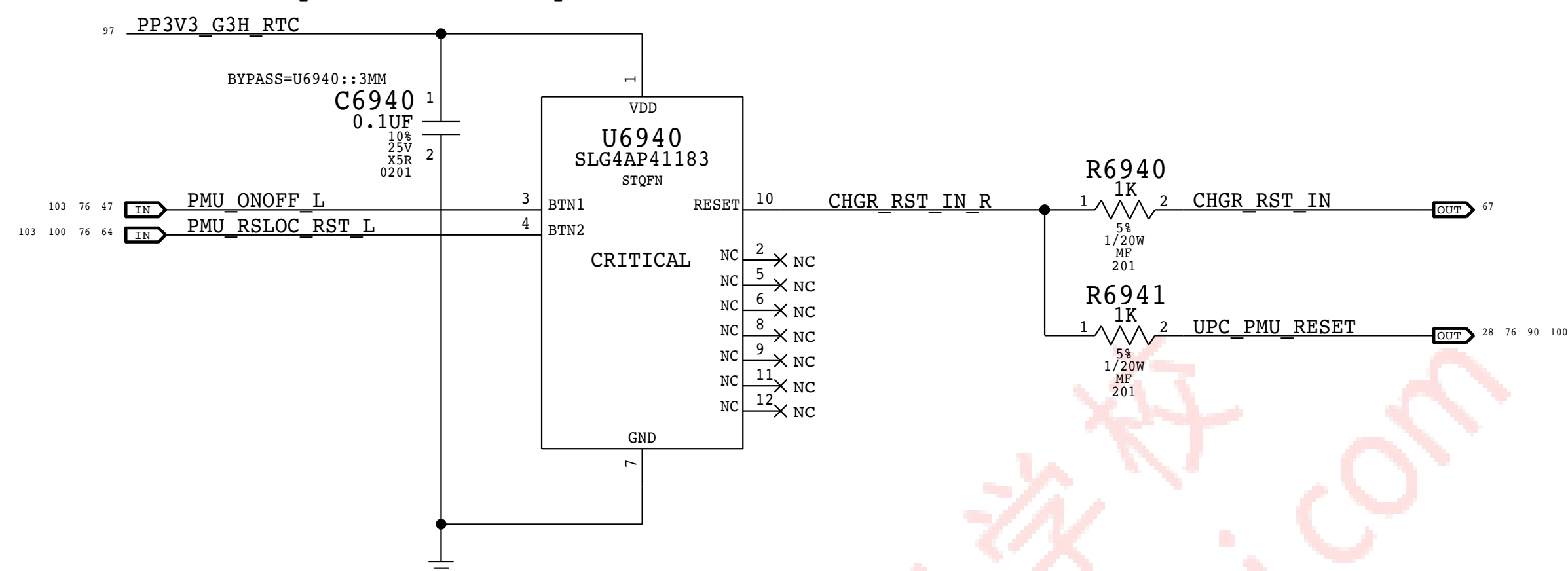
BMU POWER FLEX HOTBAR'd TO THE MLB:

| PART NUMBER | QTY | DESCRIPTION            | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|------------------------|---------------|----------|------------|
| 632-00566   | 1   | PCBA,FLEX,BMU PWR,X362 | J6950         | CRITICAL |            |



## SMC Reset Circuit

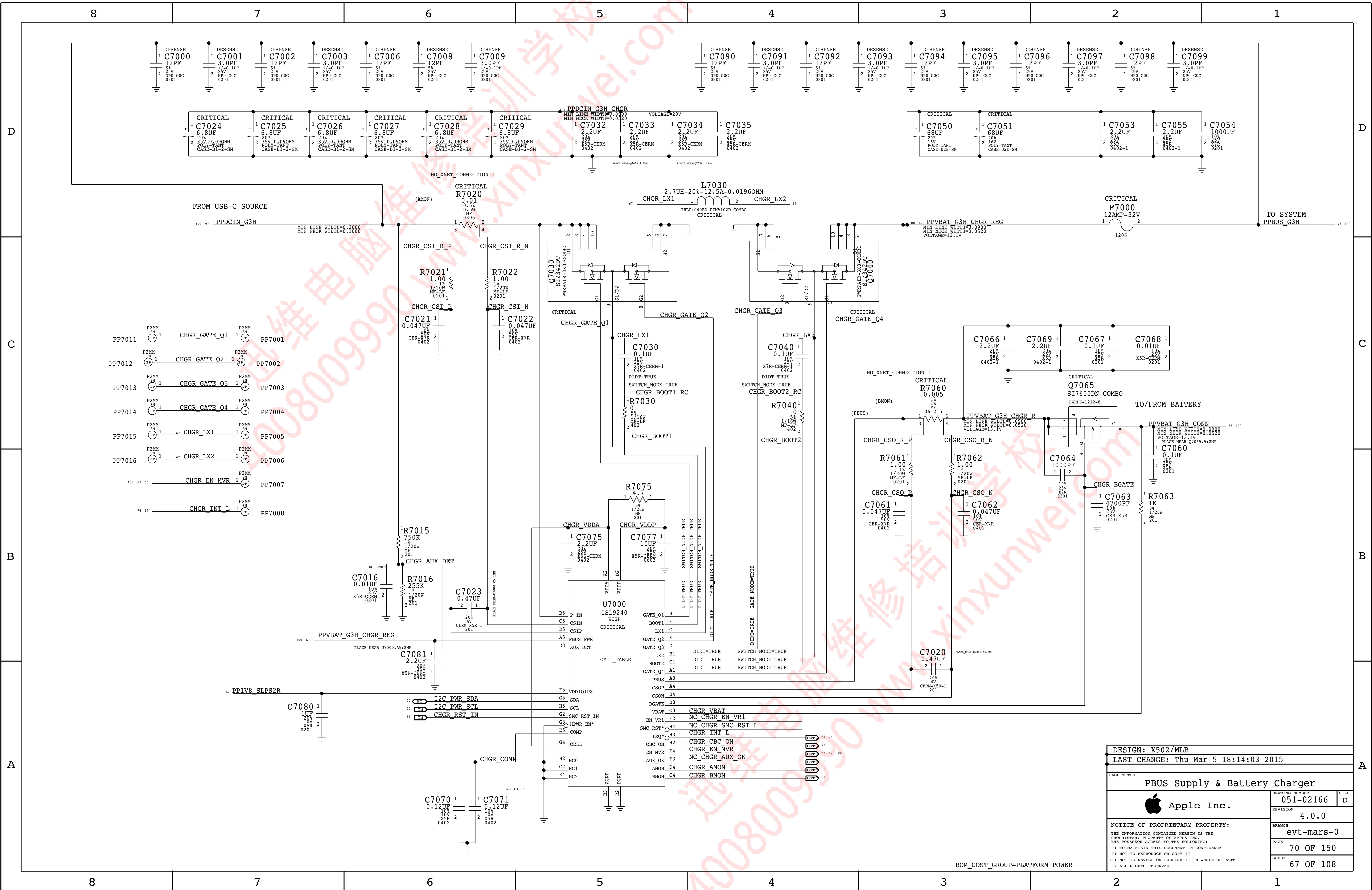
Right Shift & Left Option Control  
followed by ON OFF button press.



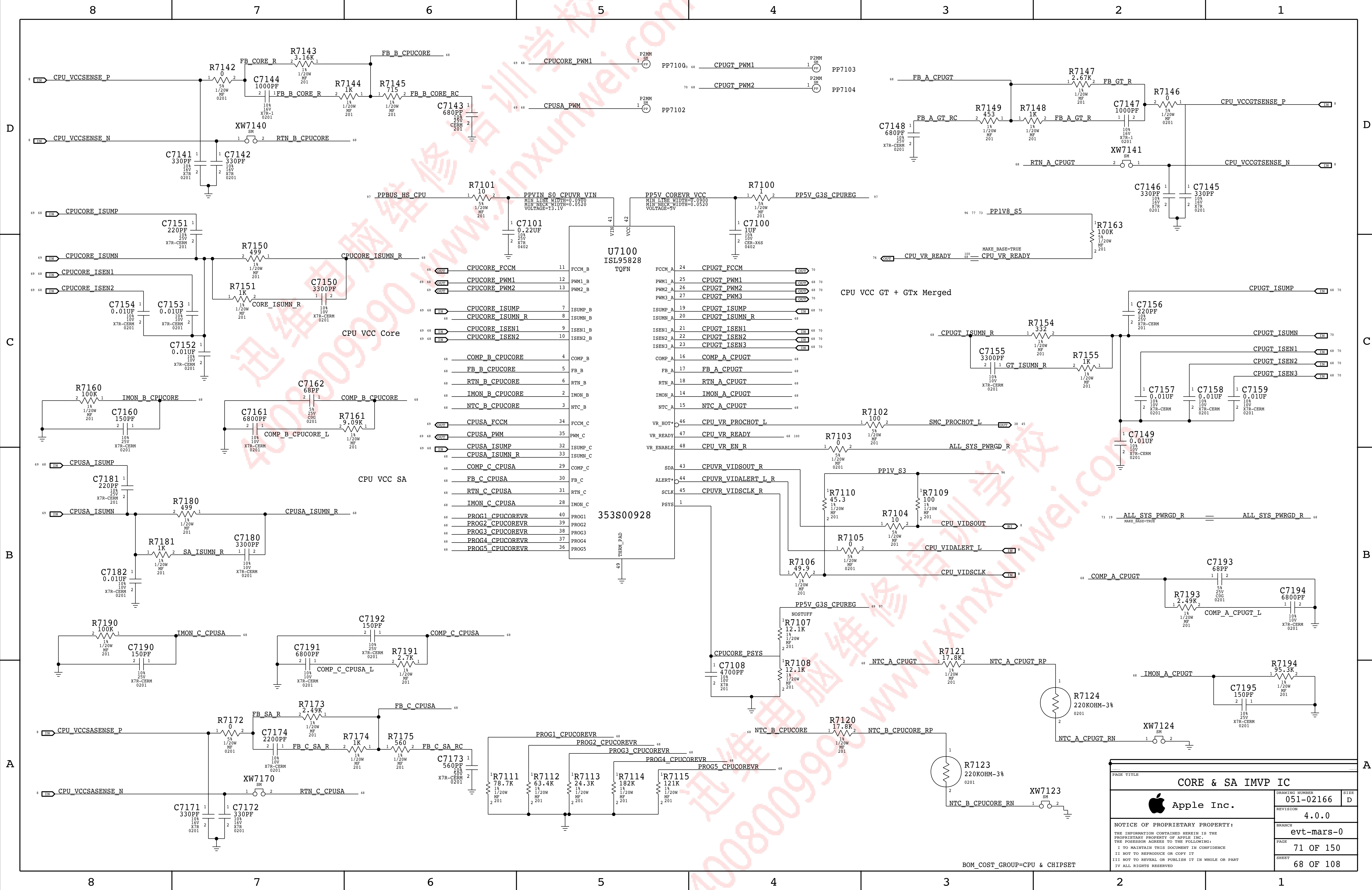
|                                                                                                                                                                                                                                                                                      |                |            |      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|------|
| PAGE TITLE                                                                                                                                                                                                                                                                           |                |            |      |
| DC-In & Battery Connectors                                                                                                                                                                                                                                                           |                |            |      |
|  Apple Inc.                                                                                                                                                                                     | DRAWING NUMBER |            | SIZE |
|                                                                                                                                                                                                                                                                                      | 051-02166      |            | D    |
|                                                                                                                                                                                                                                                                                      | REVISION       |            |      |
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BOM COST GROUP=PLATFORM POWER

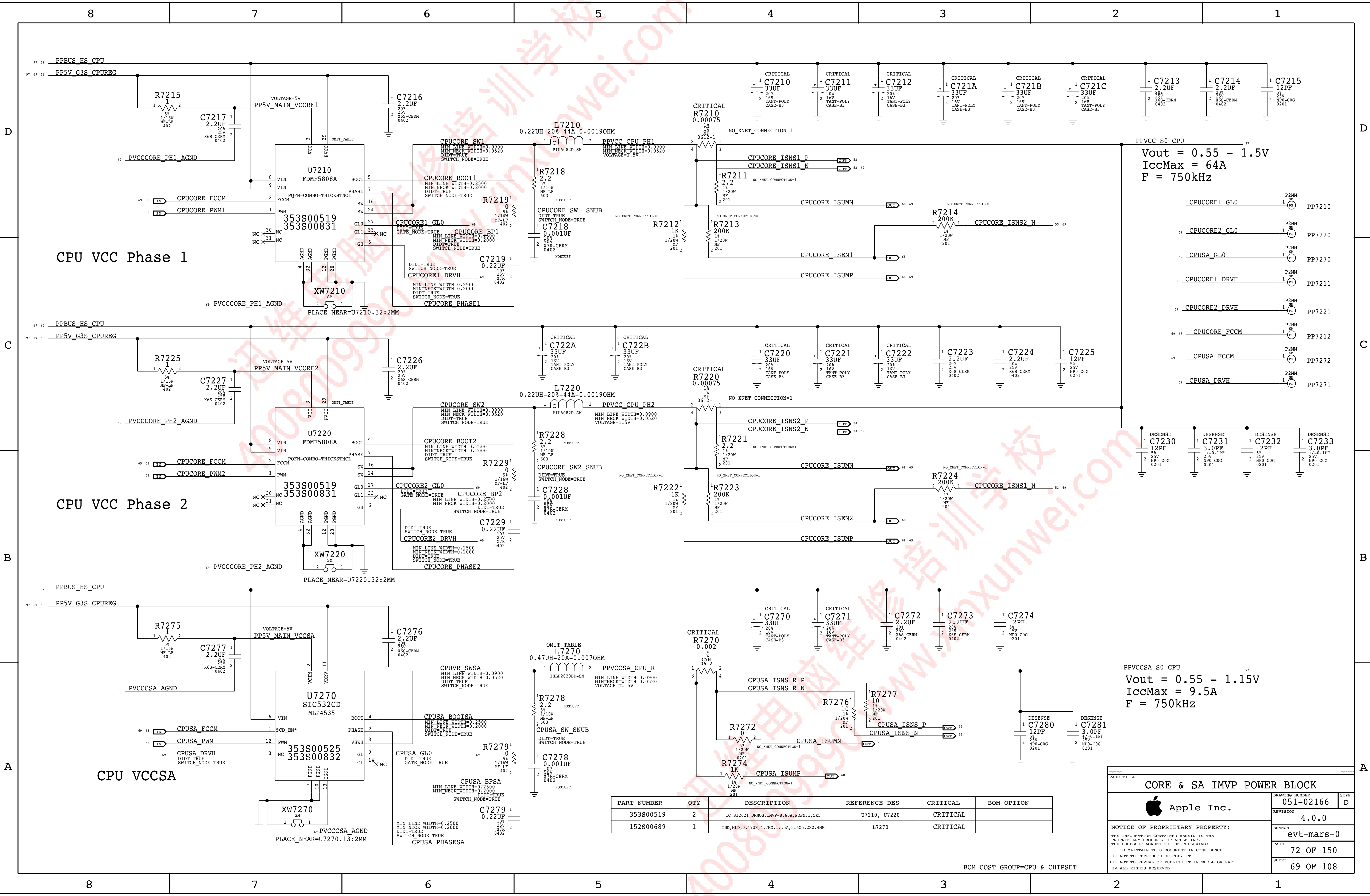












| PART NUMBER | QTY | DESCRIPTION                                   | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-----------------------------------------------|---------------|----------|------------|
| 353S00519   | 2   | IC, 8IC621, DR MOS, IMVP=8, 60A, POFN31, 5X5  | U7210, U7220  | CRITICAL |            |
| 152S00689   | 1   | IND, MLD, 0.47UH, 4.7MO, 17.5A, 5.6X5.2X2.4MM | L7270         | CRITICAL |            |

| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| CORE & SA IMVP POWER BLOCK                                                                                                                                                                                                                                                                                              |                |            |
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02166  |
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BOM\_COST\_GROUP=CPU & CHIPSET



8

7

6

5

4

3

2

1

## CPU VCCGT/GTx Phase 1

PLACE\_NEAR=U7410.32:2MM  
70 PVCCCGT\_PH1\_AGND

## CPU VCCGT/GTx Phase 2

PLACE\_NEAR=U7420.32:2MM  
70 PVCCCGT\_PH2\_AGND

## CPU VCCGT/GTx Phase 3

PLACE\_NEAR=U7430.32:2MM  
70 PVCCCGT\_PH3\_AGND

| PART NUMBER | QTY | DESCRIPTION                            | REFERENCE DES       | CRITICAL | BOM OPTION |
|-------------|-----|----------------------------------------|---------------------|----------|------------|
| 353S00497   | 3   | IC,FDMF808A,DRMOS,IMVP8,50A,PQFN31,5X5 | U7410, U7420, U7430 | CRITICAL |            |

Vout = 0.55 - 1.5V  
IccMax = 64A  
F = 750kHz

|                                                                                                                                                                                                                                                                                                                     |                |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| GT & GTX IMVP POWER BLOCK                                                                                                                                                                                                                                                                                           |                |            |
|  Apple Inc.                                                                                                                                                                                                                    | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                     | REVISION       | 4.0.0      |
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|                                                                                                                                                                                                                                                                                                                     | PAGE           | 74 OF 150  |
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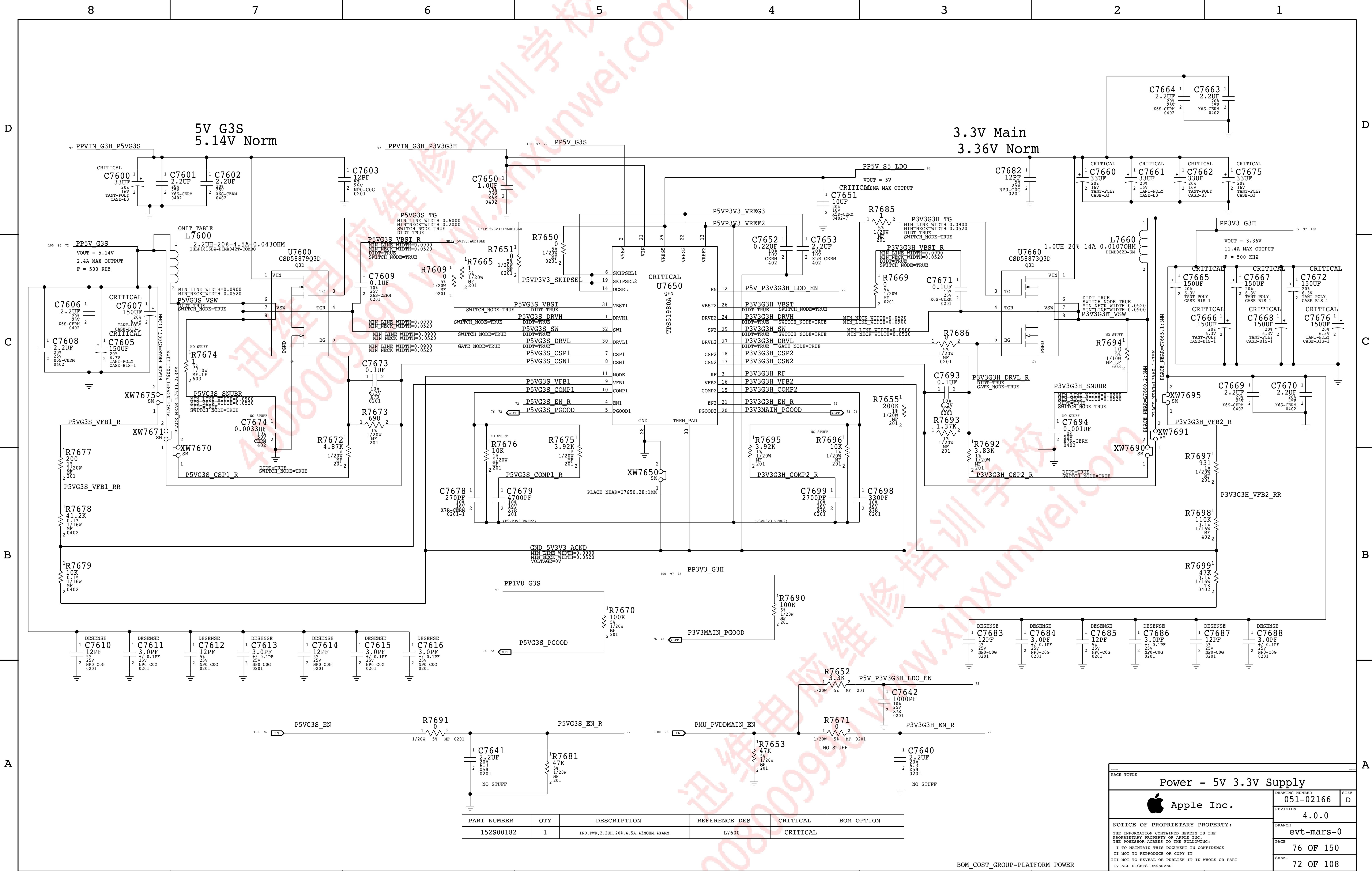
BOM\_COST\_GROUP=CPU &amp; CHIPSET



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| D |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| A |   |   |   |   |   |   |   |   |
|   | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

|                                                                                                                                                                                                                                                                                                                   |  |  |                      |            |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------|------------|--|
| SYNC MASTER=J132 MAO                                                                                                                                                                                                                                                                                              |  |  | SYNC DATE=02/09/2017 |            |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |  |  | Empty                |            |  |
|  Apple Inc.                                                                                                                                                                                                                  |  |  | DRAWING NUMBER       | 051-02166  |  |
|                                                                                                                                                                                                                                                                                                                   |  |  | REVISION             | 4.0.0      |  |
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|                                                                                                                                                                                                                                                                                                                   |  |  | PAGE                 | 75 OF 150  |  |
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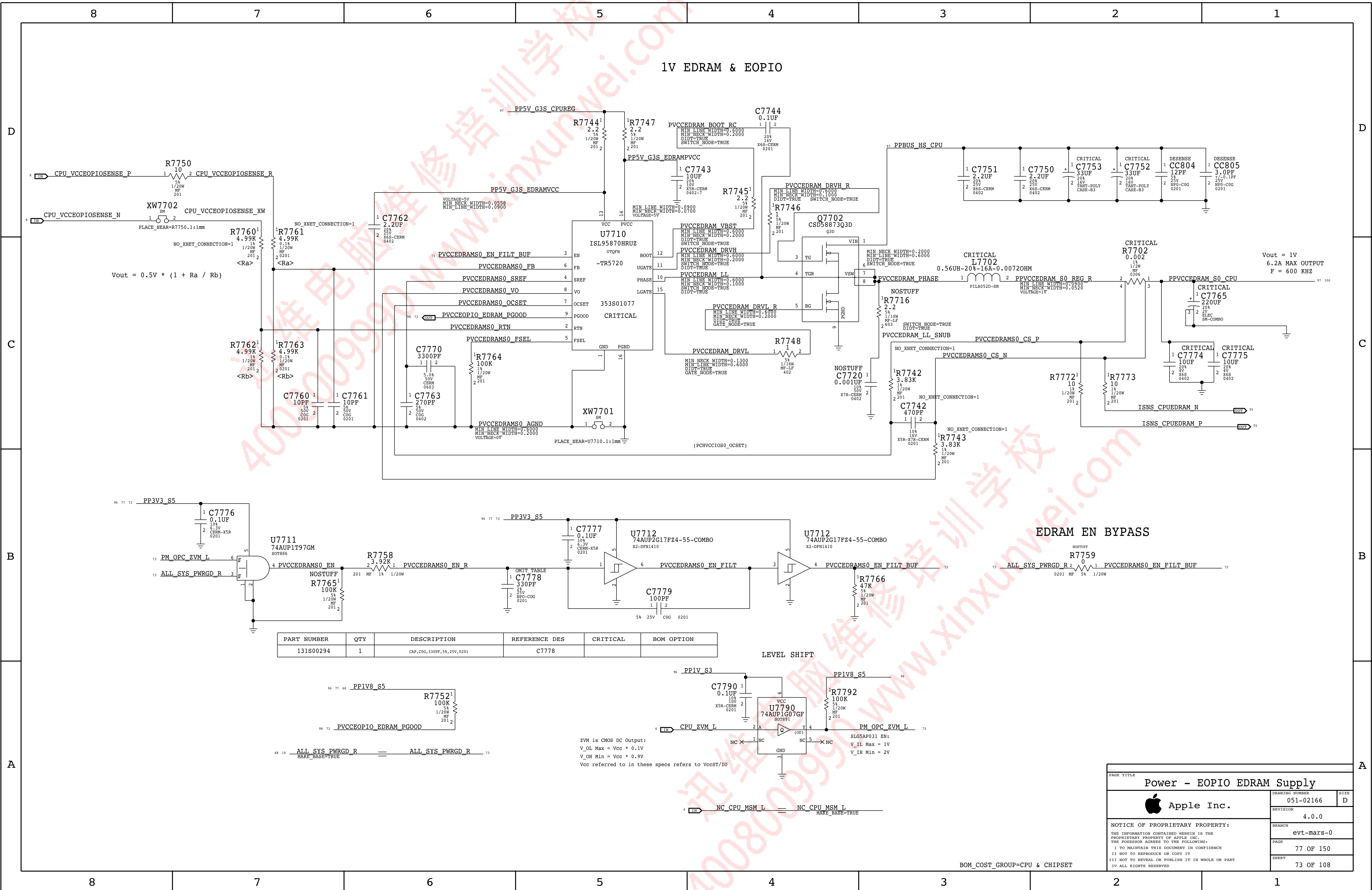


| PART NUMBER | QTY | DESCRIPTION                               | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------------------------|---------------|----------|------------|
| 152S00182   | 1   | IND, PWR, 2.2UH, 20%, 4.5A, 43MOHM, 4X4MM | L7600         | CRITICAL |            |

| PAGE TITLE                                                                                                        |                |            |
|-------------------------------------------------------------------------------------------------------------------|----------------|------------|
| Power - 5V 3.3V Supply                                                                                            |                |            |
|  Apple Inc.                  | DRAWING NUMBER | 051-02166  |
|                                                                                                                   | REVISION       | 4.0.0      |
|                                                                                                                   | BRANCH         | evt-mars-0 |
|                                                                                                                   | PAGE           | 76 OF 150  |
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BOM\_COST\_GROUP=PLATFORM POWER











D

C

B

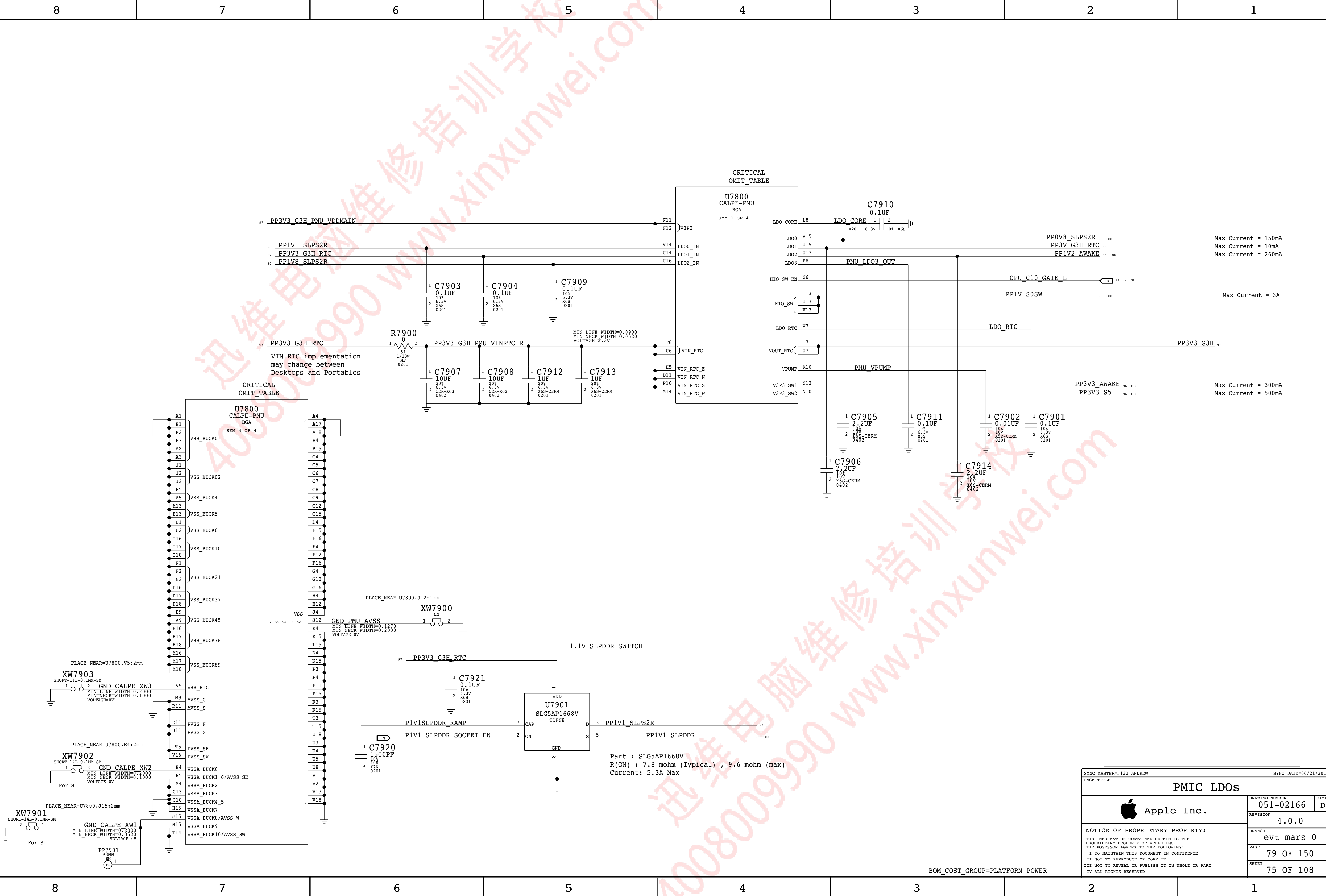
A

D

C

B

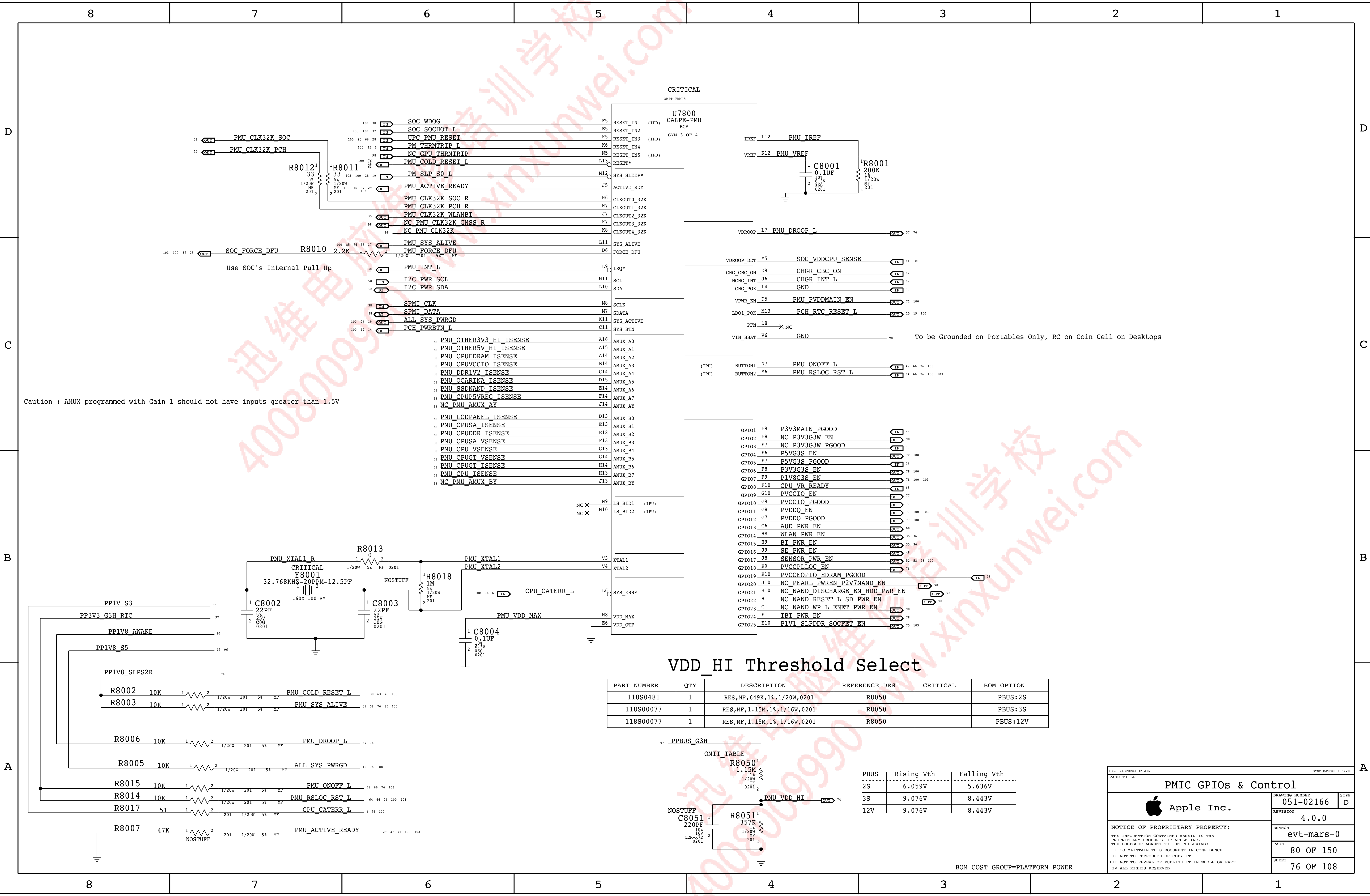
A



BOM\_COST\_GROUP=PLATFORM POWER

|                                                                                                                                                                                                                                                                                      |                |                      |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|-----------|
| SYNC_MASTER=J132 ANDREW                                                                                                                                                                                                                                                              |                | SYNC_DATE=06/21/2017 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                           |                |                      |           |
| PMIC LDOs                                                                                                                                                                                                                                                                            |                |                      |           |
|  Apple Inc.                                                                                                                                                                                     | DRAWING NUMBER |                      | SIZE      |
|                                                                                                                                                                                                                                                                                      | 051-02166      |                      | D         |
| REVISION                                                                                                                                                                                                                                                                             |                | 4.0.0                |           |
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|                                                                                                                                                                                                                                                                                      |                | PAGE                 | 79 OF 150 |
|                                                                                                                                                                                                                                                                                      |                | SHEET                | 75 OF 108 |





| PART NUMBER | QTY | DESCRIPTION                | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|----------------------------|---------------|----------|------------|
| 118S0481    | 1   | RES,MF,649K,1%,1/20W,0201  | R8050         |          | PBUS:2S    |
| 118S00077   | 1   | RES,MF,1.15M,1%,1/16W,0201 | R8050         |          | PBUS:3S    |
| 118S00077   | 1   | RES,MF,1.15M,1%,1/16W,0201 | R8050         |          | PBUS:12V   |

| PBUS | Rising Vth | Falling Vth |
|------|------------|-------------|
| 2S   | 6.059V     | 5.636V      |
| 3S   | 9.076V     | 8.443V      |
| 12V  | 9.076V     | 8.443V      |

|                                                                                                                                                                                                                                                                                                               |                |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                    |                |            |
| PMIC GPIOs & Control                                                                                                                                                                                                                                                                                          |                |            |
|                                                                                                                                                                                                                                                                                                               | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                               | REVISION       | 4.0.0      |
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|                                                                                                                                                                                                                                                                                                               | PAGE           | 80 OF 150  |
|                                                                                                                                                                                                                                                                                                               | SHEET          | 76 OF 108  |

BOM\_COST\_GROUP=PLATFORM POWER

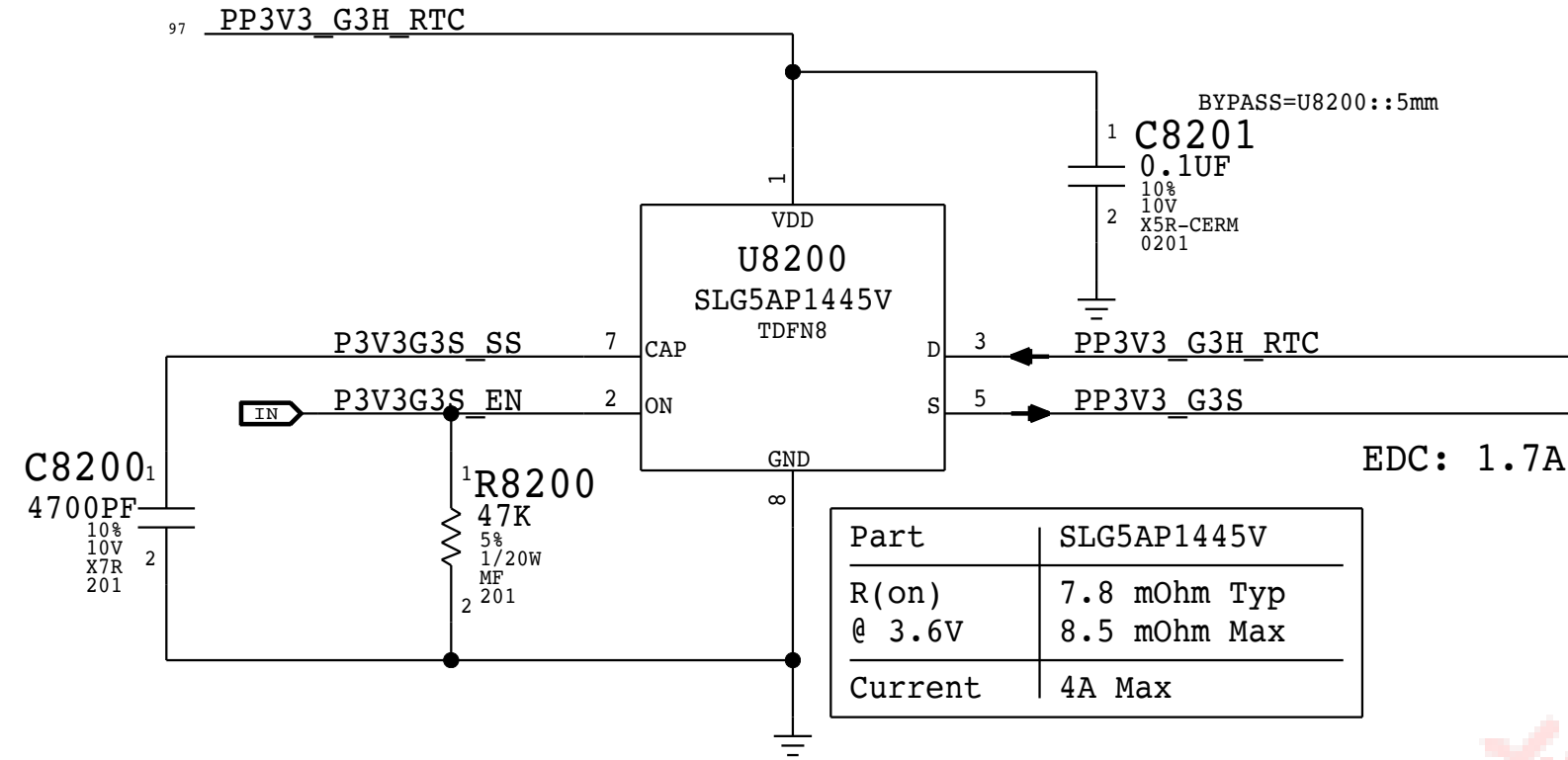




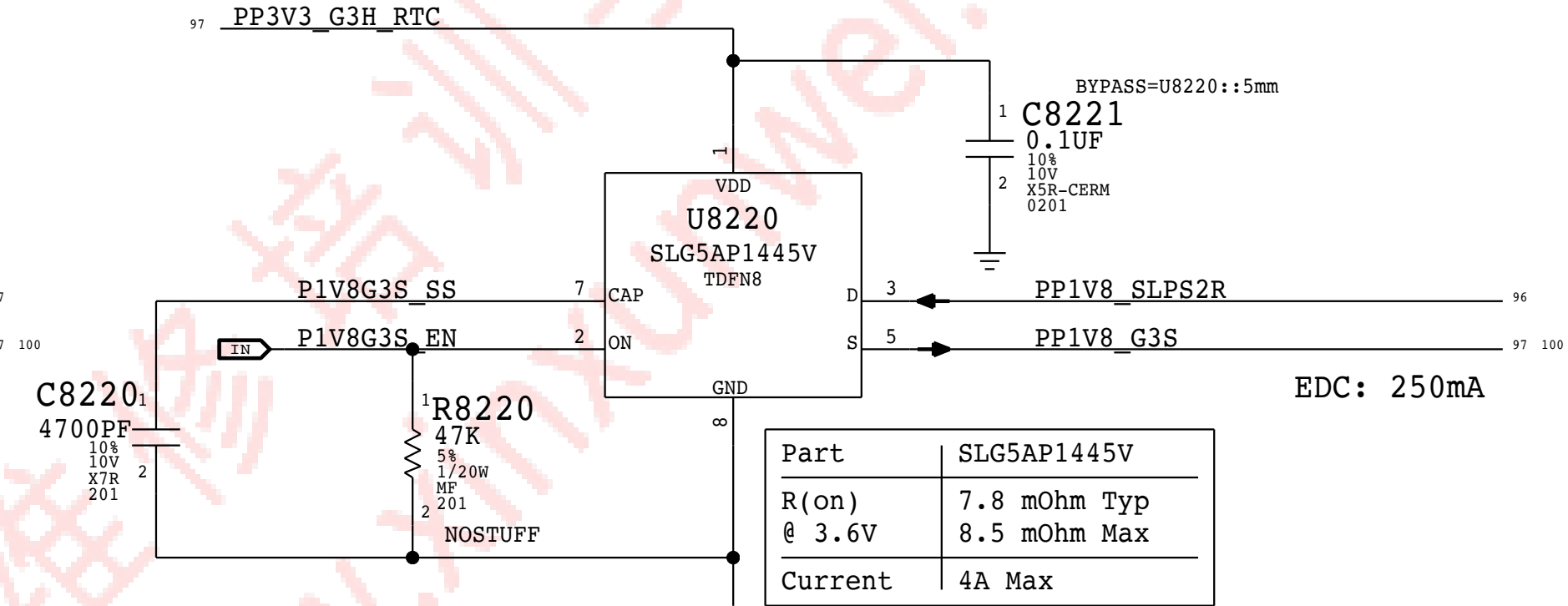


Note: Load switches may be larger than necessary

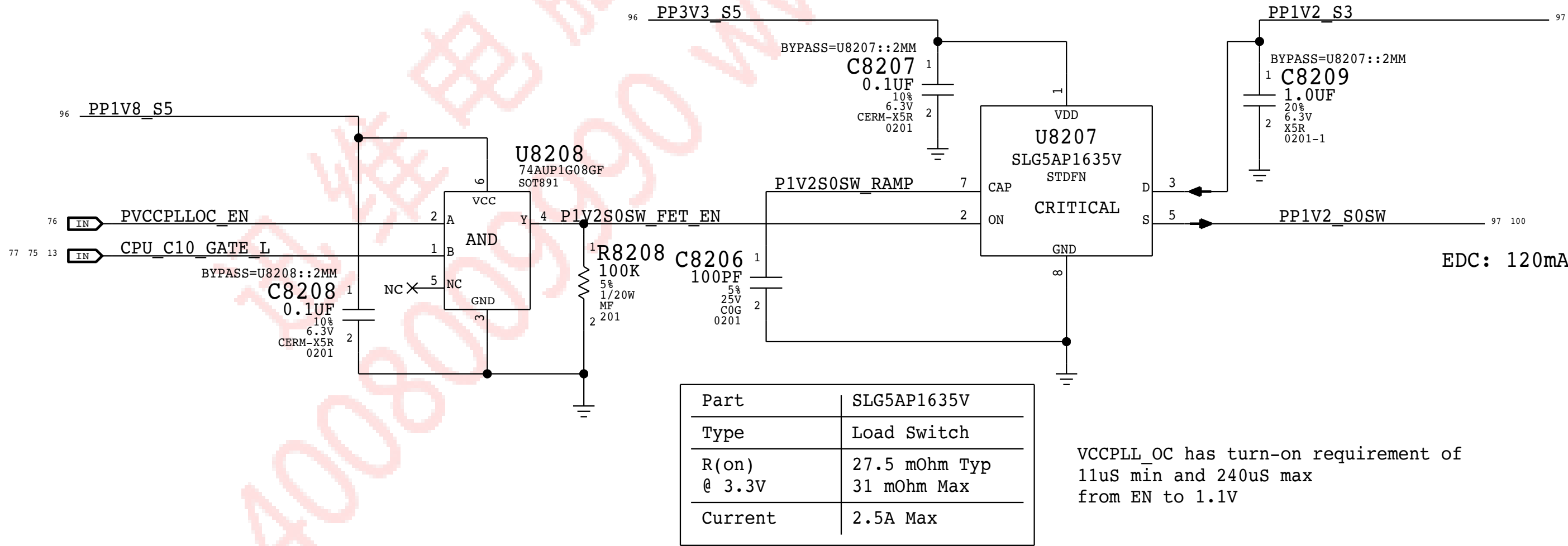
### 3.3V G3 Standby Switch



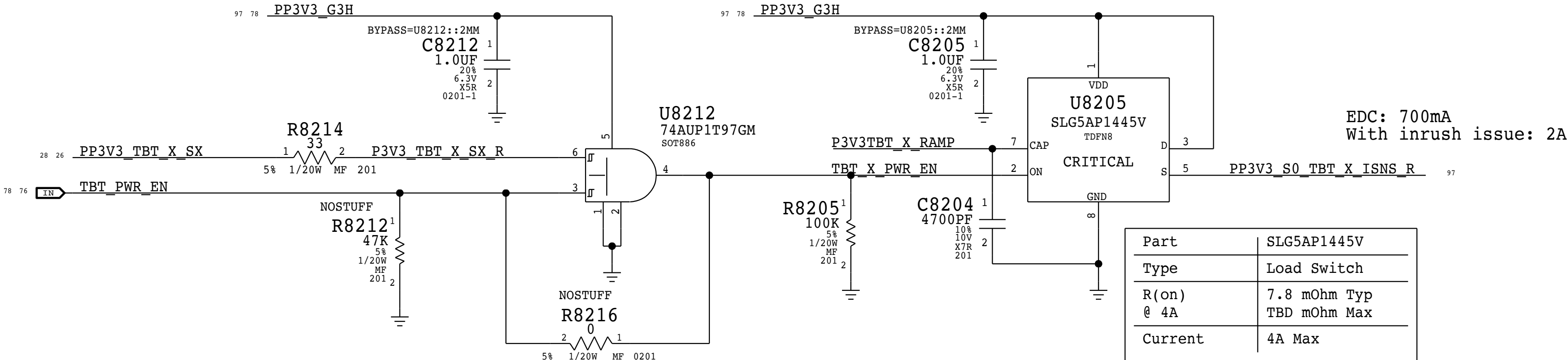
### 1.8V G3 Standby Switch



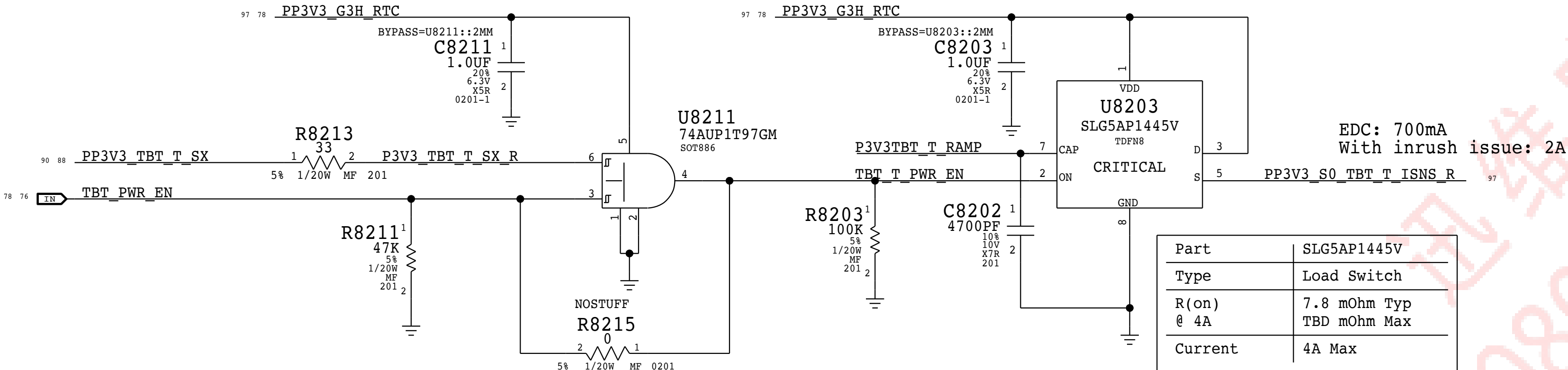
### 1.2V S0SW VCCPLL\_OC Switch



### 3.3V S0SW TBT X Switch



### 3.3V S0SW TBT T Switch

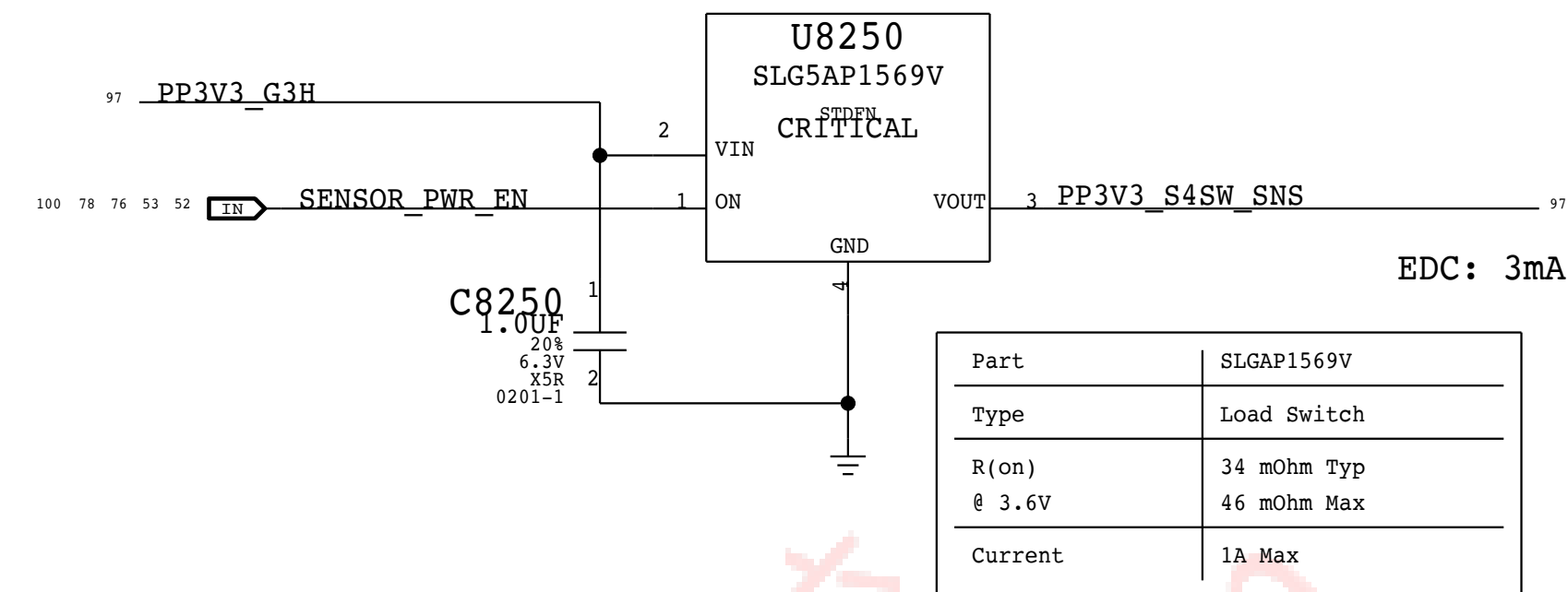


## System Power States

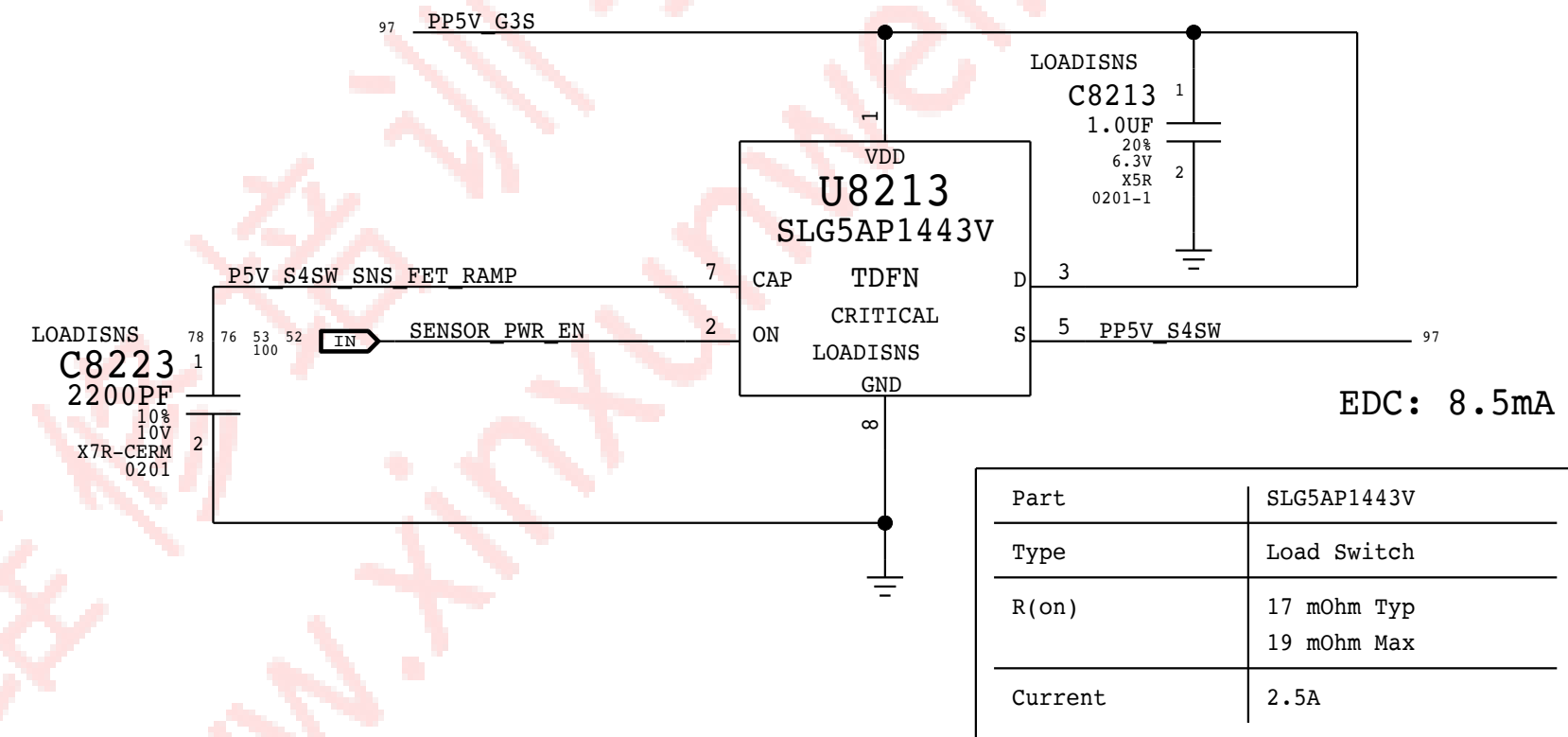
| System State:                     | Shutdown (G3H) |     | Standby (G3S)  |       | Standby (S4) |        | Sleep (S0i/S3) |        | Run (S0) |
|-----------------------------------|----------------|-----|----------------|-------|--------------|--------|----------------|--------|----------|
| CPU/PCH State:                    | Off (RTC Only) |     | Off (RTC Only) |       | Standby      |        | Sleep          |        | Run      |
| Rails                             | SoC State:     |     | S2R            | Awake | S2R          | Awake  | S2R            | Awake  | Awake    |
| PP*_S2R (0.8,1.1,1.8V)            | On             | On  | On             | On    | On           | On     | On             | On     | On       |
| PP*_DDR (0.8,0.9,1.1V)            | Off            | On  | Off            | On    | Off          | On     | Off            | On     | On       |
| PP*_AWAKE (CPU,SRAM,1.2,1.8,3.3V) | Off            | On  | Off            | On    | Off          | On     | Off            | On     | On       |
| PP3V3_G3H (VR1)                   | On             | On  | On             | On    | On           | On     | On             | On     | On       |
| PP*_G3S (1.8,3.3,5V)              | Off            | On  | On             | On    | On           | On     | On             | On     | On       |
| PP*_S5 (1.8,3.3V)                 | Off            | Off | Off            | Off   | On           | On     | On             | On     | On       |
| CPU/PCH VRs                       | Off            | Off | Off            | Off   | Off/On       | Off/On | Off/On         | Off/On | On       |

\* System: Shutdown Awake is a transition state only.  
\* SoC: SLP\_DDR is a transition state only.  
\* CPU/PCH: S4 is only used by desktops for USB wakes.  
\* CPU/PCH: S5 is a transition state. May also be used for RTC wakes.

### 3.3V Sensor Switch



### 5V Sensor Switch



|                                                                                                                                                                                                                                                                                                               |  |  |                       |            |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------|------------|------|
| SYMC PART#-1132_218                                                                                                                                                                                                                                                                                           |  |  | SYMC DATE#-09/05/2011 |            |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                    |  |  | Power FETs            |            |      |
|                                                                                                                                                                                                                                                                                                               |  |  | DRAWING NUMBER        | 051-02166  | SIZE |
|                                                                                                                                                                                                                                                                                                               |  |  | REVISION              | 4.0.0      | D    |
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BOM\_COST\_GROUP=PLATFORM POWER

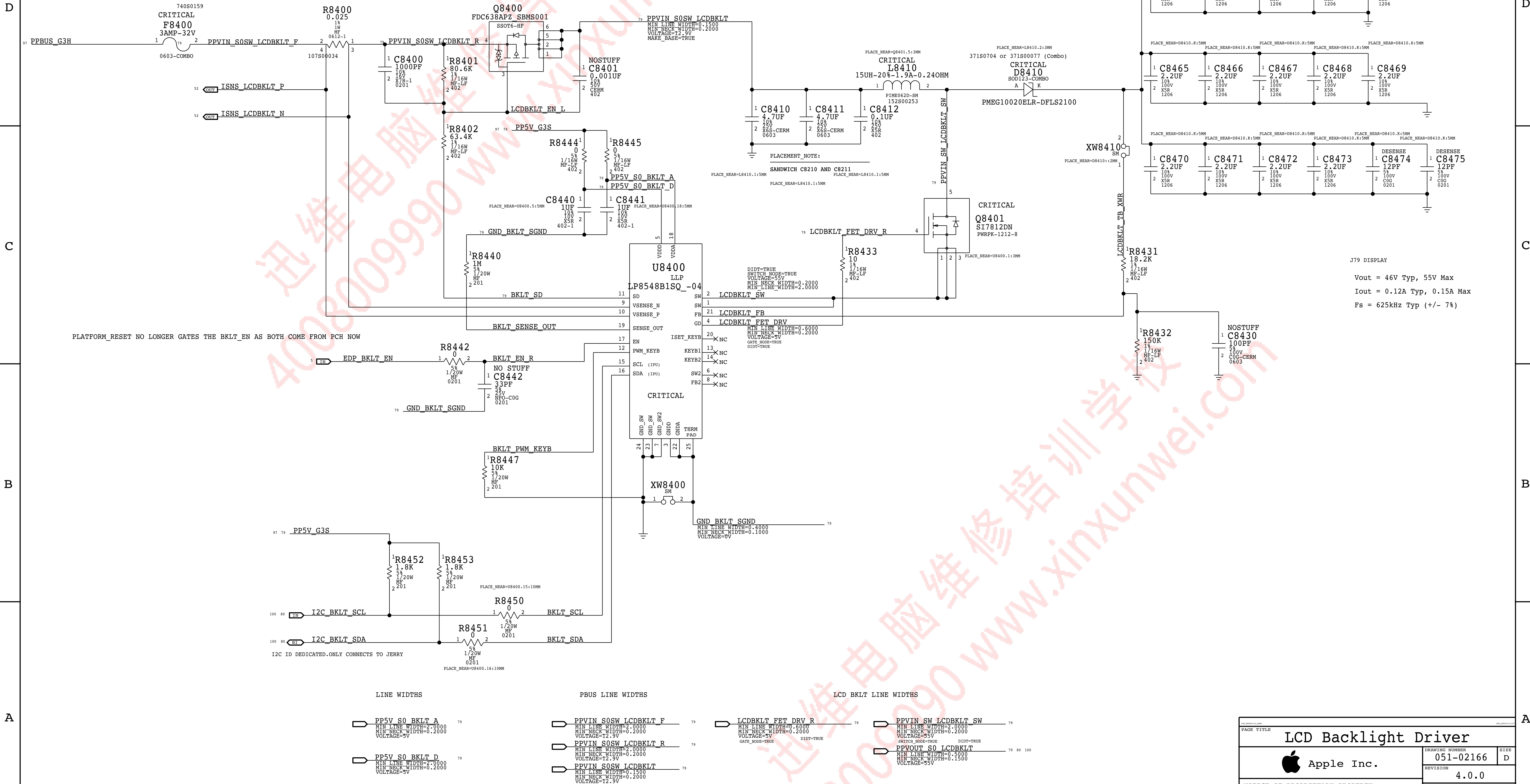


## Page Notes

Power aliases required by this page:

```
- =PPVIN_S0SN_LCDBKLTFT (9-12.6V LCD BACKLIGHT INPUT
```

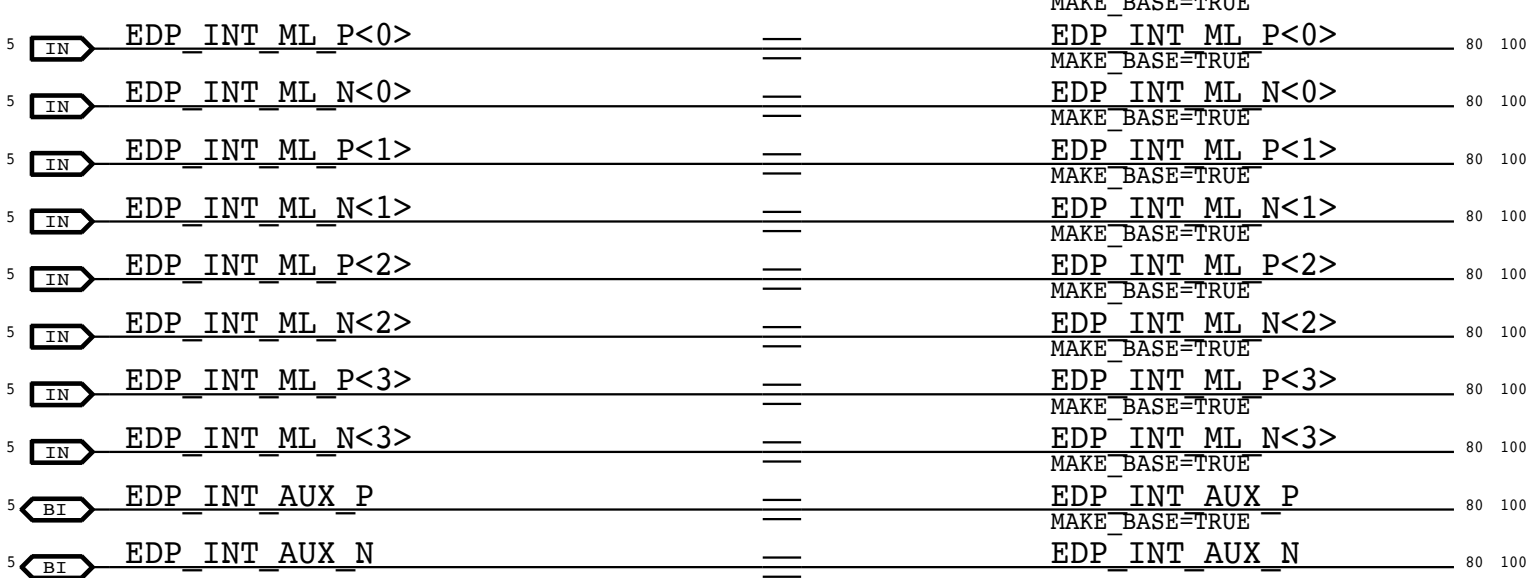
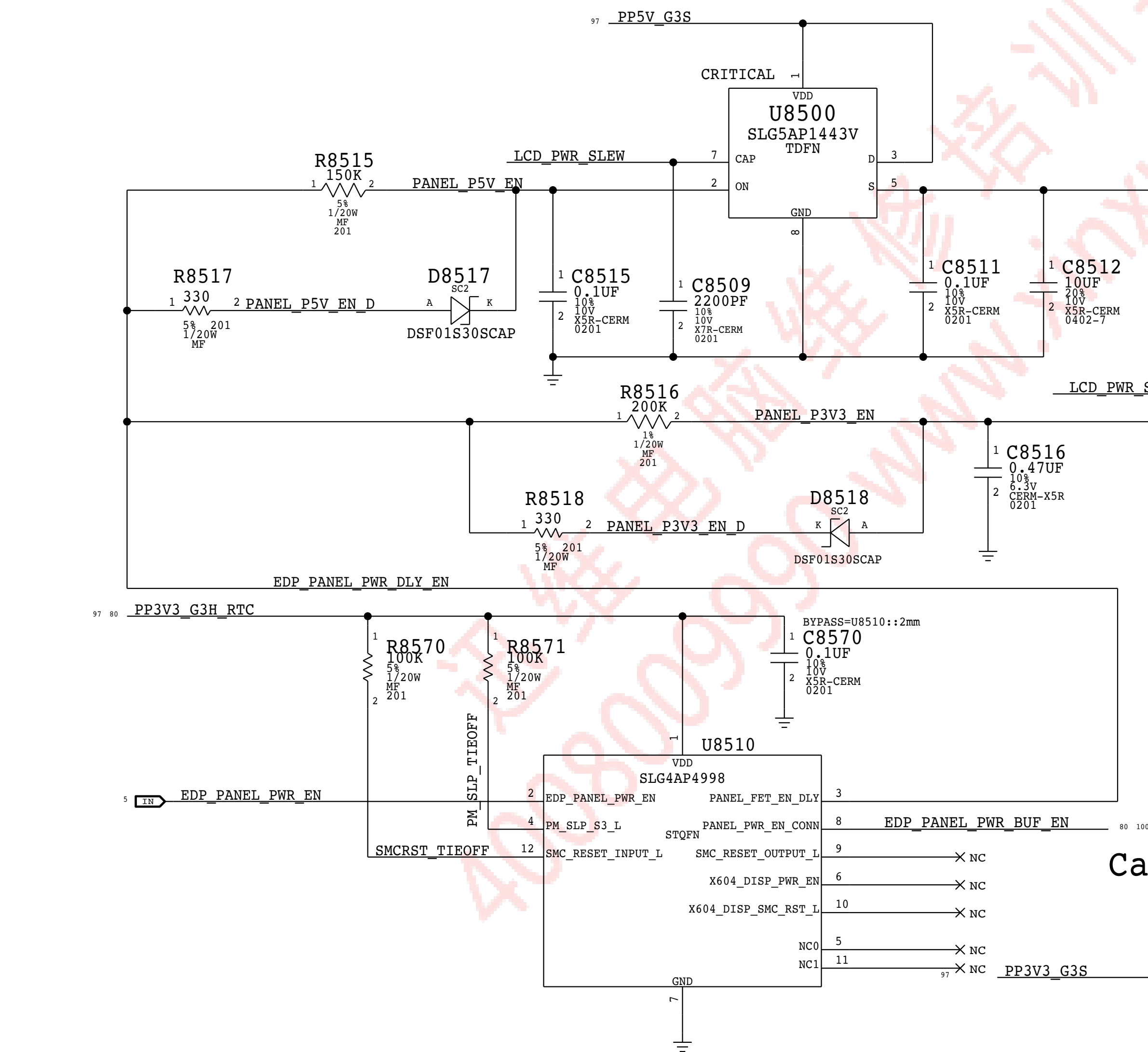
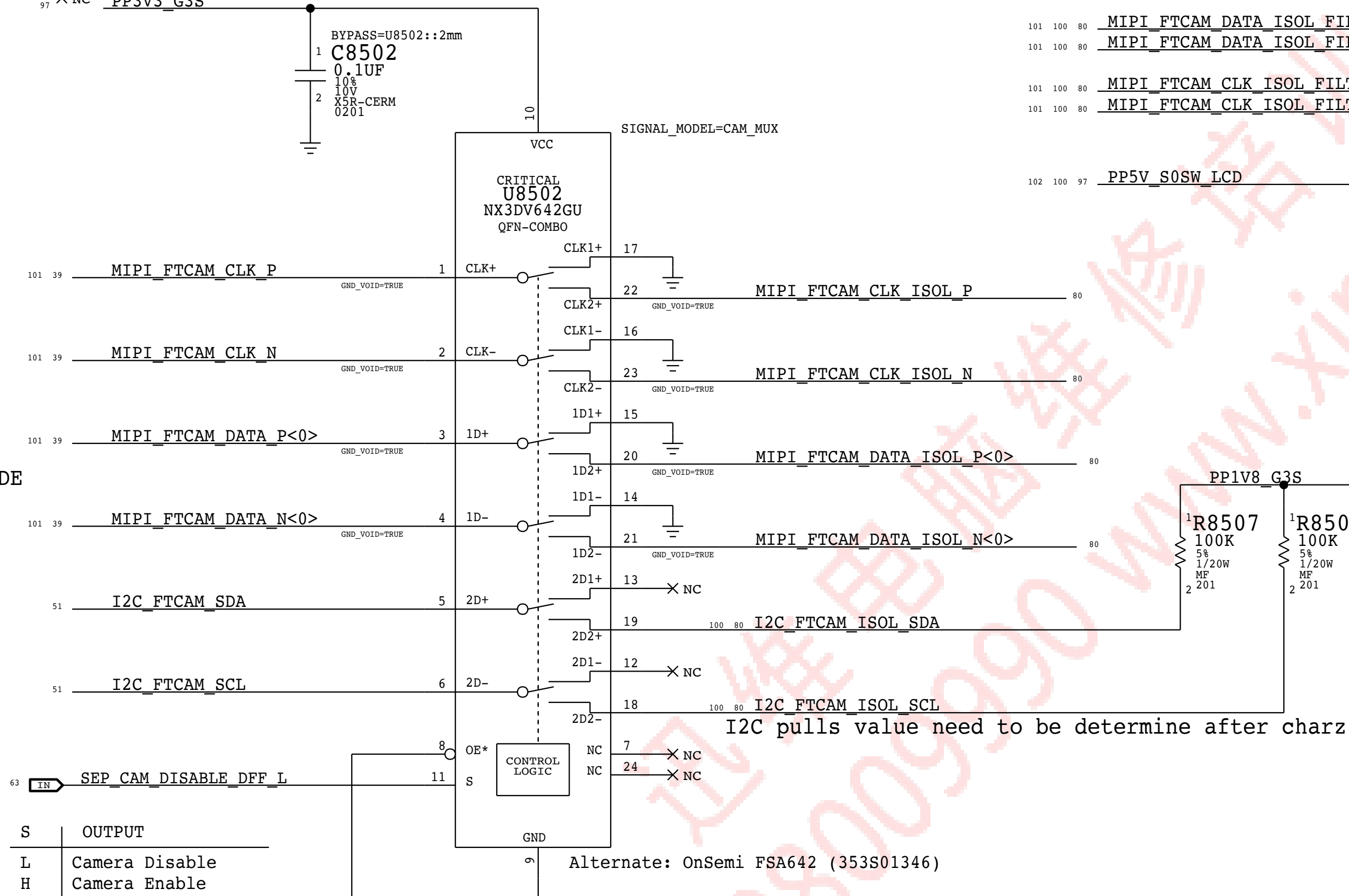
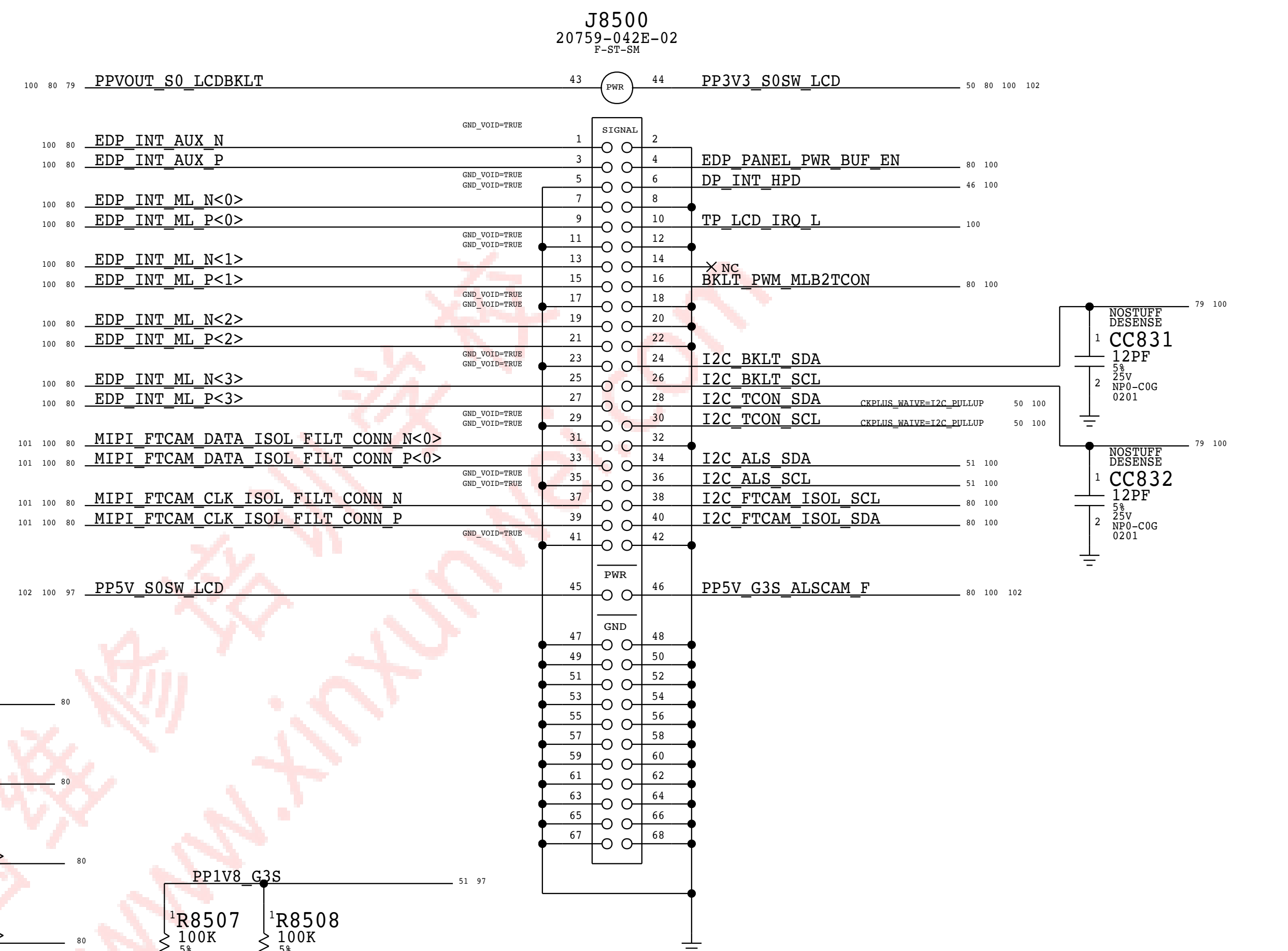
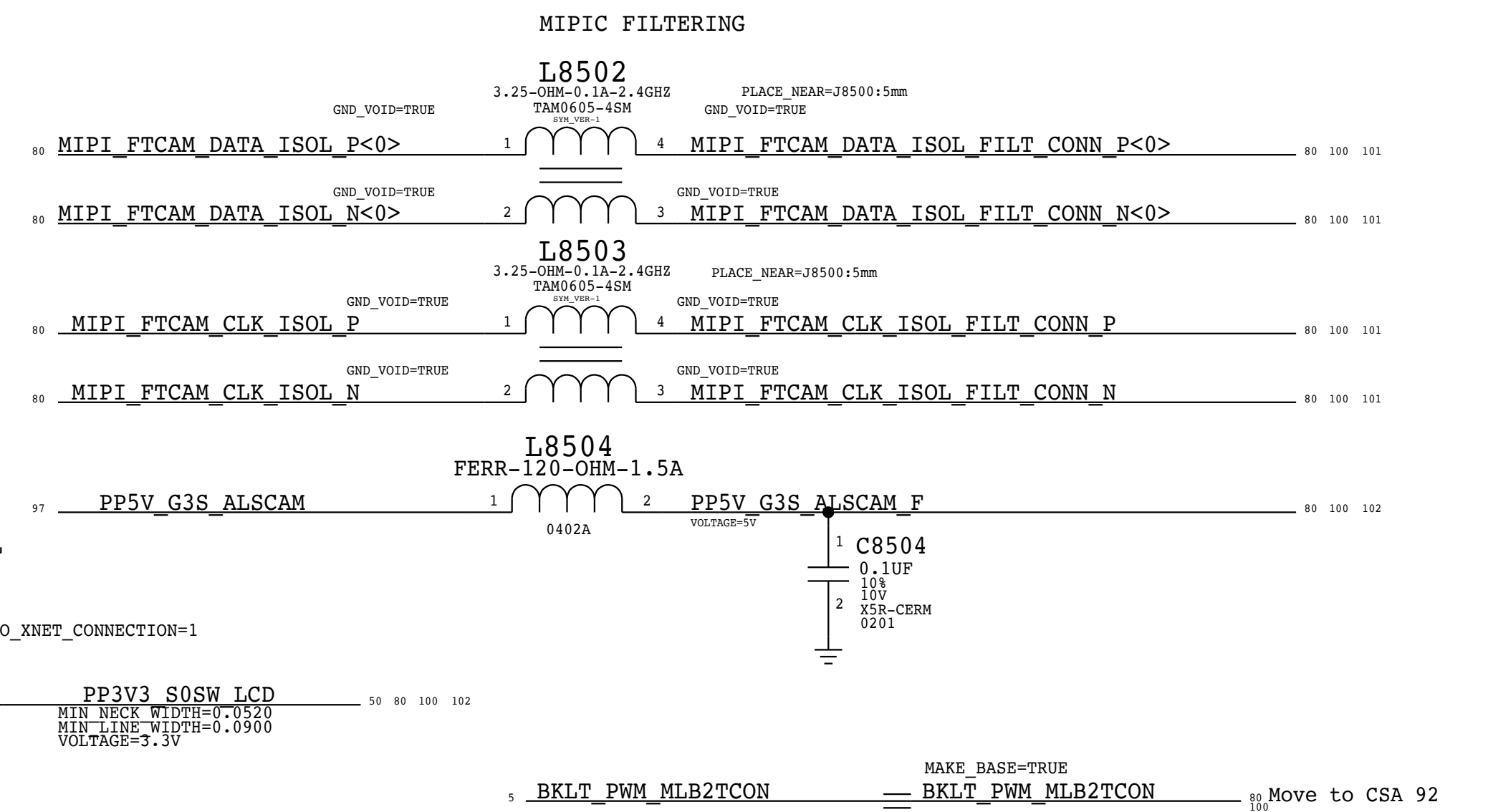
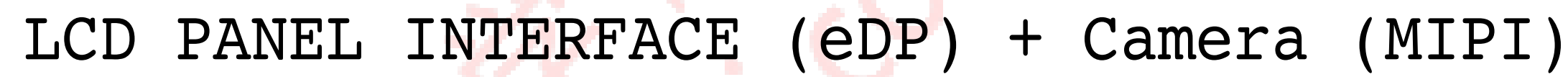
- =PP5V\_S0\_BKLT (5V BACKLIGHT DRIVER INPUT)



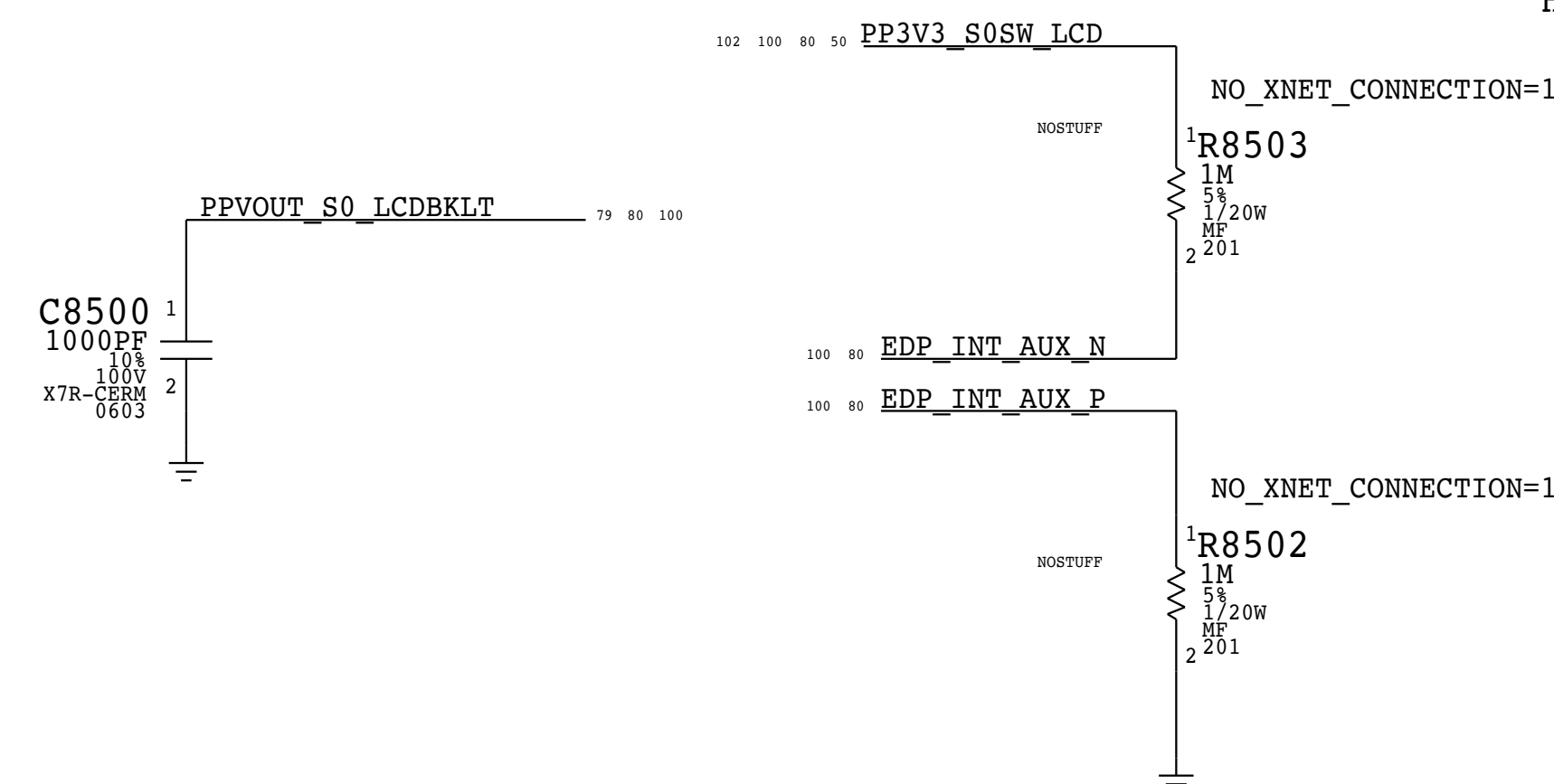
BOM COST GROUP=DISPLAY

|                                                                                                                                                                                                                                                                                      |  |                |  |            |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|--|------------|--|
| PAGE TITLE                                                                                                                                                                                                                                                                           |  | DRAWING NUMBER |  | SIZE       |  |
| LCD Backlight Driver                                                                                                                                                                                                                                                                 |  | 051-02166      |  | D          |  |
|  Apple Inc.                                                                                                                                                                                     |  | REVISION       |  | 4.0.0      |  |
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## LCD Panel HPD & AUX strapping



|                                                                                                                                                                                                                                                                                                                                         |                |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                                              |                |            |
| eDP Display Connector                                                                                                                                                                                                                                                                                                                   |                |            |
|  Apple Inc.                                                                                                                                                                                                                                        | DRAWING NUMBER | SIZE       |
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# S4E0

D

D

C

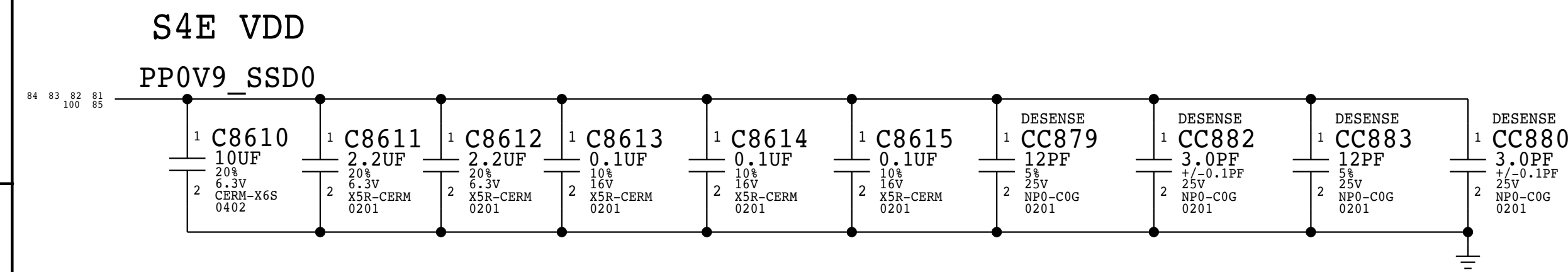
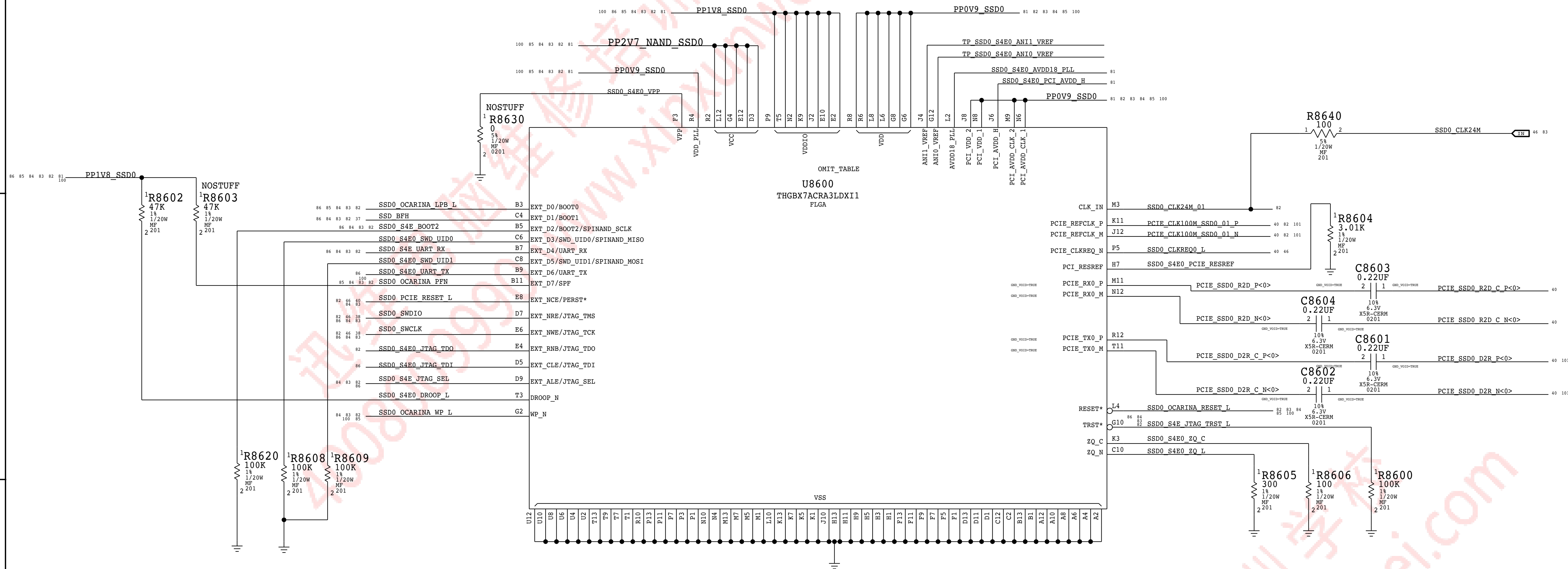
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B

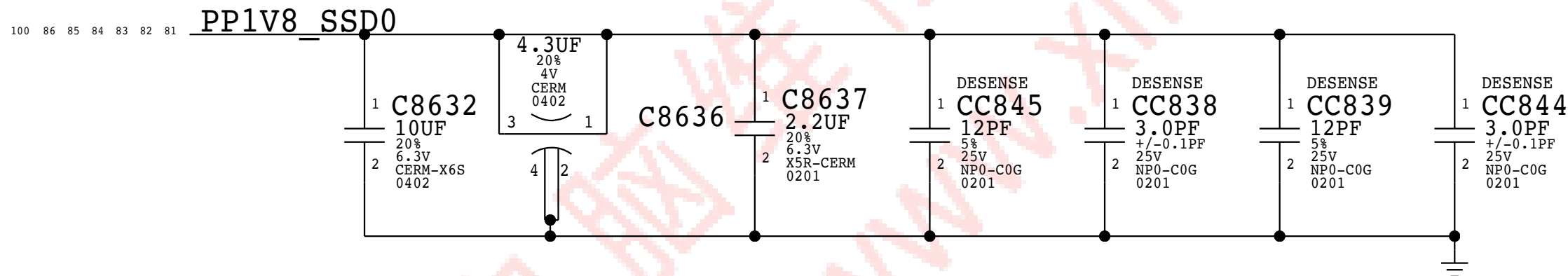
B

A

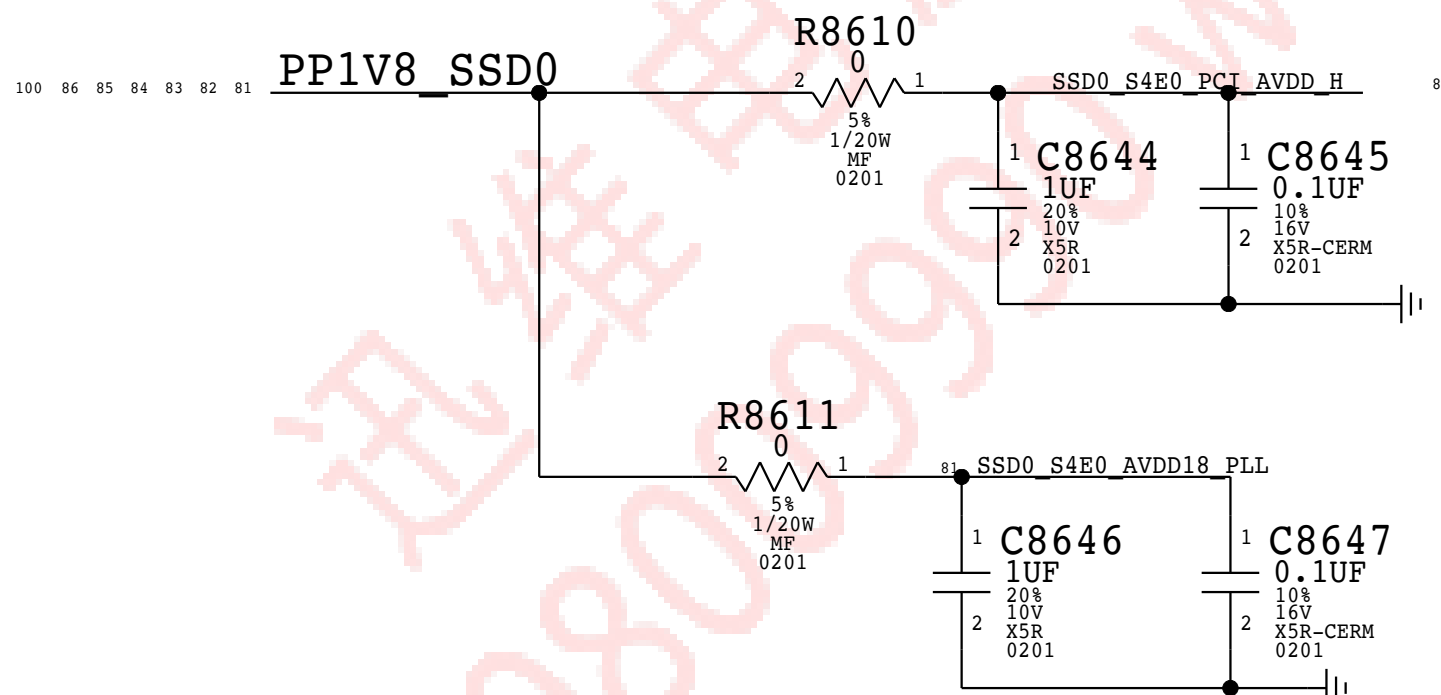
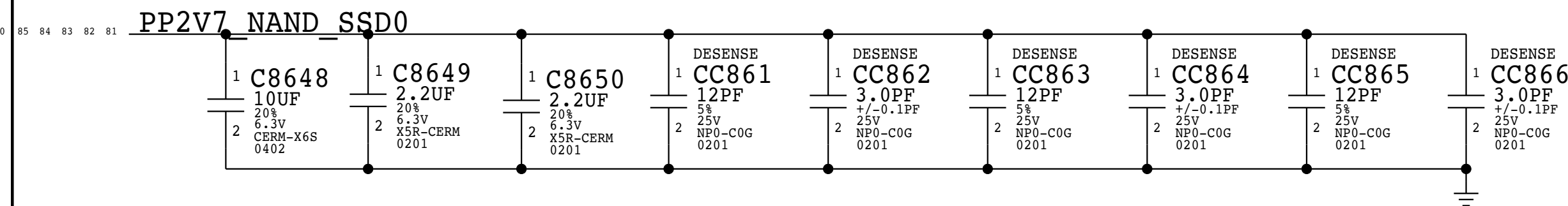
A



## S4E VDDIO



## VCC CAP



BOM\_COST\_GROUP=SSD

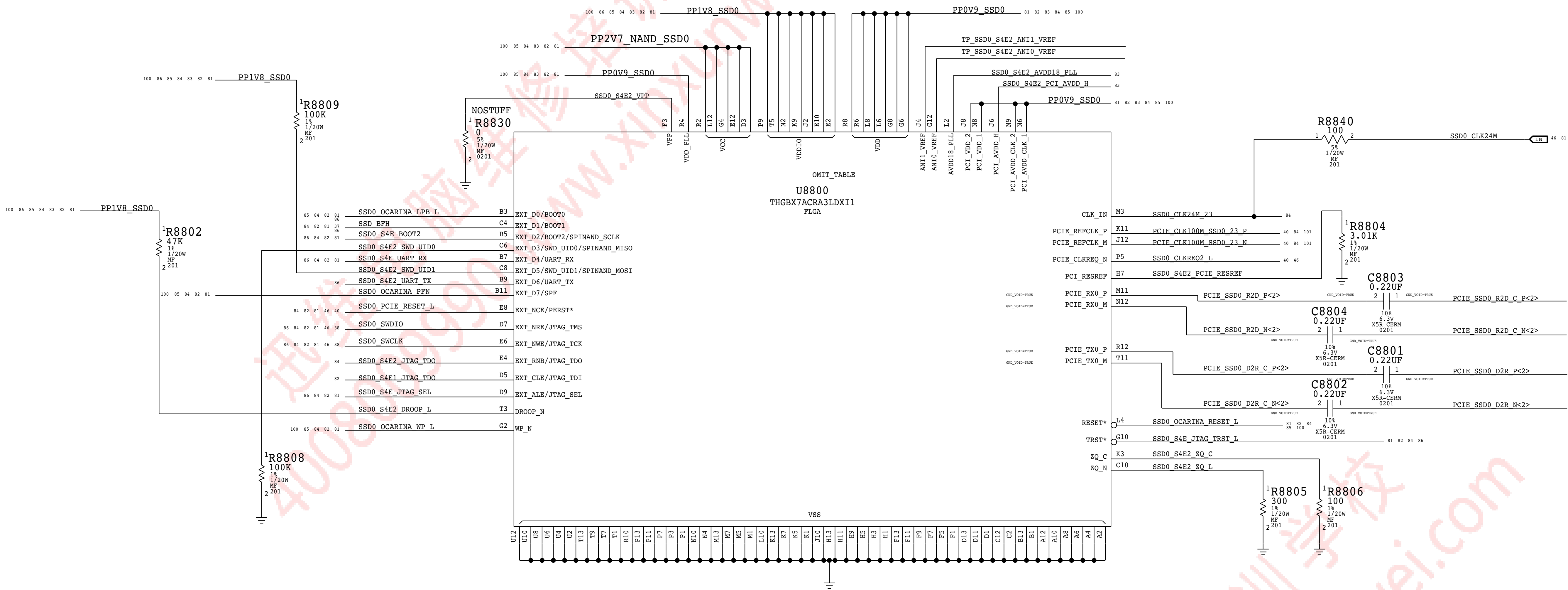
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| SYNC MASTER=J137 GS5 REDHEAD                                                                                                                                                                                                                                                                                                |  |  | SYNC DATE=02/17/2017 |  |            |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |  |                      |  |            |  |
| S4E<0>                                                                                                                                                                                                                                                                                                                      |  |  |                      |  |            |  |
|  Apple Inc.                                                                                                                                                                                                                            |  |  | DRAWING NUMBER       |  | SIZE       |  |
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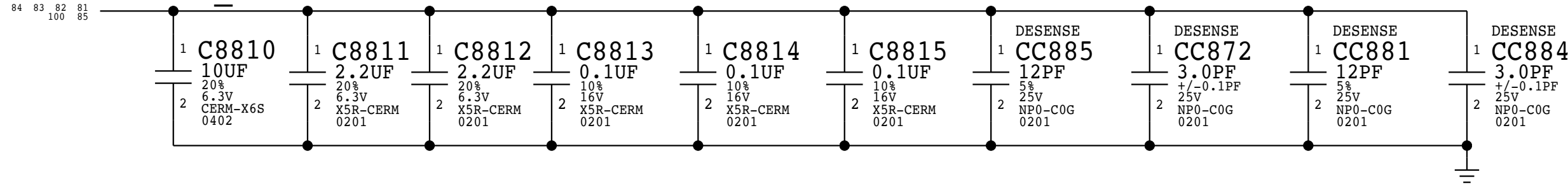


# S4E2



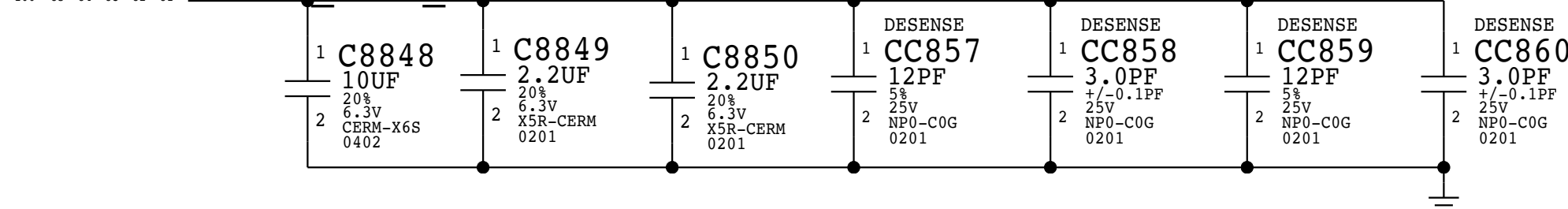
## S4E VDD

### PP0V9\_SSD0



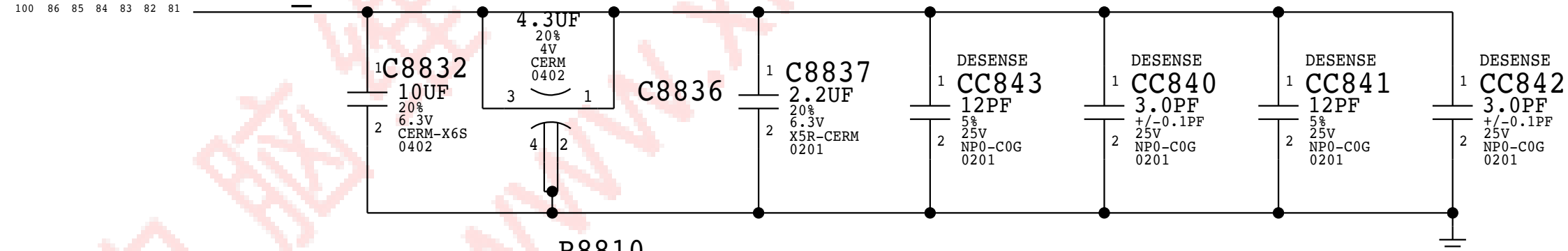
## VCC CAP

### PP2V7\_NAND\_SSD0

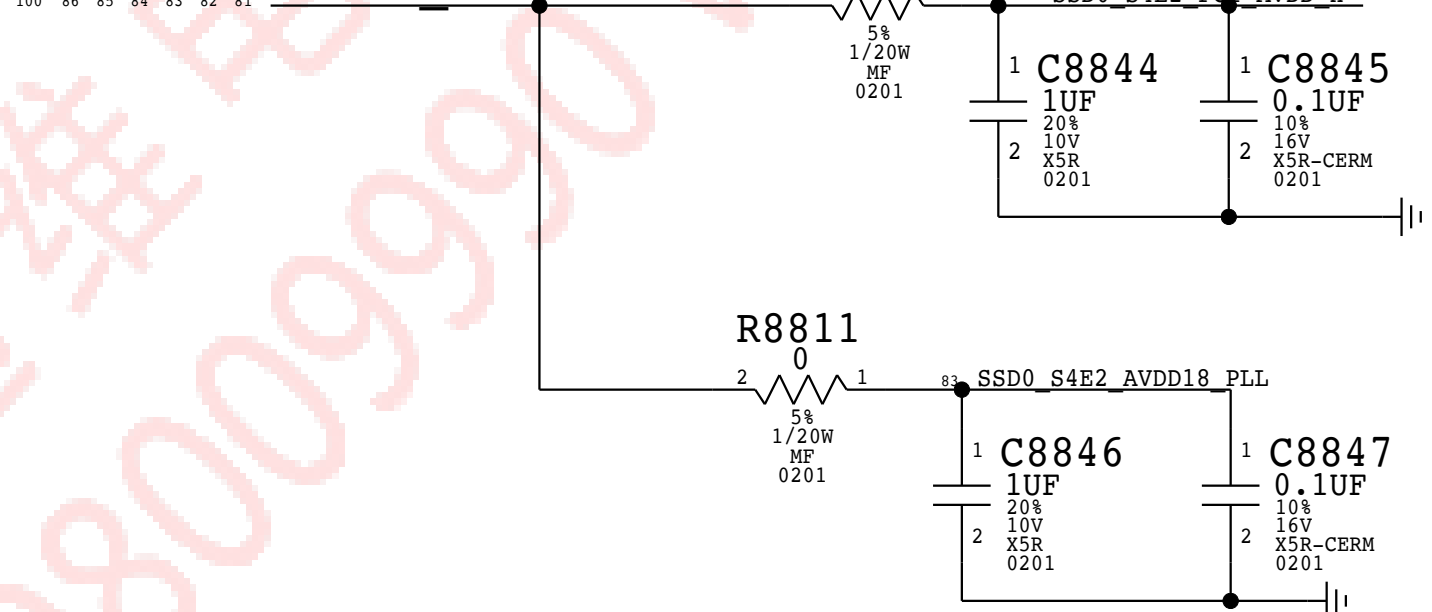


## S4E VDDIO

### PP1V8\_SSD0



### PP1V8\_SSD0

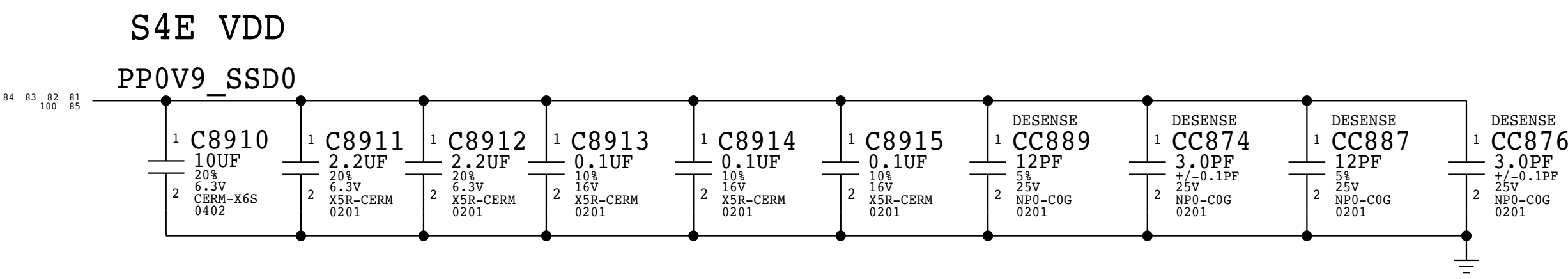
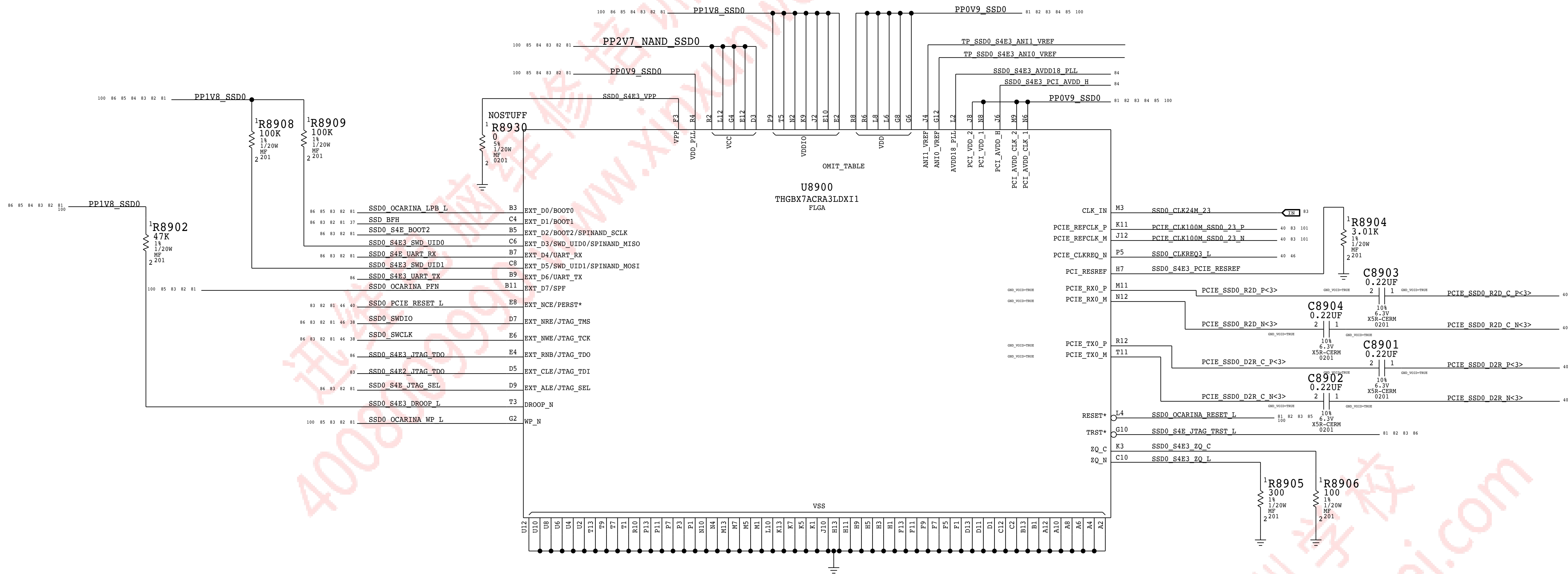


BOM\_COST\_GROUP=SSD

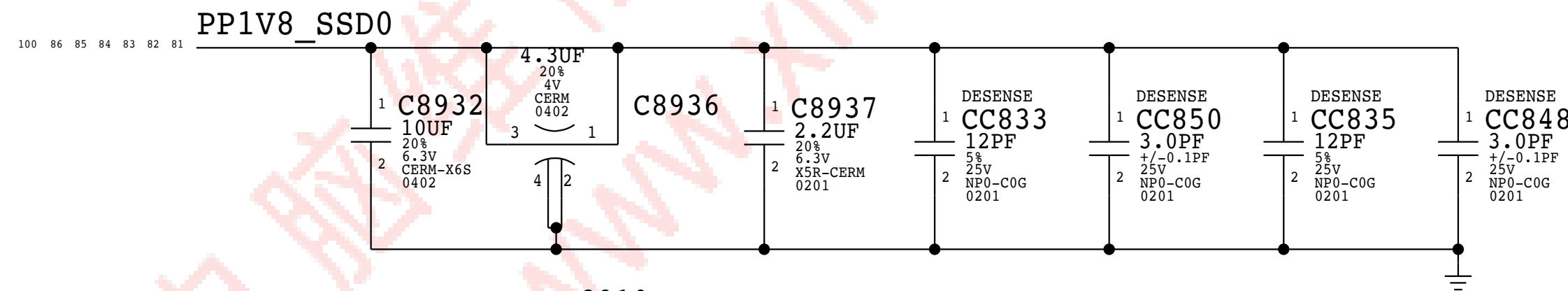
|                                                                                                                                                                                                                                                                                                                             |  |  |                      |  |            |  |
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| SYNC MASTER=J137 GS5 REDHEAD                                                                                                                                                                                                                                                                                                |  |  | SYNC DATE=02/17/2017 |  |            |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |  |                      |  |            |  |
| S4E<2>                                                                                                                                                                                                                                                                                                                      |  |  |                      |  |            |  |
|  Apple Inc.                                                                                                                                                                                                                            |  |  | DRAWING NUMBER       |  | SIZE       |  |
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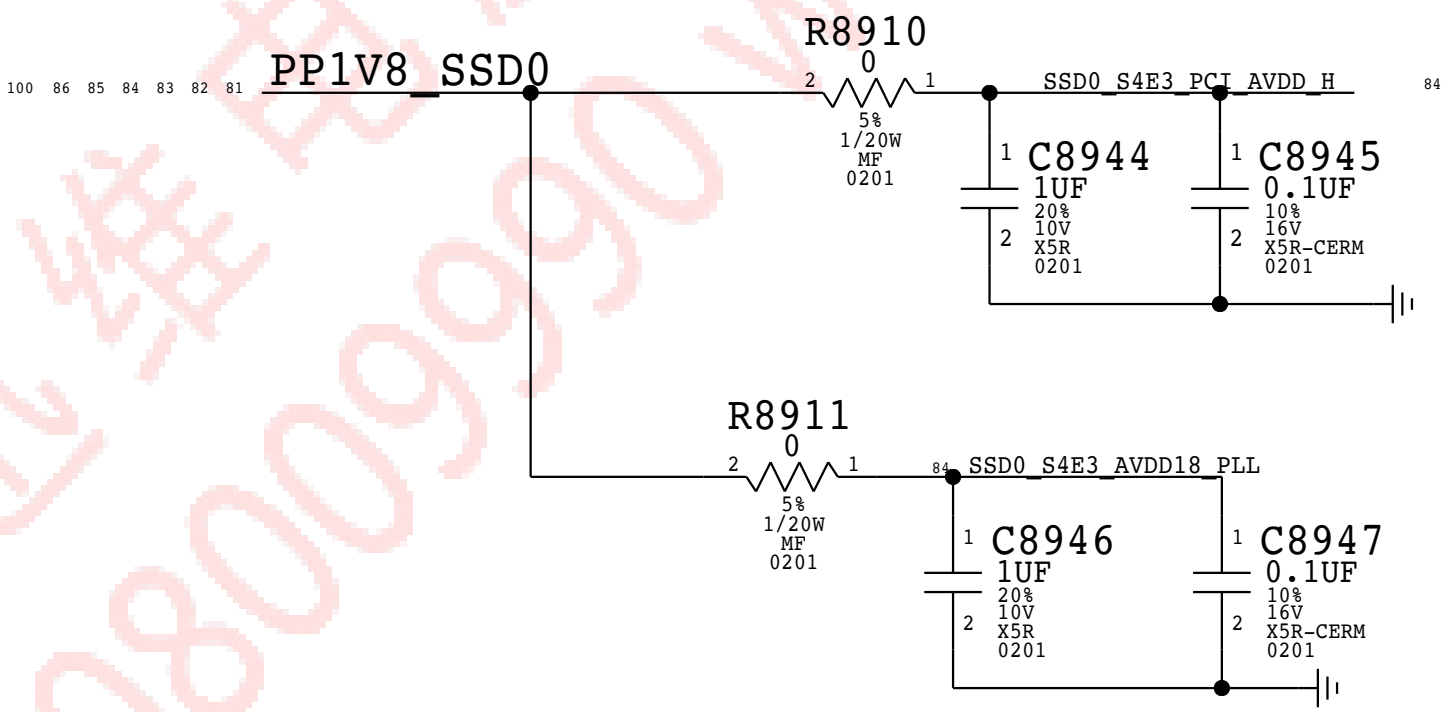
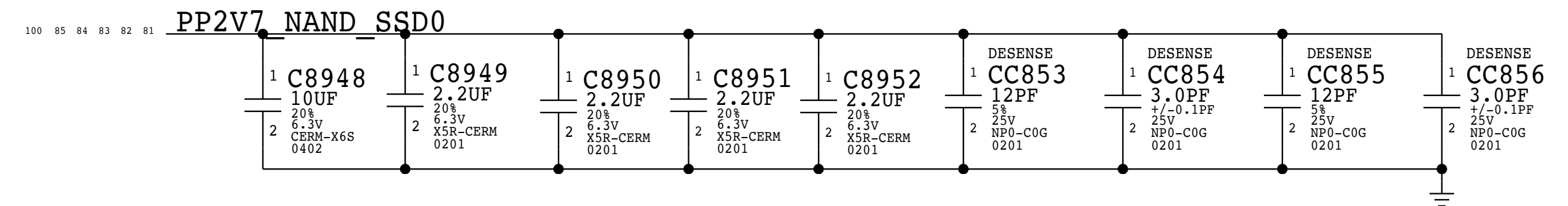
# S4E3



## S4E VDDIO



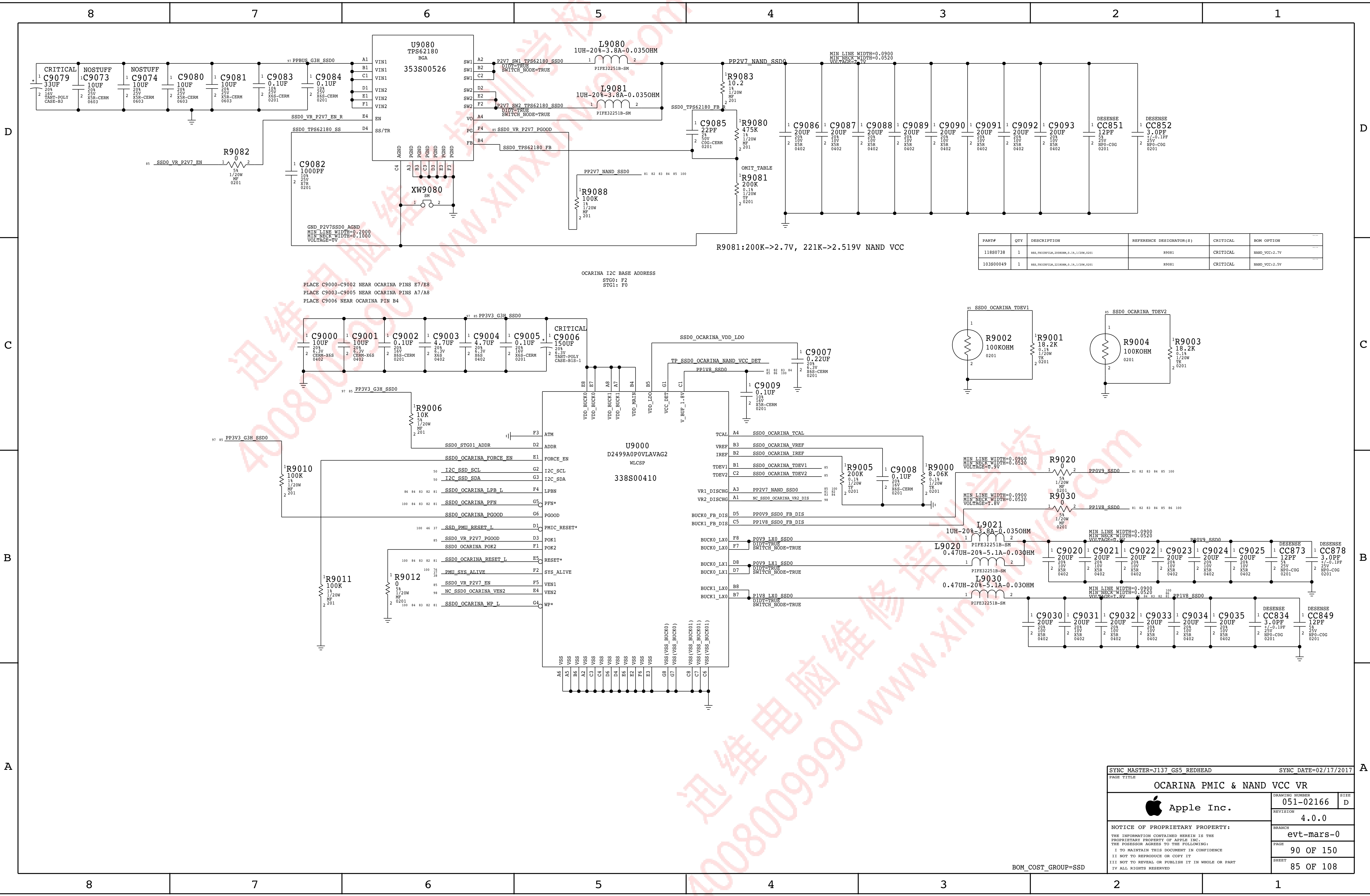
## VCC CAP



BOM\_COST\_GROUP=SSD

|                                                                                                                                                                                                                                                                                                                             |  |  |                      |  |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------|--|------------|
| SYNC MASTER=J137 GS5 REDHEAD                                                                                                                                                                                                                                                                                                |  |  | SYNC DATE=02/17/2017 |  |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |  |                      |  |            |
| S4E<3>                                                                                                                                                                                                                                                                                                                      |  |  |                      |  |            |
|  Apple Inc.                                                                                                                                                                                                                            |  |  | DRAWING NUMBER       |  | SIZE       |
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| PART#     | QTY | DESCRIPTION                           | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION    |
|-----------|-----|---------------------------------------|-------------------------|----------|---------------|
| 11880738  | 1   | RES,THICKFILM,10000HM,0.1%,1/20W,0201 | R9081                   | CRITICAL | NAND_VCC<2.7V |
| 103800049 | 1   | RES,THICKFILM,22100HM,0.1%,1/20W,0201 | R9081                   | CRITICAL | NAND_VCC<2.5V |

| PART#     | QTY | DESCRIPTION                           | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION    |
|-----------|-----|---------------------------------------|-------------------------|----------|---------------|
| 11880738  | 1   | RES,THICKFILM,10000HM,0.1%,1/20W,0201 | R9081                   | CRITICAL | NAND_VCC<2.7V |
| 103800049 | 1   | RES,THICKFILM,22100HM,0.1%,1/20W,0201 | R9081                   | CRITICAL | NAND_VCC<2.5V |

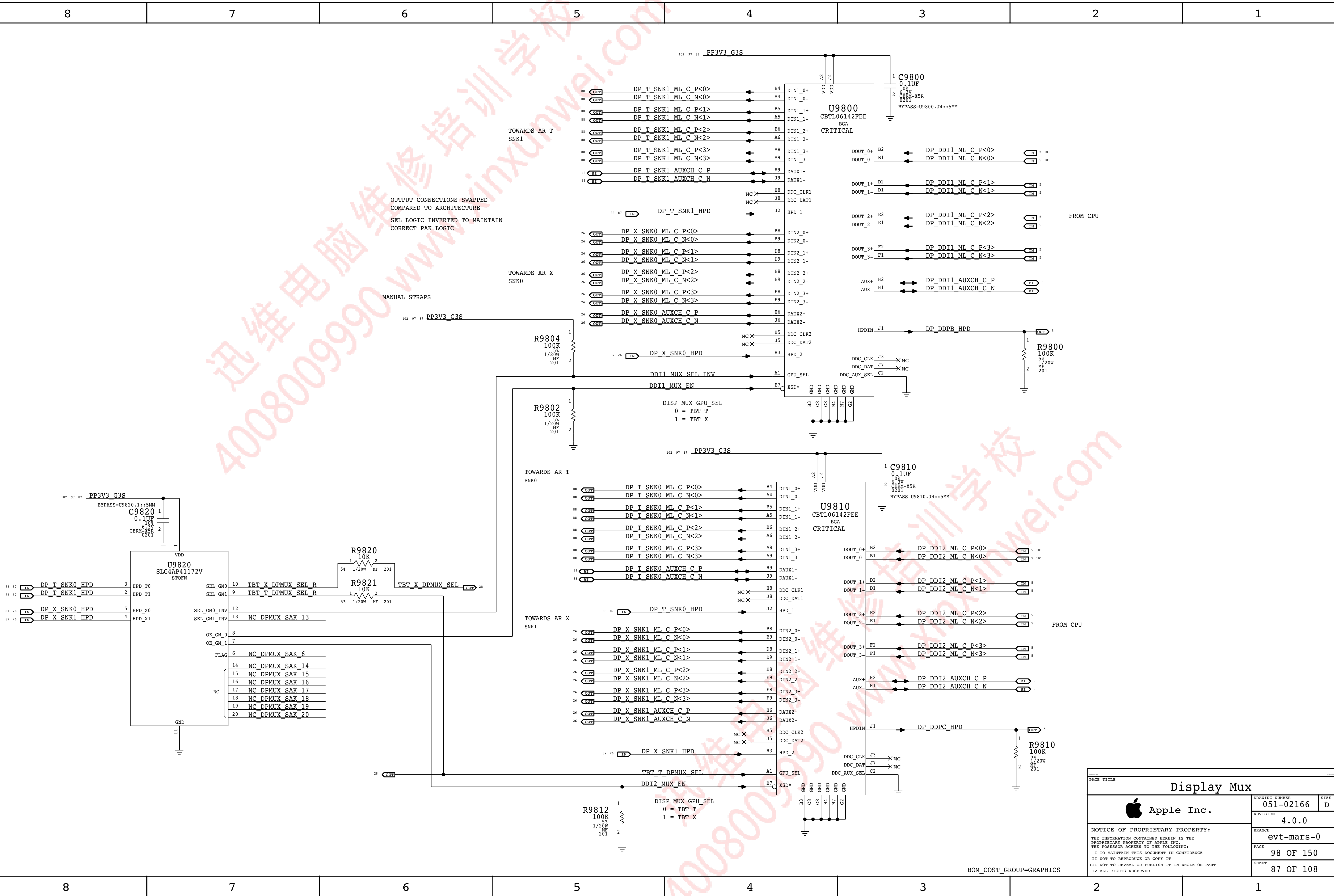
|                                                                                                                                                                                                                                                                                                                         |  |  |                            |            |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------|------------|------|
| SYNC MASTER=J137 GS5 REDHEAD                                                                                                                                                                                                                                                                                            |  |  | SYNC DATE=02/17/2017       |            |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  |  | OCARINA PMIC & NAND VCC VR |            |      |
|  Apple Inc.                                                                                                                                                                                                                        |  |  | DRAWING NUMBER             | 051-02166  | SIZE |
|                                                                                                                                                                                                                                                                                                                         |  |  | REVISION                   | 4.0.0      | D    |
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|                                                                                                                                                                                                                                                                                                                         |  |  | PAGE                       | 90 OF 150  |      |
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BOM\_COST\_GROUP=SSD

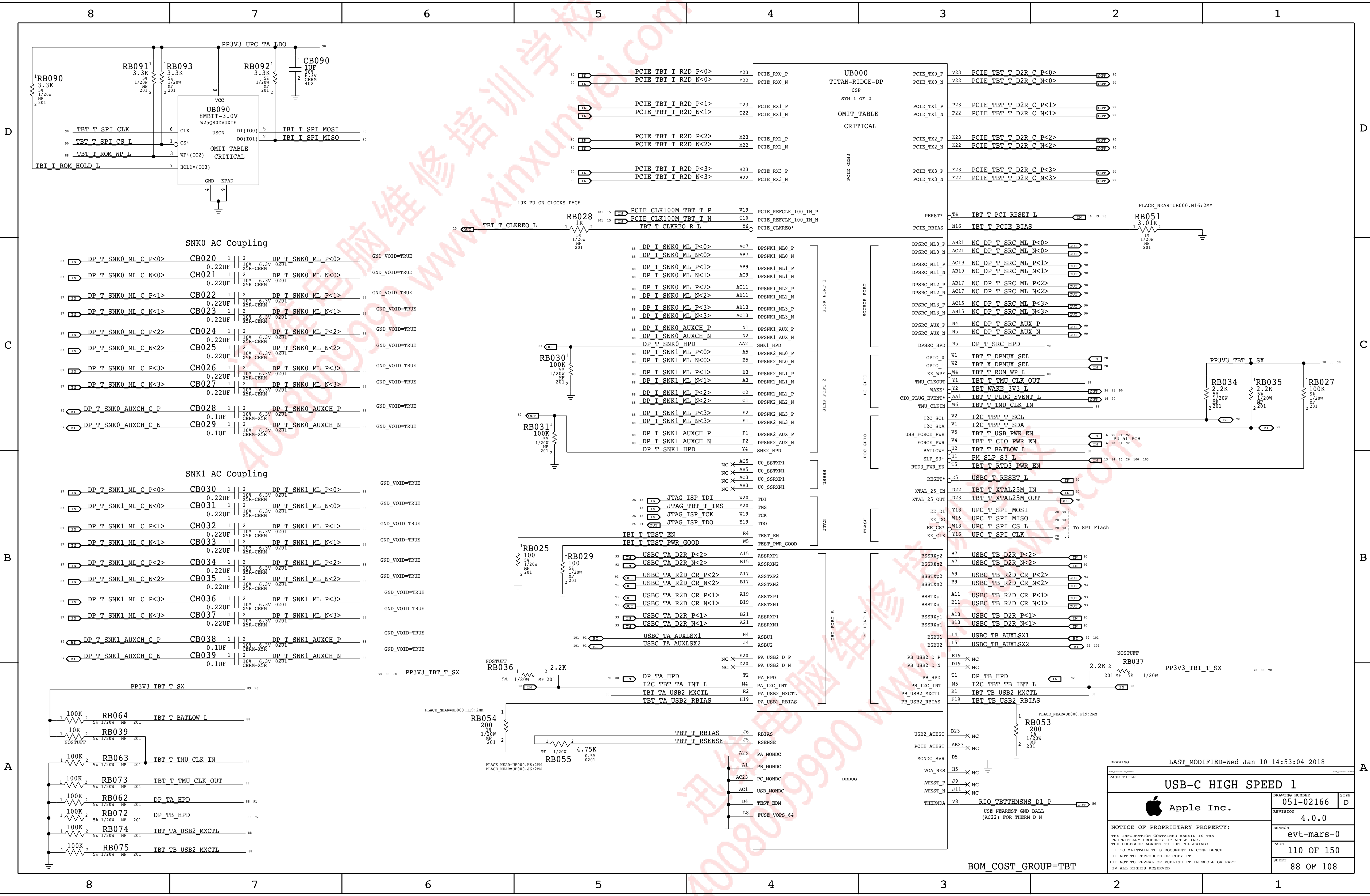












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| DRAWING                                                                                                                                                                                                                                                                                                             |  |  | LAST MODIFIED=Wed Jan 10 14:53:04 2018 |            |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                          |  |  | USB-C HIGH SPEED 1                     |            |      |
|  Apple Inc.                                                                                                                                                                                                                    |  |  | DRAWING NUMBER                         | 051-02166  | SIZE |
|                                                                                                                                                                                                                                                                                                                     |  |  | REVISION                               | 4.0.0      | D    |
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|                                                                                                                                                                                                                                                                                                                     |  |  | PAGE                                   | 110 OF 150 |      |
|                                                                                                                                                                                                                                                                                                                     |  |  | SHEET                                  | 88 OF 108  |      |



D

C

B

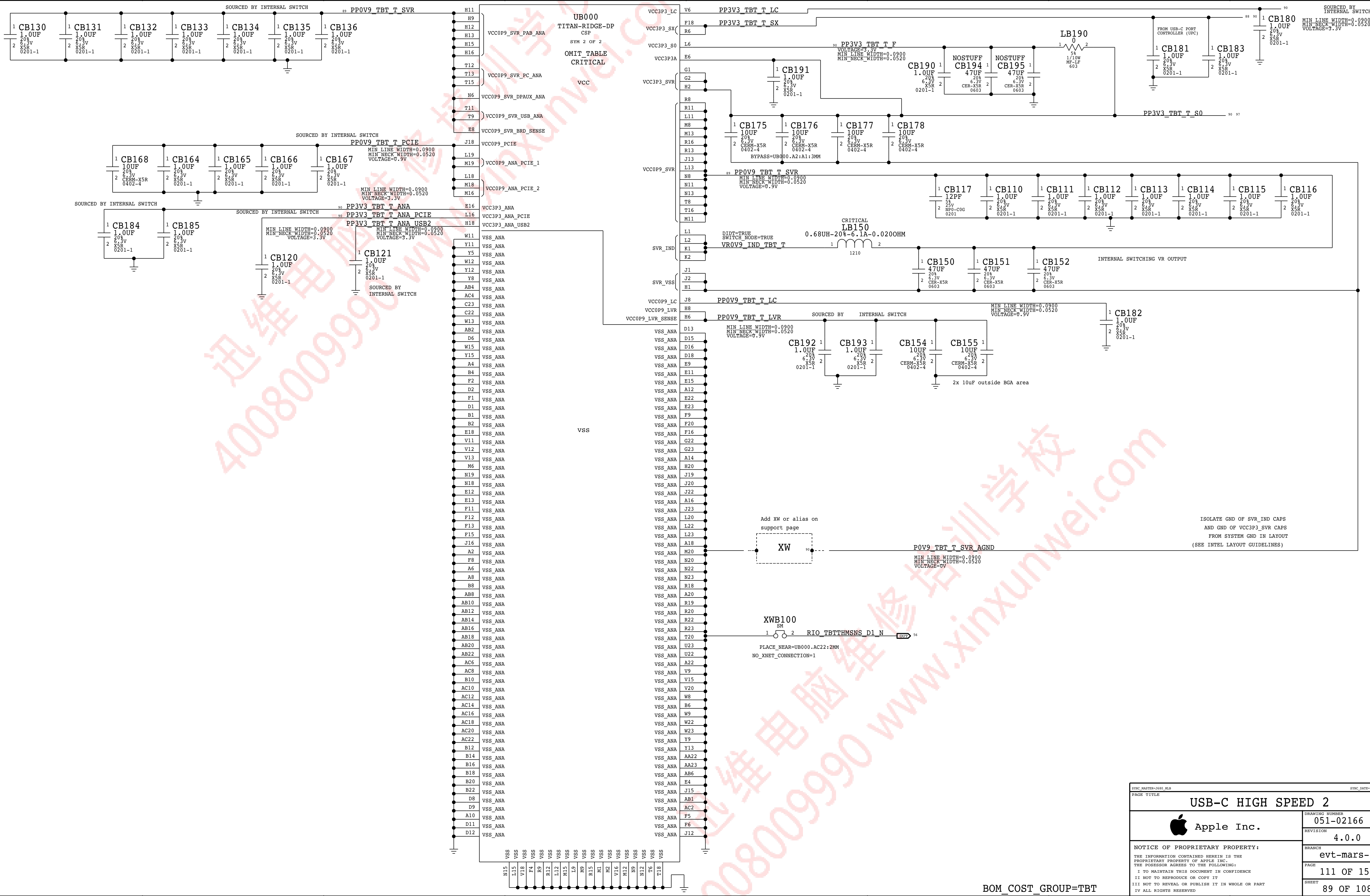
A

D

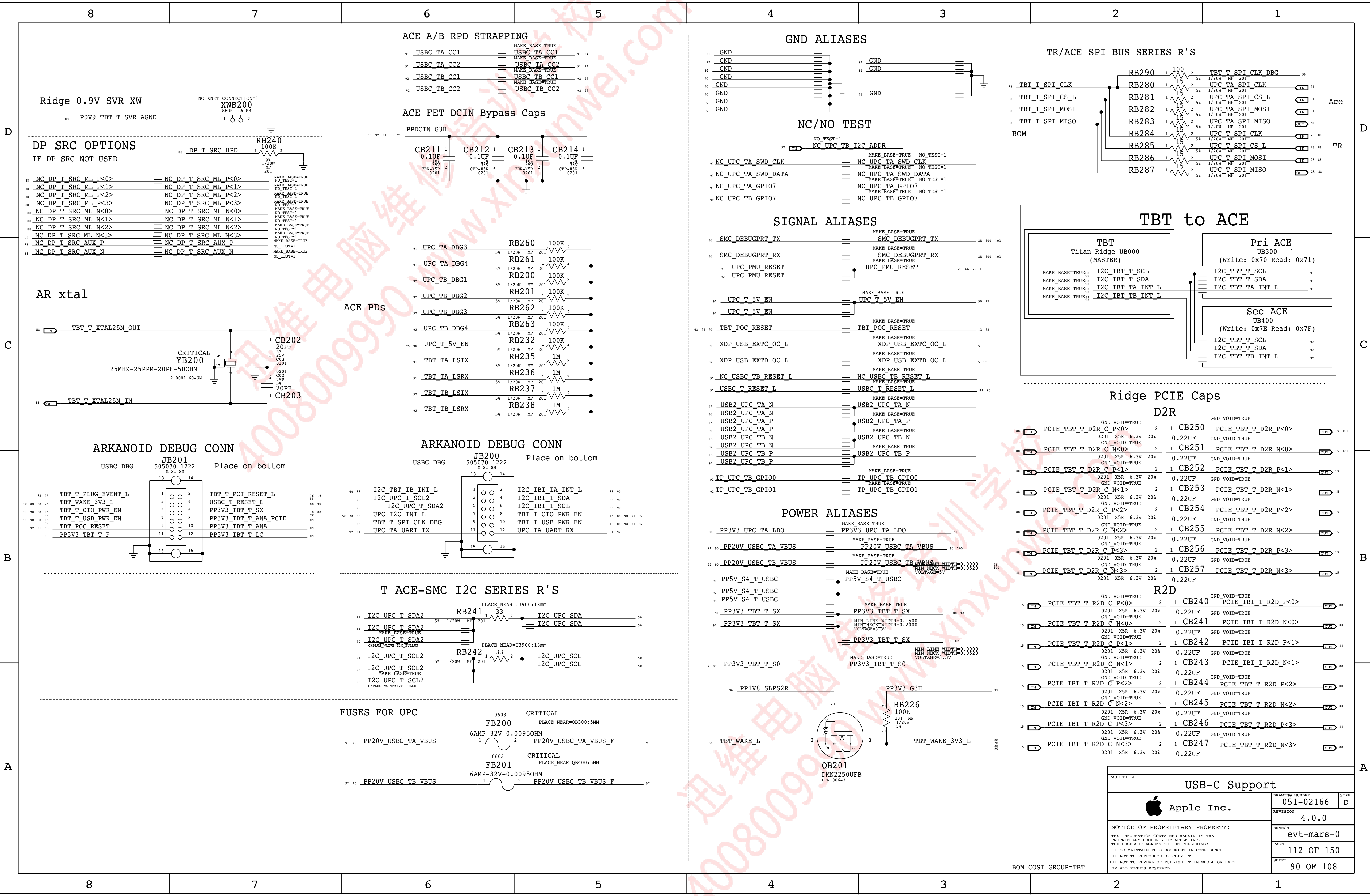
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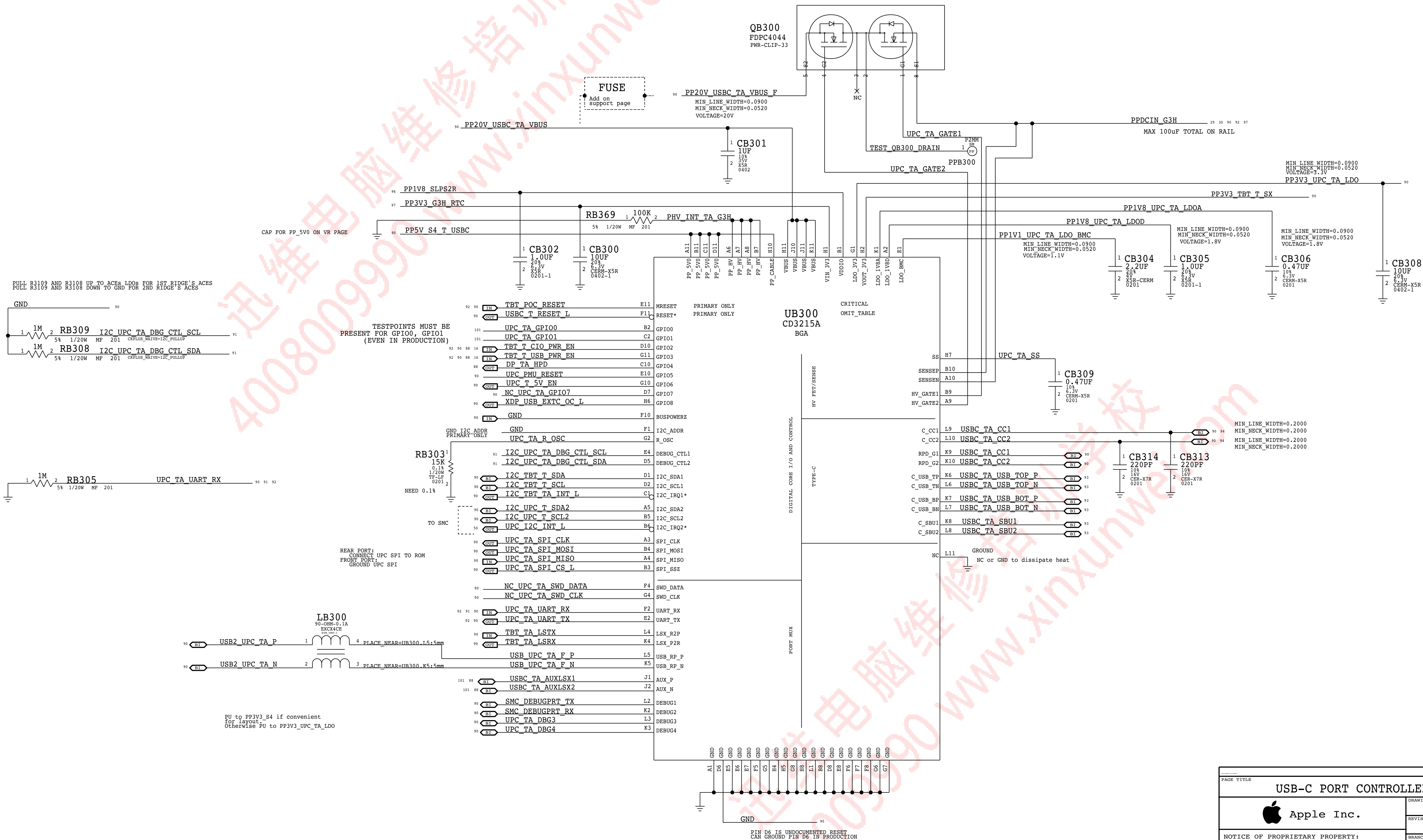
C

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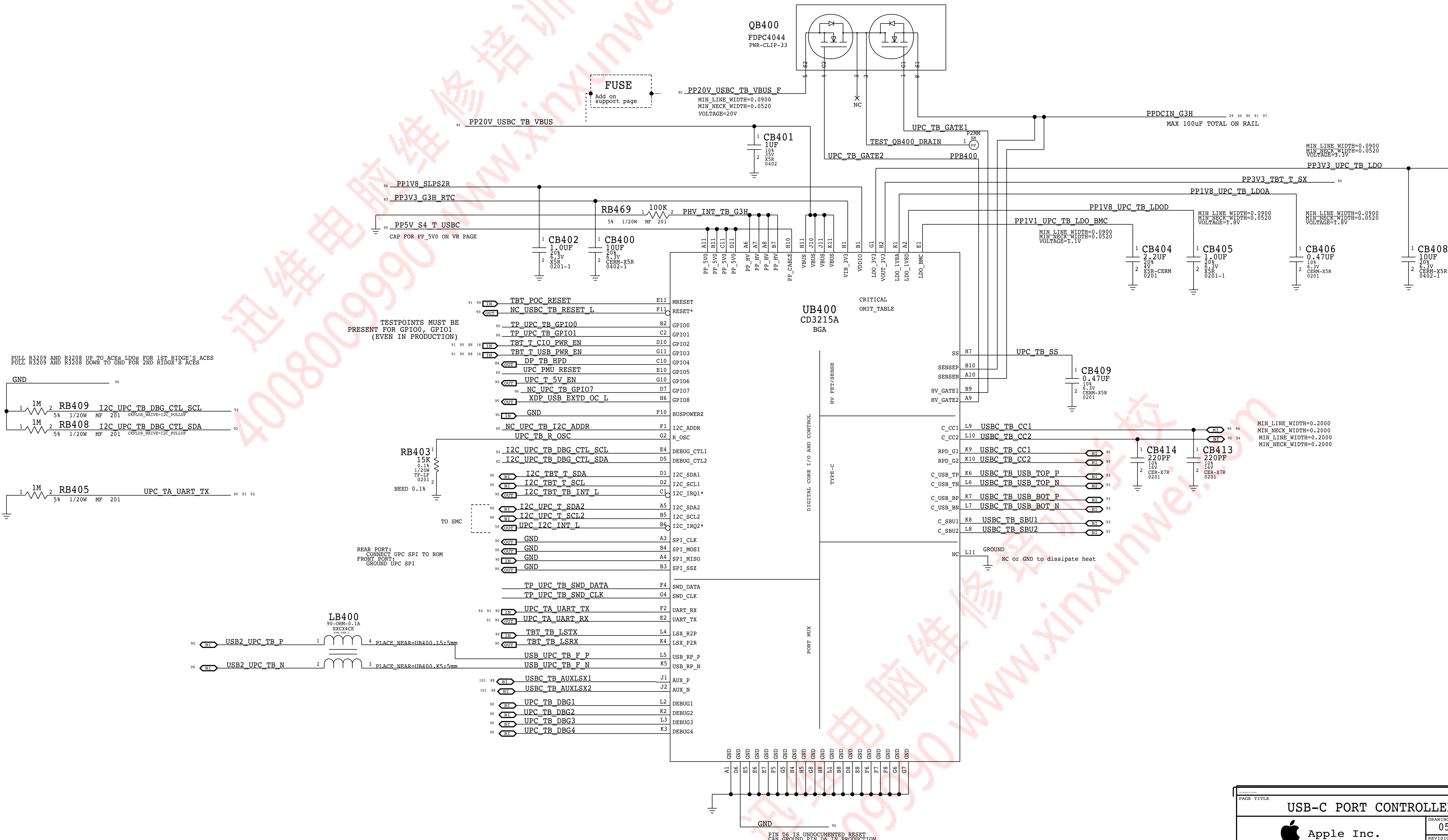


PRIMARY ACE USB-C PORT CONTROLLER (UPC)





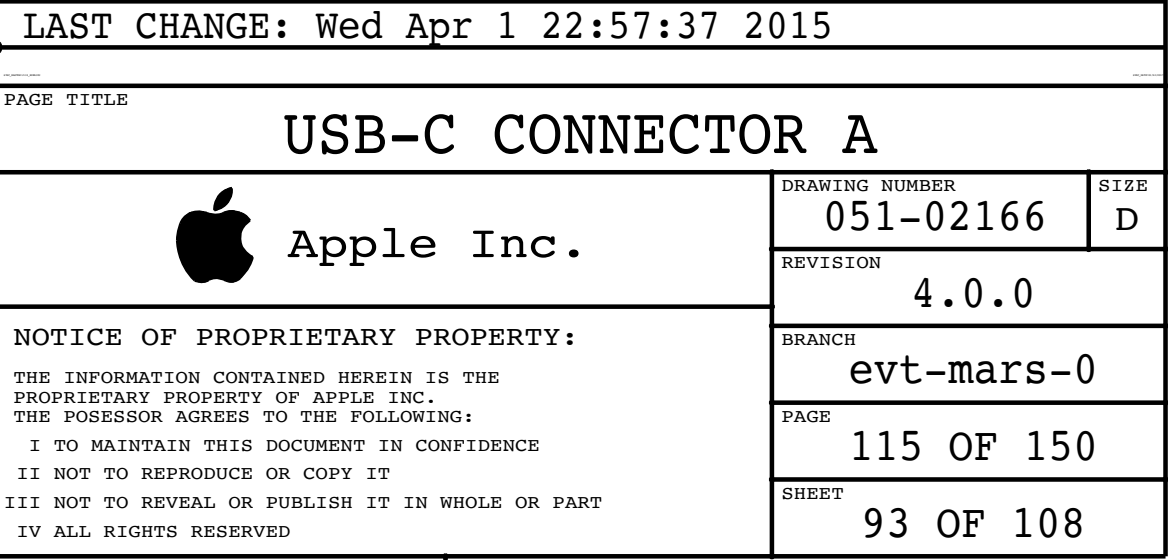
# SECONDARY ACE USB-C PORT CONTROLLER (UPC)



BOM COST GROUP=USB-C

| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| USB-C PORT CONTROLLER B                                                                                                                                                                                                                                                                                                 |                |            |
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                         | REVISION       | 4.0.0      |
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|                                                                                                                                                                                                                                                                                                                         | PAGE           | 114 OF 150 |
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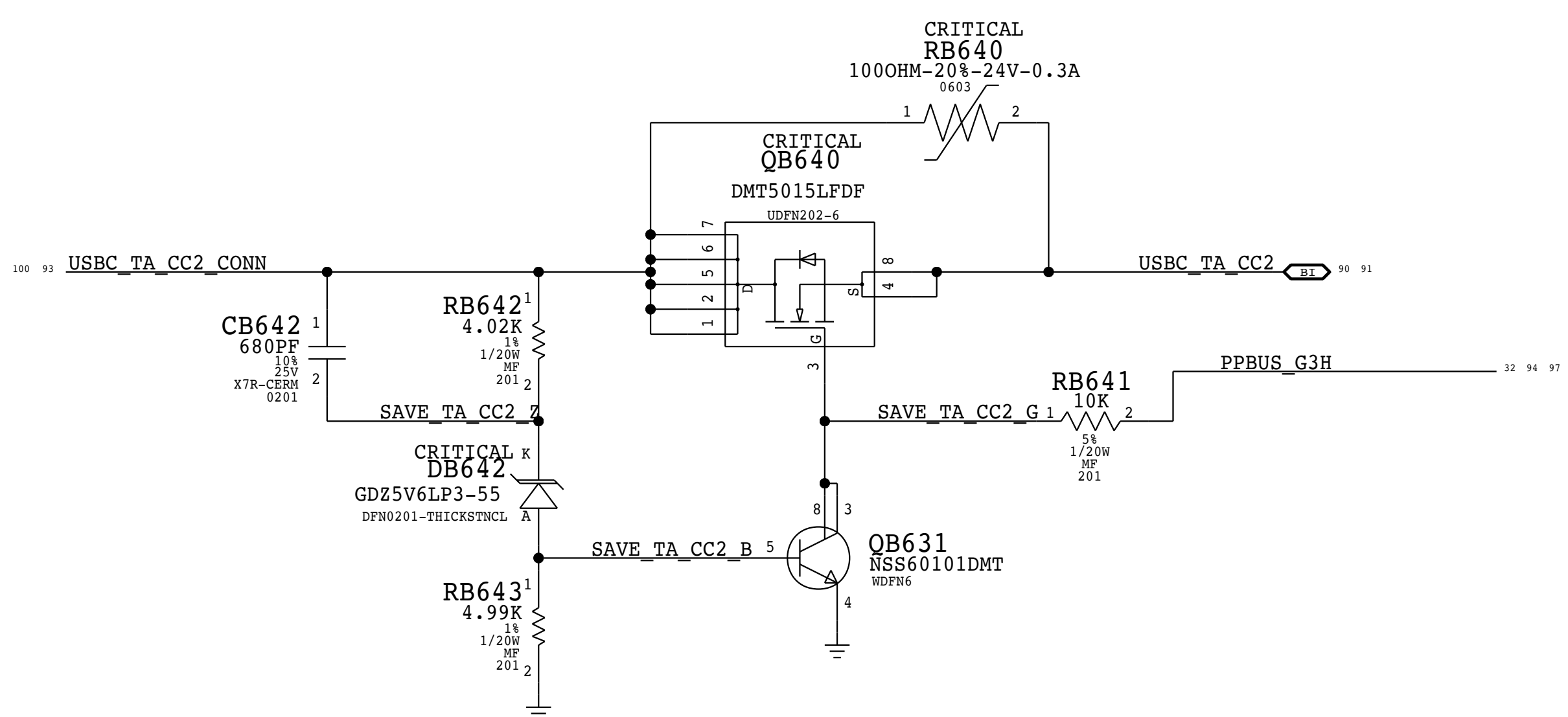
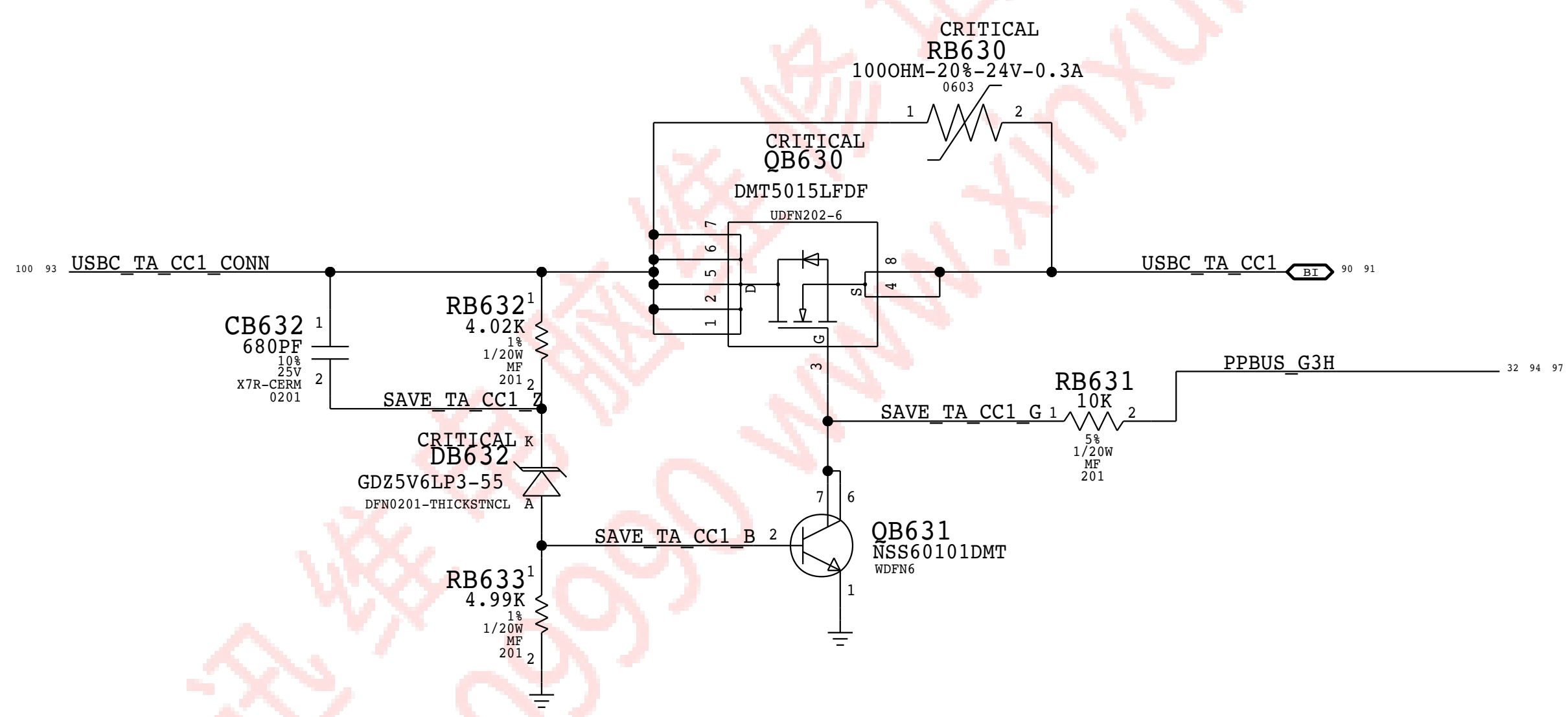
D

C

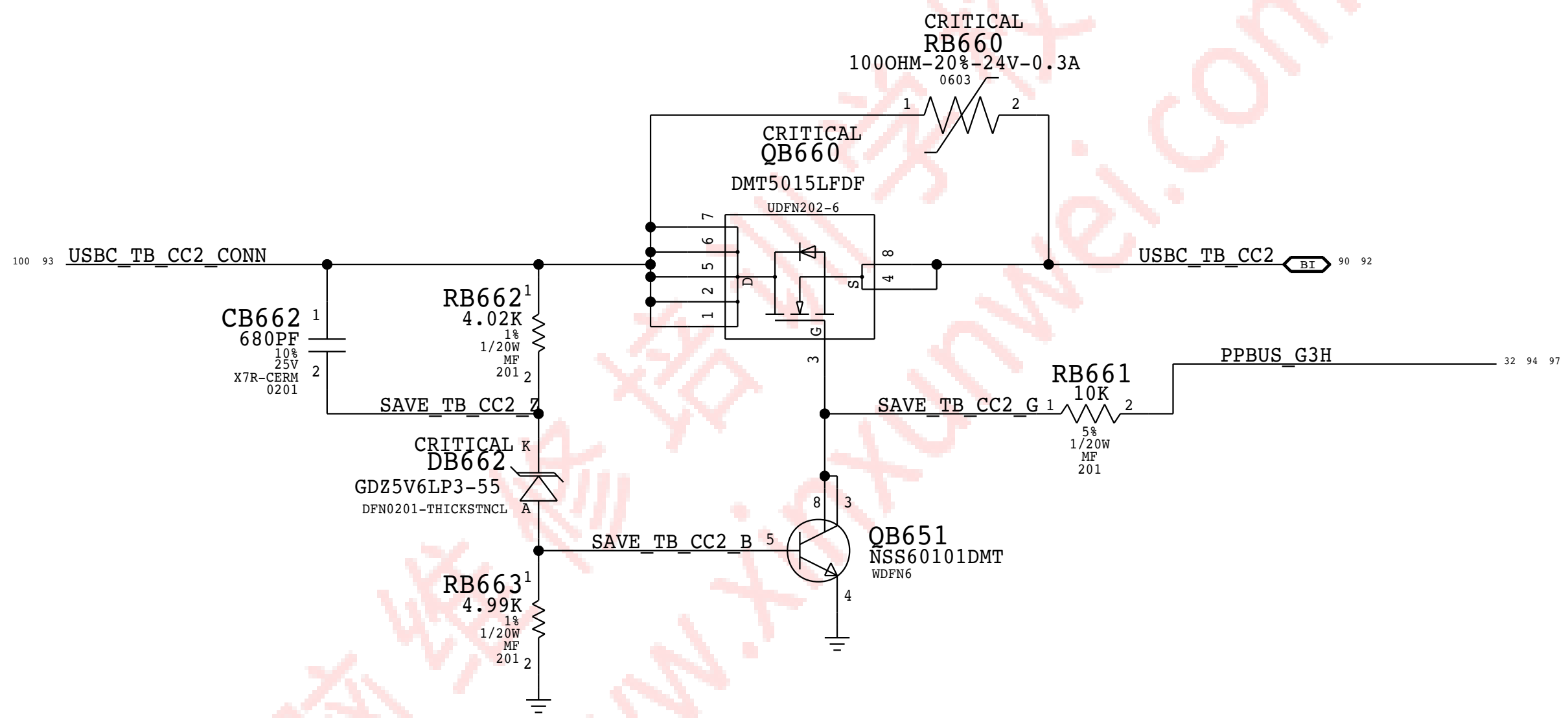
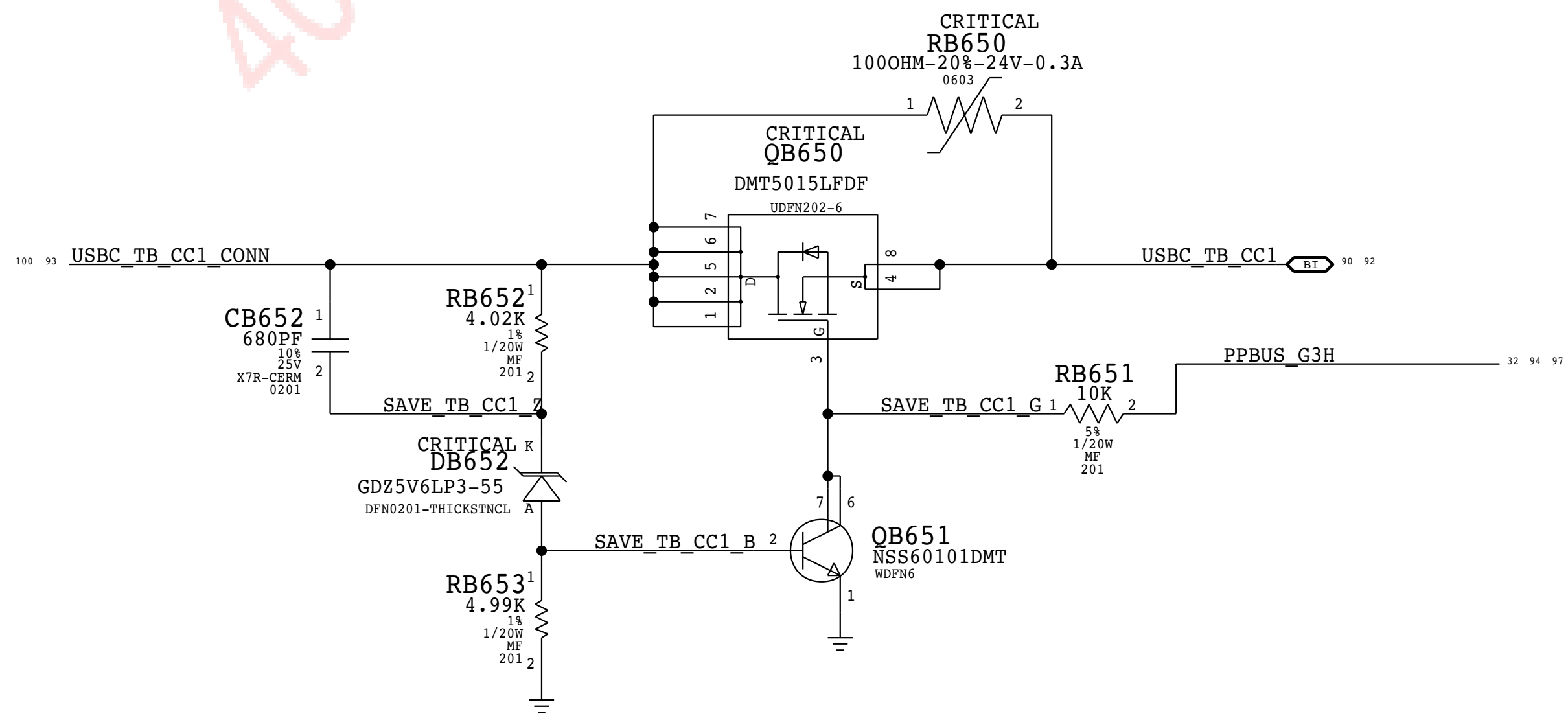
B

A

TA CC Protection



TB CC Protection



BOM\_COST\_GROUP=USB-C

|                                                                                                                                                                                                                                                                                                                         |                |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| DESIGN: X1032/MLB P4BP                                                                                                                                                                                                                                                                                                  |                |            |
| LAST CHANGE: Fri Jan 6 16:01:21 2017                                                                                                                                                                                                                                                                                    |                |            |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |            |
| USB C T Connector Support                                                                                                                                                                                                                                                                                               |                |            |
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02166  |
|                                                                                                                                                                                                                                                                                                                         | REVISION       | 4.0.0      |
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|                                                                                                                                                                                                                                                                                                                         | PAGE           | 116 OF 150 |
|                                                                                                                                                                                                                                                                                                                         | SHEET          | 94 OF 108  |

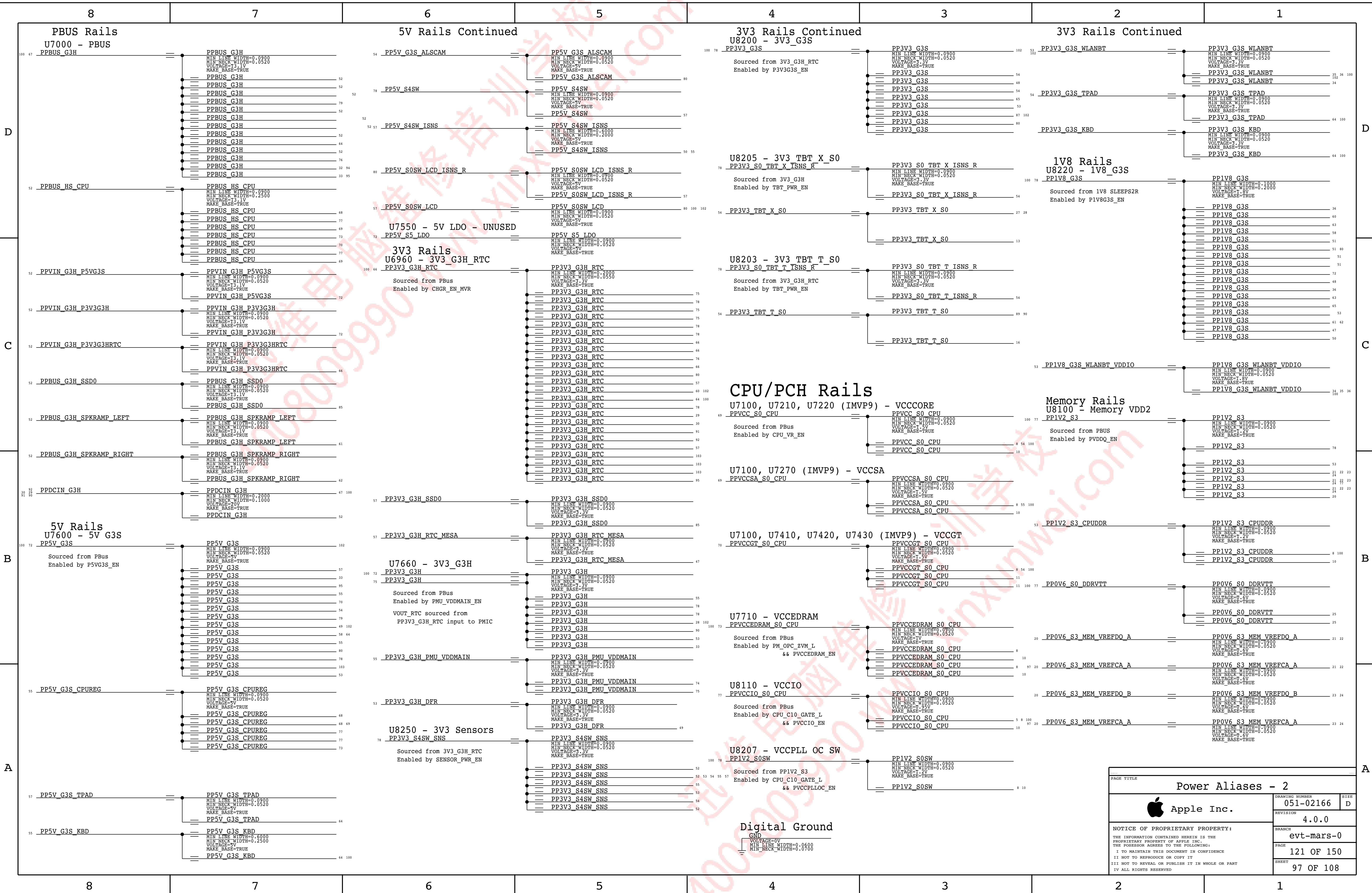














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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |  |
|  | <div>Unused SoC Signals</div> <div><div><div>38</div><div>NC_ALTIMETER_INT</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_ALTIMETER_INT</div></div><div><div>38</div><div>NC_DISP_GCON_INT_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_DISP_GCON_INT_L</div></div><div><div>38</div><div>NC_DFR_TOUCH_RSVD</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_DFR_TOUCH_RSVD</div></div><div><div>38</div><div>NC_ENET_LOW_PWR</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_ENET_LOW_PWR</div></div><div><div>38</div><div>NC_ENET_MEDIA_SENSE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_ENET_MEDIA_SENSE</div></div><div><div>40</div><div>NC_ENET_RESET_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_ENET_RESET_L</div></div><div><div>39</div><div>NC_FTCAM_CLK12M_R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_FTCAM_CLK12M_R</div></div><div><div>39</div><div>NC_FTCAM_RESET_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_FTCAM_RESET_L</div></div><div><div>37</div><div>NC_GNSS_DEV_WAKE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_GNSS_DEV_WAKE</div></div><div><div>37</div><div>NC_GNSS_HOST_TIME</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_GNSS_HOST_TIME</div></div><div><div>39</div><div>NC_I2S_CODEC1_MCLK</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_I2S_CODEC1_MCLK</div></div><div><div>39</div><div>NC_I2S_CODEC1_MCLK</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_I2S_CODEC1_MCLK</div></div><div><div>39</div><div>NC_I2S_CODEC1_R2D_R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_I2S_CODEC1_R2D_R</div></div><div><div>39</div><div>NC_I2S_HAWKING_BCLK_R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_I2S_HAWKING_BCLK_R</div></div><div><div>39</div><div>NC_I2S_HAWKING_D2R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_I2S_HAWKING_D2R</div></div><div><div>39</div><div>NC_I2S_HAWKING_LRCLK</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_I2S_HAWKING_LRCLK</div></div><div><div>38</div><div>NC_MESA_MENUKEY_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_MESA_MENUKEY_L</div></div><div><div>38</div><div>NC_PCC_EVENT</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCC_EVENT</div></div><div><div>15</div><div>NC_PCH_ENET_LOW_PWR</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCH_ENET_LOW_PWR</div></div><div><div>38</div><div>NC_SOC_PCH_GCON_INT_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SOC_PCH_GCON_INT_L</div></div><div><div>40</div><div>NC_PCIE_CLK100M_ENET_N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_ENET_N</div></div><div><div>40</div><div>NC_PCIE_CLK100M_ENET_P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_ENET_P</div></div><div><div>40</div><div>NC_PCIE_CLK100M_SSD1_01_N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_SSD1_01_N</div></div><div><div>40</div><div>NC_PCIE_CLK100M_SSD1_01_P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_SSD1_01_P</div></div><div><div>40</div><div>NC_PCIE_CLK100M_SSD1_23_N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_SSD1_23_N</div></div><div><div>40</div><div>NC_PCIE_CLK100M_SSD1_23_P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_SSD1_23_P</div></div><div><div>40</div><div>NC_PCIE_CLK100M_WLAN_N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_WLAN_N</div></div><div><div>40</div><div>NC_PCIE_CLK100M_WLAN_P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_WLAN_P</div></div><div><div>40</div><div>NC_PCIE_ENET_D2R_N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_ENET_D2R_N</div></div><div><div>40</div><div>NC_PCIE_ENET_D2R_P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_ENET_D2R_P</div></div><div><div>40</div><div>NC_PCIE_ENET_R2D_C_N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_ENET_R2D_C_N</div></div><div><div>40</div><div>NC_PCIE_ENET_R2D_C_P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_ENET_R2D_C_P</div></div><div><div>40</div><div>NC_PCIE_SSD1_D2R_N&lt;3..0&gt;</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_SSD1_D2R_N&lt;3..0&gt;</div></div><div><div>40</div><div>NC_PCIE_SSD1_D2R_P&lt;3..0&gt;</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_SSD1_D2R_P&lt;3..0&gt;</div></div><div><div>40</div><div>NC_PCIE_SSD1_R2D_C_N&lt;3..0&gt;</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_SSD1_R2D_C_N&lt;3..0&gt;</div></div><div><div>40</div><div>NC_PCIE_SSD1_R2D_C_P&lt;3..0&gt;</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_SSD1_R2D_C_P&lt;3..0&gt;</div></div><div><div>40</div><div>NC_PCIE_WLAN_D2R_N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_WLAN_D2R_N</div></div><div><div>40</div><div>NC_PCIE_WLAN_D2R_P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_WLAN_D2R_P</div></div><div><div>40</div><div>NC_PCIE_WLAN_R2D_C_N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_WLAN_R2D_C_N</div></div><div><div>40</div><div>NC_PCIE_WLAN_R2D_C_P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_WLAN_R2D_C_P</div></div><div><div>38</div><div>NC_PCIEDN_WAKE_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIEDN_WAKE_L</div></div><div><div>38</div><div>NC_PLCAM_PROX_INT_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PLCAM_PROX_INT_L</div></div><div><div>38</div><div>NC_PLCAM_ROMEO_B2B_DETECT</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PLCAM_ROMEO_B2B_DETECT</div></div><div><div>39</div><div>NC_PLCAM_RX_CLK12M_R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PLCAM_RX_CLK12M_R</div></div><div><div>39</div><div>NC_PLCAM_RX_RESET_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PLCAM_RX_RESET_L</div></div><div><div>39</div><div>NC_PLCAM_TX_CLK12M_R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PLCAM_TX_CLK12M_R</div></div><div><div>39</div><div>NC_PLCAM_TX_INT</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PLCAM_TX_INT</div></div><div><div>39</div><div>NC_PLCAM_TX_RESET_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PLCAM_TX_RESET_L</div></div><div><div>37</div><div>NC_PLCAM_TX_THROTTLE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PLCAM_TX_THROTTLE</div></div><div><div>38</div><div>NC_SDCONN_STATE_CHANGE_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SDCONN_STATE_CHANGE_L</div></div><div><div>38</div><div>NC_SMC_GFX_SELF_THROTTLE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SMC_GFX_SELF_THROTTLE</div></div><div><div>38</div><div>NC_SMC_LED_ONEWIRE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SMC_LED_ONEWIRE</div></div><div><div>38</div><div>NC_SMC_TOPBLK_SWP_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SMC_TOPBLK_SWP_L</div></div><div><div>38</div><div>NC_SPI_ALTIMETER_CS_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SPI_ALTIMETER_CS_L</div></div><div><div>38</div><div>NC_SPI_DESCRIPTOR_OVERRIDE_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SPI_DESCRIPTOR_OVERRIDE_L</div></div><div><div>40</div><div>NC_SSD1_CLK24M_R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD1_CLK24M_R</div></div><div><div>40</div><div>NC_SSD1_CLKREQ0_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD1_CLKREQ0_L</div></div><div><div>40</div><div>NC_SSD1_CLKREQ1_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD1_CLKREQ1_L</div></div><div><div>40</div><div>NC_SSD1_CLKREQ2_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD1_CLKREQ2_L</div></div><div><div>40</div><div>NC_SSD1_CLKREQ3_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD1_CLKREQ3_L</div></div><div><div>40</div><div>NC_SSD1_PCIE_RESET_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD1_PCIE_RESET_L</div></div><div><div>38</div><div>NC_SSD1_SWCLK_UART_R2D</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD1_SWCLK_UART_R2D</div></div><div><div>38</div><div>NC_SSD1_SWDIO_UART_D2R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD1_SWDIO_UART_D2R</div></div><div><div>39</div><div>NC_PCHROM_SW_EN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCHROM_SW_EN</div></div><div><div>40</div><div>NC_PCIE_DN2_R2D_CP</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_DN2_R2D_CP</div></div><div><div>40</div><div>NC_PCIE_DN2_R2D_CN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_DN2_R2D_CN</div></div><div><div>40</div><div>NC_PCIE_DN2_D2RP</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_DN2_D2RP</div></div><div><div>40</div><div>NC_PCIE_DN2_D2RN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_DN2_D2RN</div></div><div><div>40</div><div>NC_PCIE_DN3_R2D_CP</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_DN3_R2D_CP</div></div><div><div>40</div><div>NC_PCIE_DN3_R2D_CN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_DN3_R2D_CN</div></div><div><div>40</div><div>NC_PCIE_DN3_D2RP</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_DN3_D2RP</div></div><div><div>40</div><div>NC_PCIE_DN3_D2RN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_DN3_D2RN</div></div><div><div>40</div><div>NC_PCIE_CLK100M_DN2P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_DN2P</div></div><div><div>40</div><div>NC_PCIE_CLK100M_DN2N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_DN2N</div></div><div><div>40</div><div>NC_PCIEDN2_CLKREQ_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIEDN2_CLKREQ_L</div></div><div><div>40</div><div>NC_PCIEDN2_RESET_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIEDN2_RESET_L</div></div><div><div>40</div><div>NC_PCIE_CLK100M_DN3P</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_DN3P</div></div><div><div>40</div><div>NC_PCIE_CLK100M_DN3N</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIE_CLK100M_DN3N</div></div><div><div>40</div><div>NC_PCIEDN3_CLKREQ_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIEDN3_CLKREQ_L</div></div><div><div>40</div><div>NC_PCIEDN3_RESET_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PCIEDN3_RESET_L</div></div><div><div>37</div><div>NC_SOC_USB_ID</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SOC_USB_ID</div></div><div><div>41</div><div>NC_SOC_VDDSOC_SENSE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SOC_VDDSOC_SENSE</div></div><div><div>41</div><div>NC_SOC_VSSCPU_SENSE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SOC_VSSCPU_SENSE</div></div><div><div>41</div><div>NC_SOC_VSSSOC_SENSE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SOC_VSSSOC_SENSE</div></div><div><div>38</div><div>NC_TPAD_VIBE_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_TPAD_VIBE_L</div></div><div><div>39</div><div>NC_UART_GNSS_D2R_CTS_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_UART_GNSS_D2R_CTS_L</div></div><div><div>39</div><div>NC_UART_GNSS_R2D_RTS_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_UART_GNSS_R2D_RTS_L</div></div><div><div>40</div><div>NC_WLAN_CLKREQ_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_WLAN_CLKREQ_L</div></div><div><div>37</div><div>NC_WLAN_DEV_WAKE</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_WLAN_DEV_WAKE</div></div><div><div>40</div><div>NC_WLAN_PERST_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_WLAN_PERST_L</div></div></div> |   |   |   |   |   |   |   |  |
|  | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|
|  | <div>TP SoC Signals</div> <div><div><div>100</div><div>38</div><div>TP_SMC_FIXTURE_MODE_L</div><div>==</div><div>MAKP_BASE+THUR</div><div>TP_SMC_FIXTURE_MODE_L</div></div><div><div>103</div><div>100</div><div>39</div><div>TP_SOC_DEBUGPRT_RX</div><div>==</div><div>MAKP_BASE+THUR</div><div>TP_SOC_DEBUGPRT_RX</div></div><div><div>103</div><div>100</div><div>39</div><div>TP_SOC_DEBUGPRT_TX</div><div>==</div><div>MAKP_BASE+THUR</div><div>TP_SOC_DEBUGPRT_TX</div></div><div><div>37</div><div>TEST_SOC_AMUXOUT</div><div>==</div><div>MAKP_BASE+THUR</div><div>TEST_SOC_AMUXOUT</div></div><div><div>37</div><div>TEST_SOC_TST_CLKOUT</div><div>==</div><div>MAKP_BASE+THUR</div><div>TEST_SOC_TST_CLKOUT</div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |  |  |
|  | <div>SoC Aliases</div> <div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>39</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |  |  |  |  |  |  |
|  | <div>Unused PMIC Signals</div> <div><div><div>67</div><div>NC_CHGR_AUX_OK</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_CHGR_AUX_OK</div></div><div><div>76</div><div>NC_GPU_THRMTRIP</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_GPU_THRMTRIP</div></div><div><div>76</div><div>NC_NAND_DISCHARGE_EN_HDD_PWR_EN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_NAND_DISCHARGE_EN_HDD_PWR_EN</div></div><div><div>76</div><div>NC_NAND_RESET_L_SD_PWR_EN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_NAND_RESET_L_SD_PWR_EN</div></div><div><div>76</div><div>NC_NAND_WP_L_ENET_PWR_EN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_NAND_WP_L_ENET_PWR_EN</div></div><div><div>76</div><div>NC_PEARL_PWREN_P2V7NAND_EN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PEARL_PWREN_P2V7NAND_EN</div></div><div><div>76</div><div>NC_PMU_CLK32K_GNSS_R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PMU_CLK32K_GNSS_R</div></div><div><div>76</div><div>NC_P3V3G3W_EN</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_P3V3G3W_EN</div></div><div><div>76</div><div>NC_P3V3G3W_PGOOD</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_P3V3G3W_PGOOD</div></div><div><div>76</div><div>NC_PMU_CLK32K</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PMU_CLK32K</div></div><div><div>74</div><div>NC_PPBUCK8_SW1</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_PPBUCK8_SW1</div></div></div> |  |  |  |  |  |  |  |  |
|  | <div>PMIC Aliases</div> <div><div><div>76</div><div>PVCCEPIO_EDRAM_PGOOD</div><div>==</div><div>MAKP_BASE+THUR</div><div>PVCCEPIO_EDRAM_PGOOD</div></div><div><div>76</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div><div><div>76</div><div>GND</div><div>==</div><div>MAKP_BASE+THUR</div><div></div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |  |  |  |  |  |  |
|  | <div>Unused WIRELESS Signals</div> <div><div><div>35</div><div>NC_I2S_BT_D2R</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_I2S_BT_D2R</div></div><div><div>35</div><div>NC_I2S_BT_R2D</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_I2S_BT_R2D</div></div><div><div>35</div><div>NC_WLAN_PMU_TEST</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_WLAN_PMU_TEST</div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |  |  |  |  |  |  |
|  | <div>Unused SSD Signals</div> <div><div><div>85</div><div>NC_SSD0_OCARINA_VEN2</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD0_OCARINA_VEN2</div></div><div><div>85</div><div>NC_SSD0_OCARINA_VR2_DIS</div><div>==</div><div>MAKP_BASE+THUR</div><div>NO_TEST+1</div><div>NC_SSD0_OCARINA_VR2_DIS</div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  |  |  |  |  |  |  |
|  | <div>Signal Aliases</div> <div><div><div><div><div></div><div>Apple Inc.</div></div><div>051-02166</div><div>4.0.0</div></div><div><div>NOTICE OF PROPRIETARY PROPERTY:<br/>THE INFORMATION CONTAINED HEREIN IS THE<br/>PROPRIETARY PROPERTY OF APPLE INC.<br/>THE POSSESSOR AGREES TO THE FOLLOWING:<br/>I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE<br/>II NOT TO REPRODUCE OR COPY IT<br/>III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART<br/>IV ALL RIGHTS RESERVED</div><div><div>BRANCH</div><div>evt-mars-0</div></div><div><div>PAGE</div><div>122 OF 150</div></div><div><div>SHEET</div><div>98 OF 108</div></div></div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |  |  |  |  |  |  |



|   |   |   |   |                           |                |                |   |   |  |
|---|---|---|---|---------------------------|----------------|----------------|---|---|--|
|   | 8 | 7 | 6 | 5                         | 4              | 3              | 2 | 1 |  |
| D |   |   |   | Memory Bit & Byte Swizzle |                |                |   |   |  |
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| A | 8 | 7 | 6 | 5                         | 4              | 3              | 2 | 1 |  |

|                                                                                                                                                                                                                                                                                                                             |  |  |                      |  |            |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------|--|------------|--|
| SYNC_MASTER=AHARTMAN_J52                                                                                                                                                                                                                                                                                                    |  |  | SYNC_DATE=10/29/2013 |  |            |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |  |                      |  |            |  |
| LPDDR3 Bit & Byte Swizzle                                                                                                                                                                                                                                                                                                   |  |  |                      |  |            |  |
|  Apple Inc.                                                                                                                                                                                                                            |  |  | DRAWING NUMBER       |  | SIZE       |  |
|                                                                                                                                                                                                                                                                                                                             |  |  | 051-02166            |  | D          |  |
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|                                                                                                                                                                                                                                                                                                                             |  |  | BRANCH               |  | evt-mars-0 |  |
|                                                                                                                                                                                                                                                                                                                             |  |  | PAGE                 |  | 123 OF 150 |  |
|                                                                                                                                                                                                                                                                                                                             |  |  | SHEET                |  | 99 OF 108  |  |



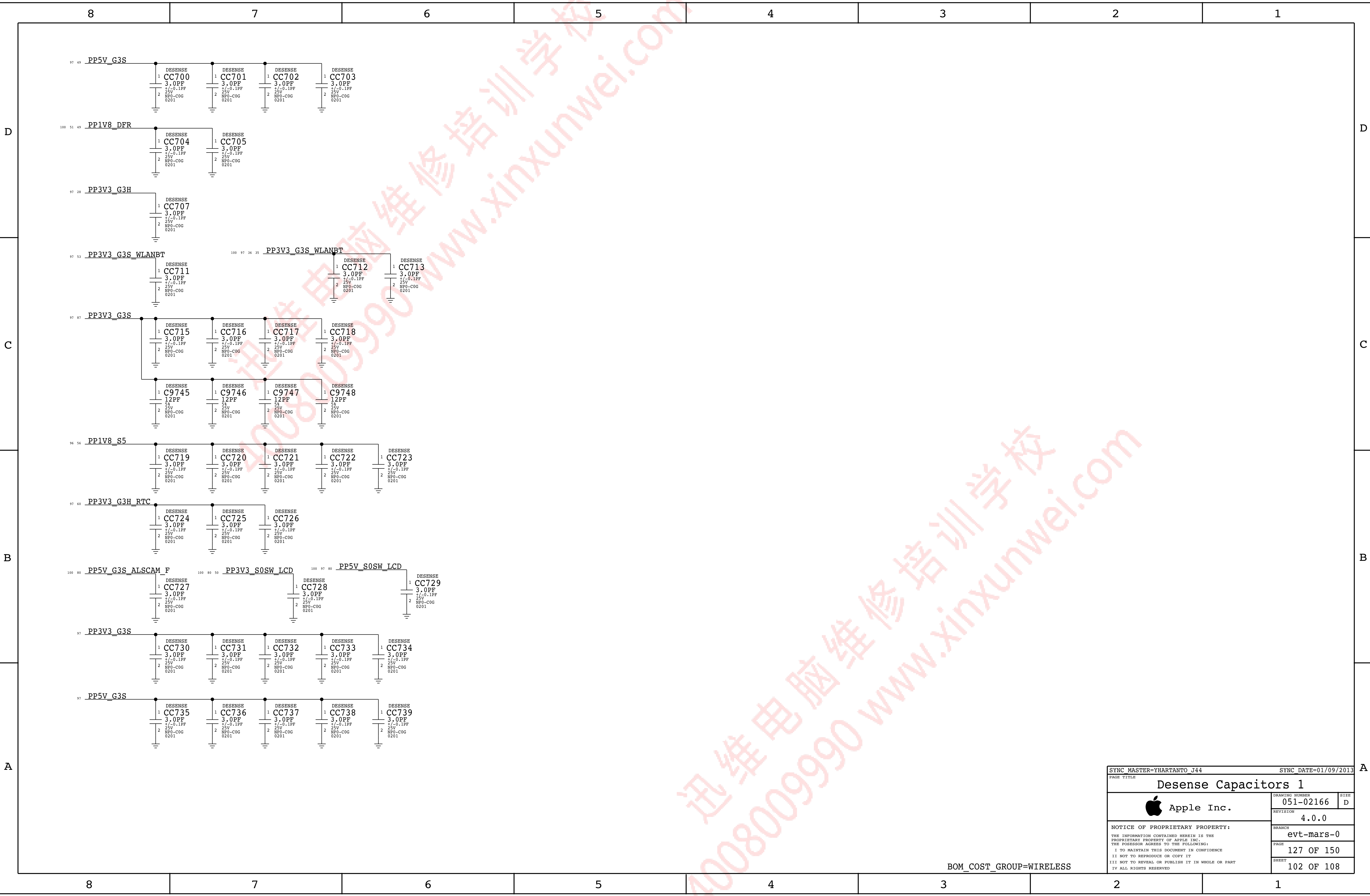
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| Functional Test Points          |   | J4900 - Mesa Connector | J8500 - eDP Connector | Probe Block Grid - DFU/SOC |   | DFU/SOC/FCT DMM/Power Sequence |   |
| J5110 - DFR Display Connector   |   |                        |                       |                            |   |                                |   |
| J5100 - DFR Touch Connector     |   |                        |                       |                            |   |                                |   |
| J6700 - Keyboard Connector      |   |                        |                       |                            |   |                                |   |
| J6701 - Trackpad Connector      |   |                        |                       |                            |   |                                |   |
| J6600 - Audio Jack Connector    |   |                        |                       |                            |   |                                |   |
| J6950 - Battery Connector       |   |                        |                       |                            |   |                                |   |
| J6951 - Battery Sense Connector |   |                        |                       |                            |   |                                |   |
|                                 |   |                        |                       |                            |   |                                |   |
|                                 |   |                        |                       |                            |   |                                |   |
|                                 |   |                        |                       |                            |   |                                |   |
|                                 |   |                        |                       |                            |   |                                |   |
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|                                 |   |                        |                       |                            |   |                                |   |
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|                                 |   |                        |                       |                            |   |                                |   |
|                                 |   |                        |                       |                            |   |                                |   |
|                                 |   |                        |                       |                            |   |                                |   |
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| SYNC MASTER-J132 CARLOS                                                                                                                                                                                                                                                                                             |  |  | SYNC DATE=06/15/2017 |            |      |
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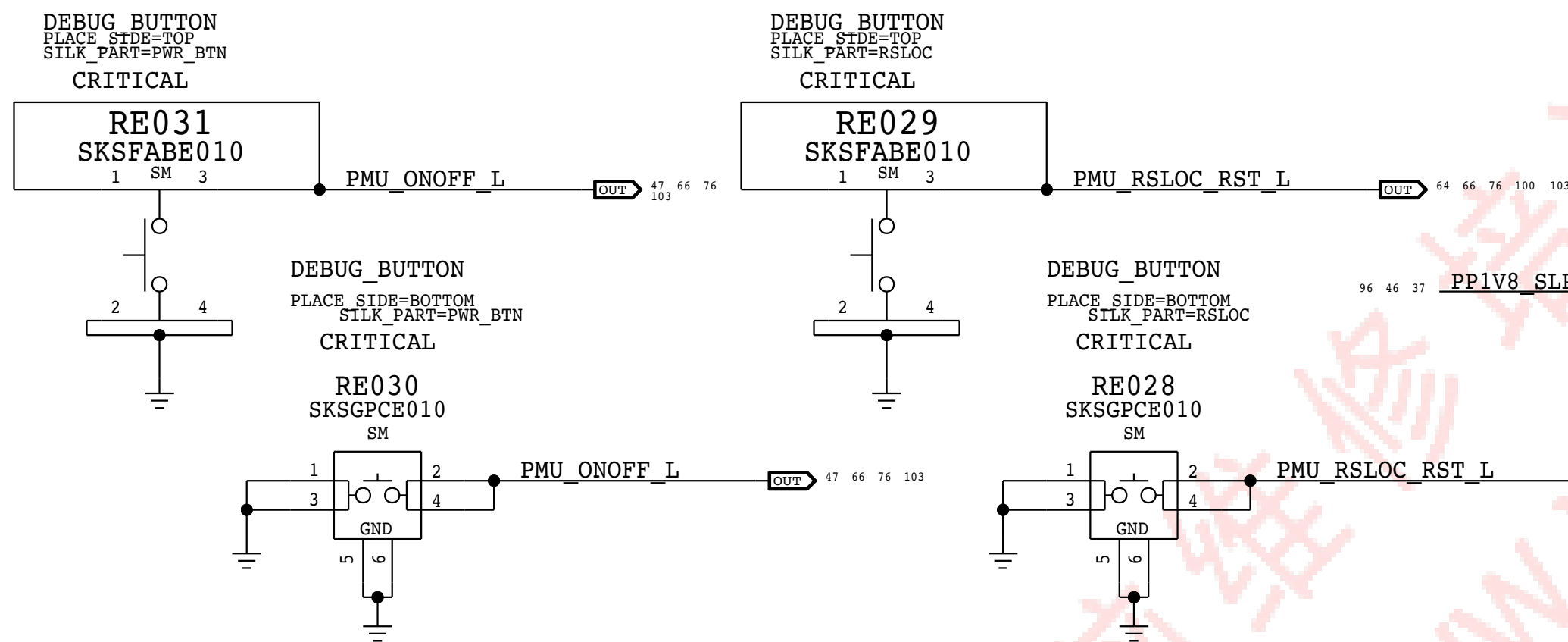


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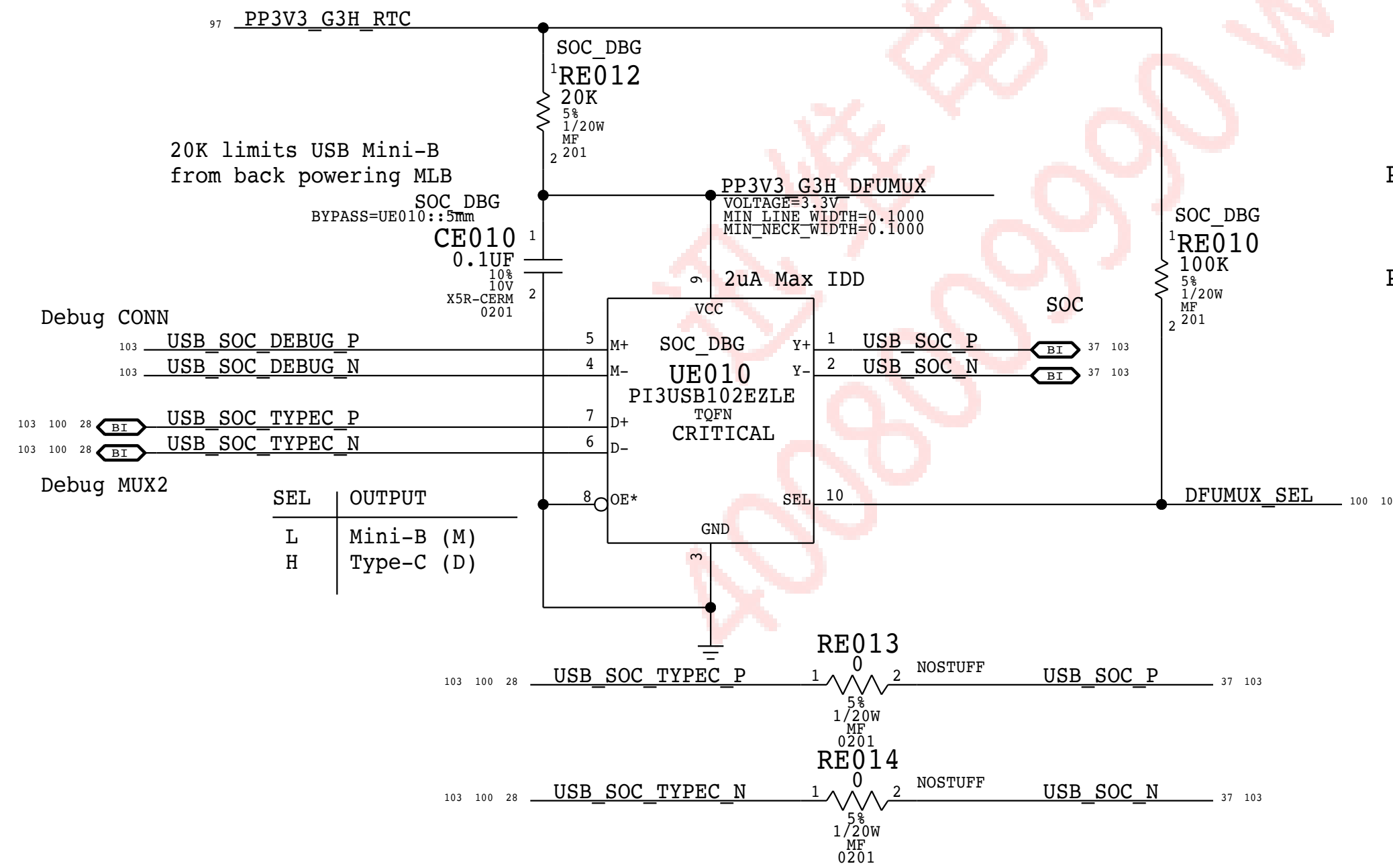
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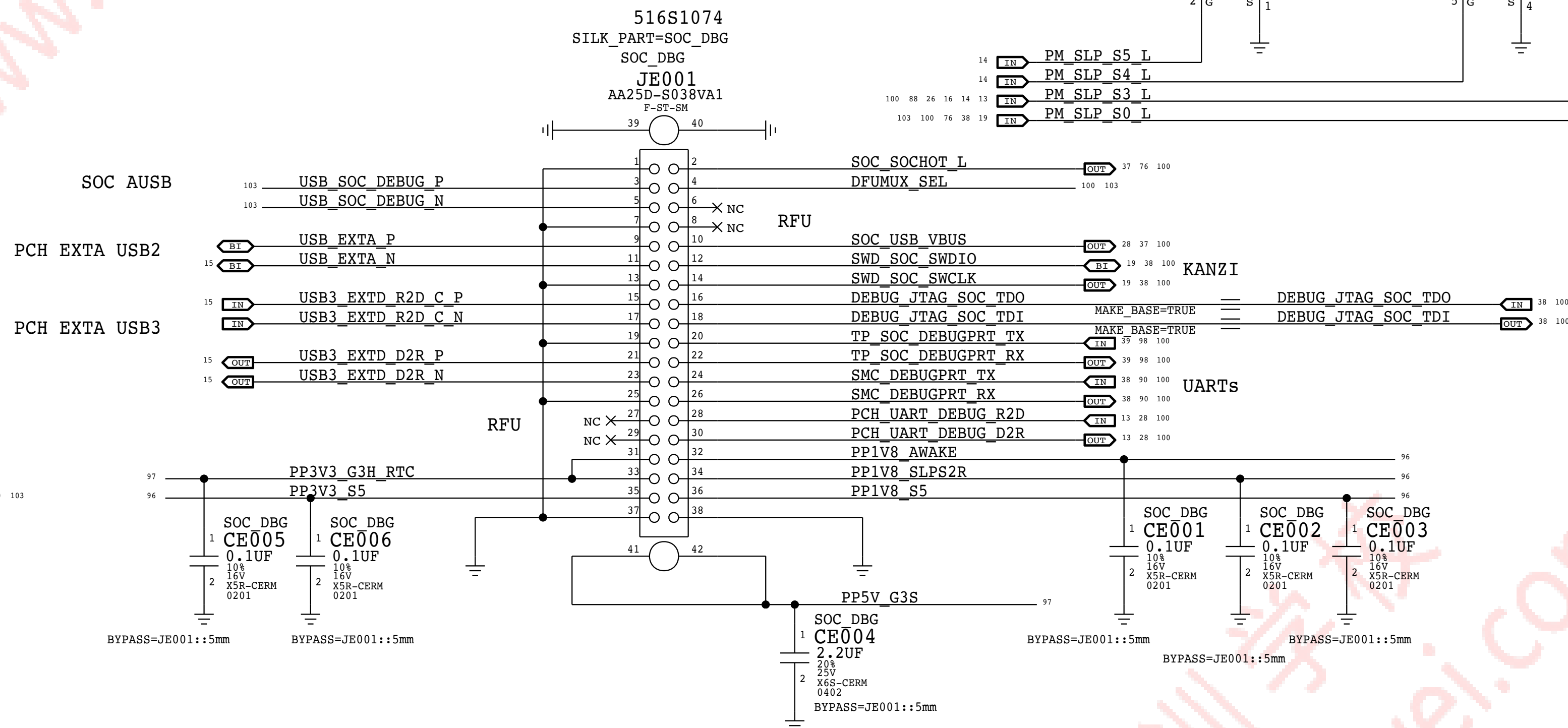
## Debug Power "Buttons"



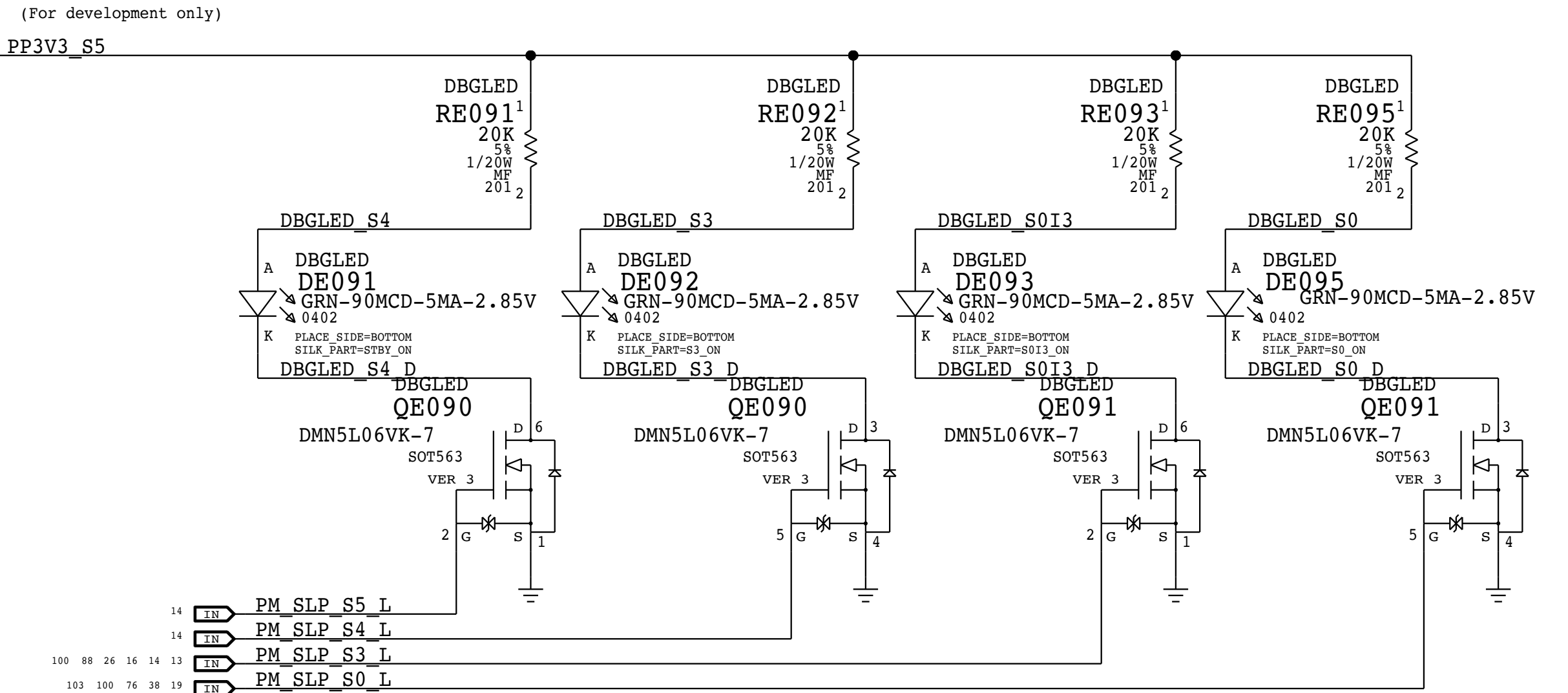
## SoC USB DFU Mux



## SOC DEBUG Connector

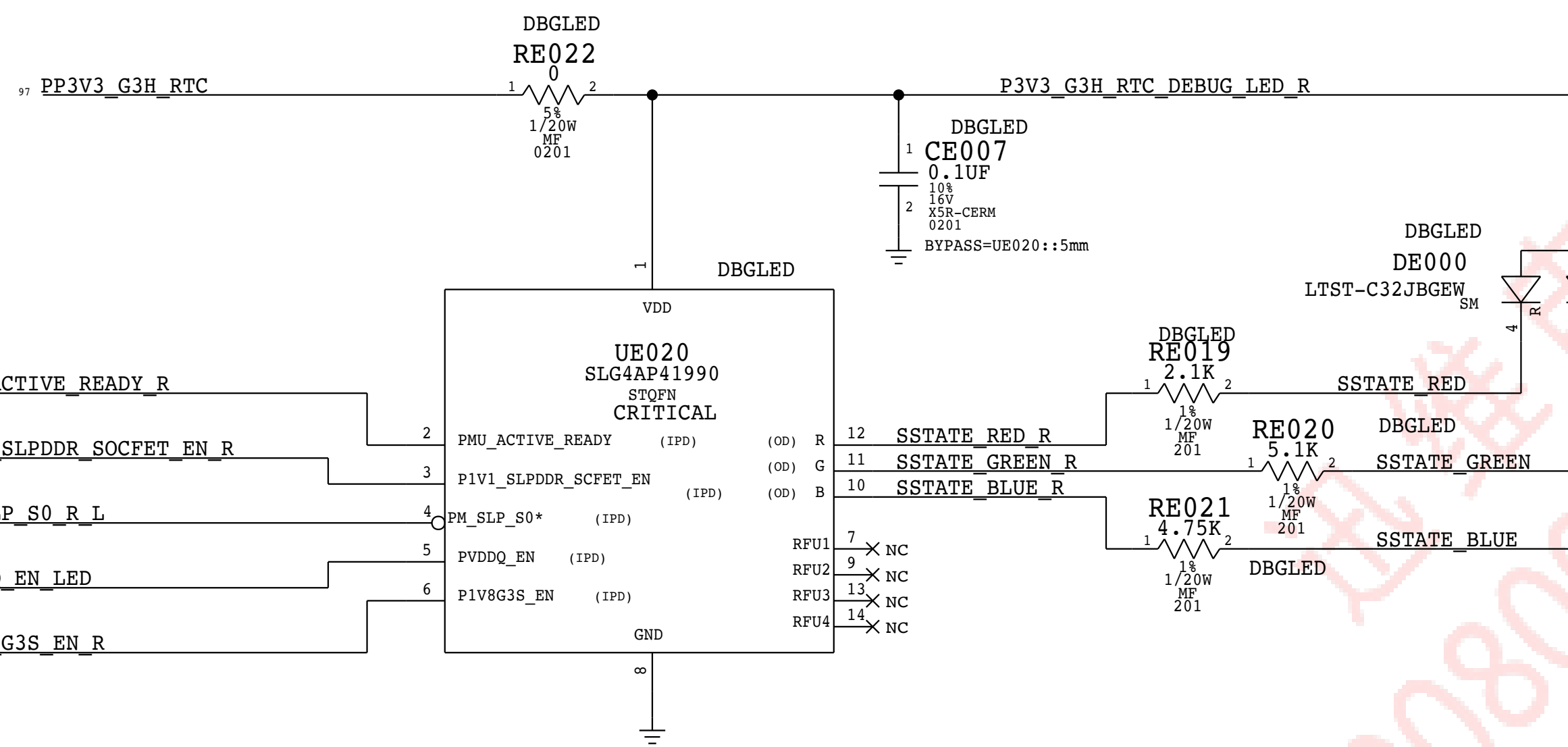


## Power State Debug LEDs

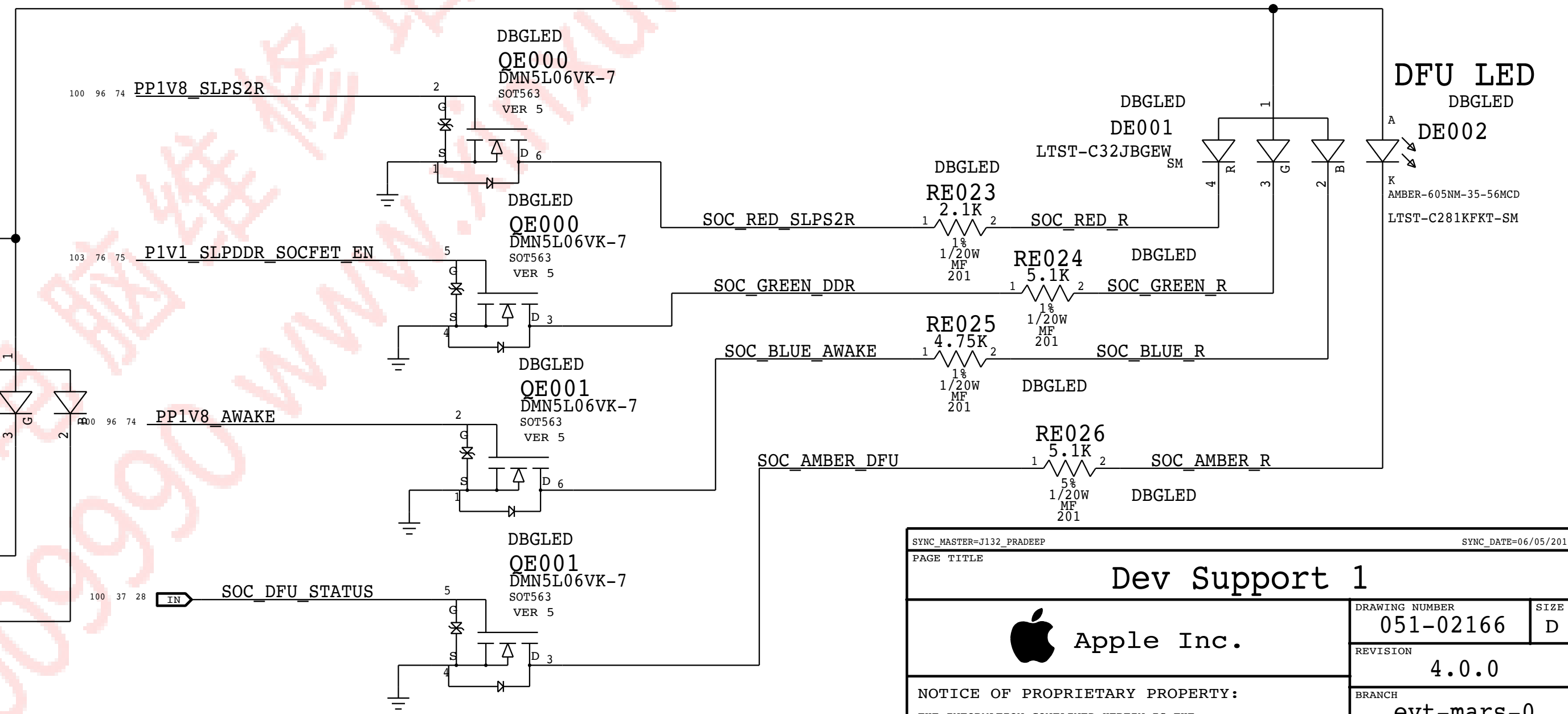


## System State LEDs

See color table on next page



## SOC State LEDs



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| D                                                                                                                                                                                                                                                                                                                    | <div>UE020 SAK Truth Table:</div> <table><tr><th colspan="5">INPUTS</th><th colspan="4">OUTPUTS (OD)</th></tr><tr><th>PMU_ACT_RDY</th><th>SLP_SCFET_EN</th><th>SLP_S0_L</th><th>VDDQ_EN</th><th>1V8G3S_EN</th><th>R</th><th>G</th><th>B</th><th>COLOR</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>BLINK</td><td>1</td><td>1</td><td>Blinking Red</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>Red</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>White</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Magenta</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Blinking Green &amp; 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                                                                                                                                                                |            |         |           |              |   | <div><div>SYNC MASTER=J79 RIO</div><div>SYNC DATE=06/18/2015</div></div> <table><tr><td colspan="3">PAGE TITLE</td></tr><tr><td colspan="3">Dev Support 2</td></tr><tr><td rowspan="2"> Apple Inc.</td><td>DRAWING NUMBER</td><td>051-02166</td></tr><tr><td>REVISION</td><td>4.0.0</td></tr><tr><td rowspan="4">NOTICE OF PROPRIETARY PROPERTY:<br/>THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.<br/>THE POSSESSOR AGREES TO THE FOLLOWING:<br/>I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE<br/>II NOT TO REPRODUCE OR COPY IT<br/>III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART<br/>IV ALL RIGHTS RESERVED</td><td>BRANCH</td><td>evt-mars-0</td></tr><tr><td>PAGE</td><td>141 OF 150</td></tr><tr><td>SHEET</td><td>104 OF 108</td></tr><tr><td colspan="2"></td></tr></table> | PAGE TITLE              |        |  | Dev Support 2 |  |  |  Apple Inc. | DRAWING NUMBER | 051-02166 | REVISION | 4.0.0       | NOTICE OF PROPRIETARY PROPERTY:<br>THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.<br>THE POSSESSOR AGREES TO THE FOLLOWING:<br>I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE<br>II NOT TO REPRODUCE OR COPY IT<br>III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART<br>IV ALL RIGHTS RESERVED | BRANCH   | evt-mars-0 | PAGE      | 141 OF 150 | SHEET | 104 OF 108 |       |   |   |   |   |   |       |   |   |              |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |         |   |   |   |   |  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| NOTICE OF PROPRIETARY PROPERTY:<br>THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.<br>THE POSSESSOR AGREES TO THE FOLLOWING:<br>I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE<br>II NOT TO REPRODUCE OR COPY IT<br>III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART<br>IV ALL RIGHTS RESERVED | BRANCH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| D                                                                                                                                                                                                                                                                                                                 | BOM Variants                    |                                                                    |                                                                         |   |   |   |   |                | D          |
|                                                                                                                                                                                                                                                                                                                   | BOM NUMBER                      | BOM NAME                                                           | BOM OPTIONS                                                             |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 685-00187                       | COMMON PARTS,MLB,X941                                              | X941_COMMON                                                             |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 985-00366                       | DEV,MLB,X941                                                       | X941_DEVEL:ENG                                                          |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 939-05022                       | PCBA,MLB,NO CPU,SS-8G,SD-256G,X941                                 | BASE_BOM,DEVEL_BOM,ALTERNATE,DRAMCFG:SS_8G,NANDCFG:SD_256G              |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 939-05023                       | PCBA,MLB,CPU SOCKET,SS-8G,SD-256G,X941                             | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:SOCKET,DRAMCFG:SS_8G,NANDCFG:SD_256G   |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 939-05373                       | PCBA,MLB,2.3G,HY-8G,SS-2T-DOE,X941                                 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:SS_2T_DOE   |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 939-05374                       | PCBA,MLB,2.3G,MI-8G,SD-256G-DOE,X941                               | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:SD_256G_DOE |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 939-05375                       | PCBA,MLB,2.3G,SS-8G,TO-256G-DOE,X941                               | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:TO_256G_DOE |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 939-05698                       | PCBA,MLB,NO CPU,RF DOE,X941                                        | SCH,PCB,RF_DOE                                                          |   |   |   |   |                |            |
| C                                                                                                                                                                                                                                                                                                                 | BOM NUMBER                      | BOM NAME                                                           | BOM OPTIONS                                                             |   |   |   |   |                | C          |
|                                                                                                                                                                                                                                                                                                                   | 639-05125                       | PCBA,MLB,2.3G,HY-8G,SD-256G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:SD_256G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05126                       | PCBA,MLB,2.3G,HY-8G,TO-256G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:TO_256G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05127                       | PCBA,MLB,2.3G,MI-8G,SD-256G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:SD_256G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05128                       | PCBA,MLB,2.3G,MI-8G,TO-256G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:TO_256G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05129                       | PCBA,MLB,2.3G,SS-8G,SD-256G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:SD_256G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05130                       | PCBA,MLB,2.3G,SS-8G,TO-256G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:TO_256G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05131                       | PCBA,MLB,2.3G,HY-16G,SD-256G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_16G,NANDCFG:SD_256G    |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05132                       | PCBA,MLB,2.3G,HY-16G,TO-256G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_16G,NANDCFG:TO_256G    |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05133                       | PCBA,MLB,2.3G,MI-16G,SD-256G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_16G,NANDCFG:SD_256G    |   |   |   |   |                |            |
| B                                                                                                                                                                                                                                                                                                                 | 639-05134                       | PCBA,MLB,2.3G,MI-16G,TO-256G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_16G,NANDCFG:TO_256G    |   |   |   |   |                | B          |
|                                                                                                                                                                                                                                                                                                                   | 639-05135                       | PCBA,MLB,2.3G,SS-16G,SD-256G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:SD_256G    |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05136                       | PCBA,MLB,2.3G,SS-16G,TO-256G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:TO_256G    |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05193                       | PCBA,MLB,2.3G,HY-8G,SD-512G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:SD_512G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05194                       | PCBA,MLB,2.3G,HY-8G,TO-512G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:TO_512G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05195                       | PCBA,MLB,2.3G,MI-8G,SD-512G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:SD_512G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05196                       | PCBA,MLB,2.3G,MI-8G,TO-512G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:TO_512G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05197                       | PCBA,MLB,2.3G,SS-8G,SD-512G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:SD_512G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05198                       | PCBA,MLB,2.3G,SS-8G,TO-512G,X941                                   | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:TO_512G     |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05199                       | PCBA,MLB,2.3G,HY-16G,SD-512G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_16G,NANDCFG:SD_512G    |   |   |   |   |                |            |
| A                                                                                                                                                                                                                                                                                                                 | 639-05200                       | PCBA,MLB,2.3G,HY-16G,TO-512G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_16G,NANDCFG:TO_512G    |   |   |   |   |                | A          |
|                                                                                                                                                                                                                                                                                                                   | 639-05201                       | PCBA,MLB,2.3G,MI-16G,SD-512G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_16G,NANDCFG:SD_512G    |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05202                       | PCBA,MLB,2.3G,MI-16G,TO-512G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_16G,NANDCFG:TO_512G    |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05203                       | PCBA,MLB,2.3G,SS-16G,SD-512G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:SD_512G    |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05204                       | PCBA,MLB,2.3G,SS-16G,TO-512G,X941                                  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:TO_512G    |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05205                       | PCBA,MLB,2.3G,HY-8G,SD-1T,X941                                     | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:SD_1T       |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05206                       | PCBA,MLB,2.3G,HY-8G,SS-1T,X941                                     | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:SS_1T       |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05207                       | PCBA,MLB,2.3G,HY-8G,TO-1T,X941                                     | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:TO_1T       |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05208                       | PCBA,MLB,2.3G,MI-8G,SD-1T,X941                                     | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:SD_1T       |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 639-05209                       | PCBA,MLB,2.3G,MI-8G,SS-1T,X941                                     | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:SS_1T       |   |   |   |   |                |            |
| 639-05210                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,MI-8G,TO-1T,X941  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:TO_1T  |                                                                         |   |   |   |   |                |            |
| 639-05211                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-8G,SD-1T,X941  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:SD_1T  |                                                                         |   |   |   |   |                |            |
| 639-05212                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-8G,SS-1T,X941  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:SS_1T  |                                                                         |   |   |   |   |                |            |
| 639-05213                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-8G,TO-1T,X941  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:TO_1T  |                                                                         |   |   |   |   |                |            |
| 639-05214                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,HY-16G,SD-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_16G,NANDCFG:SD_1T |                                                                         |   |   |   |   |                |            |
| 639-05215                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,HY-16G,SS-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_16G,NANDCFG:SS_1T |                                                                         |   |   |   |   |                |            |
| 639-05216                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,HY-16G,TO-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_16G,NANDCFG:TO_1T |                                                                         |   |   |   |   |                |            |
| 639-05217                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,MI-16G,SD-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_16G,NANDCFG:SD_1T |                                                                         |   |   |   |   |                |            |
| 639-05218                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,MI-16G,SS-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_16G,NANDCFG:SS_1T |                                                                         |   |   |   |   |                |            |
| 639-05219                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,MI-16G,TO-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_16G,NANDCFG:TO_1T |                                                                         |   |   |   |   |                |            |
| 639-05220                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-16G,SD-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:SD_1T |                                                                         |   |   |   |   |                |            |
| 639-05221                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-16G,SS-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:SS_1T |                                                                         |   |   |   |   |                |            |
| 639-05222                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-16G,TO-1T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:TO_1T |                                                                         |   |   |   |   |                |            |
| 639-05223                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,HY-8G,SS-2T,X941  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_8G,NANDCFG:SS_2T  |                                                                         |   |   |   |   |                |            |
| 639-05224                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,MI-8G,SS-2T,X941  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_8G,NANDCFG:SS_2T  |                                                                         |   |   |   |   |                |            |
| 639-05225                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-8G,SS-2T,X941  | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_8G,NANDCFG:SS_2T  |                                                                         |   |   |   |   |                |            |
| 639-05226                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,HY-16G,SS-2T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:HY_16G,NANDCFG:SS_2T |                                                                         |   |   |   |   |                |            |
| 639-05227                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,MI-16G,SS-2T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:MI_16G,NANDCFG:SS_2T |                                                                         |   |   |   |   |                |            |
| 639-05228                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-16G,SS-2T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:SS_2T |                                                                         |   |   |   |   |                |            |
| 639-05229                                                                                                                                                                                                                                                                                                         | PCBA,MLB,2.3G,SS-16G,TO-2T,X941 | BASE_BOM,DEVEL_BOM,ALTERNATE,CPU:2.3G,DRAMCFG:SS_16G,NANDCFG:TO_2T |                                                                         |   |   |   |   |                |            |
| BOM Variants 1                                                                                                                                                                                                                                                                                                    |                                 |                                                                    |                                                                         |   |   |   |   | PAGE TITLE     |            |
|  Apple Inc.                                                                                                                                                                                                                  |                                 |                                                                    |                                                                         |   |   |   |   | DRAWING NUMBER | 051-02166  |
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|                                                                                                                                                                                                                                                                                                                   |                                 |                                                                    |                                                                         |   |   |   |   |                |            |
|                                                                                                                                                                                                                                                                                                                   | 8                               | 7                                                                  | 6                                                                       | 5 | 4 | 3 | 2 | 1              |            |







| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 7                         |            | 6                       |                     | 5 |  | 4 |  | 3 |  | 2 |  | 1 |             |                           |                           |            |           |           |           |          |     |                    |           |           |           |     |                       |                     |           |           |     |                       |                     |           |           |            |                         |              |           |           |            |                   |                 |           |           |            |                   |                 |           |           |            |                         |               |           |           |     |                         |               |           |           |     |               |               |           |           |            |     |                 |           |          |     |     |                 |           |           |     |                |           |           |           |     |     |           |           |           |     |                       |           |           |          |   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          |           |     |     |           |           |           |     |     |           |           |          |     |     |                        |           |          |     |     |           |           |          |     |     |          |           |           |     |     |           |          |           |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>103S0321</td><td>103S0276</td><td></td><td>ALL</td><td></td></tr><tr><td>103S00248</td><td>103S00247</td><td></td><td>ALL</td><td>8.06K 0.1% RESISTOR</td></tr><tr><td>103S00250</td><td>103S00249</td><td></td><td>ALL</td><td>18.2K 0.1% RESISTOR</td></tr><tr><td>107S0284</td><td>107S00021</td><td>SENSOR:DEV</td><td>ALL</td><td>TFT w/ Yageo</td></tr><tr><td>107S00101</td><td>107S00005</td><td>SENSOR:DEV</td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00102</td><td>107S00017</td><td>SENSOR:DEV</td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S0276</td><td>107S00020</td><td>SENSOR:DEV</td><td>ALL</td><td>Cyntec w/ TFT</td></tr><tr><td>107S00086</td><td>107S00056</td><td>ANY</td><td>ALL</td><td>TFT w/ Cyntec</td></tr><tr><td>107S00100</td><td>107S00057</td><td></td><td>ALL</td><td>Cyntec w/ TFT</td></tr><tr><td>107S00103</td><td>107S00058</td><td>SENSOR:DEV</td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00070</td><td>107S0085</td><td></td><td>ALL</td><td>100K THERMISTOR</td></tr><tr><td>132S00202</td><td>132S00175</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00084</td><td>138S00060</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00117</td><td>138S00071</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0714</td><td>138S0713</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>138S0715</td><td>138S0732</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>138S0706</td><td>138S0739</td><td></td><td>ALL</td><td>NEC w/ Vishay</td></tr><tr><td>138S0945</td><td>138S0739</td><td></td><td>ALL</td><td>NEC w/ Rohm</td></tr><tr><td>138S00049</td><td>138S0831</td><td>ANY</td><td>ALL</td><td></td></tr><tr><td>138S00006</td><td>138S0835</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00086</td><td>138S0884</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00104</td><td>138S0978</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>138S00056</td><td>138S1100</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00369</td><td>152S00268</td><td></td><td>ALL</td><td>Cyntec w/ NEC</td></tr><tr><td>152S00403</td><td>152S00322</td><td></td><td>ALL</td><td>Murata w/ Chillisin</td></tr><tr><td>152S00864</td><td>152S00851</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00434</td><td>152S1829</td><td></td><td>ALL</td><td></td></tr><tr><td>155S0665</td><td>155S00232</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00166</td><td>155S0391</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00018</td><td>155S0664</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>155S00007</td><td>155S0667</td><td></td><td>ALL</td><td></td></tr><tr><td>155S0914</td><td>155S0897</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00190</td><td>155S0897</td><td></td><td>ALL</td><td></td></tr><tr><td>197S00046</td><td>197S00036</td><td></td><td>ALL</td><td>Epson w/ TXC</td></tr><tr><td>197S00047</td><td>197S00036</td><td></td><td>ALL</td><td>Kyocera w/ TXC</td></tr><tr><td>197S00048</td><td>197S00036</td><td></td><td>ALL</td><td>Murata w/ TXC</td></tr><tr><td>197S00053</td><td>197S00050</td><td></td><td>ALL</td><td>Kyocera w/ TXC</td></tr><tr><td>197S00054</td><td>197S00050</td><td></td><td>ALL</td><td>NDK w/ TXC</td></tr><tr><td>197S00055</td><td>197S00050</td><td></td><td>ALL</td><td>Murata w/ TXC</td></tr><tr><td>197S00120</td><td>197S00118</td><td></td><td>ALL</td><td></td></tr><tr><td>311S0426</td><td>311S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00104</td><td>311S00091</td><td>ANY</td><td>ALL</td><td></td></tr><tr><td>311S00156</td><td>311S00129</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00196</td><td>311S00195</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00060</td><td>311S0273</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00121</td><td>311S0398</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00138</td><td>311S0436</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00013</td><td>311S0508</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>335S00270</td><td>335S00203</td><td></td><td>ALL</td><td></td></tr><tr><td>335S00213</td><td>335S0888</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00878</td><td>353S00599</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00879</td><td>353S00754</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00750</td><td>353S00877</td><td></td><td>ALL</td><td></td></tr><tr><td>353S01041</td><td>353S01042</td><td></td><td>ALL</td><td></td></tr><tr><td>353S01346</td><td>353S01320</td><td></td><td>ALL</td><td></td></tr><tr><td>353S01506</td><td>353S01404</td><td>ANY</td><td>ALL</td><td></td></tr><tr><td>353S00107</td><td>353S3239</td><td>ANY</td><td>ALL</td><td>Onsemi alt to Intersil</td></tr><tr><td>353S00636</td><td>353S4037</td><td>ANY</td><td>ALL</td><td></td></tr><tr><td>353S00772</td><td>353S4068</td><td></td><td>ALL</td><td></td></tr><tr><td>371S00089</td><td>371S00085</td><td></td><td>ALL</td><td></td></tr></table> |                           |            |                         |                     |   |  |   |  |   |  |   |  |   |             | PART NUMBER               | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES   | COMMENTS: | 103S0321  | 103S0276 |     | ALL                |           | 103S00248 | 103S00247 |     | ALL                   | 8.06K 0.1% RESISTOR | 103S00250 | 103S00249 |     | ALL                   | 18.2K 0.1% RESISTOR | 107S0284  | 107S00021 | SENSOR:DEV | ALL                     | TFT w/ Yageo | 107S00101 | 107S00005 | SENSOR:DEV | ALL               | Cyntec w/ Yageo | 107S00102 | 107S00017 | SENSOR:DEV | ALL               | Cyntec w/ Yageo | 107S0276  | 107S00020 | SENSOR:DEV | ALL                     | Cyntec w/ TFT | 107S00086 | 107S00056 | ANY | ALL                     | TFT w/ Cyntec | 107S00100 | 107S00057 |     | ALL           | Cyntec w/ TFT | 107S00103 | 107S00058 | SENSOR:DEV | ALL | Cyntec w/ Yageo | 107S00070 | 107S0085 |     | ALL | 100K THERMISTOR | 132S00202 | 132S00175 |     | ALL            |           | 138S00084 | 138S00060 |     | ALL |           | 138S00117 | 138S00071 |     | ALL                   |           | 138S0714  | 138S0713 |     | ALL | Murata w/ Samsung | 138S0715 | 138S0732 |     | ALL                | Murata w/ Samsung | 138S0706 | 138S0739 |     | ALL | NEC w/ Vishay | 138S0945 | 138S0739 |     | ALL | NEC w/ Rohm | 138S00049 | 138S0831 | ANY | ALL |           | 138S00006 | 138S0835 |     | ALL |           | 138S00086 | 138S0884 |     | ALL |           | 138S00104 | 138S0978 |     | ALL | Murata w/ Taiyo | 138S00056 | 138S1100 |     | ALL |           | 152S00369 | 152S00268 |     | ALL | Cyntec w/ NEC | 152S00403 | 152S00322 |     | ALL | Murata w/ Chillisin | 152S00864 | 152S00851 |     | ALL               |           | 152S00434 | 152S1829 |     | ALL |          | 155S0665 | 155S00232 |     | ALL |          | 155S00166 | 155S0391 |     | ALL |          | 155S00018 | 155S0664 |     | ALL          | Murata w/ Taiyo | 155S00007 | 155S0667 |     | ALL              |           | 155S0914  | 155S0897 |     | ALL |           | 155S00190 | 155S0897 |     | ALL |           | 197S00046 | 197S00036 |     | ALL | Epson w/ TXC | 197S00047 | 197S00036 |     | ALL | Kyocera w/ TXC | 197S00048 | 197S00036 |     | ALL | Murata w/ TXC | 197S00053 | 197S00050 |     | ALL | Kyocera w/ TXC | 197S00054 | 197S00050 |     | ALL | NDK w/ TXC | 197S00055 | 197S00050 |     | ALL | Murata w/ TXC | 197S00120 | 197S00118 |     | ALL |           | 311S0426  | 311S00007 |     | ALL |           | 311S00104 | 311S00091 | ANY | ALL |           | 311S00156 | 311S00129 |     | ALL |           | 311S00196 | 311S00195 |     | ALL |          | 311S00060 | 311S0273 |     | ALL           |           | 311S00121 | 311S0398 |     | ALL |           | 311S00138 | 311S0436 |     | ALL |           | 311S00013 | 311S0508 |     | ALL | NXP w/ Diodes | 335S00270 | 335S00203 |     | ALL |           | 335S00213 | 335S0888 |     | ALL |           | 353S00878 | 353S00599 |     | ALL |           | 353S00879 | 353S00754 |     | ALL |           | 353S00750 | 353S00877 |     | ALL |           | 353S01041 | 353S01042 |     | ALL |           | 353S01346 | 353S01320 |     | ALL |           | 353S01506 | 353S01404 | ANY | ALL |           | 353S00107 | 353S3239 | ANY | ALL | Onsemi alt to Intersil | 353S00636 | 353S4037 | ANY | ALL |           | 353S00772 | 353S4068 |     | ALL |          | 371S00089 | 371S00085 |     | ALL |           | C        | System EE |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
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        |           |           |           |          |     |                    |           |           |           |     |                       |                     |           |           |     |                       |                     |           |           |            |                         |              |           |           |            |                   |                 |           |           |            |                   |                 |           |           |            |                         |               |           |           |     |                         |               |           |           |     |               |               |           |           |            |     |                 |           |          |     |     |                 |           |           |     |                |           |           |           |     |     |           |           |           |     |                       |           |           |          |   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          |           |     |     |           |           |           |     |     |           |           |          |     |     |                        |           |          |     |     |           |           |          |     |     |          |           |           |     |     |           |          |           |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| 152S00369                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00403                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00864                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00434                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 155S0665                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 155S00166                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 155S00018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 197S00053                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 197S00055                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 197S00120                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 311S0426                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 311S00104                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 311S00156                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 311S00196                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 353S00772                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 371S00089                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>371S00170</td><td>371S00125</td><td></td><td>ALL</td><td></td></tr><tr><td>371S00127</td><td>371S00182</td><td></td><td>ALL</td><td></td></tr><tr><td>372S00016</td><td>372S00015</td><td></td><td>ALL</td><td></td></tr><tr><td>372S0186</td><td>372S0185</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00219</td><td>376S00079</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00292</td><td>376S00294</td><td></td><td>ALL</td><td></td></tr><tr><td>376S1080</td><td>376S0820</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00074</td><td>376S0855</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00146</td><td>376S1061</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>376S00282</td><td>376S1128</td><td></td><td>ALL</td><td></td></tr><tr><td>377S00079</td><td>377S00077</td><td></td><td>ALL</td><td></td></tr><tr><td>377S0077</td><td>377S0183</td><td></td><td>ALL</td><td>Infineon w/ ST</td></tr><tr><td>740S00033</td><td>740S00002</td><td></td><td>ALL</td><td></td></tr><tr><td>740S00019</td><td>740S00007</td><td></td><td>ALL</td><td>Bourns w/ Polytronics</td></tr><tr><td>740S00003</td><td>740S0135</td><td></td><td>ALL</td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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              |                           |            |                         |                     |   |  |   |  |   |  |   |  |   | PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION                | REF DES    | COMMENTS: | 371S00170 | 371S00125 |          | ALL |                    | 371S00127 | 371S00182 |           | ALL |                       | 372S00016           | 372S00015 |           | ALL |                       | 372S0186            | 372S0185  |           | ALL        |                         | 376S00219    | 376S00079 |           | ALL        |                   | 376S00292       | 376S00294 |           | ALL        |                   | 376S1080        | 376S0820  |           | ALL        |                         | 376S00074     | 376S0855  |           | ALL |                         | 376S00146     | 376S1061  |           | ALL | NXP w/ Diodes | 376S00282     | 376S1128  |           | ALL        |     | 377S00079       | 377S00077 |          | ALL |     | 377S0077        | 377S0183  |           | ALL | Infineon w/ ST | 740S00033 | 740S00002 |           | ALL |     | 740S00019 | 740S00007 |           | ALL | Bourns w/ Polytronics | 740S00003 | 740S0135  |          | ALL |     | B                 | Audio    |          |     |                    |                   |          |          |     |     |               |          |          |     |     |             |           |          |     |     |           |           |          |     |     |           |           |          |     |     |           |           |          |     |     |                 |           |          |     |     |           |           |           |     |     |               |           |           |     |     |                     |           |           |     |                   |           |           |          |     |     |          |          |           |     |     |          |           |          |     |     |          |           |          |     |         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| PART NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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          |           |     |     |           |           |           |     |     |           |           |          |     |     |                        |           |          |     |     |           |           |          |     |     |          |           |           |     |     |           |          |           |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 371S00170                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 371S00127                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 372S00016                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 372S0186                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 376S00219                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00292                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S1080                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 376S00074                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00146                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00282                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 377S00079                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 377S0077                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 740S00033                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 740S00019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 740S00003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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          |           |     |     |           |           |           |     |     |           |           |          |     |     |                        |           |          |     |     |           |           |          |     |     |          |           |           |     |     |           |          |           |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>138S0700</td><td>138S0641</td><td></td><td>ALL</td><td>Murata w/ SS&amp;Taiyo</td></tr><tr><td>138S0660</td><td>138S0684</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0689</td><td>138S0701</td><td></td><td>ALL</td><td></td></tr><tr><td>138S1103</td><td>138S0719</td><td></td><td>ALL</td><td></td></tr><tr><td>155S0694</td><td>155S0387</td><td></td><td>ALL</td><td></td></tr><tr><td>155S0660</td><td>155S0513</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00034</td><td>155S0706</td><td></td><td>ALL</td><td>Taiyo w/ Murata</td></tr><tr><td>155S00203</td><td>155S0894</td><td></td><td>ALL</td><td></td></tr><tr><td>311S0271</td><td>311S00008</td><td></td><td>ALL</td><td></td></tr></table>                                                                                                                                                                           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REF DES    | COMMENTS: | 138S0700  | 138S0641  |          | ALL | Murata w/ SS&Taiyo | 138S0660  | 138S0684  |           | ALL |                       | 138S0689            | 138S0701  |           | ALL |                       | 138S1103            | 138S0719  |           | ALL        |                         | 155S0694     | 155S0387  |           | ALL        |                   | 155S0660        | 155S0513  |           | ALL        |                   | 155S00034       | 155S0706  |           | ALL        | Taiyo w/ Murata         | 155S00203     | 155S0894  |           | ALL |                         | 311S0271      | 311S00008 |           | ALL |               | A             | BLC       |           |            |     |                 |           |          |     |     |                 |           |           |     |                |           |           |           |     |     |           |           |           |     |                       |           |           |          |     |     |                   |          |          |     |                    |                   |          |          |     |     |               |          |          |     |     |             |           |          |     |     |           |           |          |     |     |           |           |          |     |     |           |           |          |     |     |                 |           |          |     |     |           |           |           |     |     |               |           |           |     |     |                     |           |           |     |                   |           |           |          |     |     |          |          |           |     |     |          |           |          |     |     |          |           |          |     |              |                 |           |          |     |                  |           |           |          |     |     |           |           |          |     |     |           |           |           |     |    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| 155S0660                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 155S00034                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>107S00033</td><td>107S00034</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0738</td><td>138S1101</td><td></td><td>ALL</td><td>Samsung alt to Murata</td></tr><tr><td>138S0846</td><td>138S0811</td><td></td><td>ALL</td><td>Samsung alt to Murata</td></tr><tr><td>152S00359</td><td>152S00253</td><td></td><td>ALL</td><td>Chillisin alt to Cyntec</td></tr><tr><td>353S01615</td><td>353S4160</td><td></td><td>ALL</td><td>Aizu alt to Maine</td></tr><tr><td>371S00180</td><td>371S00077</td><td></td><td>ALL</td><td>NXP alt to Diodes</td></tr><tr><td>376S1053</td><td>376S0604</td><td></td><td>ALL</td><td>Diodes alt to Fairchild</td></tr><tr><td>376S1106</td><td>376S0678</td><td></td><td>ALL</td><td>Fairchild alt to Vishay</td></tr><tr><td>740S00041</td><td>740S0159</td><td></td><td>ALL</td><td></td></tr></table>                                                          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REF DES    | COMMENTS: | 107S00033 | 107S00034 |          | ALL |                    | 138S0738  | 138S1101  |           | ALL | Samsung alt to Murata | 138S0846            | 138S0811  |           | ALL | Samsung alt to Murata | 152S00359           | 152S00253 |           | ALL        | Chillisin alt to Cyntec | 353S01615    | 353S4160  |           | ALL        | Aizu alt to Maine | 371S00180       | 371S00077 |           | ALL        | NXP alt to Diodes | 376S1053        | 376S0604  |           | ALL        | Diodes alt to Fairchild | 376S1106      | 376S0678  |           | ALL | Fairchild alt to Vishay | 740S00041     | 740S0159  |           | ALL |               | WIRELESS      |           |           |            |     |                 |           |          |     |     |                 |           |           |     |                |           |           |           |     |     |           |           |           |     |                       |           |           |          |     |     |                   |          |          |     |                    |                   |          |          |     |     |               |          |          |     |     |             |           |          |     |     |           |           |          |     |     |           |           |          |     |     |           |           |          |     |     |                 |           |          |     |     |           |           |           |     |     |               |           |           |     |     |                     |           |           |     |                   |           |           |          |     |     |          |          |           |     |     |          |           |          |     |     |          |           |          |     |              |                 |           |          |     |                  |           |           |          |     |     |           |           |          |     |     |           |           |           |     |    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        |           |           |           |          |     |                    |           |           |           |     |                       |                     |           |           |     |                       |                     |           |           |            |                         |              |           |           |            |                   |                 |           |           |            |                   |                 |           |           |            |                         |               |           |           |     |                         |               |           |           |     |               |               |           |           |            |     |                 |           |          |     |     |                 |           |           |     |                |           |           |           |     |     |           |           |           |     |                       |           |           |          |   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| 107S00033                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 138S0846                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 152S00359                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 353S01615                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 371S00180                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S1053                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 376S1106                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 740S00041                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>138S00024</td><td>138S0986</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00769</td><td>152S00659</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00770</td><td>152S0857</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00067</td><td>155S00401</td><td></td><td>ALL</td><td></td></tr><tr><td>339S00428</td><td>339S00458</td><td></td><td>ALL</td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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REF DES    | COMMENTS: | 138S00024 | 138S0986  |          | ALL |                    | 152S00769 | 152S00659 |           | ALL |                       | 152S00770           | 152S0857  |           | ALL |                       | 155S00067           | 155S00401 |           | ALL        |                         | 339S00428    | 339S00458 |           | ALL        |                   | DC-DC           |           |           |            |                   |                 |           |           |            |                         |               |           |           |     |                         |               |           |           |     |               |               |           |           |            |     |                 |           |          |     |     |                 |           |           |     |                |           |           |           |     |     |           |           |           |     |                       |           |           |          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| 152S00769                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 155S00067                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 339S00428                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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          |           |     |     |           |           |           |     |     |           |           |          |     |     |                        |           |          |     |     |           |           |          |     |     |          |           |           |     |     |           |          |           |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>107S00015</td><td>107S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00087</td><td>107S00029</td><td></td><td>ALL</td><td>TFT w/ Yageo</td></tr><tr><td>107S00071</td><td>107S00053</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00139</td><td>107S0178</td><td></td><td>ALL</td><td></td></tr><tr><td>107S0249</td><td>107S0251</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00009</td><td>128S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00070</td><td>128S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00026</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00031</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00087</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>128S0302</td><td>128S00038</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00039</td><td>128S00038</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00062</td><td>128S00067</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00065</td><td>128S00067</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00069</td><td>128S00067</td><td></td><td>ALL</td><td></td></tr><tr><td>128S0364</td><td>128S0264</td><td></td><td>ALL</td><td>Kemet w/ Panasonic</td></tr><tr><td>128S0578</td><td>128S0351</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00050</td><td>128S0351</td><td></td><td>ALL</td><td></td></tr><tr><td>132S00012</td><td>132S0401</td><td></td><td>ALL</td><td></td></tr><tr><td>132S00064</td><td>132S0409</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00077</td><td>138S00035</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00093</td><td>138S00035</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00111</td><td>138S00036</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00047</td><td>138S00073</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00229</td><td>138S00107</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0786</td><td>138S0705</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>138S00097</td><td>138S0750</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0863</td><td>138S0853</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0775</td><td>138S0860</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0789</td><td>138S0941</td><td></td><td>ALL</td><td>Murata w/ SS</td></tr><tr><td>152S00363</td><td>152S00048</td><td></td><td>ALL</td><td>Cyntec w/ Vishay</td></tr><tr><td>152S00388</td><td>152S00182</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00703</td><td>152S00182</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00383</td><td>152S00198</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00680</td><td>152S00198</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00398</td><td>152S00204</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00841</td><td>152S00238</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00765</td><td>152S00239</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00368</td><td>152S00269</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00724</td><td>152S00311</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00786</td><td>152S00344</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00725</td><td>152S00590</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00726</td><td>152S00592</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00707</td><td>152S00689</td><td></td><td>ALL</td><td></td></tr><tr><td>311S0596</td><td>311S0593</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>311S00192</td><td>311S00191</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00193</td><td>311S00191</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00831</td><td>353S00519</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00796</td><td>353S00497</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00832</td><td>353S00525</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00303</td><td>376S00012</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00204</td><td>376S00203</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00226</td><td>376S00203</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00227</td><td>376S00203</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00302</td><td>376S1038</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00281</td><td>376S1147</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00007</td><td>376S1179</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00228</td><td>376S1179</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00304</td><td>376S1187</td><td></td><td>ALL</td><td></td></tr><tr><td>740S0144</td><td>740S0118</td><td></td><td>ALL</td><td></td></tr><tr><td>740S00028</td><td>740S0118</td><td></td><td>ALL</td><td></td></tr></table> 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                                          |                           |            |                         |                     |   |  |   |  |   |  |   |  |   | PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION                | REF DES    | COMMENTS: | 107S00015 | 107S00011 |          | ALL |                    | 107S00087 | 107S00029 |           | ALL | TFT w/ Yageo          | 107S00071           | 107S00053 |           | ALL |                       | 107S00139           | 107S0178  |           | ALL        |                         | 107S0249     | 107S0251  |           | ALL        |                   | 128S00009       | 128S00007 |           | ALL        |                   | 128S00070       | 128S00007 |           | ALL        |                         | 128S00026     | 128S00011 |           | ALL |                         | 128S00031     | 128S00011 |           | ALL |               | 128S00087     | 128S00011 |           | ALL        |     | 128S0302        | 128S00038 |          | ALL |     | 128S00039       | 128S00038 |           | ALL |                | 128S00062 | 128S00067 |           | ALL |     | 128S00065 | 128S00067 |           | ALL |                       | 128S00069 | 128S00067 |          | ALL |     | 128S0364          | 128S0264 |          | ALL | Kemet w/ Panasonic | 128S0578          | 128S0351 |          | ALL |     | 128S00050     | 128S0351 |          | ALL |     | 132S00012   | 132S0401  |          | ALL |     | 132S00064 | 132S0409  |          | ALL |     | 138S00077 | 138S00035 |          | ALL |     | 138S00093 | 138S00035 |          | ALL |     | 138S00111       | 138S00036 |          | ALL |     | 138S00047 | 138S00073 |           | ALL |     | 138S00229     | 138S00107 |           | ALL |     | 138S0786            | 138S0705  |           | ALL | Murata w/ Samsung | 138S00097 | 138S0750  |          | ALL |     | 138S0863 | 138S0853 |           | ALL |     | 138S0775 | 138S0860  |          | ALL |     | 138S0789 | 138S0941  |          | ALL | Murata w/ SS | 152S00363       | 152S00048 |          | ALL | Cyntec w/ Vishay | 152S00388 | 152S00182 |          | ALL |     | 152S00703 | 152S00182 |          | ALL |     | 152S00383 | 152S00198 |           | ALL |     | 152S00680    | 152S00198 |           | ALL |     | 152S00398      | 152S00204 |           | ALL |     | 152S00841     | 152S00238 |           | ALL |     | 152S00765      | 152S00239 |           | ALL |     | 152S00368  | 152S00269 |           | ALL |     | 152S00724     | 152S00311 |           | ALL |     | 152S00786 | 152S00344 |           | ALL |     | 152S00725 | 152S00590 |           | ALL |     | 152S00726 | 152S00592 |           | ALL |     | 152S00707 | 152S00689 |           | ALL |     | 311S0596 | 311S0593  |          | ALL | NXP w/ Diodes | 311S00192 | 311S00191 |          | ALL |     | 311S00193 | 311S00191 |          | ALL |     | 353S00831 | 353S00519 |          | ALL |     | 353S00796     | 353S00497 |           | ALL |     | 353S00832 | 353S00525 |          | ALL |     | 376S00303 | 376S00012 |           | ALL |     | 376S00204 | 376S00203 |           | ALL |     | 376S00226 | 376S00203 |           | ALL |     | 376S00227 | 376S00203 |           | ALL |     | 376S00302 | 376S1038  |           | ALL |     | 376S00281 | 376S1147  |           | ALL |     | 376S00007 | 376S1179  |          | ALL |     | 376S00228              | 376S1179  |          | ALL |     | 376S00304 | 376S1187  |          | ALL |     | 740S0144 | 740S0118  |           | ALL |     | 740S00028 | 740S0118 |           | ALL |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PART NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | BOM OPTION                | REF DES    | COMMENTS:               |                     |   |  |   |  |   |  |   |  |   |             |                           |                           |            |           |           |           |          |     |                    |           |           |           |     |                       |                     |           |           |     |                       |                     |           |           |            |                         |              |           |           |            |                   |                 |           |           |            |                   |                 |           |           |            |                         |               |           |           |     |                         |               |           |           |     |               |               |           |           |            |     |                 |           |          |     |     |                 |           |           |     |                |           |           |           |     |     |           |           |           |     |                       |           |           |          |   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          |           |     |     |           |           |           |     |     |           |           |          |     |     |                        |           |          |     |     |           |           |          |     |     |          |           |           |     |     |           |          |           |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 107S00015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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  |     |                   |          |          |     |                    |                   |          |          |     |     |               |          |          |     |     |             |           |          |     |     |           |           |          |     |     |           |           |          |     |     |           |           |          |     |     |                 |           |          |     |     |           |           |           |     |     |               |           |           |     |     |                     |           |           |     |                   |           |           |          |     |     |          |          |           |     |     |          |           |          |     |     |          |           |          |     |              |                 |           |          |     |                  |           |           |          |     |     |           |           |          |     |     |           |           |           |     |     |  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          |           |     |     |           |           |           |     |     |           |           |          |     |     |                        |           |          |     |     |           |           |          |     |     |          |           |           |     |     |           |          |           |     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 107S00087                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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        |           |           |           |          |     |                    |           |           |           |     |                       |                     |           |           |     |                       |                     |           |           |            |                         |              |           |           |            |                   |                 |           |           |            |                   |                 |           |           |            |                         |               |           |           |     |                         |               |           |           |     |               |               |           |           |            |     |                 |           |          |     |     |                 |           |           |     |                |           |           |           |     |     |           |           |           |     |                       |           |           |          |   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| 107S00071                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 107S00139                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 107S0249                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 128S00026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 128S00031                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 128S00087                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 128S0302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 128S00039                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 128S00062                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 128S00069                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 128S0364                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 128S0578                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 138S00047                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00363                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00388                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00703                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00383                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00680                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00398                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00841                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00765                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00368                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00724                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00786                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00725                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00726                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 152S00707                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 311S0596                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| 311S00192                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 311S00193                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 353S00831                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 353S00832                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00204                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00226                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00227                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00228                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 376S00304                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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| D | <div>Confluence:<br/>https://confluence.apple.com/confluence/display/J79EE/Mac+HW+EE+Home</div> <div>Kismet:<br/>AFP://KISMET.APPLE.COM/KISMET-PROJECTS/J79</div> <div>Useful Wiki Links:<br/>Schematic Conventions - https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions<br/>Schematic Design Wiki - https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design</div> <div>J132 HW Radar:<br/>&lt;rdar://component/XXXXXX&gt; J132 HW EE   Proto 0<br/>&lt;rdar://component/XXXXXX&gt; J132 HW EE BOM   Proto 0<br/>&lt;rdar://component/XXXXXX&gt; J132 HW EE Characterization   Proto 0<br/>&lt;rdar://component/XXXXXX&gt; J132 HW EE Layout   Proto 0<br/>&lt;rdar://component/XXXXXX&gt; J132 HW EE Schematic   Proto 0<br/>&lt;rdar://component/XXXXXX&gt; J132 HW EE SI   Proto 0</div> <div>Other Info:<br/>Page Allocations - &lt;rdar://problem/19817053&gt; 2015 Schematic Page Allocations</div> |   |   |   |   |   |   |   | D |
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